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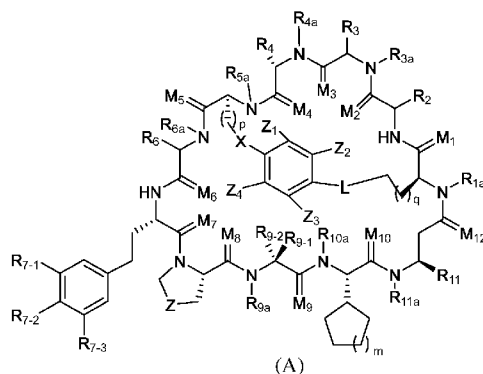
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## (54) Title: CYCLIC PEPTIDE COMPOUNDS AND COMPOSITIONS AS RAS INHIBITORS



## Cyclic Peptide Compounds and Compositions as Ras Inhibitors

### TECHNICAL FIELD

This disclosure belongs to the technical field of Medicinal Chemistry, and more particularly, this disclosure provides cyclic peptide compounds as inhibitors of Ras kinase, and compositions and use of the same in treating diseases associated with Ras kinase. The disclosure also provides degraders of Ras kinase comprising inhibitors of Ras kinase.

### BACKGROUND

Ras is a protein belonging to the small GTPase family, known as Kras, Nras, and Hras. According to its binding with GDP or GTP, Ras can be in either an inactivated or activated state. Activated Ras induces cell proliferation, survival, and differentiation by activating various downstream signals, such as the MAPK pathway, PI3K/Akt pathway, and RAL pathway, and constitutive Ras activation contributes to the occurrence and development of cancer. It is known that the Ras-RAF-MEK-ERK pathway is activated in cancer by activation of Ras upstream signals, constitutive Ras activation, and/or activating Ras mutations. These activating Ras mutations are found in many cancer types. G12, G13, and Q61 are known to be hotspots for Ras mutations, and mutations are frequently observed at G12 in Kras and at Q61 in Nras. It is also known that these mutations are associated with patient prognosis.

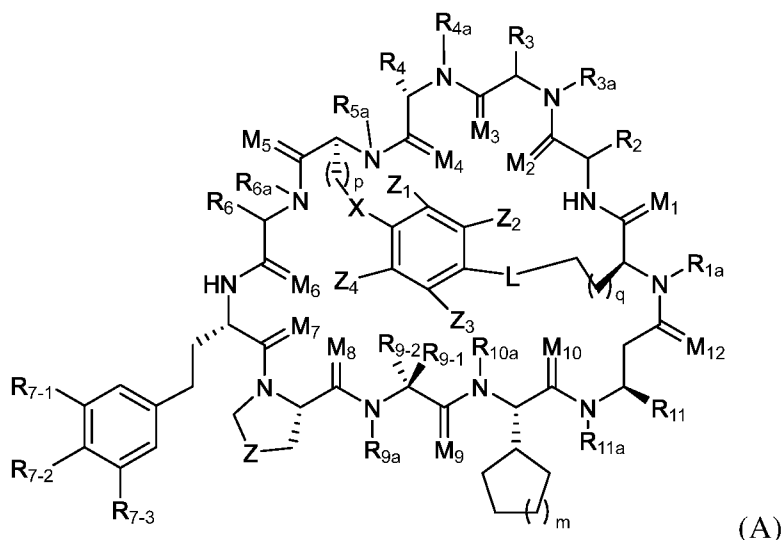
Medium-molecular weight compounds (molecular weight 500-2000), such as cyclic peptide compounds, were developed to inhibit Ras kinase.

It is desirable in the art to develop further cyclic peptide compounds as Ras inhibitors and Ras degraders hoping to increase the potency and/or lower the toxicity.

### SUMMARY

This disclosure provides cyclic peptide compounds as inhibitors of Ras kinase, and compositions and use of the same in treating diseases associated with Ras kinase. The disclosure also provides degraders of Ras kinase comprising inhibitors of Ras kinase.

In one aspect, the present disclosure provides a compound of Formula (A):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

R<sub>1a</sub>, R<sub>9a</sub>, R<sub>10a</sub>, and R<sub>11a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3a</sub>, R<sub>4a</sub>, R<sub>5a</sub>, R<sub>6a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>2-6</sub> alkenyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>2</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>3-8</sub> cycloalkyl, or 4- to 8-membered heterocyclyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

Z is O, S, or C(R<sub>8-1</sub>)(R<sub>8-2</sub>);

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

$R_{11}$  is selected from H,  $C_{1-4}$  alkyl,  $C_{1-4}$  haloalkyl, or  $C(=M_{11})N(R_{12a})(R_{12b})$ ; or  $R_{11}$  and  $R_{11a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

$R_{12a}$ , and  $R_{12b}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl; or  $R_{12a}$  and  $R_{12b}$  together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $C_{1-6}$  alkoxy,  $C_{1-6}$  haloalkoxy,  $C_{3-8}$  cycloalkyl, or 4- to 8-membered heterocyclyl;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or  $-NR_NC(=Y)-$ ;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or  $-C(=Y)NR_N-$ ;

wherein Y is O or S,  $R_N$  is selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

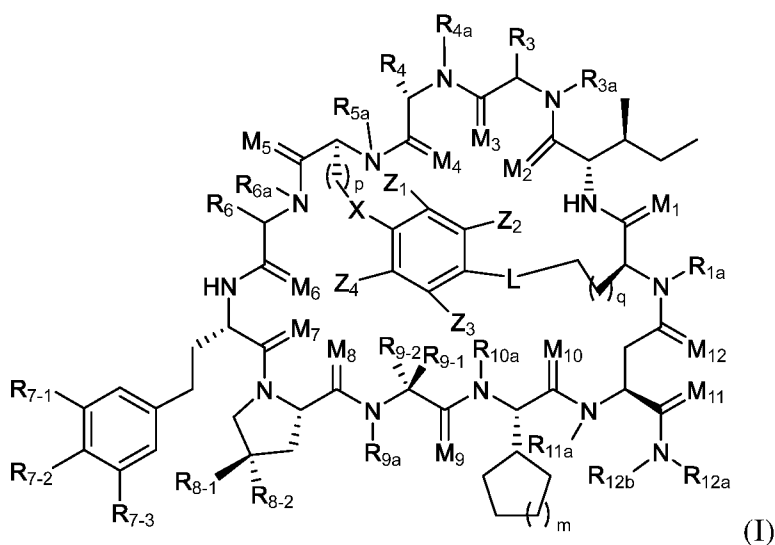
$Z_1$ ,  $Z_2$ ,  $Z_3$ , and  $Z_4$  are independently selected from H, halo,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

$p = 1, 2$ , or  $3$ ;

$q = 0, 1, 2$ , or  $3$ ;

$M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are independently selected from O or S.

In another aspect, the present disclosure provides a compound of Formula (I):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

$R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are independently selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

$R_{3a}$ ,  $R_{4a}$ ,  $R_{5a}$ ,  $R_{6a}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl;

$R_{12a}$ , and  $R_{12b}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl,

or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>12a</sub> and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or -NR<sub>N</sub>C(=Y)-;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or -C(=Y)NR<sub>N</sub>-;

wherein Y is O or S, R<sub>N</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

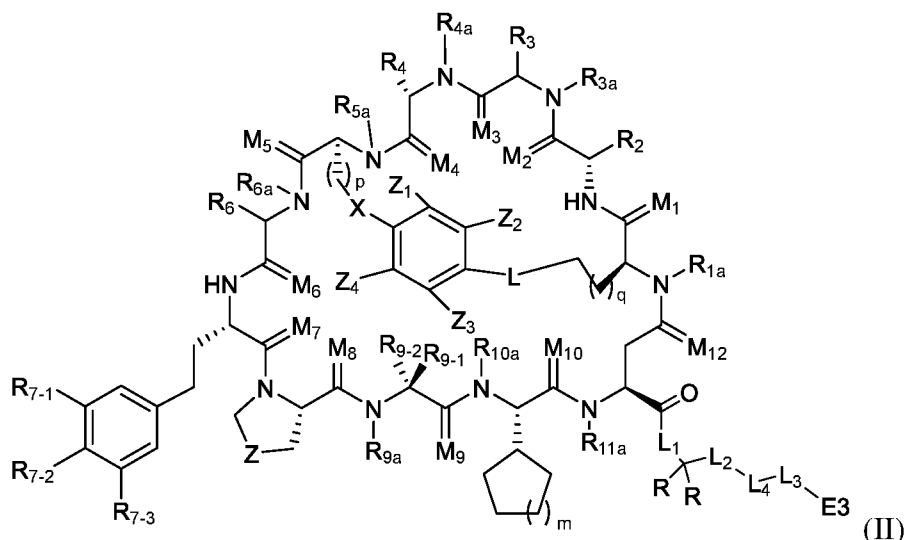
Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

p = 1, 2, or 3;

q = 0, 1, 2, or 3;

M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are independently selected from O or S.

In another aspect, the present disclosure provides a compound of Formula (II):

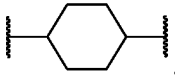
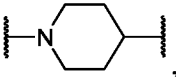
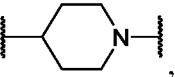
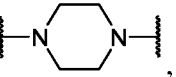
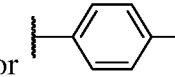


wherein,

$L_1$  is a divalent group selected from -amine group-, -diamine group-, -AA-amine group-, or -AA-diamine group-, wherein the amine group or diamine group has one amino linked to the carbonyl group on the cyclic peptide, and AA is an amino acid having the formula of - $N(R_{13a})CH(R_{13})C(=Q)-$ , wherein Q is O, or S,  $R_{13a}$  is selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl, and  $R_{13}$  is the side chain of an amino acid optionally in combination with  $R_{13a}$ , or  $R_{13}$  and  $R_{13a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

$L_2$  is  $-C_{0-8}$  alkylene-;

$L_3$  is  $-C_{0-8}$  alkylene-;

$L_4$  is selected from bond,  $-CH_2CH_2-$ ,  $-OCH_2-$ ,  $-CH_2O-$ ,  $-NHCH_2-$ ,  $-CH_2NH-$ ,  $-CH=CH-$ ,  $-C\equiv C-$ , , , , , or ;

each R is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; or CRR forms  $C=O$  or  $C=S$ ;

$E_3$  is a ligand of  $E_3$  ligase, and

and other variables are as defined in the context.

In another aspect, the present disclosure provides a pharmaceutical composition, comprising the compound, or the pharmaceutical acceptable salt, the stereoisomer, the tautomer, the stable isotopic variant, or the prodrug thereof disclosed herein; pharmaceutically acceptable excipient(s); and optionally, one or more other therapeutic agents.

In another aspect, the present disclosure provides a kit, comprising a first container which contains the compound, or the pharmaceutical acceptable salt, the stereoisomer, the tautomer, the stable isotopic variant, or the prodrug thereof disclosed herein; and optionally, a second container which contains one or more other therapeutic agents; and optionally, a third container which contains pharmaceutically acceptable excipient(s) for diluting or suspending the said compound and/or other therapeutic agent(s).

In another aspect, the present disclosure provides use of a compound, or a pharmaceutical acceptable salt, a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof disclosed herein, in the manufacture of a medicament for treating a disease mediated by Ras kinase.

5 In another aspect, the present disclosure provides the compound, or a pharmaceutical acceptable salt, a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof disclosed herein, for use in treating a disease mediated by Ras kinase.

10 In another aspect, the present disclosure provides a method of treating a disease mediated by Ras kinase in a subject in need thereof, comprising administering to the subject a compound, or a pharmaceutical acceptable salt, a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof disclosed herein.

## Definitions

### Chemical Definitions

15 Definitions of specific functional groups and chemical terms are described in more detail hereafter.

When a range of values is listed, each value and sub-range within the range are intended to be included. For example, “C<sub>1-6</sub> alkyl” is intended to include C<sub>1</sub>, C<sub>2</sub>, C<sub>3</sub>, C<sub>4</sub>, C<sub>5</sub>, C<sub>6</sub>, C<sub>1-6</sub>, C<sub>1-5</sub>, C<sub>1-4</sub>, C<sub>1-3</sub>, C<sub>1-2</sub>, C<sub>2-6</sub>, C<sub>2-5</sub>, C<sub>2-4</sub>, C<sub>2-3</sub>, C<sub>3-6</sub>, C<sub>3-5</sub>, C<sub>3-4</sub>, C<sub>4-6</sub>, C<sub>4-5</sub> and C<sub>5-6</sub> alkyl.

20 It should be understood that when described herein any of the moieties defined forth below may be substituted by a variety of substituents, and that the respective definitions are intended to include such substituted moieties within their scope as set out below. Unless otherwise stated, the term “substituted” is to be defined as set out below.

25 “C<sub>1-6</sub> alkyl” refers to a radical of a straight or branched, saturated hydrocarbon group having 1 to 6 carbon atoms. In some embodiments, C<sub>1-4</sub> alkyl is preferred. Examples of C<sub>1-6</sub> alkyl include methyl (C<sub>1</sub>), ethyl (C<sub>2</sub>), n-propyl (C<sub>3</sub>), iso-propyl (C<sub>3</sub>), n-butyl (C<sub>4</sub>), tert-butyl (C<sub>4</sub>), sec-butyl (C<sub>4</sub>), iso-butyl (C<sub>4</sub>), n-pentyl (C<sub>5</sub>), 3-pentyl (C<sub>5</sub>), pentyl (C<sub>5</sub>), neopentyl (C<sub>5</sub>), 3-methyl-2-butyl (C<sub>5</sub>), tert-pentyl (C<sub>5</sub>) and n-hexyl (C<sub>6</sub>). The term “C<sub>1-6</sub> alkyl” also includes heteroalkyl, wherein one or more (e.g., 1, 2, 3 or 4) carbon atoms are substituted with heteroatoms (e.g., 30 oxygen, sulfur, nitrogen, boron, silicon, phosphorus). Alkyl groups can be optionally substituted with one or more substituents, for example, with 1 to 5 substituents, 1 to 3 substituents or 1 substituent. Conventional abbreviations of alkyl include Me (-CH<sub>3</sub>), Et (-CH<sub>2</sub>CH<sub>3</sub>), iPr (-CH(CH<sub>3</sub>)<sub>2</sub>), nPr (-CH<sub>2</sub>CH<sub>2</sub>CH<sub>3</sub>), n-Bu (-CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>3</sub>) or i-Bu (-CH<sub>2</sub>CH(CH<sub>3</sub>)<sub>2</sub>).

35 “-C<sub>1-6</sub> alkylene-” refers to a divalent group of the “C<sub>1-6</sub> alkyl” as defined above. “-C<sub>0-6</sub> alkylene-” refers to a bond and a divalent group of the “C<sub>1-6</sub> alkyl” as defined above.

“C<sub>1-6</sub> alkylene” refers to a divalent group formed by removing another hydrogen of the C<sub>1-6</sub> alkyl, and can be a substituted or unsubstituted alkylene. In some embodiments, C<sub>1-4</sub> alkylene is particularly preferred. The unsubstituted alkylene groups include, but are not limited to, methylene (-CH<sub>2</sub>-), ethylene (-CH<sub>2</sub>CH<sub>2</sub>-), propylene (-CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>-), butylene (-

CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>-), pentylene (-CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>-), hexylene (-CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>-), etc. Examples of substituted alkylene groups, such as those substituted with one or more alkyl (methyl) groups, include, but are not limited to, substituted methylene (-CH(CH<sub>3</sub>)-, -C(CH<sub>3</sub>)<sub>2</sub>-), substituted ethylene (-CH(CH<sub>3</sub>)CH<sub>2</sub>-, -CH<sub>2</sub>CH(CH<sub>3</sub>)-, -C(CH<sub>3</sub>)<sub>2</sub>CH<sub>2</sub>-, -CH<sub>2</sub>C(CH<sub>3</sub>)<sub>2</sub>-), substituted propylene (-CH(CH<sub>3</sub>)CH<sub>2</sub>CH<sub>2</sub>-, -CH<sub>2</sub>CH(CH<sub>3</sub>)CH<sub>2</sub>-, -CH<sub>2</sub>CH<sub>2</sub>CH(CH<sub>3</sub>)-, -C(CH<sub>3</sub>)<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>-, -CH<sub>2</sub>C(CH<sub>3</sub>)<sub>2</sub>CH<sub>2</sub>-, -CH<sub>2</sub>CH<sub>2</sub>C(CH<sub>3</sub>)<sub>2</sub>-), etc.

“Halo” or “halogen” refers to fluorine (F), chlorine (Cl), bromine (Br) and iodine (I).

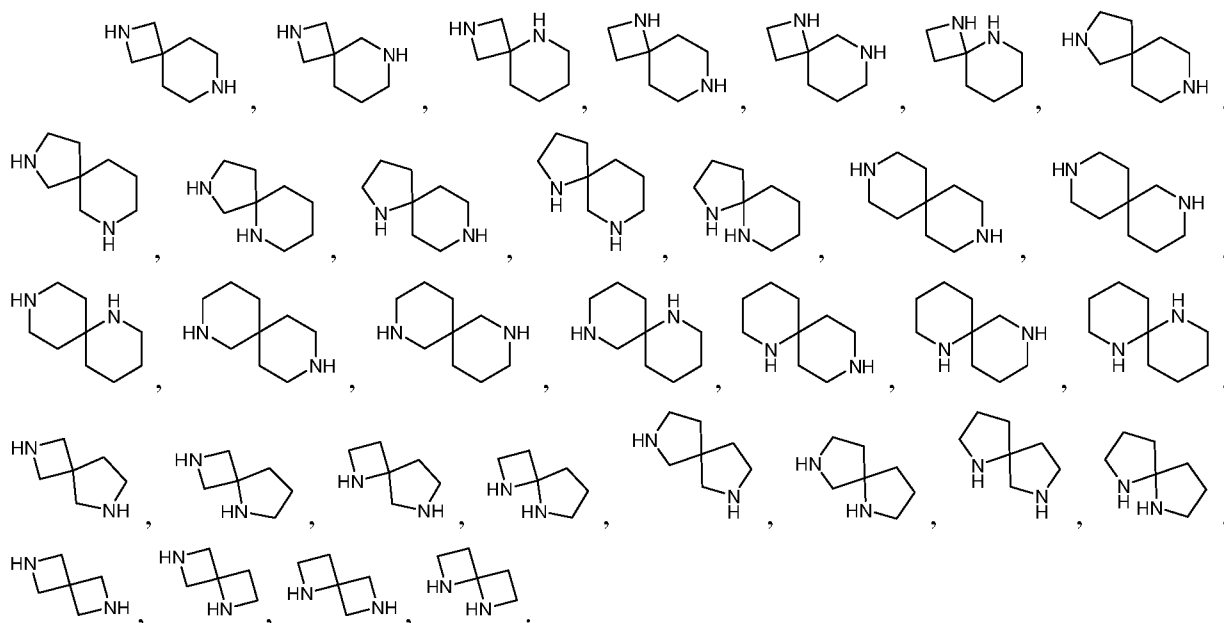
“C<sub>1-6</sub> haloalkyl” represents the “C<sub>1-6</sub> alkyl” described above, which is substituted with one or more halogen groups. Examples include the mono-, di-, poly-halogenated, including perhalogenated, alkyl. A monohalogen substituent may have one iodine, bromine, chlorine or fluorine atom in the group; a dihalogen substituent and a polyhalogen substituent may have two or more identical halogen atoms or a combination of different halogens. Examples of preferred haloalkyl groups include monofluoromethyl, difluoromethyl, trifluoromethyl, chloromethyl, dichloromethyl, trichloromethyl, pentafluoroethyl, heptafluoropropyl, difluorochloromethyl, dichlorofluoromethyl, difluoroethyl, difluoropropyl, dichloroethyl and dichloropropyl. The haloalkyl groups can be substituted at any available point of attachment, for example, with 1 to 5 substituents, 1 to 3 substituents or 1 substituent.

“C<sub>3-8</sub> cycloalkyl” refers to a radical of non-aromatic cyclic hydrocarbon group having 3 to 8 ring carbon atoms and zero heteroatoms. In some embodiments, C<sub>3-6</sub> cycloalkyl is particularly preferred, C<sub>4-6</sub> cycloalkyl is more preferred, and C<sub>5-6</sub> cycloalkyl is more preferred. The cycloalkyl also includes a ring system in which the cycloalkyl described herein is fused with one or more aryl or heteroaryl groups, wherein the point of attachment is on the cycloalkyl ring, and in such case, the number of carbon atoms continues to represent the number of carbon atoms in the cycloalkyl system. Exemplary cycloalkyl groups include, but are not limited to, cyclopropyl (C<sub>3</sub>), cyclopropenyl (C<sub>3</sub>), cyclobutyl (C<sub>4</sub>), cyclobutenyl (C<sub>4</sub>), cyclopentyl (C<sub>5</sub>), cyclopentenyl (C<sub>5</sub>), cyclohexyl (C<sub>6</sub>), cyclohexenyl (C<sub>6</sub>), cyclohexadienyl (C<sub>6</sub>), cycloheptyl (C<sub>7</sub>), cycloheptenyl (C<sub>7</sub>), cycloheptadienyl (C<sub>7</sub>), cycloheptatrienyl (C<sub>7</sub>), etc.

“3- to 11-membered heterocyclyl” refers to a radical of 3- to 11-membered non-aromatic ring system having ring carbon atoms and 1 to 5 ring heteroatoms, wherein each of the heteroatoms is independently selected from nitrogen, oxygen, sulfur, boron, phosphorus and silicon. In the heterocyclyl containing one or more nitrogen atoms, the point of attachment can be a carbon or nitrogen atom as long as the valence permits. In some embodiments, 4- to 8-membered heterocyclyl is preferred, which is a radical of a 4- to 8-membered non-aromatic ring system having ring carbon atoms and 1 to 5 ring heteroatoms. In some embodiments, 4- to 7-membered heterocyclyl is preferred, which is a radical of a 4- to 7-membered non-aromatic ring system having ring carbon atoms and 1 to 4 ring heteroatoms. 3- to 6-membered heterocyclyl is preferred, which is a radical of a 3- to 6-membered non-aromatic ring system having ring carbon atoms and 1 to 3 ring heteroatoms. 4- to 7-membered heterocyclyl is preferred, which is a radical of a 4- to 7-membered non-aromatic ring system having ring carbon atoms and 1 to 3 ring

heteroatoms. 4- to 6-membered heterocyclyl is preferred, which is a radical of a 4- to 6-membered non-aromatic ring system having ring carbon atoms and 1 to 3 ring heteroatoms. 5- to 6-membered heterocyclyl is more preferred, which is a radical of a 5- to 6-membered non-aromatic ring system having ring carbon atoms and 1 to 3 ring heteroatoms. The heterocyclyl also includes a ring system wherein the heterocyclyl described above is fused with one or more cycloalkyl groups, wherein the point of attachment is on the cycloalkyl ring, or the heterocyclyl described above is fused with one or more aryl or heteroaryl groups, wherein the point of attachment is on the heterocyclyl ring; and in such cases, the number of ring members continues to represent the number of ring members in the heterocyclyl ring system. Exemplary 3-membered heterocyclyl groups containing one heteroatom include, but are not limited to, aziridinyl, oxiranyl and thioenyl. Exemplary 4-membered heterocyclyl groups containing one heteroatom include, but are not limited to, azetidiny, oxetanyl and thietanyl. Exemplary 5-membered heterocyclyl groups containing one heteroatom include, but are not limited to, tetrahydrofuranyl, dihydrofuranyl, tetrahydrothiophenyl, dihydrothienyl, pyrrolidinyl, dihydropyrrolyl and pyrrolyl-2, 5-dione. Exemplary 5-membered heterocyclyl groups containing two heteroatoms include, but are not limited to, dioxolanyl, oxasulfuranyl, disulfuranyl, and oxazolidin-2-one. Exemplary 5-membered heterocyclyl groups containing three heteroatoms include, but are not limited to, triazoliny, oxadiazoliny, and thiadiazoliny. Exemplary 6-membered heterocyclyl groups containing one heteroatom include, but are not limited to, piperidyl, tetrahydropyranyl, dihydropyridyl and thianyl. Exemplary 6-membered heterocyclyl groups containing two heteroatoms include, but are not limited to, piperaziny, morpholiny, dithianyl and dioxanyl. Exemplary 6-membered heterocyclyl groups containing three heteroatoms include, but are not limited to, triazinanyl. Exemplary 7-membered heterocyclyl groups containing one heteroatom include, but are not limited to, azepanyl, oxepanyl and thiepanyl. Exemplary 5-membered heterocyclyl groups fused with a C<sub>6</sub> aryl (also referred to as 5,6-bicyclic heterocyclyl herein) include, but are not limited to, indoliny, isoindoliny, dihydrobenzofuranyl, dihydrobenzothiophenyl, benzoxazolinonyl, etc. Exemplary 6-membered heterocyclyl groups fused with a C<sub>6</sub> aryl (also referred as 6, 6-bicyclic heterocyclyl herein) include, but are not limited to, tetrahydroquinoliny, tetrahydroisoquinoliny, etc.

The 3- to 11-membered heterocyclyl also includes spiroheterocyclyl, that is, a group in which two rings (e.g., a heterocycle and a carbocycle) share a carbon atom, wherein at least one of the rings is a heterocyclyl as defined above. More specifically, the spiroheterocyclyl is a spiro ring formed by two 4-membered rings, two 5-membered rings, two 6-membered rings, one 4-membered ring and one 5-membered ring, one 4-membered ring and one 6-membered ring, or one 5-membered ring and one 6-membered ring, wherein at least one of the rings is a 4- to 6-membered heterocyclyl as defined above. The 4- to 6-membered heterocyclyl containing 1, 2 or 3 O, N or S heteroatoms is preferred. The 4- to 6-membered heterocyclyl containing 1 N heteroatom is more preferred. Specific spiroheterocyclyl groups include, but are not limited to:



“C<sub>6-10</sub> aryl” refers to a radical of monocyclic or polycyclic (e.g., bicyclic) 4n+2 aromatic ring system having 6-10 ring carbon atoms and zero heteroatoms (e.g., having 6 or 10 shared  $\pi$  electrons in a cyclic array). In some embodiments, the aryl group has six ring carbon atoms (“C<sub>6</sub> aryl”; for example, phenyl). In some embodiments, the aryl group has ten ring carbon atoms (“C<sub>10</sub> aryl”; for example, naphthyl, e.g., 1-naphthyl and 2-naphthyl). The aryl group also includes a ring system in which the aryl ring described above is fused with one or more cycloalkyl or heterocyclyl groups, and the point of attachment is on the aryl ring, in which case the number of carbon atoms continues to represent the number of carbon atoms in the aryl ring system.

“5- to 10-membered heteroaryl” refers to a radical of 5- to 10-membered monocyclic or bicyclic 4n+2 aromatic ring system (e.g., having 6 or 10 shared  $\pi$  electrons in a cyclic array) having ring carbon atoms and 1-4 ring heteroatoms, wherein each heteroatom is independently selected from nitrogen, oxygen and sulfur. In the heteroaryl group containing one or more nitrogen atoms, the point of attachment can be a carbon or nitrogen atom as long as the valence permits. Heteroaryl bicyclic systems may include one or more heteroatoms in one or two rings. Heteroaryl also includes ring systems wherein the heteroaryl ring described above is fused with one or more cycloalkyl or heterocyclyl groups, and the point of attachment is on the heteroaryl ring. In such case, the number of the carbon atoms continues to represent the number of carbon atoms in the heteroaryl ring system. In some embodiments, 5- to 6-membered heteroaryl groups are particularly preferred, which are radicals of 5- to 6-membered monocyclic or bicyclic 4n+2 aromatic ring systems having ring carbon atoms and 1-4 ring heteroatoms. Exemplary 5-membered heteroaryl groups containing one heteroatom include, but are not limited to, pyrrolyl, furyl and thienyl. Exemplary 5-membered heteroaryl groups containing two heteroatoms include, but are not limited to, imidazolyl, pyrazolyl, oxazolyl, isoxazolyl, thiazolyl, and isothiazolyl. Exemplary 5-membered heteroaryl groups containing three heteroatoms include, but are not limited to, triazolyl, oxadiazolyl (such as, 1, 2, 4- oxadiazolyl), and thiadiazolyl. Exemplary 5-membered heteroaryl groups containing four heteroatoms include, but are not limited to,

tetrazolyl. Exemplary 6-membered heteroaryl groups containing one heteroatom include, but are not limited to, pyridyl. Exemplary 6-membered heteroaryl groups containing two heteroatoms include, but are not limited to, pyridazinyl, pyrimidinyl, and pyrazinyl. Exemplary 6-membered heteroaryl groups containing three or four heteroatoms include, but are not limited to, triazinyl and tetrazinyl, respectively. Exemplary 7-membered heteroaryl groups containing one heteroatom include, but are not limited to, azepinyl, oxepinyl, and thiepinyl. Exemplary 5,6-bicyclic heteroaryl groups include, but are not limited to, indolyl, isoindolyl, indazolyl, benzotriazolyl, benzothiophenyl, isobenzothiophenyl, benzofuranyl, benzoisofuranyl, benzimidazolyl, benzoxazolyl, benzoisoxazolyl, benzoxadiazolyl, benzothiazolyl, benzoisothiazolyl, benzothiadiazolyl, indoliziny and purinyl. Exemplary 6, 6-bicyclic heteroaryl groups include, but are not limited to, naphthyridinyl, pteridinyl, quinolyl, isoquinolyl, cinnolinyl, quinoxaliny, phthalazinyl and quinazolinyl.

“Carbonyl”, whether used alone or in conjunction with other terms (e.g., aminocarbonyl), is represented by -C(O)-. “Thiocarbonyl”, whether used alone or in conjunction with other terms, is represented by -C(S)-.

“C<sub>1-6</sub> alkoxy” and “C<sub>1-6</sub> haloalkoxy” refers to -O-C<sub>1-6</sub> alkyl and -O-C<sub>1-6</sub> haloalkyl, respectively, and the C<sub>1-6</sub> alkyl and C<sub>1-6</sub> haloalkyl are defined above.

Alkyl, carbocyclyl, heterocyclyl, aryl, and heteroaryl groups, as defined herein, are optionally substituted groups. In general, the term “substituted”, whether preceded by the term “optionally” or not, means that at least one hydrogen present on a group (e.g., a carbon or nitrogen atom) is replaced with a permissible substituent, e.g., a substituent which upon substitution results in a stable compound, e.g., a compound which does not spontaneously undergo transformation such as by rearrangement, cyclization, elimination, or other reaction. Unless otherwise indicated, a “substituted” group has a substituent at one or more substitutable positions of the group, and when more than one position in any given structure is substituted, the substituent is either the same or different at each position. The term “substituted” is contemplated to include substitution with all permissible substituents of organic compounds, any of the substituents described herein that results in the formation of a stable compound. For purposes of this disclosure, heteroatoms such as nitrogen may have hydrogen substituents and/or any suitable substituent as described herein which satisfy the valencies of the heteroatoms and results in the formation of a stable moiety.

Exemplary substituents on carbon atoms include, but are not limited to, halogen, -CN, -NO<sub>2</sub>, -N<sub>3</sub>, -SO<sub>2</sub>H, -SO<sub>3</sub>H, -OH, -OR<sup>aa</sup>, -ON(R<sup>bb</sup>)<sub>2</sub>, -N(R<sup>bb</sup>)<sub>2</sub>, -N(R<sup>bb</sup>)<sub>3</sub><sup>+</sup>X<sup>-</sup>, -N(OR<sup>cc</sup>)R<sup>bb</sup>, -SH, -SR<sup>aa</sup>, -SSR<sup>cc</sup>, -C(=O)R<sup>aa</sup>, -CO<sub>2</sub>H, -CHO, -C(OR<sup>cc</sup>)<sub>2</sub>, -CO<sub>2</sub>R<sup>aa</sup>, -OC(=O)R<sup>aa</sup>, -OCO<sub>2</sub>R<sup>aa</sup>, -C(=O)N(R<sup>bb</sup>)<sub>2</sub>, -OC(=O)N(R<sup>bb</sup>)<sub>2</sub>, -NR<sup>bb</sup>C(=O)R<sup>aa</sup>, -NR<sup>bb</sup>CO<sub>2</sub>R<sup>aa</sup>, -NR<sup>bb</sup>C(=O)N(R<sup>bb</sup>)<sub>2</sub>, -C(=NR<sup>bb</sup>)R<sup>aa</sup>, -C(=NR<sup>bb</sup>)OR<sup>aa</sup>, -OC(=NR<sup>bb</sup>)R<sup>aa</sup>, -OC(=NR<sup>bb</sup>)OR<sup>aa</sup>, -C(=NR<sup>bb</sup>)N(R<sup>bb</sup>)<sub>2</sub>, -OC(=NR<sup>bb</sup>)N(R<sup>bb</sup>)<sub>2</sub>, -NR<sup>bb</sup>C(=NR<sup>bb</sup>)N(R<sup>bb</sup>)<sub>2</sub>, -C(=O)NR<sup>bb</sup>SO<sub>2</sub>R<sup>aa</sup>, -NR<sup>bb</sup>SO<sub>2</sub>R<sup>aa</sup>, -SO<sub>2</sub>N(R<sup>bb</sup>)<sub>2</sub>, -SO<sub>2</sub>R<sup>aa</sup>, -SO<sub>2</sub>OR<sup>aa</sup>, -OSO<sub>2</sub>R<sup>aa</sup>, -S(=O)R<sup>aa</sup>, -OS(=O)R<sup>aa</sup>, -Si(R<sup>aa</sup>)<sub>3</sub>, -OSi(R<sup>aa</sup>)<sub>3</sub>, -C(=S)N(R<sup>bb</sup>)<sub>2</sub>, -C(=O)SR<sup>aa</sup>, -C(=S)SR<sup>aa</sup>, -SC(=S)SR<sup>aa</sup>, -SC(=O)SR<sup>aa</sup>, -OC(=O)SR<sup>aa</sup>, -SC(=O)OR<sup>aa</sup>, -SC(=O)R<sup>aa</sup>,

-P(=O)<sub>2</sub>R<sup>aa</sup>, -OP(=O)<sub>2</sub>R<sup>aa</sup>, -P(=O)(R<sup>aa</sup>)<sub>2</sub>, -OP(=O)(R<sup>aa</sup>)<sub>2</sub>, -OP(=O)(OR<sup>cc</sup>)<sub>2</sub>, -P(=O)<sub>2</sub>N(R<sup>bb</sup>)<sub>2</sub>, -OP(=O)<sub>2</sub>N(R<sup>bb</sup>)<sub>2</sub>, -P(=O)(NR<sup>bb</sup>)<sub>2</sub>, -OP(=O)(NR<sup>bb</sup>)<sub>2</sub>, -NR<sup>bb</sup>P(=O)(OR<sup>cc</sup>)<sub>2</sub>, -NR<sup>bb</sup>P(=O)(NR<sup>bb</sup>)<sub>2</sub>, -P(R<sup>cc</sup>)<sub>2</sub>, -P(R<sup>cc</sup>)<sub>3</sub>, -OP(R<sup>cc</sup>)<sub>2</sub>, -OP(R<sup>cc</sup>)<sub>3</sub>, -B(R<sup>aa</sup>)<sub>2</sub>, -B(OR<sup>cc</sup>)<sub>2</sub>, -BR<sup>aa</sup>(OR<sup>cc</sup>), alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>dd</sup> groups;

or two geminal hydrogen on a carbon atom are replaced with =O, =S, =NN(R<sup>bb</sup>)<sub>2</sub>, =NNR<sup>bb</sup>C(=O)R<sup>aa</sup>, =NNR<sup>bb</sup>C(=O)OR<sup>aa</sup>, =NNR<sup>bb</sup>S(=O)<sub>2</sub>R<sup>aa</sup>, =NR<sup>bb</sup> or =NOR<sup>cc</sup> groups;

each of the R<sup>aa</sup> is independently selected from alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl, or two of the R<sup>aa</sup> groups are combined to form a heterocyclyl or heteroaryl ring, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>dd</sup> groups;

each of the R<sup>bb</sup> is independently selected from hydrogen, -OH, -OR<sup>aa</sup>, -N(R<sup>cc</sup>)<sub>2</sub>, -CN, -C(=O)R<sup>aa</sup>, -C(=O)N(R<sup>cc</sup>)<sub>2</sub>, -CO<sub>2</sub>R<sup>aa</sup>, -SO<sub>2</sub>R<sup>aa</sup>, -C(=NR<sup>cc</sup>)OR<sup>aa</sup>, -C(=NR<sup>cc</sup>)N(R<sup>cc</sup>)<sub>2</sub>, -SO<sub>2</sub>N(R<sup>cc</sup>)<sub>2</sub>, -SO<sub>2</sub>R<sup>cc</sup>, -SO<sub>2</sub>OR<sup>cc</sup>, -SOR<sup>aa</sup>, -C(=S)N(R<sup>cc</sup>)<sub>2</sub>, -C(=O)SR<sup>cc</sup>, -C(=S)SR<sup>cc</sup>, -P(=O)<sub>2</sub>R<sup>aa</sup>, -P(=O)(R<sup>aa</sup>)<sub>2</sub>, -P(=O)<sub>2</sub>N(R<sup>cc</sup>)<sub>2</sub>, -P(=O)(NR<sup>cc</sup>)<sub>2</sub>, alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl, or two R<sup>bb</sup> groups are combined to form a heterocyclyl or a heteroaryl ring, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>dd</sup> groups;

each of the R<sup>cc</sup> is independently selected from hydrogen, alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl, or two R<sup>cc</sup> groups are combined to form a heterocyclyl or a heteroaryl ring, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>dd</sup> groups;

each of the R<sup>dd</sup> is independently selected from halogen, -CN, -NO<sub>2</sub>, -N<sub>3</sub>, -SO<sub>2</sub>H, -SO<sub>3</sub>H, -OH, -OR<sup>ee</sup>, -ON(R<sup>ff</sup>)<sub>2</sub>, -N(R<sup>ff</sup>)<sub>2</sub>, -N(R<sup>ff</sup>)<sub>3</sub><sup>+</sup>X<sup>-</sup>, -N(OR<sup>ee</sup>)R<sup>ff</sup>, -SH, -SR<sup>ee</sup>, -SSR<sup>ee</sup>, -C(=O)R<sup>ee</sup>, -CO<sub>2</sub>H, -CO<sub>2</sub>R<sup>ee</sup>, -OC(=O)R<sup>ee</sup>, -OCO<sub>2</sub>R<sup>ee</sup>, -C(=O)N(R<sup>ff</sup>)<sub>2</sub>, -OC(=O)N(R<sup>ff</sup>)<sub>2</sub>, -NR<sup>ff</sup>C(=O)R<sup>ee</sup>, -NR<sup>ff</sup>CO<sub>2</sub>R<sup>ee</sup>, -NR<sup>ff</sup>C(=O)N(R<sup>ff</sup>)<sub>2</sub>, -C(=NR<sup>ff</sup>)OR<sup>ee</sup>, -OC(=NR<sup>ff</sup>)R<sup>ee</sup>, -OC(=NR<sup>ff</sup>)OR<sup>ee</sup>, -C(=NR<sup>ff</sup>)N(R<sup>ff</sup>)<sub>2</sub>, -OC(=NR<sup>ff</sup>)N(R<sup>ff</sup>)<sub>2</sub>, -NR<sup>ff</sup>C(=NR<sup>ff</sup>)N(R<sup>ff</sup>)<sub>2</sub>, -NR<sup>ff</sup>SO<sub>2</sub>R<sup>ee</sup>, -SO<sub>2</sub>N(R<sup>ff</sup>)<sub>2</sub>, -SO<sub>2</sub>R<sup>ee</sup>, -SO<sub>2</sub>OR<sup>ee</sup>, -OSO<sub>2</sub>R<sup>ee</sup>, -S(=O)R<sup>ee</sup>, -Si(R<sup>ee</sup>)<sub>3</sub>, -OSi(R<sup>ee</sup>)<sub>3</sub>, -C(=S)N(R<sup>ff</sup>)<sub>2</sub>, -C(=O)SR<sup>ee</sup>, -C(=S)SR<sup>ee</sup>, -SC(=S)SR<sup>ee</sup>, -P(=O)<sub>2</sub>R<sup>ee</sup>, -P(=O)(R<sup>ee</sup>)<sub>2</sub>, -OP(=O)(R<sup>ee</sup>)<sub>2</sub>, -OP(=O)(OR<sup>ee</sup>)<sub>2</sub>, alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl, heteroaryl, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>gg</sup> groups, or two geminal R<sup>dd</sup> substituents can be combined to form =O or =S;

each of the R<sup>ee</sup> is independently selected from alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, aryl, heterocyclyl, and heteroaryl, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>gg</sup> groups;

each of the R<sup>ff</sup> is independently selected from hydrogen, alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl, or two R<sup>ff</sup> groups are combined to form a

heterocyclyl or a heteroaryl ring, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>gg</sup> groups;

each of the R<sup>gg</sup> is independently selected from halogen, -CN, -NO<sub>2</sub>, -N<sub>3</sub>, -SO<sub>2</sub>H, -SO<sub>3</sub>H, -OH, -OC<sub>1-6</sub> alkyl, -ON(C<sub>1-6</sub> alkyl)<sub>2</sub>, -N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -N(C<sub>1-6</sub> alkyl)<sub>3</sub><sup>+</sup>X<sup>-</sup>, -NH(C<sub>1-6</sub> alkyl)<sub>2</sub><sup>+</sup>X<sup>-</sup>, -NH<sub>2</sub>(C<sub>1-6</sub> alkyl)<sup>+</sup>X<sup>-</sup>, -NH<sub>3</sub><sup>+</sup>X<sup>-</sup>, -N(OC<sub>1-6</sub> alkyl)(C<sub>1-6</sub> alkyl), -N(OH)(C<sub>1-6</sub> alkyl), -NH(OH), -SH, -SC<sub>1-6</sub> alkyl, -SS(C<sub>1-6</sub> alkyl), -C(=O)(C<sub>1-6</sub> alkyl), -CO<sub>2</sub>H, -CO<sub>2</sub>(C<sub>1-6</sub> alkyl), -OC(=O)(C<sub>1-6</sub> alkyl), -OCO<sub>2</sub>(C<sub>1-6</sub> alkyl), -C(=O)NH<sub>2</sub>, -C(=O)N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -OC(=O)NH(C<sub>1-6</sub> alkyl), -NHC(=O)(C<sub>1-6</sub> alkyl), -N(C<sub>1-6</sub> alkyl)C(=O)(C<sub>1-6</sub> alkyl), -NHCO<sub>2</sub>(C<sub>1-6</sub> alkyl), -NHC(=O)N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -NHC(=O)NH(C<sub>1-6</sub> alkyl), -NHC(=O)NH<sub>2</sub>, -C(=NH)O(C<sub>1-6</sub> alkyl), -OC(=NH)(C<sub>1-6</sub> alkyl), -OC(=NH)OC<sub>1-6</sub> alkyl, -C(=NH)N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -C(=NH)NH(C<sub>1-6</sub> alkyl), -C(=NH)NH<sub>2</sub>, -OC(=NH)N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -OC(NH)NH(C<sub>1-6</sub> alkyl), -OC(NH)NH<sub>2</sub>, -NHC(NH)N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -NHC(=NH)NH<sub>2</sub>, -NHSO<sub>2</sub>(C<sub>1-6</sub> alkyl), -SO<sub>2</sub>N(C<sub>1-6</sub> alkyl)<sub>2</sub>, -SO<sub>2</sub>NH(C<sub>1-6</sub> alkyl), -SO<sub>2</sub>NH<sub>2</sub>, -SO<sub>2</sub>C<sub>1-6</sub> alkyl, -SO<sub>2</sub>OC<sub>1-6</sub> alkyl, -OSO<sub>2</sub>C<sub>1-6</sub> alkyl, -SOC<sub>1-6</sub> alkyl, -Si(C<sub>1-6</sub> alkyl)<sub>3</sub>, -OSi(C<sub>1-6</sub> alkyl)<sub>3</sub>, -C(=S)N(C<sub>1-6</sub> alkyl)<sub>2</sub>, C(=S)NH(C<sub>1-6</sub> alkyl), C(=S)NH<sub>2</sub>, -C(=O)S(C<sub>1-6</sub> alkyl), -C(=S)SC<sub>1-6</sub> alkyl, -SC(=S)SC<sub>1-6</sub> alkyl, -P(=O)<sub>2</sub>(C<sub>1-6</sub> alkyl), -P(=O)(C<sub>1-6</sub> alkyl)<sub>2</sub>, -OP(=O)(C<sub>1-6</sub> alkyl)<sub>2</sub>, -OP(=O)(OC<sub>1-6</sub> alkyl)<sub>2</sub>, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>2</sub>-C<sub>6</sub> alkenyl, C<sub>2</sub>-C<sub>6</sub> alkynyl, C<sub>3</sub>-C<sub>7</sub> carbocyclyl, C<sub>6</sub>-C<sub>10</sub> aryl, C<sub>3</sub>-C<sub>7</sub> heterocyclyl, C<sub>5</sub>-C<sub>10</sub> heteroaryl; or two geminal R<sup>gg</sup> substituents may combine to form =O or =S; wherein X<sup>-</sup> is a counter-ion.

Exemplary substituents on nitrogen atoms include, but are not limited to, hydrogen, -OH, -OR<sup>aa</sup>, -N(R<sup>cc</sup>)<sub>2</sub>, -CN, -C(=O)R<sup>aa</sup>, -C(=O)N(R<sup>cc</sup>)<sub>2</sub>, -CO<sub>2</sub>R<sup>aa</sup>, -SO<sub>2</sub>R<sup>aa</sup>, -C(=NR<sup>bb</sup>)R<sup>aa</sup>, -C(=NR<sup>cc</sup>)OR<sup>aa</sup>, -C(=NR<sup>cc</sup>)N(R<sup>cc</sup>)<sub>2</sub>, -SO<sub>2</sub>N(R<sup>cc</sup>)<sub>2</sub>, -SO<sub>2</sub>R<sup>cc</sup>, -SO<sub>2</sub>OR<sup>cc</sup>, -SOR<sup>aa</sup>, -C(=S)N(R<sup>cc</sup>)<sub>2</sub>, -C(=O)SR<sup>cc</sup>, -C(=S)SR<sup>cc</sup>, -P(=O)<sub>2</sub>R<sup>aa</sup>, -P(=O)(R<sup>aa</sup>)<sub>2</sub>, -P(=O)<sub>2</sub>N(R<sup>cc</sup>)<sub>2</sub>, -P(=O)(NR<sup>cc</sup>)<sub>2</sub>, alkyl, haloalkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl, or two R<sup>cc</sup> groups attached to a nitrogen atom combine to form a heterocyclyl or a heteroaryl ring, wherein each of the alkyl, alkenyl, alkynyl, carbocyclyl, heterocyclyl, aryl and heteroaryl is independently substituted with 0, 1, 2, 3, 4 or 5 R<sup>dd</sup> groups, and wherein R<sup>aa</sup>, R<sup>bb</sup>, R<sup>cc</sup> and R<sup>dd</sup> are as described herein.

#### Other Definitions

The term “natural amino acid” is the basic structural unit of protein, which is the basis for later modification of protein by organisms. There are 20 kinds of natural amino acids in total. In addition, on the basis of these basic amino acids, organisms can also synthesize some kinds of derived amino acids such as hydroxyproline, hydroxylysine, and the like. These biosynthetic amino acids are collectively referred to as “natural amino acids”. Natural amino acids are generally L-form. The 20 most common natural amino acids are shown in the Table below:

|     | Abbreviation | Name    | Branch      | Molecular Weight | Isoelectric Point | R group          |
|-----|--------------|---------|-------------|------------------|-------------------|------------------|
| Gly | G            | Glycine | Hydrophilic | 75.07            | 5.97              | -H               |
| Ala | A            | Alanine | Hydrophobic | 89.09            | 6.02              | -CH <sub>3</sub> |

|     |   |               |             |        |       |                                                                |
|-----|---|---------------|-------------|--------|-------|----------------------------------------------------------------|
| Val | V | Valine        | Hydrophobic | 117.15 | 6.48  | -CH-(CH <sub>3</sub> ) <sub>2</sub>                            |
| Leu | L | Leucine       | Hydrophobic | 131.17 | 5.98  | -CH <sub>2</sub> -CH(CH <sub>3</sub> ) <sub>2</sub>            |
| Ile | I | Isoleucine    | Hydrophobic | 131.17 | 6.05  | -CH(CH <sub>3</sub> )-CH <sub>2</sub> -CH <sub>3</sub>         |
| Phe | F | Phenylalanine | Hydrophobic | 165.19 | 5.49  | -CH <sub>2</sub> -C <sub>6</sub> H <sub>5</sub>                |
| Trp | W | Tryptophan    | Hydrophobic | 204.23 | 5.89  | -C <sub>8</sub> NH <sub>6</sub>                                |
| Tyr | Y | Tyrosine      | Hydrophilic | 181.19 | 5.64  | -CH <sub>2</sub> -C <sub>6</sub> H <sub>4</sub> -OH            |
| Asp | D | Aspartic acid | Acidic      | 133.10 | 2.85  | -CH <sub>2</sub> -COOH                                         |
| Asn | N | Asparagine    | Hydrophilic | 132.12 | 5.41  | -CH <sub>2</sub> -CONH <sub>2</sub>                            |
| Glu | E | Glutamic acid | Acidic      | 147.13 | 3.15  | -(CH <sub>2</sub> ) <sub>2</sub> -COOH                         |
| Lys | K | Lysine        | Alkaline    | 146.19 | 9.60  | -(CH <sub>2</sub> ) <sub>4</sub> -NH <sub>2</sub>              |
| Gln | Q | Glutamine     | Hydrophilic | 146.15 | 5.65  | -(CH <sub>2</sub> ) <sub>2</sub> -CONH <sub>2</sub>            |
| Met | M | Methionine    | Hydrophobic | 149.21 | 5.74  | -(CH <sub>2</sub> ) <sub>2</sub> -S-CH <sub>3</sub>            |
| Ser | S | Serine        | Hydrophilic | 105.09 | 5.68  | -CH <sub>2</sub> -OH                                           |
| Thr | T | Threonine     | Hydrophilic | 119.12 | 5.60  | -CH(CH <sub>3</sub> )-OH                                       |
| Cys | C | Cysteine      | Hydrophilic | 121.16 | 5.05  | -CH <sub>2</sub> -SH                                           |
| Pro | P | Proline       | Hydrophobic | 115.13 | 6.30  | -C <sub>3</sub> H <sub>6</sub> - linked to amino               |
| His | H | Histidine     | Alkaline    | 155.16 | 7.60  | -CH <sub>2</sub> -C <sub>3</sub> H <sub>3</sub> N <sub>2</sub> |
| Arg | R | Arginine      | Alkaline    | 174.20 | 10.76 | -(CH <sub>2</sub> ) <sub>3</sub> -NHC(NH)NH <sub>2</sub>       |

The term “amino acid” includes “natural amino acid” as mentioned above and unnatural amino acid, such as D-form amino acids.

The term “treating” as used herein relates to reversing, alleviating or inhibiting the progression or prevention of the disorders or conditions to which the term applies, or of one or more symptoms of such disorders or conditions. The noun “treatment” as used herein relates to the action of treating, which is a verb, and the latter is as just defined.

The term “pharmaceutically acceptable” as used herein refers to the substance, which are suitable for the contact with patients’ tissues within a reliable medical judgment, and do not produce inappropriate toxicity, irritation, allergy, etc. They are commensurate with a reasonable benefit/risk ratio, and are effective for their intended use. The term includes, if possible, the zwitterionic form of the compounds of the disclosure.

The term “salt” refers to a relatively non-toxic addition salt of inorganic and organic acids to the compounds of the present disclosure. These salts can be prepared in situ during the final separation and purification of the compounds, or by isolating salts produced by separately reacting the purified compound in the free base form with a suitable organic or inorganic acid.

The pharmaceutically acceptable base addition salts are formed with metals or amines, such as alkali metal and alkaline earth metal hydroxides or organic amines. Examples of the metals used as cations include sodium, potassium, magnesium, calcium, etc. Examples of suitable amines are N, N’-dibenzylethylenediamine, chloroprocaine, choline, diethanolamine, ethylenediamine, N-methylglucamine and procaine.

The salts can be prepared from the inorganic acids, which include sulfates, pyrosulfates, bisulfates, sulfites, bisulfites, nitrates, phosphates, monohydrogen phosphates, dihydrogen phosphates, metaphosphates, pyrophosphates, chlorides, bromides and iodides. Examples of the acids include hydrochloric acid, nitric acid, sulfuric acid, hydrobromic acid, hydroiodic acid, phosphoric acid, etc. The representative salts include hydrobromide, hydrochloride, sulfate, bisulfate, nitrate, acetate, oxalate, valerate, oleate, palmitate, stearate, laurate, borate, benzoate, lactate, phosphate, tosylate, citrate, maleate, fumarate, succinate, tartrate, naphthalate, methanesulfonate, glucoheptanate, lactobionate, lauryl sulfonate, isethionate, etc. The salts can also be prepared from the organic acids, which include aliphatic monocarboxylic and dicarboxylic acids, phenyl-substituted alkanoic acids, hydroxyalkanoic acids, alkanedioic acid, aromatic acids, aliphatic and aromatic sulfonic acids, etc. The representative salts include acetate, propionate, octanoate, isobutyrate, oxalate, malonate, succinate, suberate, sebacate, fumarate, maleate, mandelate, benzoate, chlorobenzoate, methyl benzoate, dinitrobenzoate, naphthoate, besylate, tosylate, phenylacetate, citrate, lactate, maleate, tartrate, methanesulfonate, etc. The pharmaceutically acceptable salts can include cations based on alkali metals and alkaline earth metals, such as sodium, lithium, potassium, calcium, magnesium, etc., as well as non-toxic ammonium, quaternary ammonium, and amine cations including, but not limited to, ammonium, tetramethylammonium, tetraethylammonium, methylamine, dimethylamine, trimethylamine, triethylamine, ethylamine, etc. Salts of amino acids are also included, such as arginine salts, gluconates, galacturonates, etc. (for example, see Berge S. M. et al., "Pharmaceutical Salts," J. Pharm. Sci., 1977; 66: 1- 19 for reference).

"Subjects" to which administration is contemplated include, but are not limited to, humans (e.g., males or females of any age group, e.g., pediatric subjects (e.g., infants, children, adolescents) or adult subjects (e.g., young adults, middle-aged adults or older adults) and/or non-human animals, such as mammals, e.g., primates (e.g., cynomolgus monkeys, rhesus monkeys), cattle, pigs, horses, sheep, goats, rodents, cats and/or dogs. In some embodiments, the subject is a human. In some embodiments, the subject is a non-human animal. The terms "human", "patient" and "subject" can be used interchangeably herein.

"Disease," "disorder," and "condition" can be used interchangeably herein.

Unless indicated, otherwise the term "treatment" as used herein includes the effect on a subject who is suffering from a particular disease, disorder, or condition, which reduces the severity of the disease, disorder, or condition, or delays or slows the progression of the disease, disorder or condition ("therapeutic treatment"). The term also includes the effect that occurs before the subject begins to suffer from a specific disease, disorder or condition ("prophylactic treatment").

Generally, the "effective amount" of a compound refers to an amount sufficient to elicit a target biological response. As understood by those skilled in the art, the effective amount of the compound of the disclosure can vary depending on the following factors, such as the desired biological endpoint, the pharmacokinetics of the compound, the diseases being treated, the mode

of administration, and the age, health status and symptoms of the subjects. The effective amount includes therapeutically effective amount and prophylactically effective amount.

Unless indicated, otherwise the “therapeutically effective amount” of the compound as used herein is an amount sufficient to provide therapeutic benefits in the course of treating a disease, disorder or condition, or to delay or minimize one or more symptoms associated with the disease, disorder or condition. The therapeutically effective amount of a compound refers to the amount of the therapeutic agent that, when used alone or in combination with other therapies, provides a therapeutic benefit in the treatment of a disease, disorder or condition. The term “therapeutically effective amount” can include an amount that improves the overall treatment, reduces or avoids the symptoms or causes of the disease or condition, or enhances the therapeutic effect of other therapeutic agents.

Unless indicated, otherwise the “prophylactically effective amount” of the compound as used herein is an amount sufficient to prevent a disease, disorder or condition, or an amount sufficient to prevent one or more symptoms associated with a disease, disorder or condition, or an amount sufficient to prevent the recurrence of a disease, disorder or condition. The prophylactically effective amount of a compound refers to the amount of a therapeutic agent that, when used alone or in combination with other agents, provides a prophylactic benefit in the prevention of a disease, disorder or condition. The term “prophylactically effective amount” can include an amount that improves the overall prevention, or an amount that enhances the prophylactic effect of other preventive agents.

“Combination” and related terms refer to the simultaneous or sequential administration of the compounds of the present disclosure and other therapeutic agents. For example, the compounds of the present disclosure can be administered simultaneously or sequentially in separate unit dosage with other therapeutic agents, or simultaneously in a single unit dosage with other therapeutic agents.

The compounds of the present disclosure may include one or more asymmetric centers, and thus may exist in a variety of stereoisomeric forms, for example, enantiomers and/or diastereomers. For example, the compounds of the present disclosure may be in the form of an individual enantiomer, diastereomer or geometric isomer (e.g., cis- and trans-isomers), or may be in the form of a mixture of stereoisomers, including racemic mixture and a mixture enriched in one or more stereoisomers. The isomers can be separated from the mixture by the methods known to those skilled in the art, including chiral high pressure liquid chromatography (HPLC) and the formation and crystallization of chiral salts; or preferred isomers can be prepared by asymmetric synthesis.

It should be understood, unless otherwise specified, that compounds with asymmetric center(s) include all optical isomers and mixtures thereof. Furthermore, unless otherwise specified, all isomer compounds and carbon-carbon double bonds included in the present disclosure may be in the form of Z and E. Compounds which exist in different tautomeric forms,

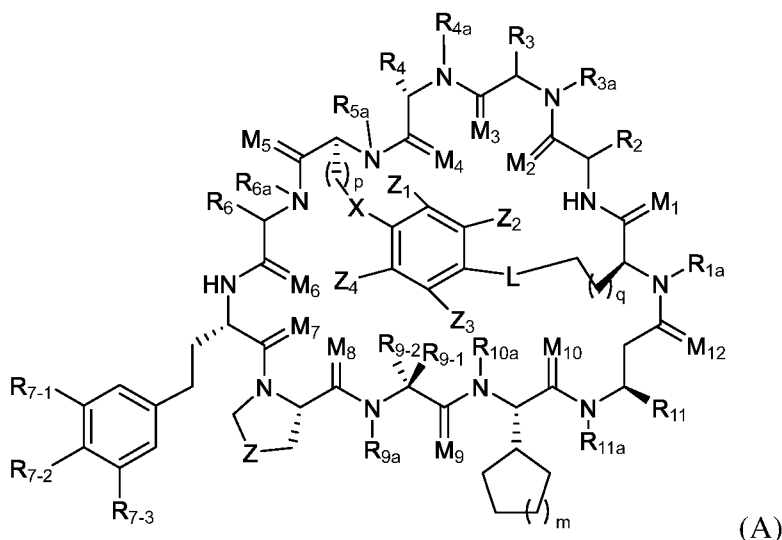
one of which is not limited to any particular tautomer, but is intended to cover all tautomeric forms.

The present disclosure also comprises compounds that are labeled with isotopes (isotopic variants, such as stable isotopic variants), which are equivalent to those described in formula (I), but one or more atoms are replaced by atoms having an atom mass or mass number that are different from that of atoms that are common in nature. Examples of isotopes which may be introduced into the compounds of the disclosure include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur, fluorine and chlorine, such as  $^2\text{H}$ ,  $^3\text{H}$ ,  $^{13}\text{C}$ ,  $^{11}\text{C}$ ,  $^{14}\text{C}$ ,  $^{15}\text{N}$ ,  $^{18}\text{O}$ ,  $^{17}\text{O}$ ,  $^{31}\text{P}$ ,  $^{32}\text{P}$ ,  $^{35}\text{S}$ ,  $^{18}\text{F}$  and  $^{36}\text{Cl}$ , respectively. Compounds of the present disclosure that comprise the above isotopes and/or other isotopes of other atoms, prodrugs thereof and pharmaceutically acceptable salts of said compounds or prodrugs all are within the scope of the present disclosure. Certain isotope-labeled compounds of the present disclosure, such as those incorporating radioactive isotopes (e.g.,  $^3\text{H}$  and  $^{14}\text{C}$ ), can be used for the measurement of the distribution of drug and/or substrate in tissue. Tritium, which is  $^3\text{H}$  and carbon-14, which is  $^{14}\text{C}$  isotope, are particularly preferred, because they are easy to prepare and detect. Furthermore, replaced by heavier isotopes, such as deuterium, which is  $^2\text{H}$ , may provide therapeutic benefits due to the higher metabolic stability, such as prolonging the half-life *in vivo* or decreasing the dosage requirements, and thus may be preferred in some cases. Isotope-labeled compounds of formula (I) of the present disclosure and prodrugs thereof can be prepared generally by using readily available isotope-labeled reagents to replace non-isotope-labeled reagents in the following schemes and/or the procedures disclosed in the examples.

In addition, prodrugs are also included within the context of the present disclosure. The term “prodrug” as used herein refers to a compound that is converted into an active form that has medical effects *in vivo* by, for example, hydrolysis in blood. Pharmaceutically acceptable prodrugs are described in T. Higuchi and V. Stella, Prodrugs as Novel Delivery Systems, A.C.S. Symposium Series, Vol. 14, Edward B. Roche, ed., Bioreversible Carriers in Drug Design, American Pharmaceutical Association and Pergamon Press, 1987, and D. Fleisher, S. Ramon and H. Barbra “Improved oral drug delivery: solubility limitations overcome by the use of prodrugs”, Advanced Drug Delivery Reviews (1996) 19(2) 115-130, each of which are incorporated herein by reference.

## DETAILED DESCRIPTION

This disclosure discloses a compound of Formula (A):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

R<sub>1a</sub>, R<sub>9a</sub>, R<sub>10a</sub>, and R<sub>11a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3a</sub>, R<sub>4a</sub>, R<sub>5a</sub>, R<sub>6a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>2-6</sub> alkenyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>2</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>3-8</sub> cycloalkyl, or 4- to 8-membered heterocyclyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

Z is O, S, or C(R<sub>8-1</sub>)(R<sub>8-2</sub>);

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

$R_{11}$  is selected from H,  $C_{1-4}$  alkyl,  $C_{1-4}$  haloalkyl, or  $C(=M_{11})N(R_{12a})(R_{12b})$ ; or  $R_{11}$  and  $R_{11a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

$R_{12a}$ , and  $R_{12b}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl; or  $R_{12a}$  and  $R_{12b}$  together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $C_{1-6}$  alkoxy,  $C_{1-6}$  haloalkoxy,  $C_{3-8}$  cycloalkyl, or 4- to 8-membered heterocyclyl;

$X$  is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or  $-NR_NC(=Y)-$ ;

$L$  is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or  $-C(=Y)NR_N-$ ;

wherein  $Y$  is O or S,  $R_N$  is selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

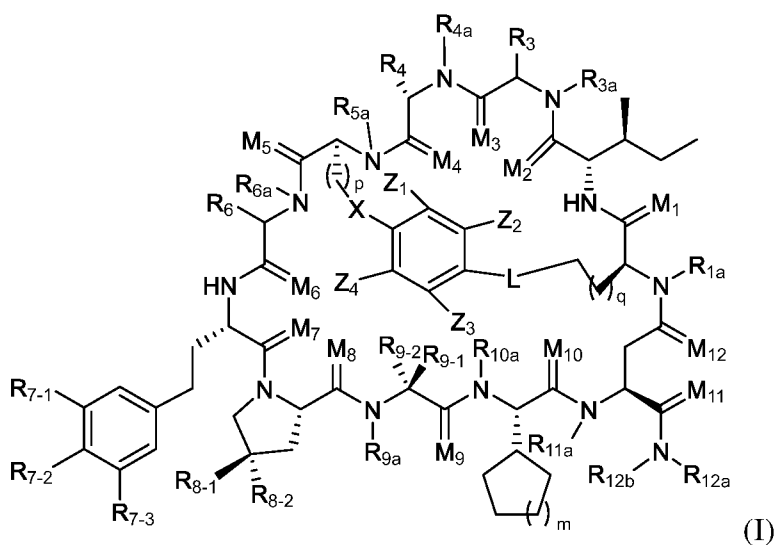
$Z_1$ ,  $Z_2$ ,  $Z_3$ , and  $Z_4$  are independently selected from H, halo,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

$p = 1, 2$ , or  $3$ ;

$q = 0, 1, 2$ , or  $3$ ;

$M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are independently selected from O or S.

This disclosure also discloses a compound of Formula (I):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

$R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are independently selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

$R_{3a}$ ,  $R_{4a}$ ,  $R_{5a}$ ,  $R_{6a}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl;

$R_{12a}$ , and  $R_{12b}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl,

or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>12a</sub> and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or -NR<sub>N</sub>C(=Y)-;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or -C(=Y)NR<sub>N</sub>-;

wherein Y is O or S, R<sub>N</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

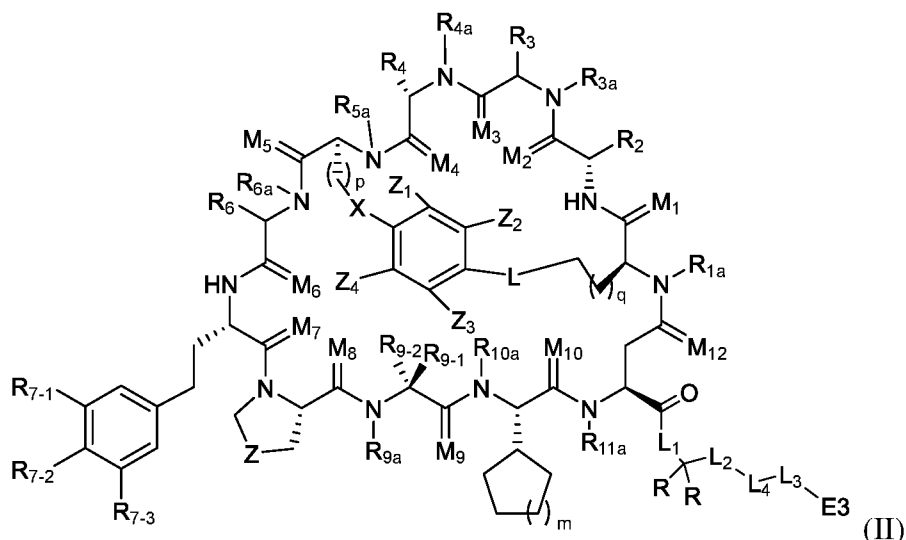
Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

p = 1, 2, or 3;

q = 0, 1, 2, or 3;

M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are independently selected from O or S.

This disclosure also discloses a compound of Formula (II):



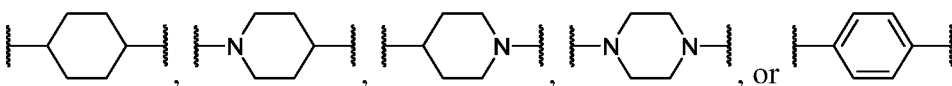
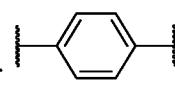
wherein,

$L_1$  is a divalent group selected from -amine group-, -diamine group-, -AA-amine group-, or -AA-diamine group-, wherein the amine group or diamine group has one amino linked to the carbonyl group on the cyclic peptide, and AA is an amino acid having the formula of - $N(R_{13a})CH(R_{13})C(=Q)-$ , wherein Q is O, or S,  $R_{13a}$  is selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl, and  $R_{13}$  is the side chain of an amino acid optionally in combination with  $R_{13a}$ , or  $R_{13}$  and  $R_{13a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocycle;

$L_2$  is  $-C_{0-8}$  alkylene-;

$L_3$  is  $-C_{0-8}$  alkylene-;

$L_4$  is selected from bond,  $-CH_2CH_2-$ ,  $-OCH_2-$ ,  $-CH_2O-$ ,  $-NHCH_2-$ ,  $-CH_2NH-$ ,  $-CH=CH-$ ,

$-C\equiv C-$ , , or ;

each R is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; or CRR forms  $C=O$  or  $C=S$ ;

E3 is a ligand of E3 ligase, and

and variables are as defined in the context.

### **Detailed Description of the Embodiments**

$R_{1a}$ ,  $R_{3a}$ ,  $R_{4a}$ ,  $R_{5a}$ ,  $R_{6a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$

In some embodiments,  $R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are H; in some embodiments,  $R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are  $C_{1-6}$  alkyl, such as  $C_{1-4}$  alkyl; in some embodiments,  $R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are  $C_{1-6}$  haloalkyl; in some embodiments,  $R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are Me; in some embodiments,  $R_{1a}$ ,  $R_{3a}$ ,  $R_{4a}$ ,  $R_{6a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are Me, and  $R_{5a}$  is Et, n-Pr, i-Amyl, or 3,3,3-trifluoropropyl; in some embodiments,  $R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are selected from  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl.

In some embodiments,  $R_{3a}$  is  $C_{1-6}$  alkyl or  $C_{1-6}$  haloalkyl, such as  $C_{1-4}$  alkyl or  $C_{1-4}$  haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl; in some embodiments,  $R_{3a}$  is phenyl; in some embodiments,  $R_{3a}$  is benzyl; in some embodiments,  $R_{3a}$  is selected from cyclopropyl; in some embodiments,  $R_{3a}$  is selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $C_{3-8}$

cycloalkyl, 4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; in some embodiments, R<sub>3a</sub> is selected from C<sub>1-4</sub> alkyl, C<sub>1-4</sub> haloalkyl, C<sub>3-6</sub> cycloalkyl, or -C<sub>1-6</sub> alkylene-phenyl; in some embodiments, R<sub>3a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as H, Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl; in some embodiments, R<sub>3a</sub> is selected from H, or C<sub>1-4</sub> alkyl, such as Me; in some embodiments, R<sub>3a</sub> is selected from cyclopropyl, phenyl, or benzyl; in some embodiments, R<sub>3a</sub> is selected from cyclopropyl, or benzyl.

In some embodiments, R<sub>4a</sub> is C<sub>1-6</sub> alkyl or C<sub>1-6</sub> haloalkyl, such as C<sub>1-4</sub> alkyl or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl; in some embodiments, R<sub>4a</sub> is selected from Me, Et, n-Pr, or 3,3,3-trifluoropropyl; in some embodiments, R<sub>4a</sub> is selected from cyclopropyl, phenyl, or benzyl; in some embodiments, R<sub>4a</sub> is cyclopropyl; in some embodiments, R<sub>4a</sub> is phenyl; in some embodiments, R<sub>4a</sub> is benzyl.

In some embodiments, R<sub>5a</sub> is H; in some embodiments, R<sub>5a</sub> is C<sub>1-6</sub> alkyl or C<sub>1-6</sub> haloalkyl, such as C<sub>1-4</sub> alkyl or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, i-Amyl, or 3,3,3-trifluoropropyl; in some embodiments, R<sub>5a</sub> is phenyl; in some embodiments, R<sub>5a</sub> is benzyl.

In some embodiments, R<sub>6a</sub> is H; in some embodiments, R<sub>6a</sub> is C<sub>1-6</sub> alkyl or C<sub>1-6</sub> haloalkyl, such as C<sub>1-4</sub> alkyl or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl; in some embodiments, R<sub>6a</sub> is selected from Me, Et, or 3,3,3-trifluoropropyl; in some embodiments, R<sub>6a</sub> is cyclopropyl; in some embodiments, R<sub>6a</sub> is phenyl; in some embodiments, R<sub>6a</sub> is benzyl.

## R<sub>2</sub>

In some embodiments, R<sub>2</sub> is H; in some embodiments, R<sub>2</sub> is C<sub>1-6</sub> alkyl; in some embodiments, R<sub>2</sub> is C<sub>1-6</sub> haloalkyl; in some embodiments, R<sub>2</sub> is C<sub>3-8</sub> cycloalkyl; in some embodiments, R<sub>2</sub> is 4- to 8-membered heterocyclyl; in some embodiments, R<sub>2</sub> is Ile, Val, Val(S), Gly(cPent), and Gly(cHex), preferably Ile.

## R<sub>12a</sub>, and R<sub>12b</sub>

In some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are H; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are C<sub>1-6</sub> alkyl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are C<sub>1-6</sub> haloalkyl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; in some embodiments, R<sub>12a</sub>, and R<sub>12b</sub> are optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl.

## R<sub>3</sub>

In some embodiments, R<sub>3</sub> is H; in some embodiments, R<sub>3</sub> is C<sub>1-6</sub> alkyl; in some embodiments, R<sub>3</sub> is C<sub>1-6</sub> haloalkyl; in some embodiments, R<sub>3</sub> is Me; in some embodiments, R<sub>3</sub>

is selected from H, or C<sub>1-4</sub> alkyl; in some embodiments, R<sub>3</sub> is -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl; in some embodiments, R<sub>3</sub> is -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl; in some embodiments, R<sub>3</sub> is -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl; in some embodiments, R<sub>3</sub> is -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; in some embodiments, R<sub>3</sub> has a S chirality; in some embodiments, R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl.

#### R<sub>4</sub>

In some embodiments, R<sub>4</sub> is H; in some embodiments, R<sub>4</sub> is C<sub>1-6</sub> alkyl; in some embodiments, R<sub>4</sub> is C<sub>1-6</sub> haloalkyl; in some embodiments, R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl, which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl; in some embodiments, R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4-membered heterocyclyl, such as azetidiny.

#### R<sub>6</sub>

In some embodiments, R<sub>6</sub> is H; in some embodiments, R<sub>6</sub> is C<sub>1-4</sub> alkyl; in some embodiments, R<sub>6</sub> is C<sub>1-4</sub> haloalkyl; in some embodiments, R<sub>6</sub> has a R chirality.

#### R<sub>7-1</sub> and R<sub>7-3</sub>

In some embodiments, R<sub>7-1</sub> and R<sub>7-3</sub> are H; in some embodiments, R<sub>7-1</sub> and R<sub>7-3</sub> are F; in some embodiments, R<sub>7-1</sub> is F and R<sub>7-3</sub> is H; in some embodiments, R<sub>7-1</sub> is Cl and R<sub>7-3</sub> is H; in some embodiments, R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, F, or Cl; in some embodiments, R<sub>7-1</sub> and/or R<sub>7-3</sub> is C<sub>1-6</sub> alkoxyl, such as OCH<sub>3</sub>; in some embodiments, R<sub>7-1</sub> and/or R<sub>7-3</sub> is C<sub>1-6</sub> haloalkoxyl.

#### R<sub>7-2</sub>

In some embodiments, R<sub>7-2</sub> is C<sub>1-6</sub> alkyl; in some embodiments, R<sub>7-2</sub> is C<sub>1-6</sub> haloalkyl, such as CF<sub>3</sub>.

#### Z

In some embodiments, Z is O; in some embodiments, Z is S; in some embodiments, Z is C(R<sub>8-1</sub>)(R<sub>8-2</sub>).

#### R<sub>8-1</sub> and R<sub>8-2</sub>

In some embodiments, R<sub>8-1</sub> is H; in some embodiments, R<sub>8-1</sub> is halo, such as F; in some embodiments, R<sub>8-1</sub> is C<sub>1-6</sub> alkoxyl, such as C<sub>2</sub> alkoxyl; in some embodiments, R<sub>8-1</sub> is C<sub>1-6</sub> haloalkoxyl.

In some embodiments, R<sub>8-2</sub> is H; in some embodiments, R<sub>8-2</sub> is halo, such as F; in some embodiments, R<sub>8-2</sub> is C<sub>1-6</sub> alkoxyl, such as C<sub>2</sub> alkoxyl; in some embodiments, R<sub>8-2</sub> is C<sub>1-6</sub> haloalkoxyl.

R<sub>9-1</sub> and R<sub>9-2</sub>

In some embodiments, R<sub>9-1</sub> is C<sub>1-6</sub> alkyl; in some embodiments, R<sub>9-1</sub> is C<sub>1-6</sub> haloalkyl.

In some embodiments, R<sub>9-2</sub> is H; in some embodiments, R<sub>9-2</sub> is C<sub>1-4</sub> alkyl; in some  
5 embodiments, R<sub>9-2</sub> is C<sub>1-4</sub> haloalkyl.

In some embodiments, R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected  
form a C<sub>3-6</sub> cycloalkyl; in some embodiments, R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they  
are connected form a 4- to 7-membered heterocyclyl.

10 m

In some embodiments, m = 0; in some embodiments, m = 1; in some embodiments, m =  
2; in some embodiments, m = 3.

R<sub>11</sub>

15 In some embodiments, R<sub>11</sub> is H; in some embodiments, R<sub>11</sub> is C<sub>1-4</sub> alkyl; in some  
embodiments, R<sub>11</sub> is C<sub>1-4</sub> haloalkyl; in some embodiments, R<sub>11</sub> is C(=M<sub>11</sub>)N(R<sub>12a</sub>)(R<sub>12b</sub>); in some  
embodiments, R<sub>11</sub> and R<sub>11a</sub> together with the atoms to which they are connected form a 4- to 7-  
membered heterocyclyl.

20 X

In some embodiments, X is a bond; in some embodiments, X is O; in some embodiments,  
X is S; in some embodiments, X is NH; in some embodiments, X is a 5- to 6-membered  
heteroarylene; in some embodiments, X is -NR<sub>N</sub>C(=Y)-.

25 L

In some embodiments, L is a bond; in some embodiments, L is O; in some embodiments,  
L is S; in some embodiments, L is NH; in some embodiments, L is a phenylene; in some  
embodiments, L is a 5- to 6-membered heteroarylene; in some embodiments, L is -C(=Y)NR<sub>N</sub>-.

In some embodiments, Y is O; in some embodiments, Y is S; in some embodiments, R<sub>N</sub>  
30 is H; in some embodiments, R<sub>N</sub> is C<sub>1-6</sub> alkyl; in some embodiments, R<sub>N</sub> is C<sub>1-6</sub> haloalkyl.

Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub>

In some embodiments, Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are H; in some embodiments, Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub>  
are halo, such as F; in some embodiments, Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are C<sub>1-6</sub> alkyl; in some embodiments,  
35 Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are C<sub>1-6</sub> haloalkyl; in some embodiments, Z<sub>1</sub>, and Z<sub>4</sub> are Cl, and Z<sub>2</sub>, and Z<sub>3</sub> are  
F; in some embodiments, Z<sub>1</sub>, and Z<sub>4</sub> are F, and Z<sub>2</sub>, and Z<sub>3</sub> are Cl.

p

In some embodiments, p = 1; in some embodiments, p = 2; in some embodiments, p = 3.

q

In some embodiments,  $q = 0$ ; in some embodiments,  $q = 1$ ; in some embodiments,  $q = 2$ ; in some embodiments,  $q = 3$ .

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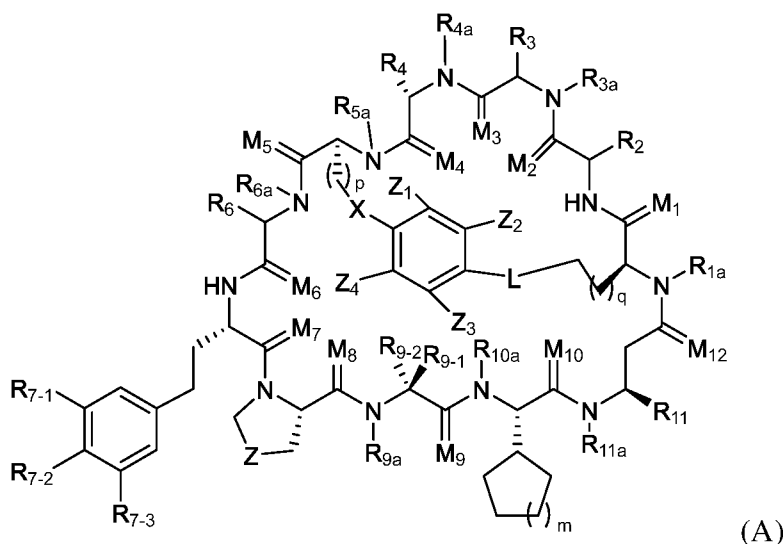
$M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$

In some embodiments,  $M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are O; in some embodiments,  $M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are S; in some embodiments,  $M_1$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are O, and  $M_2$  is S; in some  
 10 embodiments,  $M_1$ ,  $M_2$ ,  $M_4$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are O, and  $M_3$  is S; in some  
 embodiments,  $M_1$ ,  $M_2$ ,  $M_3$ ,  $M_5$ ,  $M_6$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are O, and  $M_4$  is S; in some  
 embodiments,  $M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$ ,  $M_7$ ,  $M_8$ ,  $M_9$ ,  $M_{10}$ ,  $M_{11}$ , and  $M_{12}$  are O, and  $M_6$  is S.

This disclosure discloses the following solutions:

15

Solution 1. A compound of Formula (A):



(A)

or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

$R_{1a}$ ,  $R_{9a}$ ,  $R_{10a}$ , and  $R_{11a}$  are independently selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

20

$R_{3a}$ ,  $R_{4a}$ ,  $R_{5a}$ ,  $R_{6a}$  are independently selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $C_{2-6}$  alkenyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl; which is optionally substituted with 1, or more halo,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl;

$R_2$  is selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $C_{3-8}$  cycloalkyl, or 4- to 8-membered  
 25 heterocyclyl;

$R_3$  is selected from H,  $C_{1-6}$  alkyl,  $C_{1-6}$  haloalkyl,  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-

membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

Z is O, S, or C(R<sub>8-1</sub>)(R<sub>8-2</sub>);

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

R<sub>11</sub> is selected from H, C<sub>1-4</sub> alkyl, C<sub>1-4</sub> haloalkyl, or C(=M<sub>11</sub>)N(R<sub>12a</sub>)(R<sub>12b</sub>); or R<sub>11</sub> and R<sub>11a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>12a</sub>, and R<sub>12b</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>12a</sub> and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>1-6</sub> alkoxy, C<sub>1-6</sub> haloalkoxy, C<sub>3-8</sub> cycloalkyl, or 4- to 8-membered heterocyclyl;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or -NR<sub>N</sub>C(=Y)-;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or -C(=Y)NR<sub>N</sub>-;

wherein Y is O or S, R<sub>N</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

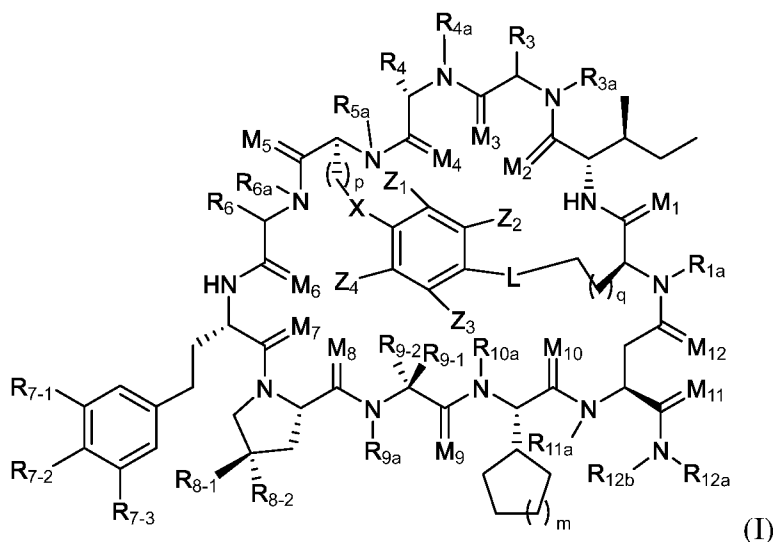
Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

p = 1, 2, or 3;

q = 0, 1, 2, or 3;

M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are independently selected from O or S.

Solution 2. A compound of Formula (I):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

R<sub>1a</sub>, R<sub>9a</sub>, R<sub>10a</sub>, and R<sub>11a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3a</sub>, R<sub>4a</sub>, R<sub>5a</sub>, R<sub>6a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl;

R<sub>12a</sub>, and R<sub>12b</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>12a</sub> and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or -NR<sub>N</sub>C(=Y)-;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or -C(=Y)NR<sub>N</sub>-;

wherein Y is O or S, R<sub>N</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

p = 1, 2, or 3;

q = 0, 1, 2, or 3;

M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are independently selected from O or S.

Solution 3. The compound according to solution 1 or 2, or a pharmaceutically acceptable salt, a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, wherein R<sub>1a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

Solution 4. The compound according to any one of solutions 1 to 3, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>3a</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>3-8</sub> cycloalkyl, 4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; preferably, C<sub>1-4</sub> alkyl, C<sub>1-4</sub> haloalkyl, C<sub>3-6</sub> cycloalkyl, or -C<sub>1-6</sub> alkylene-phenyl; preferably, H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

Solution 5. The compound according to any one of solutions 1 to 3, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>3a</sub> is selected from cyclopropyl, phenyl, or benzyl; preferably, cyclopropyl, or benzyl.

Solution 6. The compound according to any one of solutions 1 to 5, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>4a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl, preferably, Me, Et, n-Pr, or 3,3,3-trifluoropropyl.

Solution 7. The compound according to any one of solutions 1 to 5, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>4a</sub> is selected from cyclopropyl, phenyl, or benzyl; preferably, cyclopropyl, or benzyl.

Solution 8. The compound according to any one of solutions 1 to 7, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>5a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, i-Amyl, or 3,3,3-trifluoropropyl; preferably, n-Pr.

Solution 9. The compound according to any one of solutions 1 to 7, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>5a</sub> is selected from phenyl, or benzyl.

Solution 10. The compound according to any one of solutions 1 to 9, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof,

wherein  $R_{6a}$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl; preferably, Me, Et, or 3,3,3-trifluoropropyl; preferably, Me.

Solution 11. The compound according to any one of solutions 1 to 9, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{6a}$  is selected from cyclopropyl, phenyl, or benzyl; preferably, cyclopropyl.

Solution 12. The compound according to any one of solutions 1 to 11, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{9a}$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably,  $C_{1-4}$  alkyl, such as Me.

Solution 13. The compound according to any one of solutions 1 to 11, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{9a}$  is H.

Solution 14. The compound according to any one of solutions 1 to 13, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{10a}$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably,  $C_{1-4}$  alkyl, such as Me.

Solution 15. The compound according to any one of solutions 1 to 14, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{11a}$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably,  $C_{1-4}$  alkyl, such as Me.

Solution 16. The compound according to any one of solutions 1 to 15, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{11}$  is  $C(=M_{11})N(R_{12a})(R_{12b})$ .

Solution 17. The compound according to any one of solutions 1 to 16, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{12a}$ , and  $R_{12b}$  are independently selected from  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably,  $C_{1-4}$  alkyl, such as Me or Et; or  $R_{12a}$  and  $R_{12b}$  together with the atom to which they are connected form a 5- to 6-membered heterocyclyl, such as piperidinyl, pyrrolidine, or morpholine.

Solution 18. The compound according to any one of solutions 1 to 17, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_3$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably, H, or  $C_{1-4}$  alkyl, such as Me; preferably,  $R_3$  has a S chirality.

Solution 19. The compound according to any one of solutions 1 to 17, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_3$  is selected from  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl; or  $R_3$  and  $R_{3a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl.

Solution 20. The compound according to any one of solutions 1 to 19, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_4$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably, H, or  $C_{1-4}$  alkyl, such as Me; or  $R_4$  and  $R_{4a}$  together with the atoms to which they are connected form a 4- to 6-membered

heterocyclyl, which is optionally substituted with -OC<sub>1-4</sub> alkyl, or -OC<sub>1-4</sub> haloalkyl; preferably, R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form an azetidiny.

Solution 21. The compound according to any one of solutions 1 to 20, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>6</sub> is selected from H, or C<sub>1-4</sub> alkyl; preferably, R<sub>6</sub> has a R chirality; preferably, R<sub>6</sub> is H.

Solution 22. The compound according to any one of solutions 1 to 21, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, F, or Cl; preferably, F.

Solution 23. The compound according to any one of solutions 1 to 21, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from C<sub>1-6</sub> alkoxy, or C<sub>1-6</sub> haloalkoxy.

Solution 24. The compound according to any one of solutions 1 to 23, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>7-2</sub> is selected from C<sub>1-4</sub> haloalkyl, such as CF<sub>3</sub>.

Solution 25. The compound according to any one of solutions 1 to 24, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein Z is O, or S.

Solution 26. The compound according to any one of solutions 1 to 24, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein Z is C(R<sub>8-1</sub>)(R<sub>8-2</sub>).

Solution 27. The compound according to any one of solutions 1 to 26, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>8-1</sub> and R<sub>8-2</sub> are H.

Solution 28. The compound according to any one of solutions 1 to 26, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>8-1</sub> and R<sub>8-2</sub> are F.

Solution 29. The compound according to any one of solutions 1 to 26, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>8-1</sub> is OEt and R<sub>8-2</sub> is H; or R<sub>8-1</sub> is H and R<sub>8-2</sub> is OEt.

Solution 30. The compound according to any one of solutions 1 to 29, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>9-1</sub> is selected from C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, or isopropyl; and R<sub>9-2</sub> is H, or C<sub>1-4</sub> alkyl, such as Me.

Solution 31. The compound according to any one of solutions 1 to 30, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, such as cyclopropyl, cyclobutyl, or cyclopentyl; preferably, cyclobutyl; preferably, cyclopentyl.

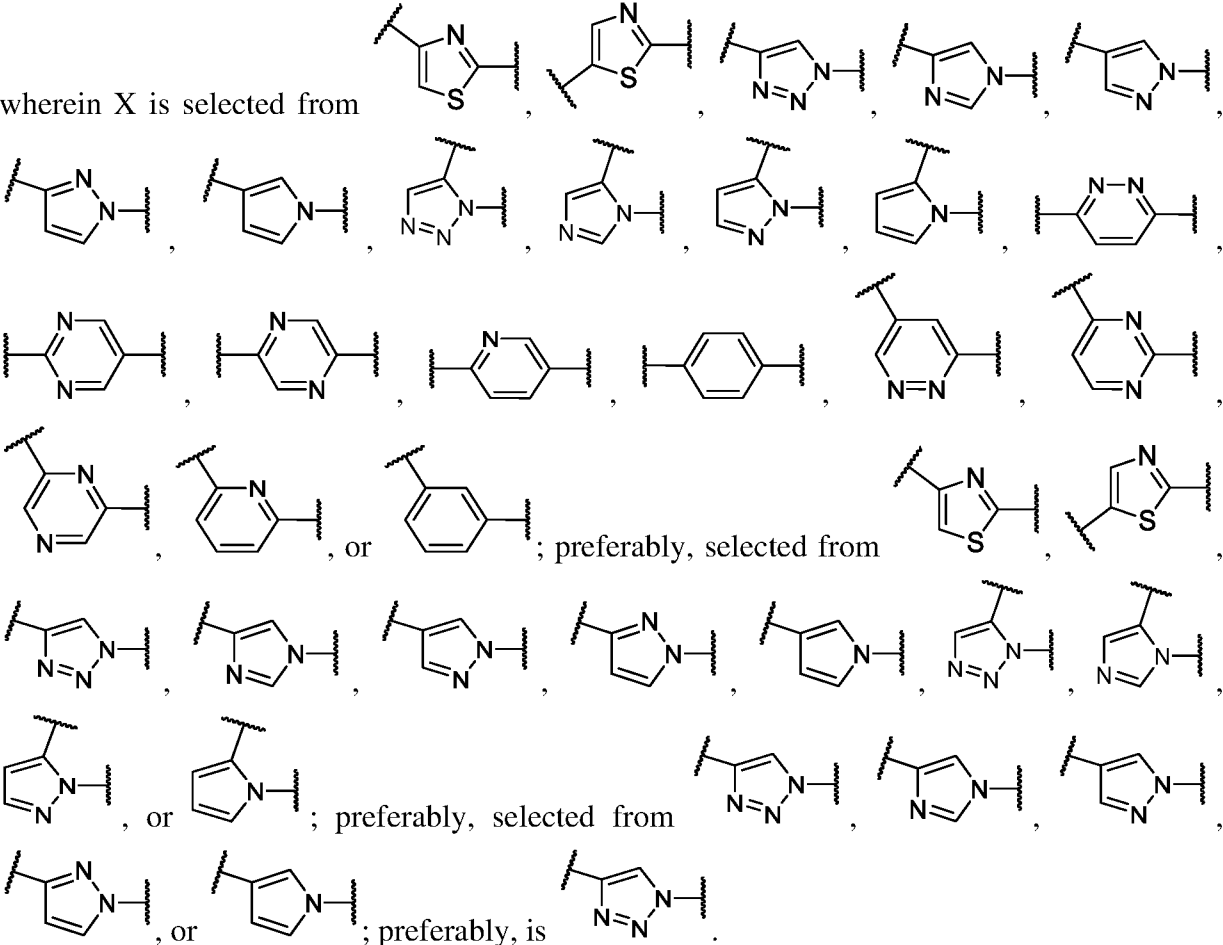
Solution 32. The compound according to any one of solutions 1 to 31, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $m = 1$ .

Solution 33. The compound according to any one of solutions 1 to 32, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein X is a bond, or S; preferably, X is a bond; preferably, X is S.

Solution 34. The compound according to any one of solutions 1 to 32, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein X is  $\text{-NHC(=O)-}$ , or  $\text{-NHC(=S)-}$ .

Solution 35. The compound according to any one of solutions 1 to 32, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof,

wherein X is selected from



preferably, selected from

or

preferably, selected from

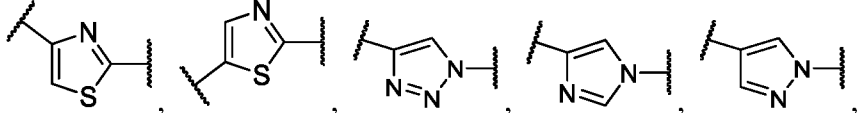
or

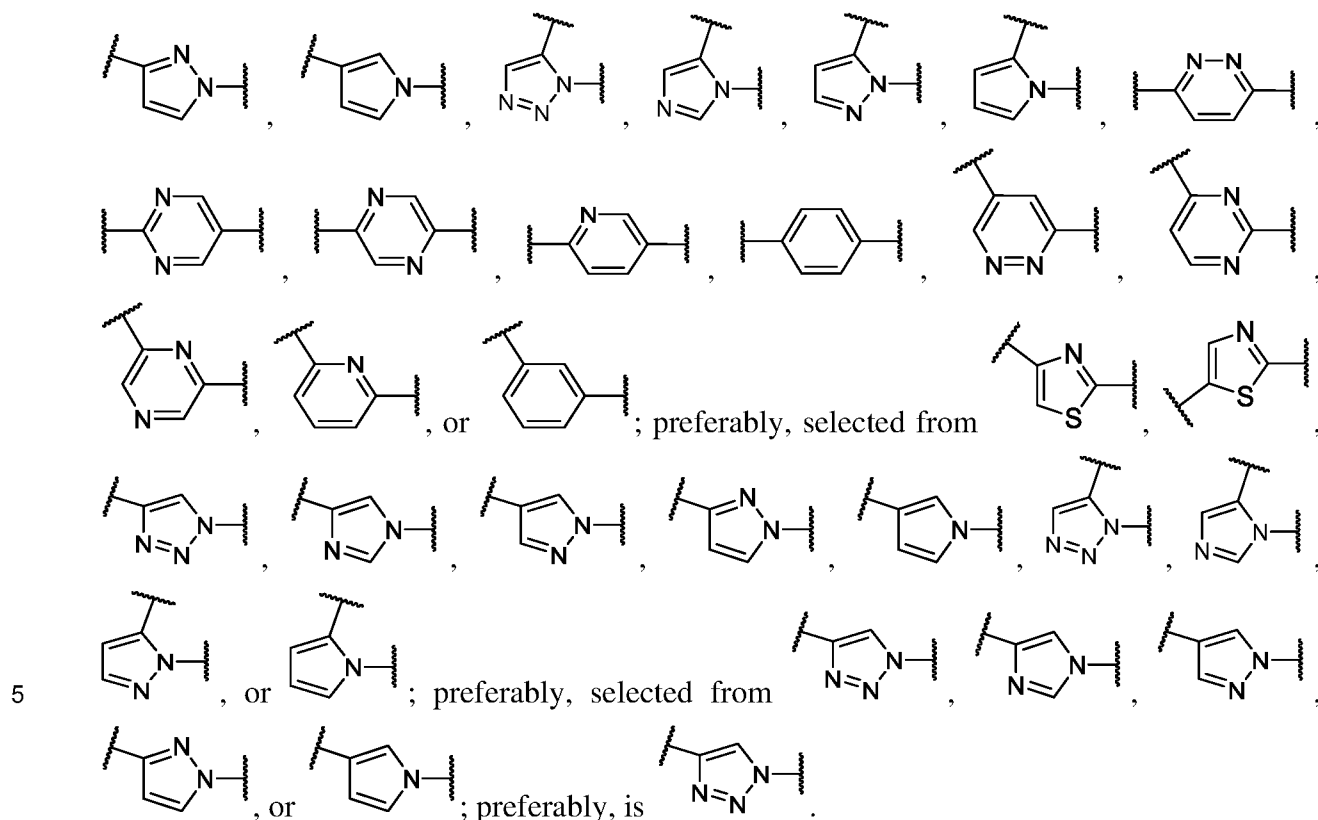
preferably, is

Solution 36. The compound according to any one of solutions 1 to 35, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L is  $\text{-C(=O)NH-}$ , or  $\text{-C(=S)NH-}$ ; preferably,  $\text{-C(=S)NH-}$ .

Solution 37. The compound according to any one of solutions 1 to 35, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof,

wherein L is selected from





Solution 38. The compound according to any one of solutions 1 to 35, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L is a bond, or S.

10 Solution 39. The compound according to any one of solutions 1 to 38, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $Z_1$ ,  $Z_2$ ,  $Z_3$ , and  $Z_4$  are independently selected from H, or halo, such as F, Cl, Br; preferably, F.

15 Solution 40. The compound according to any one of solutions 1 to 39, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $Z_1$ , and  $Z_4$  are Cl, and  $Z_2$ , and  $Z_3$  are F; or  $Z_1$ , and  $Z_4$  are F, and  $Z_2$ , and  $Z_3$  are Cl.

Solution 41. The compound according to any one of solutions 1 to 39, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $p = 1$ .

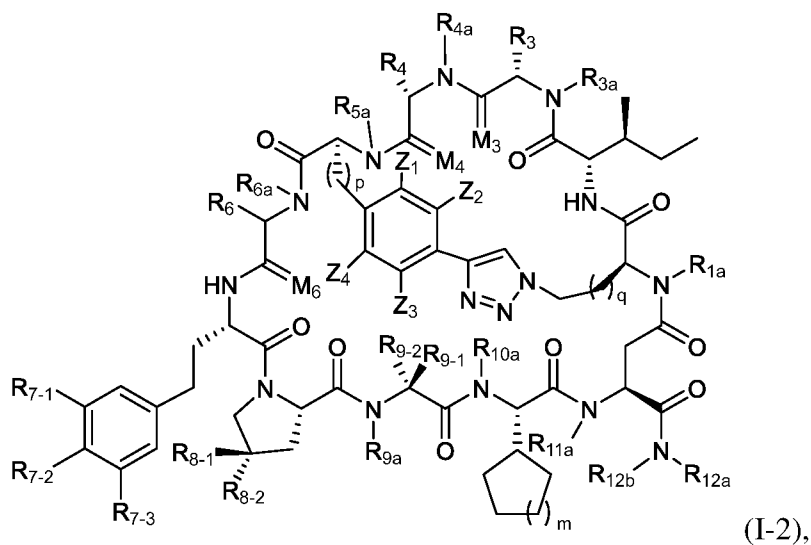
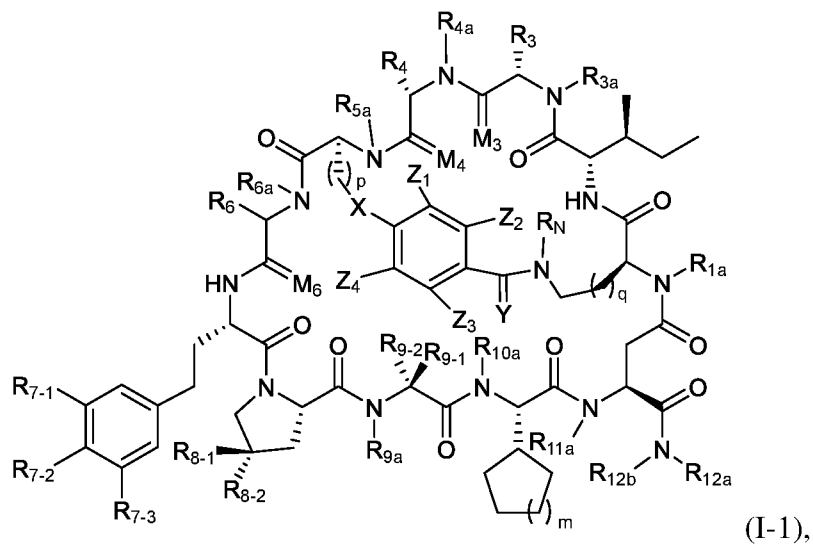
20 Solution 42. The compound according to any one of solutions 1 to 41, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $q = 1, 2$ , or  $3$ ; preferably,  $q = 1$ .

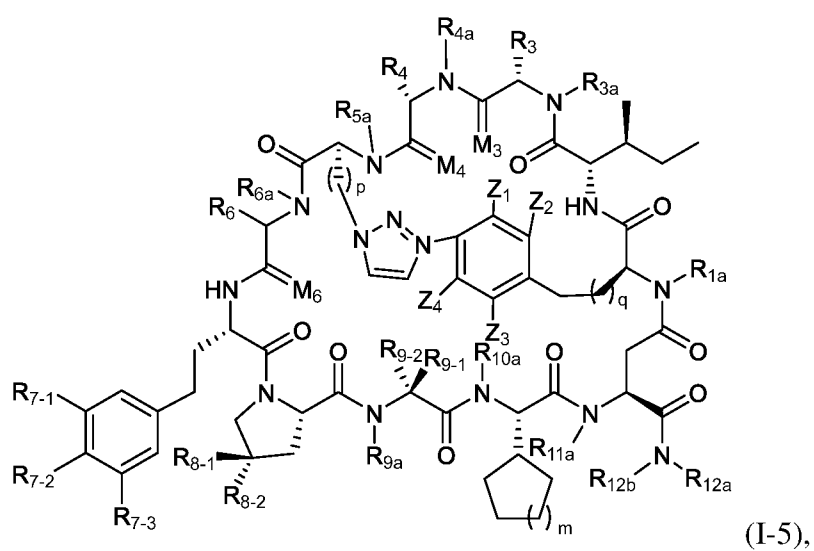
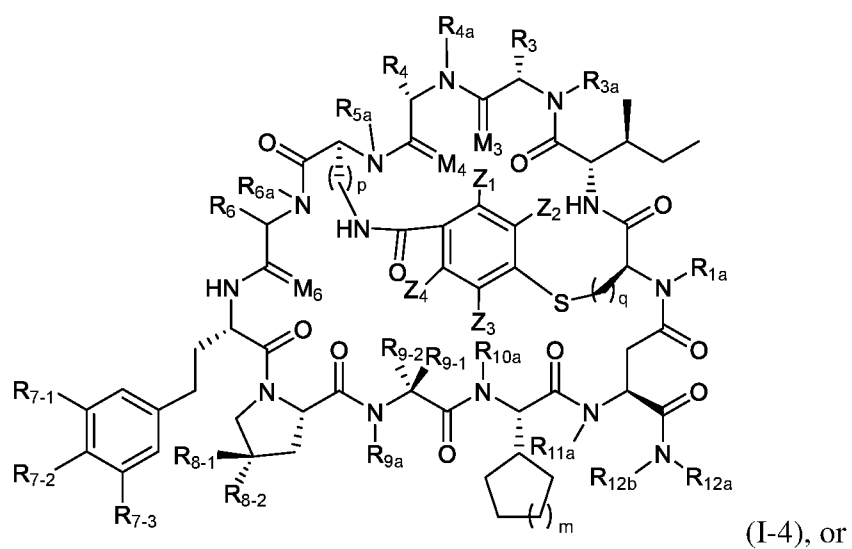
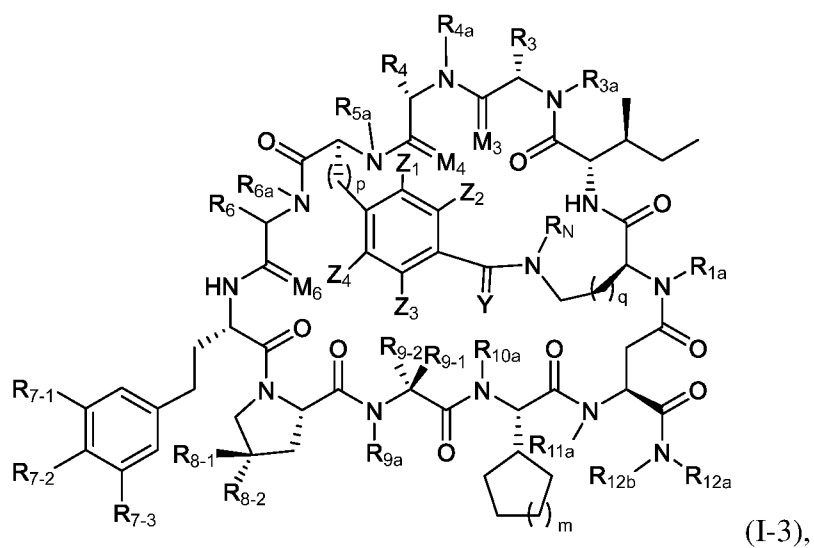
25 Solution 43. The compound according to any one of solutions 1 to 42, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $M_1, M_2, M_3, M_4, M_5, M_6, M_7, M_8, M_9, M_{10}, M_{11}$ , and  $M_{12}$  are O.

Solution 44. The compound according to any one of solutions 1 to 42, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein one or more  $M_3, M_4, M_6$  are S.

Solution 45. The compound according to any one of solutions 1 to 42, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein one or more  $M_2$  is S.

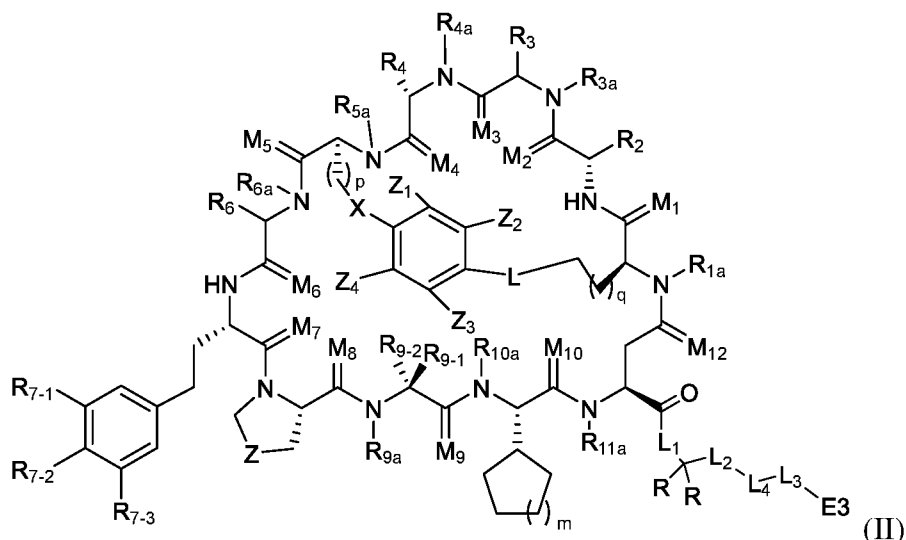
46. The compound according to any one of solutions 1 to 45, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the compound has a formula selected from the following:





wherein the variables are as defined in any one of solutions 1 to 45.

Solution 47. A compound of Formula (II):

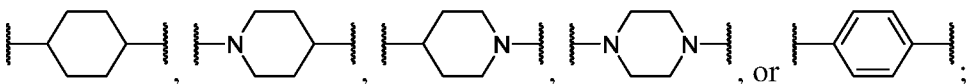


wherein,

$L_1$  is a divalent group selected from -amine group-, -diamine group-, -AA-amine group-, or -AA-diamine group-, wherein the amine group or diamine group has one amino linked to the carbonyl group on the cyclic peptide, and AA is an amino acid having the formula of - $N(R_{13a})CH(R_{13})C(=Q)-$ , wherein Q is O, or S,  $R_{13a}$  is selected from H,  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl, and  $R_{13}$  is the side chain of an amino acid optionally in combination with  $R_{13a}$ , or  $R_{13}$  and  $R_{13a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

$L_2$  is  $-C_{0-8}$  alkylene-;

$L_3$  is  $-C_{0-8}$  alkylene-;

$L_4$  is selected from bond,  $-CH_2CH_2-$ ,  $-OCH_2-$ ,  $-CH_2O-$ ,  $-NHCH_2-$ ,  $-CH_2NH-$ ,  $-CH=CH-$ ,  $-C\equiv C-$ , ;

each R is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; or CRR forms  $C=O$  or  $C=S$ ;

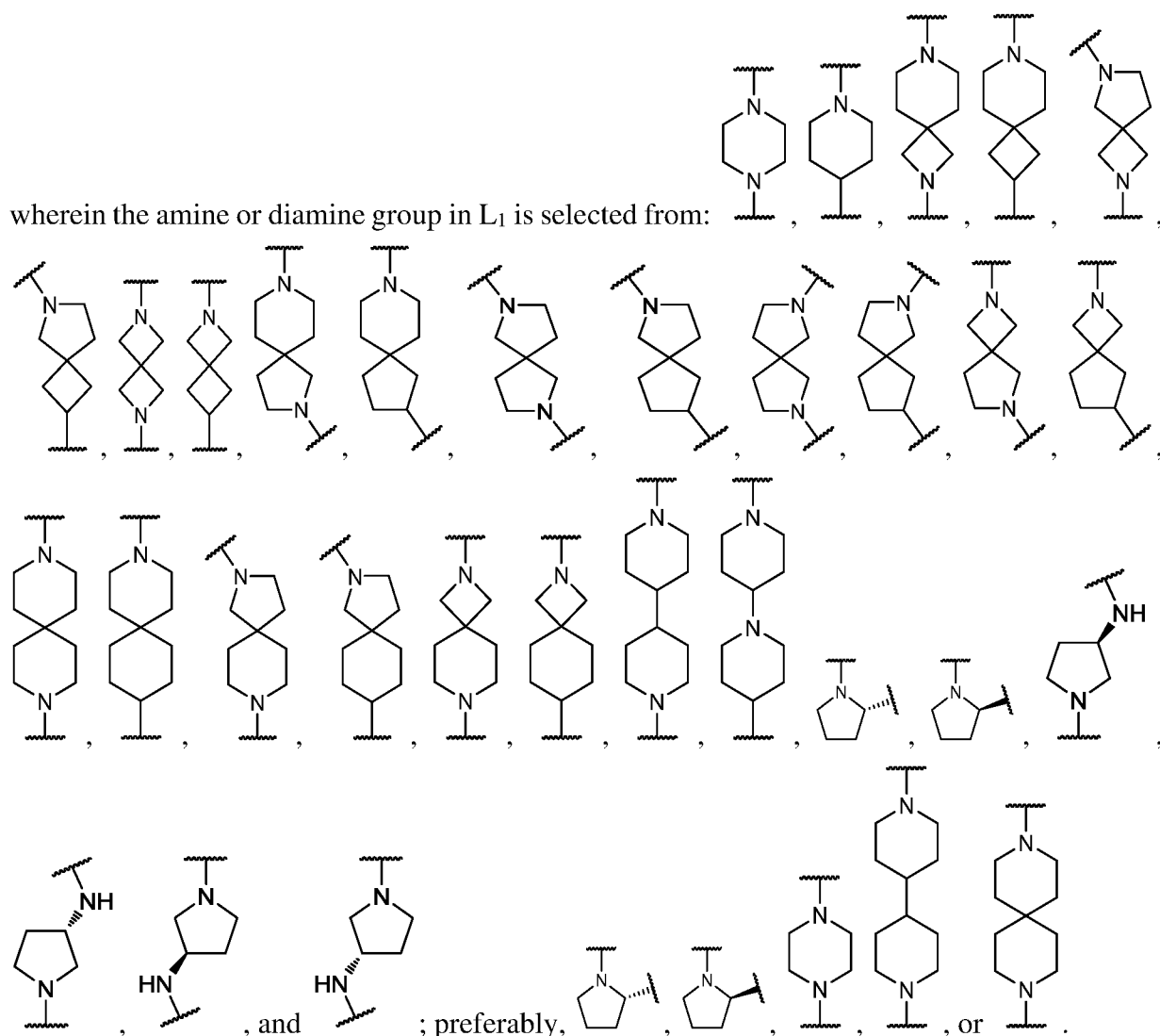
E3 is a ligand of E3 ligase, and

and variables are as defined in any one of solutions 1 to 45.

Solution 48. The compound according to solution 47, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $L_1$  is -amine group-, or -diamine group-.

Solution 49. The compound according to solution 47 or 48, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $L_1$  is -AA-amine group-, or -AA-diamine group-.

Solution 50. The compound according to any one of solutions 47 to 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof,



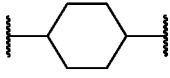
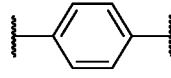
5       Solution 51. The compound according to any one of solutions 47 to 50, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the amine group in L<sub>1</sub> is -N(R<sub>N1</sub>)-, wherein R<sub>N1</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl.

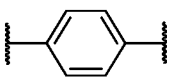
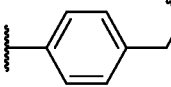
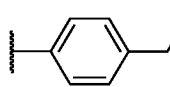
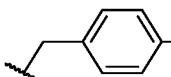
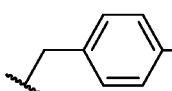
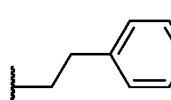
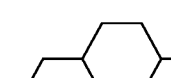
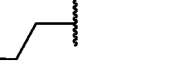
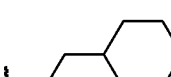
10       Solution 52. The compound according to any one of solutions 47 and 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein in -N(R<sub>13a</sub>)CH(R<sub>13</sub>)C(=Q)- of AA, Q is O; R<sub>13a</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl, such as Me; and R<sub>13</sub> is H, or Me; or R<sub>13</sub> and R<sub>13a</sub> together with the atoms to which they are connected form a 4- to 6-membered heterocycl, such as azetidiny, piperazinyl, or pyrrolidinyl.

15       Solution 53. The compound according to any one of solutions 47 and 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein in -N(R<sub>13a</sub>)CH(R<sub>13</sub>)C(=Q)- of AA, Q is O or S; R<sub>13a</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl, such as Me; and R<sub>13</sub> is H.

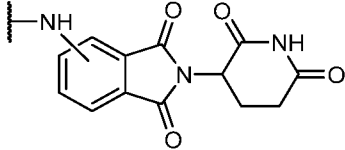
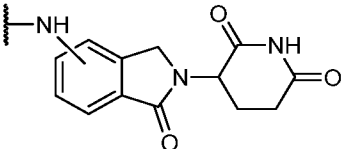
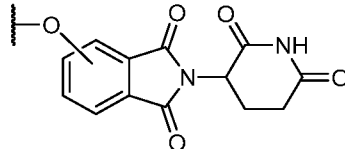
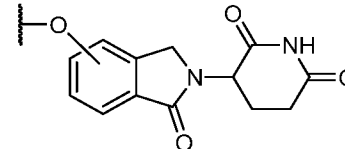
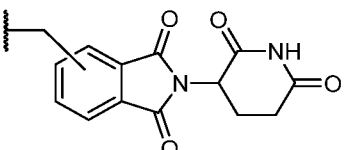
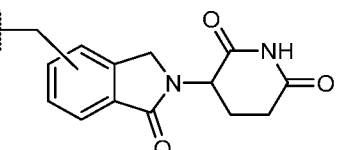
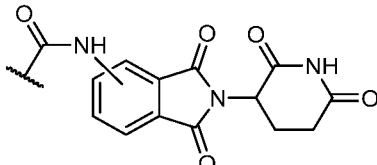
20       Solution 54. The compound according to any one of solutions 47 and 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable

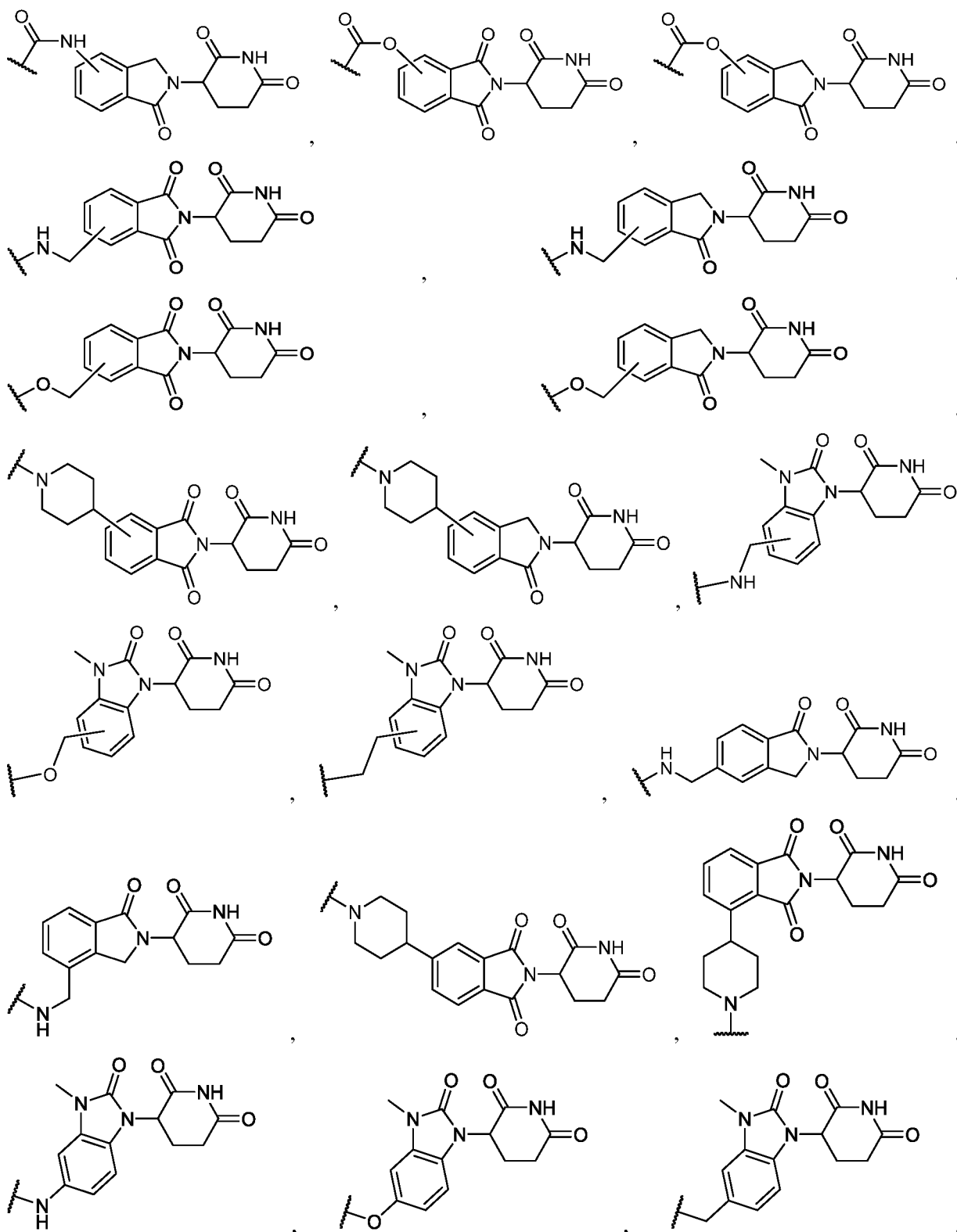
salt thereof, wherein in  $-N(R_{13a})CH(R_{13})C(=Q)-$  of AA, Q is O or S;  $R_{13}$  is the side chain of a natural amino acid (optionally in combination with  $R_{13a}$ ) selected from Gly, Ala, Val, Leu, Ile, Phe, Trp, Tyr, Asp, Asn, Glu, Lys, Gln, Met, Ser, Thr, Cys, Pro, His, and Arg; preferably Pro in combination with  $R_{13a}$ .

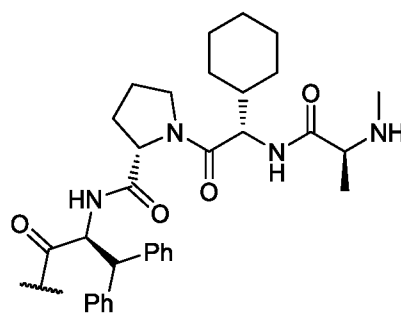
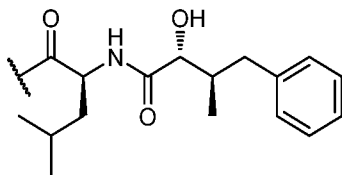
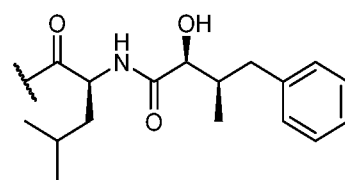
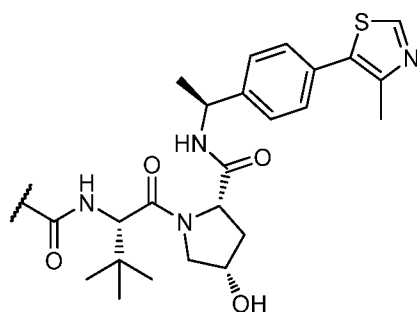
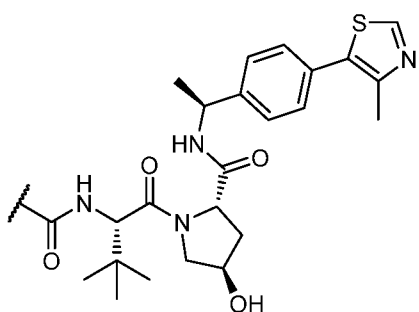
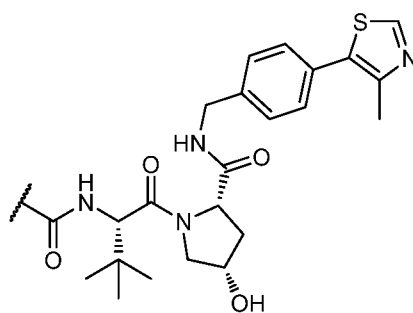
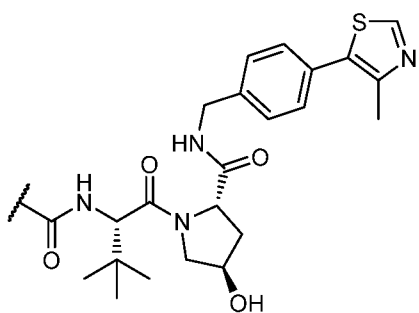
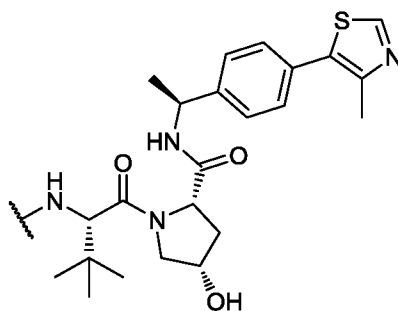
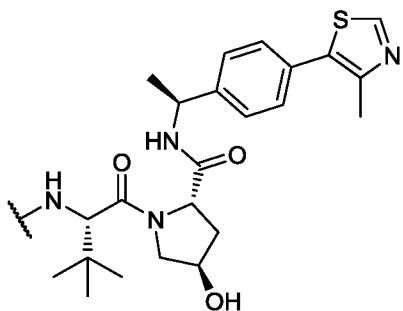
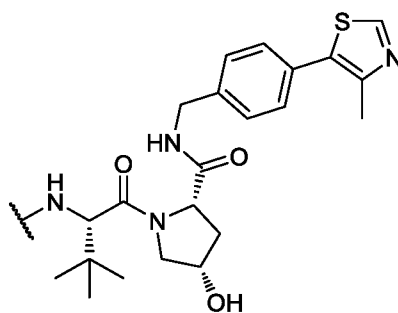
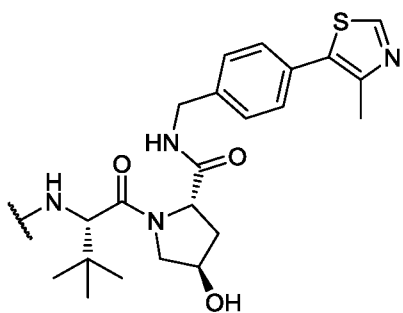
5 Solution 55. The compound according to any one of solutions 47 to 54, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $L_4$  is selected from  $-\text{CH}_2\text{CH}_2-$ ,  $-\text{NHCH}_2-$ ,  $-\text{CH}=\text{CH}-$ , , or ; preferably,  $-\text{CH}_2\text{CH}_2-$  or  $-\text{NHCH}_2-$ .

10 Solution 56. The compound according to any one of solutions 47 to 55, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $-L_2-L_4-L_3-$  is selected from bond,  $-(\text{CH}_2)_4-$ ,  $-(\text{CH}_2)_5-$ ,  $-(\text{CH}_2)_6-$ ,  $-\text{NH}(\text{CH}_2)_4-$ ,  $-\text{NH}(\text{CH}_2)_5-$ ,  $-\text{NH}(\text{CH}_2)_6-$ ,  $-\text{CH}=\text{CH}-(\text{CH}_2)_2-$ ,  $-\text{CH}=\text{CH}-(\text{CH}_2)_3-$ ,  $-\text{CH}=\text{CH}-(\text{CH}_2)_4-$ ,  $-(\text{CH}_2)_2-\text{CH}=\text{CH}-$ ,  $-(\text{CH}_2)_3-\text{CH}=\text{CH}-$ ,  $-(\text{CH}_2)_4-\text{CH}=\text{CH}-$ , , , , , , , , , , and ; preferably,  $-(\text{CH}_2)_4-$ ,  $-(\text{CH}_2)_5-$ , or  $-(\text{CH}_2)_6-$ ; preferably,  $-\text{NH}(\text{CH}_2)_4-$ ,  $-\text{NH}(\text{CH}_2)_5-$ ,  $-\text{NH}(\text{CH}_2)_6-$ ; preferably,  $-(\text{CH}_2)_5-$ , or  $-(\text{CH}_2)_6-$ ; preferably,  $-(\text{CH}_2)_6-$ .

Solution 57. The compound according to any one of solutions 47 to 56, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof,

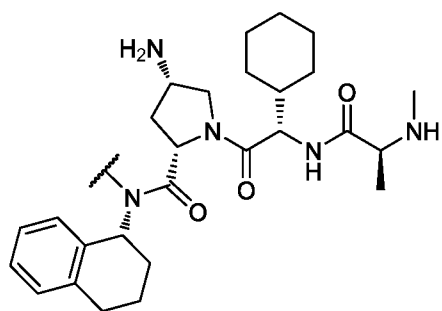
20 wherein the ligand of E3 ligase is selected from , , , , , , .



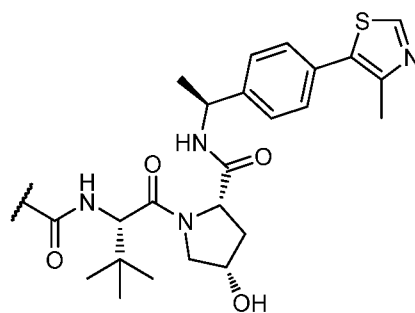
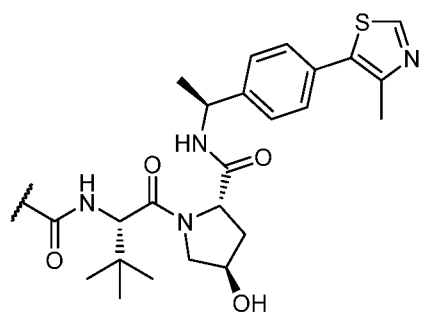
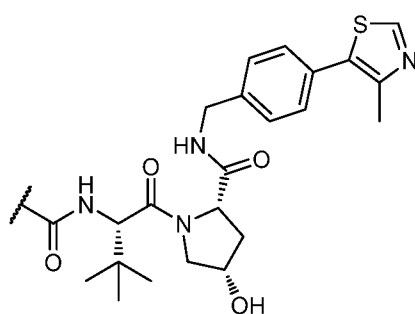
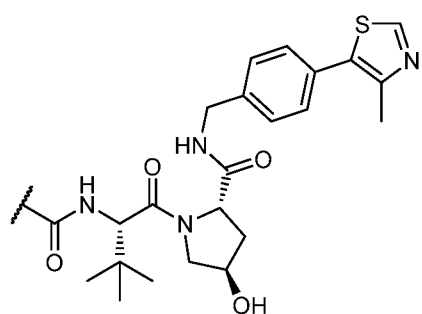
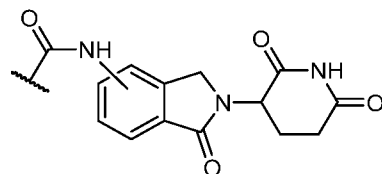
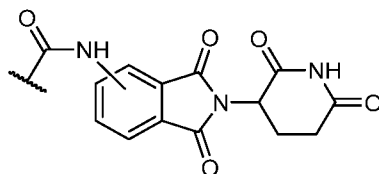
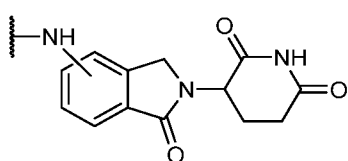
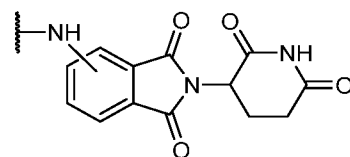


5

and

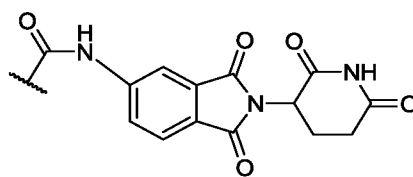
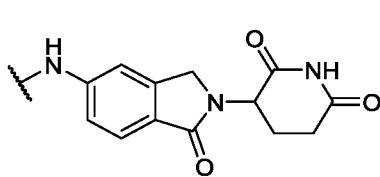
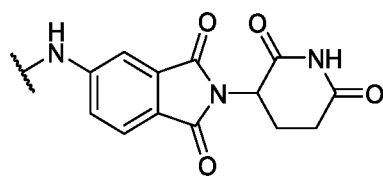


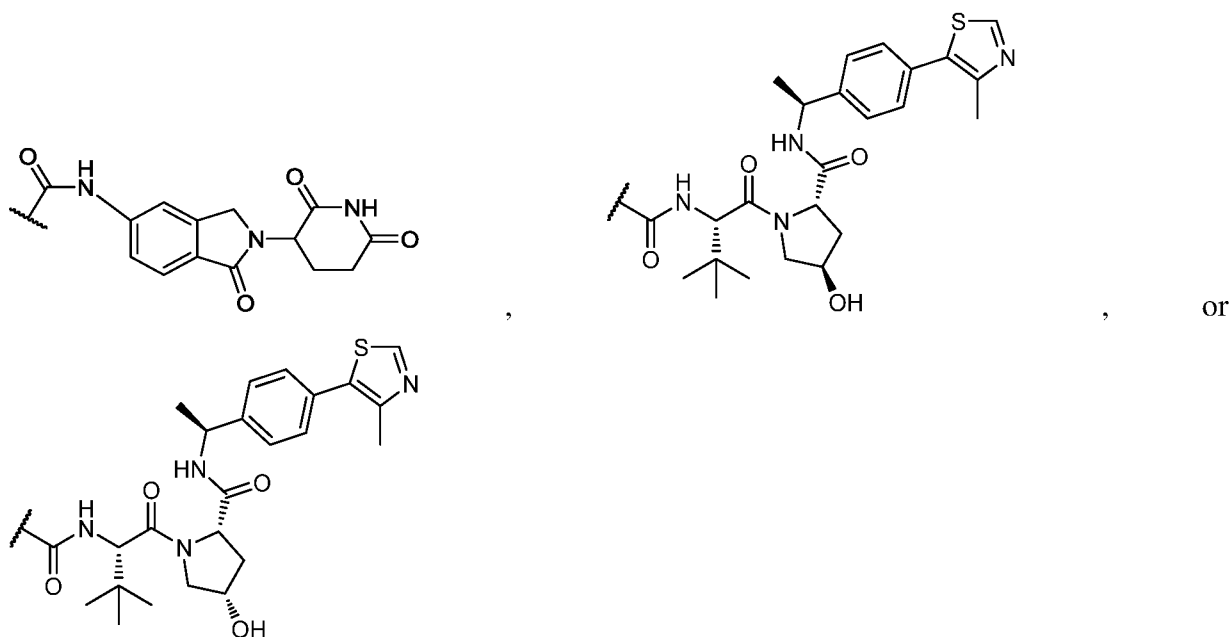
; preferably,



, or

; preferably,





Solution 58. The compound according to any one of solutions 47 to 57, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein CRR forms C=O.

Solution 59. The compound according to any one of solutions 1 to 58, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is selected from compounds 1001 to 1231, 2001 to 2173, and 3001 to 3003.

Solution 60. A pharmaceutical composition, comprising:

the compound, or the stereoisomer, the tautomer, the stable isotopic variant, or the prodrug thereof, or the pharmaceutical acceptable salt thereof according to any one of solutions 1 to 59;

pharmaceutically acceptable excipient(s); and

optionally, one or more other therapeutic agents, such as RAS inhibitors, RTK inhibitors, EGFR inhibitors, immune checkpoint inhibitors for example PD-1 or PD-L1 inhibitors, and the like.

Solution 61. A kit, comprising:

a first container which contains the compound, or the stereoisomer, the tautomer, the stable isotopic variant, or the prodrug thereof, or the pharmaceutical acceptable salt thereof according to any one of solutions 1 to 59; and

optionally, a second container which contains one or more other therapeutic agents; and

optionally, a third container which contains pharmaceutically acceptable excipient(s) for diluting or suspending the said compound and/or other therapeutic agent(s), such as RAS inhibitors, RTK inhibitors, EGFR inhibitors, immune checkpoint inhibitors for example PD-1 or PD-L1 inhibitors, and the like.

Solution 62. Use of a compound, or a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof, or a pharmaceutical acceptable salt thereof according to any one of solutions 1 to 59, in the manufacture of a medicament for treating a disease mediated by Ras kinase.

Solution 63. The compound, or a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof, or a pharmaceutical acceptable salt thereof according to any one of solutions 1 to 59, for use in treating a disease mediated by Ras kinase.

Solution 64. A method of treating a disease mediated by Ras kinase in a subject in need thereof, comprising administering to the subject a compound, or a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof, or a pharmaceutical acceptable salt thereof according to any one of solutions 1 to 59.

Solution 65. The use of solution 62, or the compound for use of solution 63, or the method of solution 64, wherein the disease mediated by Ras kinase is cancer.

Solution 66. The use, or the compound for use, or the method of solution 65, wherein the cancer is solid cancer or blood cancer.

Solution 67. The use, or the compound for use, or the method of solution 65 or 66, wherein the cancer is selected from lung cancer, esophageal cancer, stomach cancer, colon cancer, uterine cancer, ovarian cancer, pancreatic cancer, bladder cancer, breast cancer, brain cancer, glioblastoma (GBM), kidney cancer, liver cancer, mesothelioma, prostate cancer, sarcoma, testicular cancer, thyroid cancer, uterine cancer, cholangiocarcinoma, thyroid cancer, skin cancer, uveal melanoma; leukemia, malignant lymphoma, and multiple myeloma.

Solution 68. The use, or the compound for use, or the method of solution 65 or 66, wherein the cancer is head and neck cancer, biliary tract cancer, Nasopharyngeal cancer, non-small cell lung cancer, small cell lung cancer, colon cancer, rectal cancer, endometrial cancer, cervical cancer, AML (acute myeloid leukemia), CML (chronic myelogenous leukemia), ALL (acute lymphocytic leukemia), CLL (chronic lymphocytic leukemia), Hodgkin lymphoma, and Non-Hodgkin lymphoma.

Solution 69. The use, or the compound for use, or the method of any one of solutions 65 to 68, wherein the cancer is associated with Ras gene abnormalities or RAS pathway overactivation.

Solution 70. The use, or the compound for use, or the method of any one of solutions 65 to 69, wherein the cancer is resistant to known KRAS inhibitors, such as sotorasib or adagrasib.

Solution 71. The use, or the compound for use, or the method of any one of solutions 62 to 70, which comprises using other therapeutic agent(s), such as RAS inhibitors, RTK inhibitors, EGFR inhibitors, immune checkpoint inhibitors for example PD-1 or PD-L1 inhibitors, and the like.

### **Pharmaceutical Compositions and Administration**

In another aspect, the present disclosure provides a pharmaceutical composition comprising a compound of the present disclosure (also referred to as the “active ingredient”) and pharmaceutically acceptable excipients. In some embodiments, the pharmaceutical composition comprises an effective amount of the compound of the present disclosure. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of the compound of the present disclosure. In some embodiments, the pharmaceutical composition comprises a prophylactically effective amount of the compound of the present disclosure.

Pharmaceutically acceptable excipients for use in the present disclosure refer to the non-toxic carriers, adjuvants or vehicles, which do not destroy the pharmacological activity of the compounds formulated together. Pharmaceutically acceptable carriers, adjuvants, or vehicles that can be used in the compositions of the present disclosure include (but are not limited to) ion exchangers, alumina, aluminum stearate, lecithin, serum proteins (such as human serum proteins), buffer substances (such as phosphate), glycine, sorbic acid, potassium sorbate, mixture of partial glycerides of saturated plant fatty acids, water, salts or electrolytes (such as protamine sulfate), disodium hydrogen phosphate, potassium hydrogen phosphate, sodium chloride, zinc salts, silica gel, magnesium trisilicate, polyvinyl pyrrolidone, cellulose-based substance, polyethylene glycol, sodium carboxymethyl cellulose, polyacrylate, wax, polyethylene-polyoxypropylene block polymers, polyethylene glycol and lanolin.

The present disclosure also includes kits (e.g., pharmaceutical packs). The kits provided may include a compound of the present disclosure, other therapeutic agent(s), and a first and a second containers (e.g., vials, ampoules, bottles, syringes, and/or dispersible packages or other suitable containers) containing the compound of the present disclosure and other therapeutic agent(s). In some embodiments, the provided kits can also optionally include a third container containing a pharmaceutically acceptable excipient for diluting or suspending the compound of the present disclosure and/or other therapeutic agent(s). In some embodiments, the compound of the present disclosure provided in the first container and other therapeutic agent(s) provided in the second container are combined to form a unit dosage form.

The pharmaceutical composition provided by the present disclosure can be administered by a variety of routes including, but not limited to, oral administration, parenteral administration, inhalation administration, topical administration, rectal administration, nasal administration, buccal administration, vaginal administration, administration by implant or other means of administration. For example, parenteral administration as used herein includes subcutaneous administration, intradermal administration, intravenous administration, intramuscular administration, intra-articular administration, intra-arterial administration, intrasynovial administration, intrasternal administration, intracerebroventricular administration, intralesional administration, and intracranial injection or infusion techniques.

Generally, the compounds provided herein are administered in an effective amount. The amount of the compound actually administered will typically be determined by a physician, in the light of the relevant circumstances, including the condition to be treated, the route of administration selected, the actual compound administered, the age, weight and response of the individual patient, the severity of the patient's symptoms, etc.

### **Indications and Methods of Treatment**

The cyclic peptide compounds disclosed herein can be used in treating cancer.

As used herein, "cancer" and "cancerous" refer to or describe the physiological condition in mammals that is typically characterized by unregulated cell growth/proliferation. Examples of cancer include solid tumors and blood cancers. Further examples of cancer include, but are not limited to, carcinoma, lymphoma, blastoma, sarcoma, and leukemia. More specific examples of such cancers are lung cancer, esophageal cancer, thyroid cancer, stomach cancer, liver cancer, colon cancer, uterine cancer, ovarian cancer, pancreatic cancer, kidney cancer, bladder cancer, thyroid cancer, skin cancer, leukemia, malignant lymphoma, and multiple myeloma. More specific examples of such cancers are brain tumor, squamous cell carcinoma, small cell lung cancer, non-small cell lung carcinoma, adenocarcinoma of the lung, squamous cell carcinoma of the lung, carcinoma of the peritoneum, hepatocellular carcinoma cancer, gastrointestinal cancer, glioma, cervical cancer, ovarian cancer, liver cancer, bladder cancer, hepatoma, breast cancer, colon cancer, rectal cancer, endometrial cancer, endometrial cancer, cervical cancer, salivary gland cancer, kidney cancer, liver cancer, prostate cancer, vulvar cancer, thyroid cancer, hepatocellular carcinoma, AML (acute myelogenous leukemia), CML (chronic myelogenous leukemia), ALL (acute lymphocytic leukemia), CLL (chronic lymphocytic leukemia), Hodgkin lymphoma, Non-Hodgkin lymphoma and other lymphoproliferative disorders, and various types of head and neck cancer. In one aspect, the "cancer" may be any cancer other than pancreatic cancer.

As used herein, "cell proliferative disorder" and "proliferative disorder" refer to disorders associated with some degree of abnormal cell proliferation. In one aspect, the cell proliferative disorder is cancer.

As used herein, "tumor" refers to all neoplastic cell growth and proliferation and all precancerous and cancerous cells and tissues, whether malignant or benign.

The terms "cancer", "cancerous", "cell proliferative disorder", "proliferative disorder" and "tumor" as used herein are not mutually exclusive.

In one aspect, the cancer includes cancers associated with Ras gene aberrations and/or Ras pathway overactivation. Cancers associated with abnormalities in the Ras gene include, for example, cancer cells with mutations in the coding region of the Ras gene and/or amplification of the copy number of the Ras gene, resulting in increased Ras activity (including an increase in production of Ras protein). A mutation in the Ras gene coding region refers to, for example, one or more base deletions, additions or substitutions in the Ras gene coding region. Amplification

of Ras gene copy number refers to the presence of a Ras gene with a higher copy number than the copy number of the Ras gene normally present on the chromosome. A large number of copies means a large number compared to normal cells, and a copy number of 4 or more can be said to be an amplified copy number. The cancer also includes cancers that are associated with RAS pathway overactivation due to overactivation of upstream signaling pathways like RTK pathways or abnormalities in genes involved in RAS pathway regulation.

Whether cancer is associated with Ras gene abnormality can be determined by confirming whether a subject suffering from cancer has Ras gene abnormality. For example, by identifying a mutation in the nucleotide sequence of the coding region of the Ras gene in a biological sample obtained from a subject suffering from cancer.

In addition, whether or not a cancer is a cancer associated with an abnormality of the Ras gene can be determined by confirming whether or not a subject suffering from cancer has copy number amplification of the Ras gene. For example, by confirming that the copy number of the Ras gene is amplified and/or the production of the Ras protein is increased in a biological sample obtained from a subject suffering from cancer, it is possible to determine whether the cancer is caused by an abnormality of the Ras gene. Increased production of Ras protein can also be referred to as overexpression of Ras protein. Whether the production of Ras protein is increased can be determined by, for example, comparing the expression level of Ras protein in general, the expression level of Ras protein in a subject not suffering from the cancer, or the expression level of Ras protein in a subject before suffering from cancer.

Any method can be used as long as it can detect a mutation in the coding region of the Ras gene or an amplification of the copy number of the Ras gene in a biological sample of a subject suffering from cancer. For example, a Ras gene may be isolated from a biological sample obtained from a cancer-affected subject, amplified by PCR or the like, and the nucleotide sequence thereof may be directly determined using a sequencer or the like. Alternatively, for example, commercially available in vitro diagnostic agents that can detect Ras gene mutations may be used.

As isoforms of Ras protein, three types of Kras protein, Nras protein, and Hras protein are known. In one aspect, the Ras gene is one or more Ras genes selected from the group consisting of Kras, Nras and Hras genes. Preferably, the Ras gene is a Kras gene.

In one aspect, cancers include cancers associated with mutated Ras protein production and/or increased Ras protein production. The production of mutant Ras proteins and/or increased production of Ras proteins usually results from the aforementioned Ras gene abnormalities.

In one aspect, the mutant Ras protein is one or more mutant Ras proteins selected from the group consisting of mutant Kras proteins, mutant Nras proteins, and mutant Hras proteins. Preferably, the mutant Ras protein is a mutant Kras protein.

In one aspect, the compounds disclosed herein can be used as therapeutic or prophylactic agents for diseases, preferably cancer, in subjects. In a further aspect, compounds disclosed herein are provided for use in the treatment or prevention of disease, preferably cancer.

In one aspect, the disclosure provides a method of treating or preventing a disease, preferably cancer. In one aspect, the method comprises administering to a subject with such disease, preferably cancer, an effective amount of a compound disclosed herein.

In one aspect, the disclosure provides use of a compound disclosed herein in the manufacture or preparation of a medicament. In one aspect, the medicament is for the treatment or prevention of disease, preferably cancer. In a further aspect, the medicament is of use in a method of treating or preventing a disease, preferably cancer, comprising administering to a subject with cancer an effective amount of a compound disclosed herein.

In one aspect, the compounds disclosed herein can be used as inhibitors of Ras in a subject. In one embodiment, inhibition of Ras can include inhibiting binding between Ras and other signaling molecules (eg, binding of Ras to RAF or SOS) and inhibiting intracellular signaling. In one embodiment, degradation of Ras can include decreasing the effective amount and concentration of overexpressed Ras.

In one aspect, the disclosure provides methods for inhibiting Ras in a subject. In one aspect, the method comprises decreasing the amount and concentration of Ras in a subject. In one aspect, the method comprises administering to the subject an effective amount of a compound disclosed herein to inhibit Ras. In one aspect, the method comprises administering to the subject an effective amount of a compound disclosed herein to degrade Ras. In one embodiment, inhibition of Ras can include, for example, inhibition of binding of Ras to SOS or RAF.

In a further aspect, the present disclosure provides compounds disclosed herein for use in inhibiting Ras and/or degrading Ras. In certain embodiments, the disclosure provides the compounds disclosed herein for inhibition of Ras and/or degrading Ras in a subject, comprising administering to said subject an effective amount of a compound disclosed herein.

A "subject" according to any of the above aspects is preferably a human.

Examples of Ras inhibition include inhibition of binding between Ras and SOS or other effector proteins such as RAF, PI3K or SHANK3.

### **Drug combinations**

The compounds disclosed herein can be used either alone or in combination with other agents. In one aspect, the above-described embodiments further comprise administering to the subject an effective amount of at least one additional therapeutic agent. For example, the compounds disclosed herein may be co-administered with at least one additional therapeutic agent.

Combination therapy, as described above, includes combined administration (where two or more therapeutic agents are included in the same or separate formulations), and separate

administration (where the compounds disclosed herein are administered separately from one or more therapeutic agents). Administration of the compound disclosed herein may be prior to, concurrently with, and/or after the administration of the additional therapeutic agent. In one aspect, administration of a compound disclosed herein and administration of an additional therapeutic agent are within about 1 month, or within about 1, 2, or 3 weeks, or about 1, 2, 3, 4, 5 or 6 days. The compounds disclosed herein can also be used in combination with radiation therapy.

The compounds disclosed herein (and any additional therapeutic agents) can be administered with any of the following routes, including oral, parenteral, pulmonary, nasal, and intralesional administration if local treatment is desired. Parenteral injection includes intramuscular, intravenous, intraarterial, intraperitoneal, or subcutaneous administration. Dosing may be by any suitable route, eg, by injection, such as intravenous or subcutaneous injection, depending in part on whether the administration is in a short or long term. Various dosing schedules are contemplated herein, including, but not limited to, single dose or repeated doses over various time points, bolus doses, and pulse infusions.

The compounds disclosed herein are formulated, dosed, and administered in a manner consistent with good medical practice. Factors to be considered in this regard are the particular disorder being treated, the particular mammal being treated, the clinical symptoms of the individual patient, the cause of the disorder, the site of delivery of the agent, the method of administration, the schedule, and other factors known to health care practitioners. The compounds are optionally, but not necessarily, formulated with one or more agents currently used to prevent or treat the disorder in question.

Effective amounts of such other agents will depend on the amount of compound present in the formulation, the type of disorder or treatment, and other factors discussed above.

An “effective amount” of an agent, a compound disclosed herein, and a medicament disclosed herein is an amount effective to achieve the desired therapeutic or prophylactic result.

## EXAMPLES

### Compound Preparation

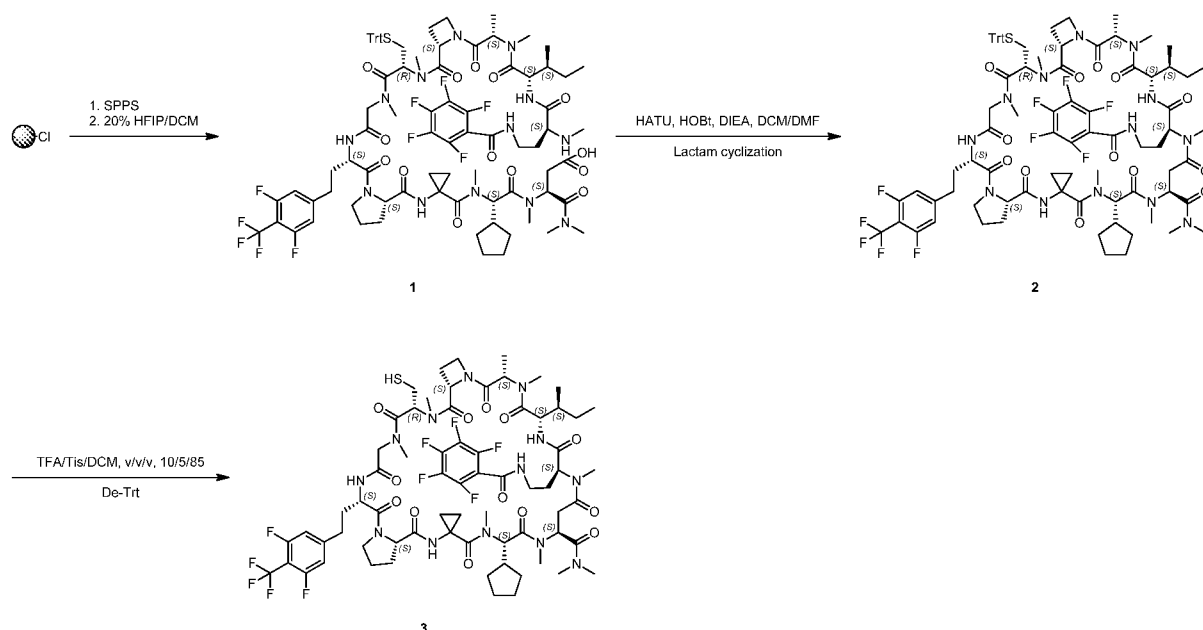
The compounds disclosed herein could be prepared by solid phase synthesis, which involves several difficult amide couplings. Procedures for completing difficult couplings such as these in the following scheme are described in the following references:

Journal of Medicinal Chemistry 2022 65 (19), 13401-13412

Journal of the American Chemical Society 2023 145 (30), 16610-16620

Journal of the American Chemical Society 2023 145 (44), 24035-24051

The general procedure to prepare compounds 1001, 1002, 1003, 1004, 1006, 1007, 1010, 1012, and 1018 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (20 mL) was added to the vessel containing 2-CTC Resin (0.20 mmol, 0.67 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Then the resin was drained. A solution of Fmoc-*N*-MeAsp-NMe<sub>2</sub> (80.0 mg, 0.20 mmol, 1.00 equiv.) in DCM (20 mL) was added to the resin, followed by addition of DIEA (103 mg, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Then MeOH (0.7 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (20 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was washed with DMF (20 mL \* 5). The de-protection reaction was monitored by ninhydrin test or chloranil test, completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: a solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.), and DIC (5.20 equiv.) in 50% NMP/DMF (20 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test, if the resin showed colorless, the coupling was complete. The reaction was then drained and washed with DMF (20 mL \* 5).
- 4) Steps 2 and 3 were repeated for the following amino acid couplings: (Entries 3~11)
- 5) De-Alloc: A solution of PhSiH<sub>3</sub> (10.00 equiv.) and Pd(PPh<sub>3</sub>)<sub>4</sub> (0.10 equiv.) in DCM (20 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 15 min \* 3 at 20 °C. The resin was washed with DMF (20 mL \* 5). The de-Alloc reaction was monitored by ninhydrin test, if it showed blue or brownish red, the reaction was completed.
- 6) Coupling: A solution of 2,3,4,5,6-pentafluorobenzoic acid (3.00 mmol, 3.00 equiv.), DIC (3.00 mmol, 3.00 equiv.) in DMF (20 mL) was added to the resin with N<sub>2</sub> bubbling for 30.0

min at 20 °C. The coupling reaction was monitored by ninhydrin test, if it showed colorless, the coupling was completed. The resin was then washed with DMF (20 mL \* 5).

- 7) After the last position completed, a solution of 2% DBU in DMF (20 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> flow for 10 min at 25 °C. The resin was washed with DMF (20 mL \* 5), MeOH (20 mL \* 3). The resin was drained and dried under vacuum.

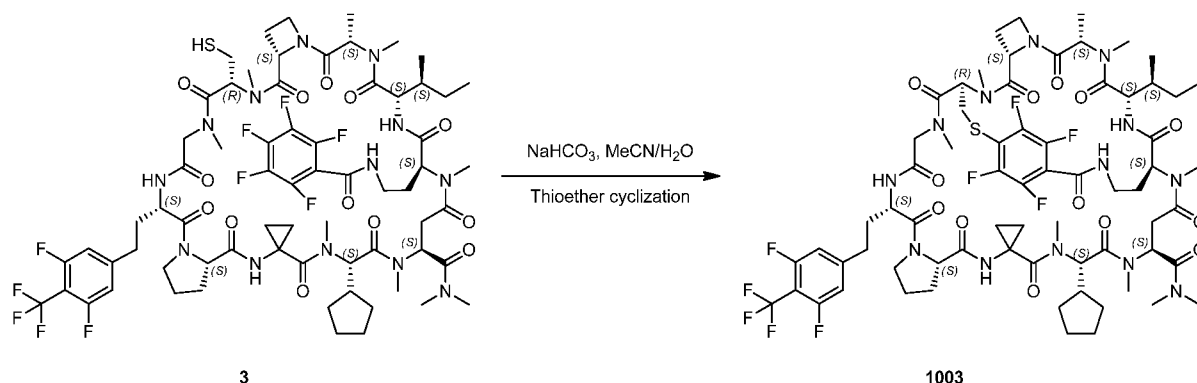
The scale of the synthesis was 0.20 mmol. For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| #  | Materials                                                             | Coupling reagents                        |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 1  | Fmoc- <i>N</i> -MeAsp-NMe <sub>2</sub> (1.00 equiv.)                  | DIEA (4.00 equiv.)                       |
| 2  | Fmoc- <i>N</i> -Me-Gly(cPent)-OH (3.60 equiv.)                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3  | Fmoc-1-NH <sub>2</sub> -1-cPro-OH (3.60 equiv.)                       | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc- <i>N</i> -Me-Cys(Trt)-OH (3.60 equiv.)                          | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 8  | Fmoc-Aze-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc- <i>N</i> -Me-Ala-OH (3.60 equiv.)                               | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc- <i>N</i> -Me-Dab(Alloc)-OH (1.50 equiv.)                        | HOAt (0.96 equiv.) and DIC (2.17 equiv.) |

Peptide cleavage, lactam cyclization and de-tritylation were conducted according to the following steps:

- 1) Cleavage cocktail (20% HFIP/DCM, 20 mL) was added to the flask containing the side chain protected peptide at room temperature and the mixture was bubbled with N<sub>2</sub> for 10 min (20 mL \* 3).
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure.
- 3) The residue was lyophilized to give the linear peptide compound **1** (0.32 g, crude) as a white solid.
- 4) Compound **1** (0.32 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (180 mL). Next, HATU (136 mg, 0.35 mmol, 2.00 equiv.), HOBt (49.0 mg, 0.35 mmol, 2.00 equiv.), and DIEA (93.0 mg, 0.71 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h.
- 5) The mixture was concentrated under reduced pressure to afford crude compound **2**.
- 6) The crude compound **2** was dissolved in a solution of TFA/Tis/DCM (v/v/v, 10/5/85, 20 mL), and the mixture was stirred at 25 °C for 20 min.
- 7) The mixture was concentrated under reduced pressure directly to give residue.
- 8) The residue was precipitated in isopropyl ether (cold, 50 mL) and centrifuged (3 mins at 3000 rpm). The resulting solid was dried under reduced pressure.

9) The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to give compound **3** (50.0 mg, 95.06% purity) as a white solid. The structure was confirmed by LCMS analysis (Rt = 0.600 min, [M+H]<sup>+</sup> = 1518.7).



To complete the synthesis, a solution of compound **3** (50.0 mg, 27.8  $\mu$ mol, 1.00 equiv.) was prepared using 50% ACN/H<sub>2</sub>O (28 mL) to which NaHCO<sub>3</sub> was added until the pH reached ~8. Next, the mixture was stirred at 25 °C for 2 h. LCMS showed compound **3** was consumed and one main peak with the desired MS (Rt = 0.543 min, [M+H]<sup>+</sup> = 1499.6) was detected. The reaction mixture was dried by lyophilization to give a residue. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford **1003** (11.9 mg, 98.8% purity, 3.3% yield) as a white solid.

**Purification Conditions:**

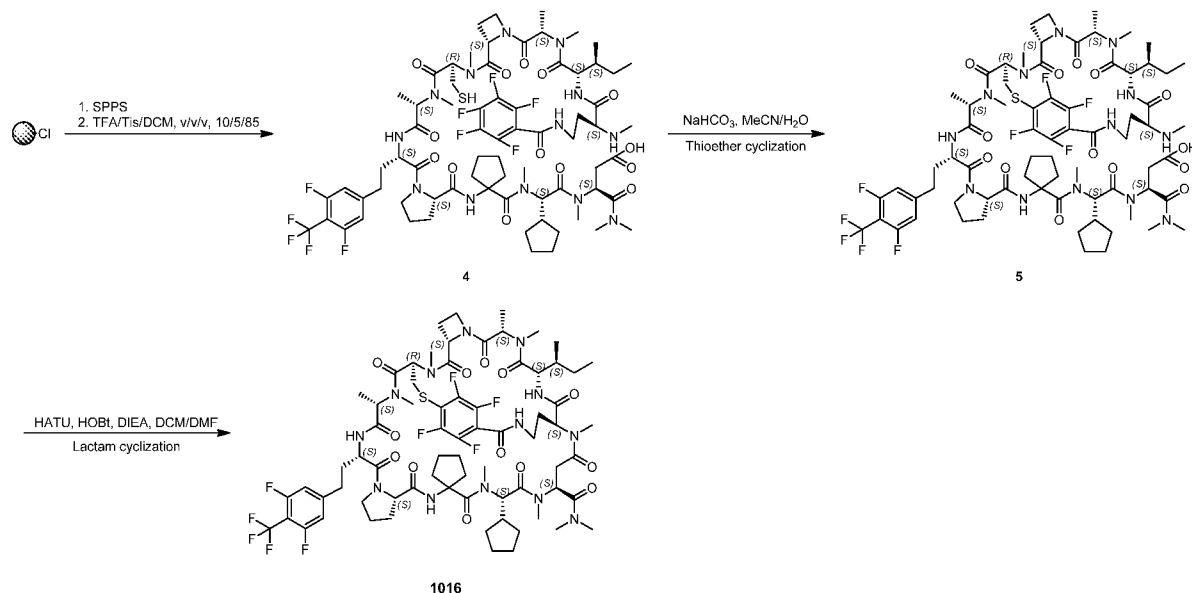
| Purification conditions |                                         |
|-------------------------|-----------------------------------------|
| Sample Preparation      | Dissolve in ACN/Water=2/5               |
| Instrument              | AUNO LC-2000                            |
| Mobile Phase            | A: 0.1% TFA in water                    |
|                         | B: ACN                                  |
| Gradient                | 38-90%-48 min. Retention time: 23.9 min |
| Column                  | YMC-Actus Triart C8, 30*150 mm, 10 μm   |
| Flow Rate               | 25 mL/min                               |
| Wavelength              | 220/254 nm                              |
| Oven Temperature        | Room temperature                        |

The same general procedure was used to prepare compounds 1001, 1002, 1004, 1006, 1007, 1010, 1012, and 1018 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity(%) | State | Color |
|-------------|-------------|-----------|-------|-------|
| <b>1007</b> | 3.3 mg      | 90.2      | solid | white |
| <b>1006</b> | 12.7 mg     | 97.9      | solid | white |
| <b>1002</b> | 12.2 mg     | 92.0      | solid | white |
| <b>1004</b> | 10.5 mg     | 94.4      | solid | white |
| <b>1001</b> | 8.0 mg      | 97.6      | solid | white |

|             |         |      |       |       |
|-------------|---------|------|-------|-------|
| <b>1018</b> | 1.5 mg  | 96.3 | solid | white |
| <b>1010</b> | 11.1 mg | 98.8 | solid | white |
| <b>1012</b> | 5.5 mg  | 94.4 | solid | white |

The general procedure to prepare compounds 1016, 1005, 1008, 1011, 1020, 1021, 1013, 1014, 1022, and 1023 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (20 mL) was added to the vessel containing 2-CTC Resin (0.20 mmol, 0.67 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-Me<sub>2</sub> (80.0 mg, 0.20 mmol, 1.00 equiv.) in DCM (20 mL) was added to the resin, followed by addition of DIEA (103 mg, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.7 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (20 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (20 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was washed with DMF (20 mL \* 5). The de-protection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.) and DIC (5.20 equiv.) in 50% NMP/DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was complete. The reaction was then drained and washed with DMF (20 mL \* 5).
- 4) Steps 2 and 3 were repeated for the following amino acid couplings: (Entries 3~11)
- 5) Alloc Deprotection: A solution of PhSiH<sub>3</sub> (10.00 equiv.) and Pd(PPh<sub>3</sub>)<sub>4</sub> (0.10 equiv.) in DCM (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 15 min \* 3 at 25 °C.

The resin was washed with DMF (20 mL \* 5). The de-Alloc reaction was monitored by ninhydrin test. If it showed blue or brownish red, the reaction was completed.

- 6) Coupling: A solution of 2,3,4,5,6-pentafluorobenzoic acid (0.60 mmol, 3.00 equiv.), DIC (0.60 mmol, 3.00 equiv.) in DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 30.0 min at 25 °C. The coupling reaction was monitored by ninhydrin test. If it showed colorless, the coupling was completed. The resin was then washed with DMF (20 mL \* 5).
- 7) After the last coupling was complete, a solution of 2% DBU in DMF (20 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> flow for 10 min at 25 °C. The resin was washed with DMF (20 mL \* 5), and MeOH (20 mL \* 3). The resin was drained and dried under vacuum.

For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

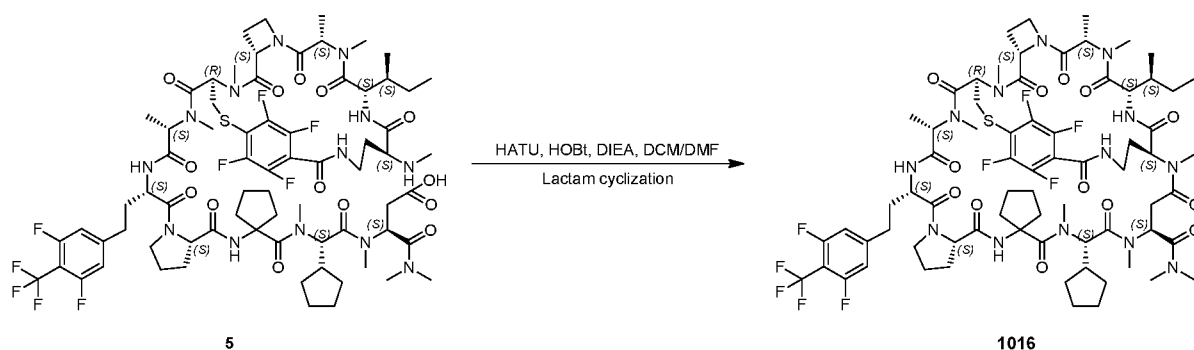
| #  | Materials                                                             | Coupling Reagents                        |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 1  | Fmoc- <i>N</i> -Me-Asp-NMe <sub>2</sub> (1.00 equiv.)                 | DIEA (4.00 equiv.)                       |
| 2  | Fmoc- <i>N</i> -Me-Gly(cPent)-OH (3.60 equiv.)                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3  | Fmoc- <i>c</i> Leu-OH (3.60 equiv.)                                   | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc- <i>N</i> -Me-Ala-OH (3.60 equiv.)                               | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc- <i>N</i> -Me-Cys(Trt)-OH (3.60 equiv.)                          | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 8  | Fmoc-Aze-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc- <i>N</i> -Me-Ala-OH (3.60 equiv.)                               | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc- <i>N</i> -Me-Dab(Alloc)-OH (1.50 equiv.)                        | HOAt (0.96 equiv.) and DIC (2.17 equiv.) |
| 12 | 2,3,4,5,6-pentafluorobenzoic acid (3.00 equiv.)                       | DIC (3.00 equiv.)                        |

Peptide cleavage and thioether cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (TFA/Tis/DCM, v/v/v, 10/5/85, 20 mL) was added to the flask containing the side chain protected peptide and the mixture was stirred at 25 °C for 10 min.
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give residue.
- 3) The residue was precipitated in isopropyl ether (cold, 50 mL) and centrifuged (3 mins at 3000 rpm). The solid was dried under reduced pressure to give the linear peptide compound **4** (0.10 g, crude) as a white solid.

To a solution of compound **4** (0.10 g, 64.2 μmol, 1.00 equiv.) in 50% ACN/H<sub>2</sub>O (65 mL) was added NaHCO<sub>3</sub> until the pH reached ~8. Next, the mixture was stirred at 25 °C for 16 h. LCMS showed compound **4** was consumed and one main peak with desired the MS (*R*<sub>t</sub> = 0.431 min, [M+H]<sup>+</sup> = 1558.9) was detected. To the reaction mixture was added 1 N HCl until the pH reached ~6. The mixture solution was dried by lyophilization to give a residue. Next, the residue was dissolved in DCM (100 mL). The mixture was washed with H<sub>2</sub>O (100 mL), brine (100 mL), and dried over Na<sub>2</sub>SO<sub>4</sub>. After filtration, the mixture was evaporated under reduced pressure to afford

compound **5** (0.21 g, crude) as a yellow oil. The residue was used in the next step directly without purification.



To complete the synthesis, compound **5** (0.21 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (140 mL). Next, HATU (106 mg, 0.42 mmol, 2.00 equiv.), HOBt (37.8 mg, 0.28 mmol, 2.00 equiv.), and DIEA (72.4 mg, 0.56 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h. LCMS showed compound **5** was consumed and one main peak with the desired MS ( $R_t = 0.380$  min,  $[M+H]^+ = 1563.6$ ) was detected. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford **1016** (14.7 mg, 92.8% purity, 4.4% yield) as a white solid.

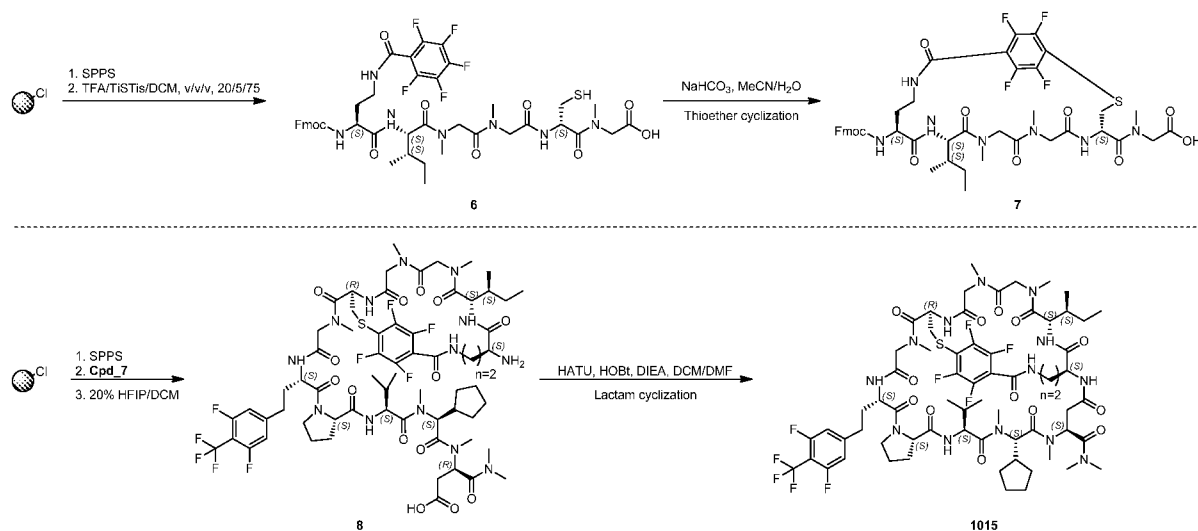
| Purification Conditions |                                         |
|-------------------------|-----------------------------------------|
| Sample Preparation      | Dissolve in DMF                         |
| Instrument              | AUNO LC-2000                            |
| Mobile Phase            | A: 0.1% TFA in water                    |
|                         | B: ACN                                  |
| Gradient                | 45-90%-48 min. Retention time: 27.5 min |
| Column                  | Phenomenex Luna 5um C8 100A 250*21.2mm  |
| Flow Rate               | 20 mL/min                               |
| Wavelength              | 220/254 nm                              |
| Oven Temp               | Room temperature                        |

The same general procedure was used to prepare compounds 1005, 1008, 1011, 1020, 1021,  
15 1013, 1014, 1022, and 1023 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity(%) | State | Color |
|-------------|-------------|-----------|-------|-------|
| <b>1005</b> | 11.2        | 90.3      | solid | white |
| <b>1008</b> | 11.0        | 93.7      | solid | white |
| <b>1011</b> | 11.8        | 87.8      | solid | white |
| <b>1020</b> | 8.34        | 91.7      | solid | white |
| <b>1021</b> | 26.4        | 96.4      | solid | white |
| <b>1013</b> | 16.4        | 91.2      | solid | white |

|             |      |      |       |       |
|-------------|------|------|-------|-------|
| <b>1014</b> | 19.8 | 95.1 | solid | white |
| <b>1022</b> | 10.7 | 90.1 | solid | white |
| <b>1023</b> | 1.19 | 95.2 | solid | white |

The general procedure to prepare compounds 1015, 1019 and 1017 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (50 mL) was added to the vessel containing 2-ChloroTrityl (2-CTC) Resin (2.50 mmol, 5.00 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-Sar-OH (778 mg, 2.50 mmol, 1.00 equiv.) in DCM (50 mL) was added to the resin, followed by addition of DIEA (0.43 mL, 2.50 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with  $\text{N}_2$  flow at 25 °C for 2 h. MeOH (5 mL) was added to the mixture and bubbled with  $\text{N}_2$  for 30 min. The resin was drained and washed with DMF (50 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 20% piperidine in DMF (10 mL) was added to the resin and the mixture was bubbled with  $\text{N}_2$  for 30 min at 25 °C. The resin was washed with DMF (50 mL \* 5). The de-protection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-Cys (Trt)-OH (3.00 equiv.), HATU (2.85 equiv.) and DIEA (6.00 equiv.) in DMF (40 mL) was added to the resin with  $\text{N}_2$  bubbling for 1 h at 20 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was complete. The reaction was drained and washed with DMF (40 mL \* 5).
- 4) Steps 2 and 3 were repeated for the following amino acid couplings: (Entries 3~4).
- 5) De-Alloc: A solution of  $\text{PhSiH}_3$  (10.0 equiv.) and  $\text{Pd}(\text{PPh}_3)_4$  (0.10 equiv.) in DCM (20 mL) was added to the resin. The mixture was bubbled with  $\text{N}_2$  for 15 min \* 3 at 20 °C. The resin was washed with DMF (50 mL \* 5). The de-Alloc reaction was monitored by ninhydrin test. If it showed blue or brownish red, the reaction was complete.

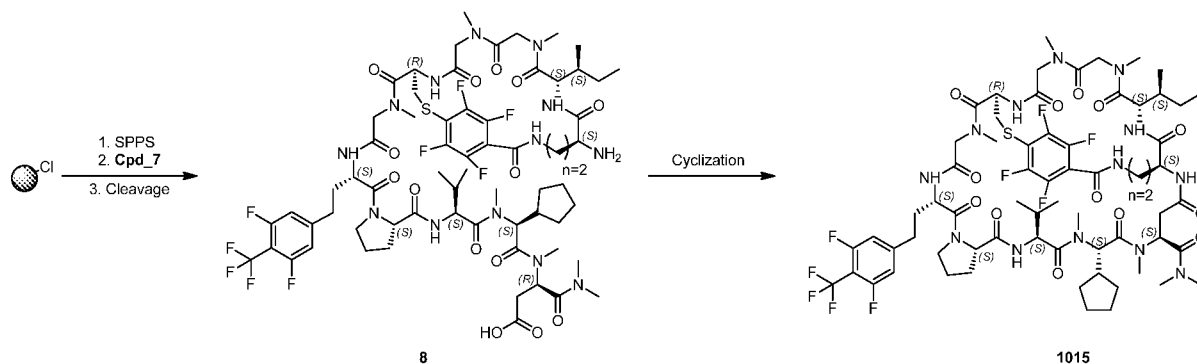
- 6) Coupling: A solution of 2,3,4,5,6-pentafluorobenzoic acid (3.00 mmol, 3.00 equiv.), DIC (3.00 mmol, 3.00 equiv.) in DMF (40 mL) was added to the resin with N<sub>2</sub> bubbling for 30 min at 20 °C. The coupling reaction was monitored by ninhydrin test. If it showed colorless, the coupling was complete. The resin was washed with DMF (40 mL \* 5), MeOH (40 mL \* 3) and then dried under vacuum.

The scale of the synthesis was 2.50 mmol. For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| # | Materials                                       | Coupling reagents                                                                      |
|---|-------------------------------------------------|----------------------------------------------------------------------------------------|
| 1 | Fmoc-Sar-OH (1.00 equiv.)                       | DIEA (4.00 equiv.)                                                                     |
| 2 | Fmoc-Cys(Trt)-OH (3.00 equiv.)                  | HATU (2.85 equiv.) and DIEA (6.00 equiv.)                                              |
| 3 | Fmoc-Sar-Sar-OH (3.00 equiv.)                   | HATU (2.85 equiv.) and DIEA (6.00 equiv.)                                              |
| 4 | Fmoc-Dab(Alloc)-OH (3.00 equiv.)                | HATU (2.85 equiv.) and DIEA (6.00 equiv.)                                              |
| 5 | De-Alloc                                        | PhSiH <sub>3</sub> (10.00 equiv.) and Pd(PPh <sub>3</sub> ) <sub>4</sub> (0.10 equiv.) |
| 6 | 2,3,4,5,6-pentafluorobenzoic acid (3.00 equiv.) | DIC (3 equiv.)                                                                         |

Peptide cleavage and macrocyclization were conducted according to the following steps:

- 1) Cleavage cocktail (TFA/Tis/DCM, v/v/v, 20/5/75) was added to the flask containing the side chain protected peptide at room temperature and stirred for 20 min.
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure.
- 3) The residue was precipitated in cold petroleum ether and dried under reduced pressure to give the linear peptide compound **6** (2.5 g, crude) as a white solid.
- 4) Compound **6** (2.5 g, crude) was dissolved in 30% ACN/ H<sub>2</sub>O (2.5 L). Next, the pH value of the mixture was adjusted to ~8 by addition of NaHCO<sub>3</sub>. The mixture was stirred at 20 °C for 5 h. LCMS showed the compound **6** (2.5 g, crude) was consumed and desired mass (R<sub>t</sub> = 1.586 min) was detected.
- 5) The reaction was quenched with 2 M HCl until pH = ~6. The mixture was filtered, and the filtrate was purified by RPLC (C18 Flash, A: 0.075% TFA in H<sub>2</sub>O, B: ACN) to afford compound **7** (500 mg, 99.1% purity, 21.2% yield) as a white solid.  
 LCMS: R<sub>t</sub> = 1.586 min, MS cal.: 943.3, MS observed: [M+H]<sup>+</sup> = 944.4  
 LCMS: R<sub>t</sub> = 1.520 min, MS cal.: 943.3, MS observed: [M+H]<sup>+</sup> = 944.4  
 HPLC: R<sub>t</sub> = 16.945 min, purity: 99.1%



To complete the synthesis, the remaining portion of the peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (5 mL) was added to the vessel containing 2-CTC resin (0.10 mmol, 0.33 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-isoAsp(NMe<sub>2</sub>)-OH (0.04 g, 0.10 mmol, 1.00 equiv.) in DCM (5 mL) was added to the resin, followed by addition of DIEA (52.0 mg, 0.40 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.3 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (5 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (5 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was washed with DMF (5 mL \* 5). The de-protection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.) and DIC (5.20 equiv.) in 50% NMP/DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was complete. The reaction was then drained and washed with DMF (5 mL \* 5).
- 4) Steps 2 and 3 were repeated for the following amino acid couplings: (Entries 3~6)
- 5) After coupling of the last amino acid, a solution of 2% DBU in DMF (5 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> flow for 10 min at 25 °C. The resin was washed with DMF (10 mL \* 5), MeOH (10 mL \* 3). The resin was drained and then dried under vacuum.

The scale of the synthesis was 0.10 mmol. For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| # | Materials                                                             | Coupling Reagents                        |
|---|-----------------------------------------------------------------------|------------------------------------------|
| 1 | Fmoc- <i>N</i> -Me-Asp-OAll (1.00 equiv.)                             | DIEA (4.00 equiv.)                       |
| 2 | Fmoc- <i>N</i> -Me-Gly(cPent)-OH (3.60 equiv.)                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3 | Fmoc-Val-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4 | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5 | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |

|   |                                 |                                          |
|---|---------------------------------|------------------------------------------|
| 6 | Compound <b>5</b> (1.50 equiv.) | HOAt (0.96 equiv.) and DIC (2.17 equiv.) |
|---|---------------------------------|------------------------------------------|

Peptide cleavage and macrocyclization were conducted according to the following steps:

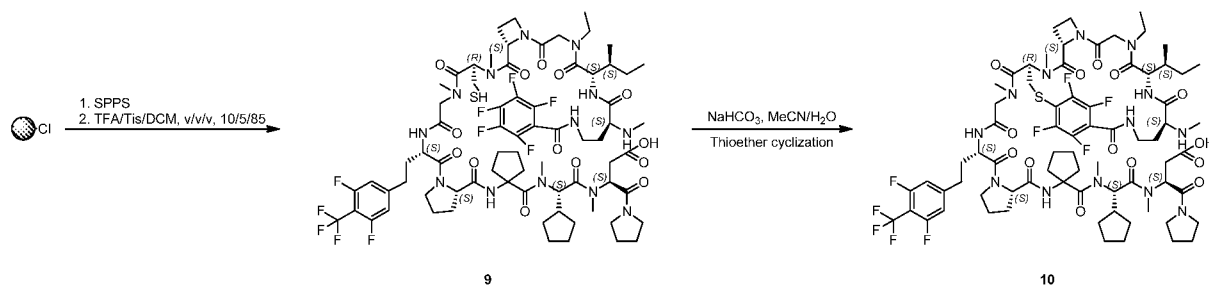
- 1) Cleavage cocktail (20% HFIP/DCM, 20 mL) was added to the flask containing the side chain protected peptide at room temperature and the mixture was bubbled under N<sub>2</sub> for 10 min (20 mL\*3).
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure.
- 3) The residue was lyophilized to give compound **8** (0.10 g, crude) as a white solid.
- 4) Compound **6** (0.10 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (65 mL). Next, HATU (49.4 mg, 0.13 mmol, 2.00 equiv.), HOBt (17.6 mg, 0.13 mmol, 2.00 equiv.), and DIEA (33.6 mg, 0.26 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h.
- 5) The mixture was concentrated under reduced pressure.
- 6) The residue was purified by prep-HPLC (A: 10 mmol NH<sub>4</sub>HCO<sub>3</sub> in H<sub>2</sub>O, B: ACN) to give **1015** (31.6 mg, 95.9% purity, 20.7% yield) as a white solid.

| Purification Conditions |                                                                |
|-------------------------|----------------------------------------------------------------|
| Sample Preparation      | Dissolve in ACN/Water=2/5                                      |
| Instrument              | AUNO LC-2000                                                   |
| Mobile Phase            | A: 0.1% TFA in water                                           |
|                         | B: ACN                                                         |
| Gradient                | 46-90%-48 min. Retention time: 25.9 min                        |
| Column                  | YMC-Actus C18,30 × 150 mm, 5 μm + Chenqiao BP-C18,5um,30×150mm |
| Flow Rate               | 25 mL/min                                                      |
| Wavelength              | 220/254 nm                                                     |
| Oven Temp               | Room temperature                                               |

The same general procedure was used to prepare compounds 1019 and 1017 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity(%) | State | Color |
|-------------|-------------|-----------|-------|-------|
| <b>1017</b> | 10.4 mg     | 90.1      | solid | white |
| <b>1019</b> | 28.3 mg     | 95.4      | solid | white |

The general procedure to prepare compounds 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1031, 1032, 1033, 1034 and 1035 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

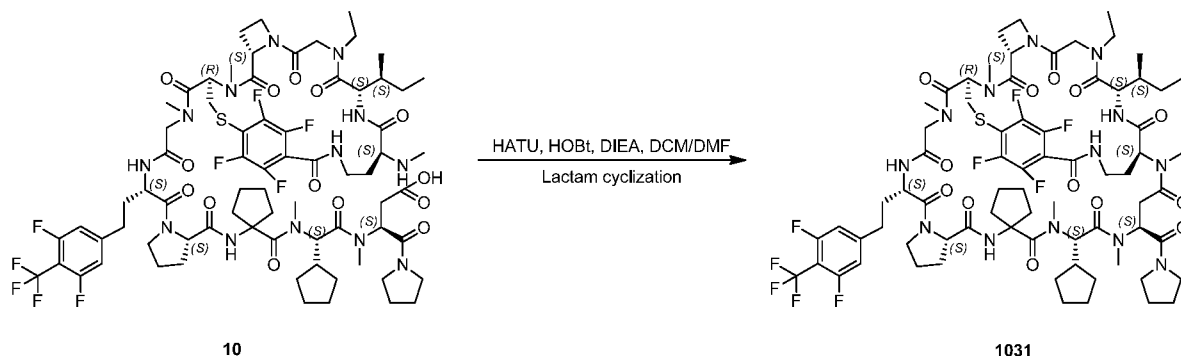
- 1) Resin preparation: DCM (20 mL) was added to the vessel containing 2-CTC Resin (0.20 mmol, 0.67 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-pyrrolidine (85.0 mg, 0.20 mmol, 1.00 equiv.) in DCM (20 mL) was added to the resin, followed by addition of DIEA (103 mg, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.7 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (20 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (20 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was washed with DMF (20 mL \* 5). The deprotection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.) and DIC (5.20 equiv.) in 50% NMP/DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was judged to be complete. The reaction was then drained and washed with DMF (20 mL \* 5).
- 4) Steps 2~3 was repeated for the following amino acids elongation: Numbers # 3-11, in the Table below.
- 5) De-Alloc: A solution of PhSiH<sub>3</sub> (10.00 equiv.) and Pd(PPh<sub>3</sub>)<sub>4</sub> (0.10 equiv.) in DCM (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 15 min \* 3 at 25 °C. The resin was washed with DMF (20 mL \* 5). The de-Alloc reaction was monitored by ninhydrin test. If the resin showed blue or brownish red, the reaction was judged to be complete.
- 6) Coupling: A solution of 2,3,4,5,6-pentafluorobenzoic acid (0.60 mmol, 3.00 equiv.), DIC (0.60 mmol, 3.00 equiv.) in DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 30.0 min at 25 °C. The coupling reaction was monitored by ninhydrin test. If the resin showed colorless, the coupling was judged to be complete. The resin was then washed with DMF (20 mL \* 5).
- 7) After the last amino acid was coupled, a solution of 2% DBU in DMF (20 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> flow for 10 min at 25 °C. The resin was then washed with DMF (20 mL \* 5), and MeOH (20 mL \* 3). The resin was drained and dried under vacuum.

For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| #  | Materials                                                             | Coupling Reagents                        |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 1  | Fmoc-N-Me-Asp-pyrrolidine (1.00 equiv.)                               | DIEA (4.00 equiv.)                       |
| 2  | Fmoc-N-Me-Gly(cPent)-OH (3.60 equiv.)                                 | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3  | Fmoc-cLeu-OH (3.60 equiv.)                                            | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc-N-Me-Cys(Trt)-OH (3.60 equiv.)                                   | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 8  | Fmoc-Aze-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc-N-Et-Gly-OH (2.00 equiv.)                                        | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc-N-Me-Dab(Alloc)-OH (1.50 equiv.)                                 | HOAt (0.96 equiv.) and DIC (2.17 equiv.) |
| 12 | 2,3,4,5,6-pentafluorobenzoic acid (3.00 equiv.)                       | DIC (3.00 equiv.)                        |

Peptide cleavage and thioether cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (TFA/TIS/DCM, v/v/v, 10/5/85, 20 mL) was added to the flask containing the side chain protected peptide and the mixture was stirred at 25 °C for 10 min.
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give a residue.
- 3) The residue was precipitated in cold isopropyl ether (50 mL) and centrifuged (3 mins at 3000 rpm). The solid was dried under reduced pressure to give the linear peptide compound **9** (0.10 g, crude) as an off-white solid.
- 4) To a solution of compound **9** (0.10 g, 62.8 μmol, 1.00 equiv.) in 50% ACN/H<sub>2</sub>O (63 mL) was added NaHCO<sub>3</sub> until the pH reached ~8. The mixture was then stirred at 25 °C for 16 h. LCMS showed compound **9** was consumed and one main peak with the desired MS (Rt = 0.433 min) was detected. To the reaction mixture was added 1 N HCl until the pH reached ~6. The mixture solution was dried by lyophilization to give crude compound **10**.
- 5) The crude compound **10** was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **10** (40.0 mg, 10.6% yield, 83.7% purity) as a white solid.



To complete the synthesis, Compound **10** (40.0 mg, 1.00 equiv.) was dissolved in 50% DCM/DMF (25 mL). Next, HATU (19.5 mg, 0.05 mmol, 2.00 equiv.), HOBT (6.90 mg, 0.05

mmol, 2.00 equiv.), and DIEA (13.0 mg, 0.10 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h. LCMS showed compound **10** was consumed and two peaks with the desired MS (Peak1: Rt = 0.416 min, Peak2: Rt = 0.536 min) were detected. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford **1031 (Peak1)** (19.9 mg, 92.9% purity, 5.9% yield) as a white solid. LCMS: Rt= 1.331 min, MS cal.: 1552.67, MS observed: [M+H]<sup>+</sup> = 1552.8. HPLC: Rt= 8.867 min, purity: 92.9%.

| Purification Conditions |                                         |
|-------------------------|-----------------------------------------|
| Sample Preparation      | ACN/Water = 4/5                         |
| Instrument              | AUNO LC-2000                            |
| Mobile Phase            | A: 0.1% TFA in water                    |
|                         | B: ACN                                  |
| Gradient                | 41-90%-43 min. Retention time: 26.7 min |
| Column                  | YMC-Actus Triart C8, 30*150 mm, 10 μm   |
| Flow Rate               | 25 mL/min                               |
| Wavelength              | 220/254 nm                              |
| Oven Temp.              | Ambient                                 |

The same general procedure was used to prepare compounds 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1032, 1033, 1034 and 1035.

HPLC Method for 1024, 1025, 1026, 1027, 1033 and 1034:

| HPLC Method                |                                                    |                   |                   |
|----------------------------|----------------------------------------------------|-------------------|-------------------|
| Instrument                 | Thermo Scientific VANQUISH CORE                    |                   |                   |
| Mobile Phase               | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                            | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (55-85-20 min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                            | 0                                                  | 45                | 55                |
|                            | 20                                                 | 15                | 85                |
|                            | 20.1                                               | 10                | 90                |
|                            | 22                                                 | 10                | 90                |
| Column                     | Gemini 5um C18 110A 150*4.6mm                      |                   |                   |
| Flow Rate                  | 1 mL/min                                           |                   |                   |
| Wavelength                 | 220&254 nm                                         |                   |                   |
| Oven Temp.                 | 50 °C                                              |                   |                   |

HPLC Method for 1028, 1030, 1032 and 1035:

| HPLC Method |
|-------------|
|-------------|

|                            |                                                    |                   |                   |
|----------------------------|----------------------------------------------------|-------------------|-------------------|
| Instrument                 | Thermo Scientific VANQUISH CORE                    |                   |                   |
| Mobile Phase               | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                            | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (65-95-20 min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                            | 0                                                  | 35                | 65                |
|                            | 20                                                 | 5                 | 95                |
|                            | 20.1                                               | 10                | 90                |
|                            | 22                                                 | 10                | 90                |
| Column                     | Gemini 5um C18 110A 150*4.6mm                      |                   |                   |
| Flow Rate                  | 1 mL/min                                           |                   |                   |
| Wavelength                 | 220&254 nm                                         |                   |                   |
| Oven Temp.                 | 50 °C                                              |                   |                   |

HPLC Method for 1029:

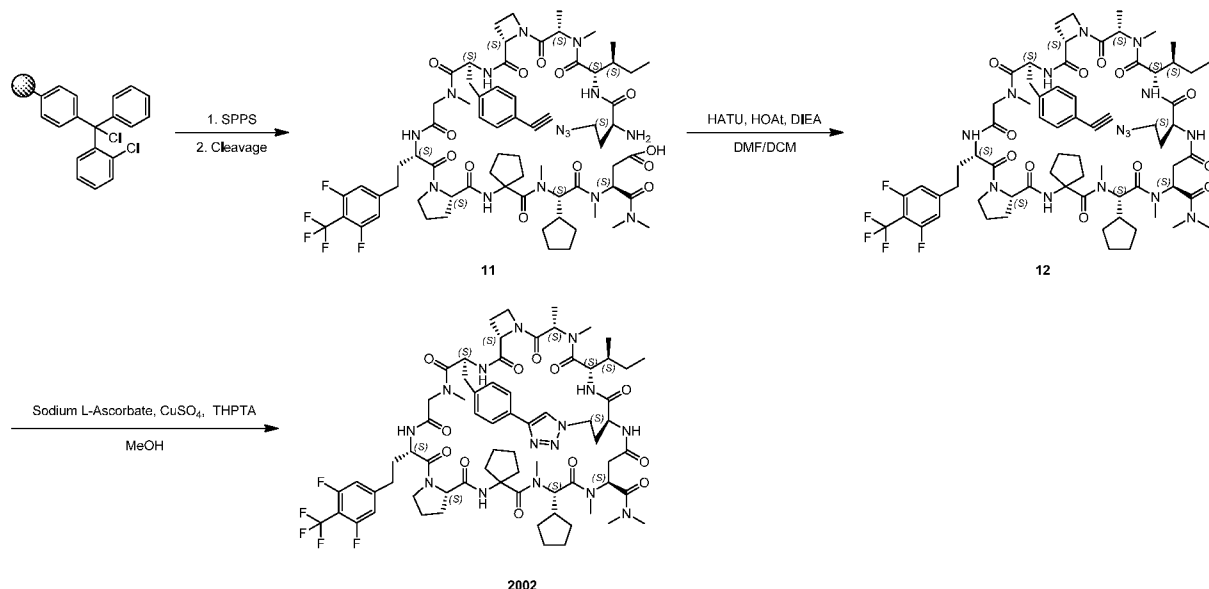
| HPLC Method                |                                                    |                   |                   |
|----------------------------|----------------------------------------------------|-------------------|-------------------|
| Instrument                 | Thermo Scientific VANQUISH CORE                    |                   |                   |
| Mobile Phase               | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                            | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (65-95-20 min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                            | 0                                                  | 35                | 65                |
|                            | 20                                                 | 5                 | 95                |
|                            | 20.1                                               | 10                | 90                |
|                            | 22                                                 | 10                | 90                |
| Column                     | Gemini 5um C18 110A 150*4.6mm                      |                   |                   |
| Flow Rate                  | 1 mL/min                                           |                   |                   |
| Wavelength                 | 220 & 254 nm                                       |                   |                   |
| Oven Temp.                 | 50 °C                                              |                   |                   |

The same general procedure was used to prepare compounds 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1032, 1033, 1034 and 1035 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity(%) | State | Color |
|-------------|-------------|-----------|-------|-------|
| <b>1024</b> | 12.6        | 93.6      | solid | white |
| <b>1025</b> | 2.04        | 85.2      | solid | white |
| <b>1026</b> | 15.4        | 98.6      | solid | white |
| <b>1027</b> | 5.53        | 86.9      | solid | white |
| <b>1028</b> | 3.69        | 95.8      | solid | white |
| <b>1029</b> | 2.82        | 87.7      | solid | white |
| <b>1030</b> | 3.88        | 85.2      | solid | white |
| <b>1032</b> | 11.0        | 96.7      | solid | white |
| <b>1033</b> | 14.2        | 96.3      | solid | white |

|             |      |      |       |       |
|-------------|------|------|-------|-------|
| <b>1034</b> | 19.1 | 95.2 | solid | white |
| <b>1035</b> | 13.2 | 97.5 | solid | white |

The general procedure to prepare compounds 2001, 2002, 2004, 2005, 2007, 2009, 2012, 2016, 2017, and 2019 is as follows:



- 5 The peptide was synthesized using standard Fmoc chemistry according to the following steps.
  - 1) Resin preparation: DCM (15 mL) was added to the vessel containing 2-chlorotrityl (2-CTC) resin (0.20 mmol, 0.18 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-NMe<sub>2</sub> (79.3 mg, 0.20 mmol, 1.00 equiv.) in DCM (15 mL) was added to the resin, followed by addition of DIEA (0.13 mL, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.2 mL) was added to the mixture and the solution was bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (15 mL \* 5) before proceeding to the next step.
  - 10 2) Deprotection: 2% DBU in DMF (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (15 mL \* 5). The deprotection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
  - 15 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.), and DIC (5.20 equiv.) in 50% NMP/DMF (15 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was complete. The reaction was then drained and washed with DMF (15 mL \* 5).
  - 20 4) Steps 2 and 3 were repeated for the following amino acid couplings: (Entries 3~11)
  - 25 5) After coupling of the last amino acid, a solution of 2% DBU in DMF (15 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then

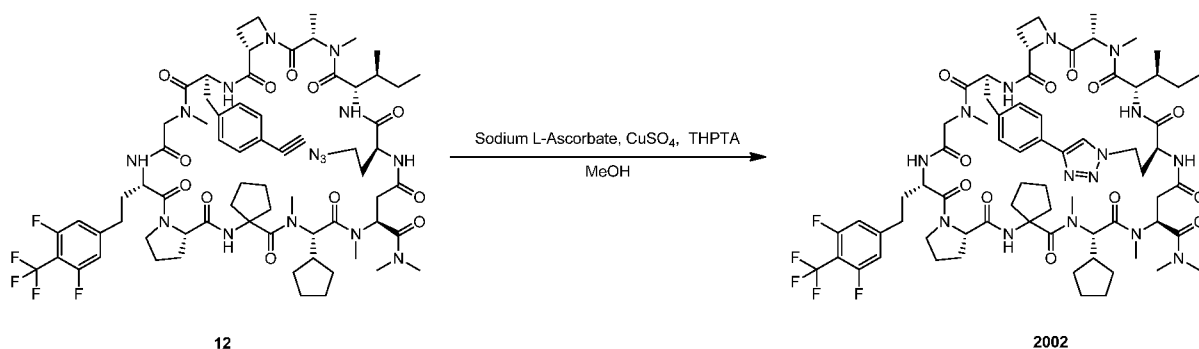
washed with DMF (30 mL \* 5), and MeOH (30 mL \* 3). The resin was drained and then dried under vacuum.

The scale of the synthesis was 0.20 mmol. For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| #  | Materials                                                             | Coupling Reagents                        |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 1  | Fmoc-N-Me-Asp-NMe <sub>2</sub> (1.00 equiv.)                          | DIEA (4.00 equiv.)                       |
| 2  | Fmoc-N-Me-Gly(cPent)-OH (3.60 equiv.)                                 | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3  | Fmoc-cLeu-OH (3.60 equiv.)                                            | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc-Phe(4-Acetylene)-OH (3.60 equiv.)                                | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 8  | Fmoc-Aze-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc-N-Me-Ala-OH (3.60 equiv.)                                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc-Dab(N <sub>3</sub> )-OH (3.60 equiv.)                            | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |

Peptide cleavage and macrocyclization were conducted according to the following steps:

- 1) Cleavage cocktail (20% HFIP/DCM, 15 mL) was added to the flask containing the side chain protected peptide at room temperature and the mixture was bubbled with N<sub>2</sub> for 10 min (15 mL\*3).
- 2) The mixture was then filtered. The filtrate was collected and concentrated under reduced pressure to give compound **11** (0.195 g, crude) as a white solid.
- 3) Compound **11** (0.195 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (135 mL). Next, HATU (103 mg, 0.271 mmol, 2.00 equiv.), HOBT (36.6 mg, 0.271 mmol, 2.00 equiv.), and DIEA (35.0 mg, 0.543 mmol, 4.00 equiv.) were added to the mixture. The mixture was then stirred at 25 °C for 2 h.
- 4) The mixture was concentrated under reduced pressure directly to give crude compound **12**.
- 5) The crude compound **12** was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **12** (50.0 mg, 97.8% purity, 17.6% yield) as a white solid.



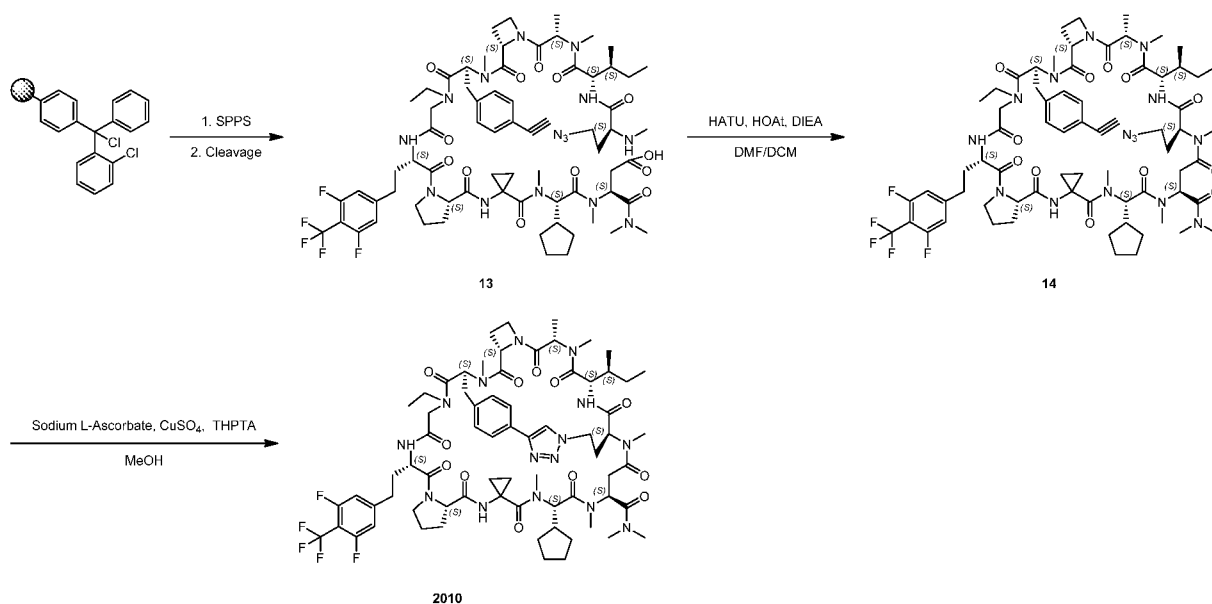
To complete the synthesis, a fresh mixture of sodium *L*-Ascorbate (0.40 M, 1.76 mL, 20.00 equiv.), CuSO<sub>4</sub> (0.40 M, 440 μL, 5.00 equiv.), and THPTA (76.5 mg, 176 μmol, 5.00 equiv.) was added to a solution of compound **12** (50.0 mg, 35.2 μmol, 1.00 equiv.) in MeOH (35 mL) under N<sub>2</sub> atmosphere at 0 °C. Next, the mixture was stirred at 0 °C for 1 h. LCMS indicated compound **12** was consumed and the desired MS (*R*<sub>t</sub> = 0.335 min, [M+2H]<sup>2+</sup> = 710.3) was detected. The mixture was then purified directly by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford **2002** (7.80 mg, 5.26 μmol, 14.9% yield, 91.0% purity) as a white solid.

| Purification Conditions |                                         |
|-------------------------|-----------------------------------------|
| Sample Preparation      | Dissolve in DMF                         |
| Instrument              | AUNO LC-2000                            |
| Mobile Phase            | A: 0.1% TFA in water                    |
|                         | B: ACN                                  |
| Gradient                | 47-90%-48 min. Retention time: 28.3 min |
| Column                  | YMC C8 250*30mm 5um 110A                |
| Flow Rate               | 25 mL/min                               |
| Wavelength              | 220/254 nm                              |
| Oven Temp               | Room temperature                        |

The same general procedure was used to prepare compounds 2001, 2004, 2005, 2007, 2009, 2012, 2016, 2017, and 2019 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity (%) | State | Color |
|-------------|-------------|------------|-------|-------|
| <b>2019</b> | 10.4        | 99.4       | solid | white |
| <b>2009</b> | 13.0        | 98.1       | solid | white |
| <b>2012</b> | 14.9        | 98.2       | solid | white |
| <b>2001</b> | 4.6         | 90.0       | solid | white |
| <b>2004</b> | 7.4         | 97.8       | solid | white |
| <b>2005</b> | 8.8         | 95.1       | solid | white |
| <b>2016</b> | 6.4         | 97.1       | solid | white |
| <b>2017</b> | 10.6        | 96.2       | solid | white |
| <b>2007</b> | 4.1         | 95.0       | solid | white |

The general procedure to prepare compounds 2011, 2013, 2015, 2020, 2003, 2014, 2006, 2008, 2021, 2010, 2018, 2031, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029 and 2030 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

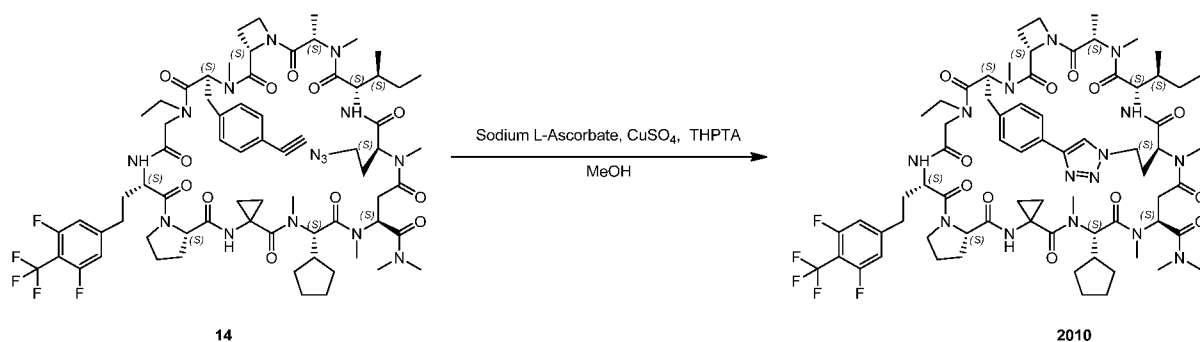
- 1) Resin preparation: DCM (15 mL) was added to the vessel containing 2-CTC Resin (0.20 mmol, 0.18 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-NMe<sub>2</sub> (79.3 mg, 0.20 mmol, 1.00 equiv.) in DCM (15 mL) was added to the resin, followed by addition of DIEA (0.13 mL, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.2 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (15 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (15 mL \* 5). The deprotection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.), and DIC (5.20 equiv.) in 50% NMP/DMF (15 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was complete. The reaction was then drained and washed with DMF (15 mL \* 5).
- 4) Steps 2 and 3 were repeated for the following amino acid coupling: Entries 3-11 in the table below.
- 5) After coupling of the last amino acid, a solution of 2% DBU in DMF (15 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (30 mL \* 5), and MeOH (30 mL \* 3). The resin was drained and then dried under vacuum.

| # | Materials                                             | Coupling Reagents                        |
|---|-------------------------------------------------------|------------------------------------------|
| 1 | Fmoc- <i>N</i> -Me-Asp-NMe <sub>2</sub> (1.00 equiv.) | DIEA (4.00 equiv.)                       |
| 2 | Fmoc- <i>N</i> -Me-Gly(cPent)-OH (3.60 equiv.)        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |

|    |                                                                       |                                          |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 3  | Fmoc-1-NH <sub>2</sub> -1-cPro-Acid-OH (3.60 equiv.)                  | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc-N-Et-Gly-OH (3.60 equiv.)                                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc-N-Me-Phe(4-Acetylene)-OH (3.60 equiv.)                           | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 8  | Fmoc-Aze-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc-N-Me-Ala-OH (3.60 equiv.)                                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc-N-Me-Dab(N <sub>3</sub> )-OH (2.00 equiv.)                       | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |

Peptide cleavage and cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (20% HFIP/DCM, 15 mL) was added to the flask containing the side chain protected peptide at room temperature and the mixture was bubbled with N<sub>2</sub> for 10 min (15 mL\*3).
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give compound **13** (0.180 g, crude) as a white solid.
- 3) Compound **13** (0.180 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (125 mL). Next, HATU (94.3 mg, 0.248 mmol, 2.00 equiv.), HOBT (33.5 mg, 0.248 mmol, 2.00 equiv.), and DIEA (64.0 mg, 0.496 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h.
- 4) The mixture was concentrated under reduced pressure directly to give crude compound **14**.
- 5) The crude compound **14** was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **14** (70.0 mg, 39.4% yield, 83.8% purity) as a white solid.



To complete the synthesis, a fresh mixture of sodium *L*-Ascorbate (0.40 M, 2.44 mL, 20.0 equiv.), CuSO<sub>4</sub> (0.40 M, 611 μL, 5.00 equiv.), and THPTA (106 mg, 244 μmol, 5.00 equiv.) was added to a solution of compound **14** (70.0 mg, 48.8 μmol, 1.00 equiv.) in MeOH (50 mL) under N<sub>2</sub> atmosphere at 0 °C. Next, the mixture was stirred at 0 °C for 1 h. LCMS indicated compound **14** was consumed and the desired mass (*R*<sub>t</sub> = 0.345 min, [M+H]<sup>+</sup> = 1433.0) was detected. The mixture was concentrated under reduced pressure. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) directly to afford **2010** (15.5 mg, 10.8 μmol, 22.1% yield, 96.2% purity) as a white solid.

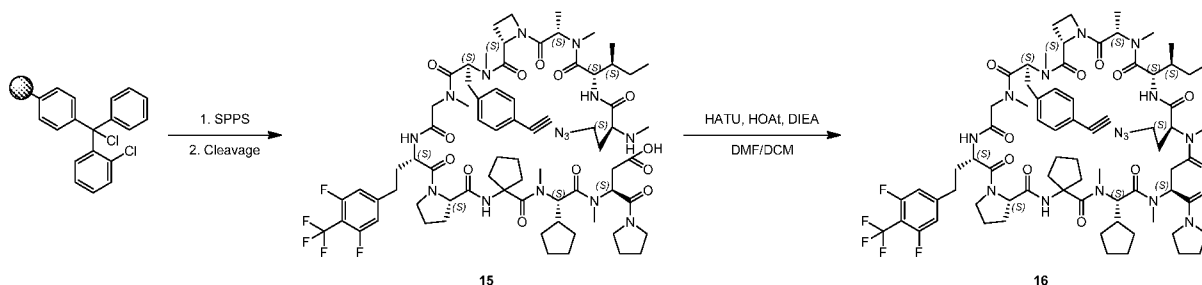
| Purification Conditions |                                         |    |    |
|-------------------------|-----------------------------------------|----|----|
| Instrument              | AUNO LC-2000                            |    |    |
| Column                  | Phenomenex Luna C8, 21.2 × 250 mm, 5 μm |    |    |
| Wavelength              | 220 nm&254 nm                           |    |    |
| Column Oven Temperature | Ambient                                 |    |    |
| Flow Rate               | 20.0 mL/min                             |    |    |
| Injector Volume         | 8 mL                                    |    |    |
| Diluent                 | ACN/Water=1/4                           |    |    |
| Mobile Phase A          | 0.1% TFA in water                       |    |    |
| Mobile Phase B          | ACN                                     |    |    |
| Gradient Program        | Time (min)                              | A% | B% |
|                         | 0                                       | 64 | 36 |
|                         | 40                                      | 34 | 66 |
|                         | 40.1                                    | 10 | 90 |
|                         | 45                                      | 10 | 90 |
|                         | 45.1                                    | 90 | 10 |
|                         | 50                                      | 90 | 10 |
| Retention Time          | 31 min                                  |    |    |

The same general procedure was used to prepare compounds compounds 2011, 2013, 2015, 2020, 2003, 2014, 2006, 2008, 2021, 2018, 2031, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029 and 2030 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity (%) | State | Color |
|-------------|-------------|------------|-------|-------|
| <b>2011</b> | 19.9        | 99.3       | solid | white |
| <b>2013</b> | 14.8        | 99.3       | solid | white |
| <b>2015</b> | 11.7        | 99.5       | solid | white |
| <b>2020</b> | 3.00        | 66.9       | solid | white |
| <b>2003</b> | 1.51        | 96.4       | solid | white |
| <b>2014</b> | 19.0        | 98.3       | solid | white |
| <b>2006</b> | 2.94        | 98.5       | solid | white |
| <b>2008</b> | 1.95        | 98.6       | solid | white |
| <b>2021</b> | 15.6        | 96.3       | solid | white |
| <b>2018</b> | 14.3        | 97.8       | solid | white |
| <b>2031</b> | 11.4        | 96.8       | solid | white |
| <b>2022</b> | 11.0        | 97.8       | solid | white |
| <b>2023</b> | 16.2        | 96.9       | solid | white |
| <b>2024</b> | 15.7        | 97.7       | solid | white |
| <b>2025</b> | 15.2        | 99.1       | solid | white |
| <b>2026</b> | 14.9        | 96.3       | solid | white |
| <b>2027</b> | 12.8        | 96.3       | solid | white |
| <b>2028</b> | 13.5        | 98.5       | solid | white |

|      |      |      |       |       |
|------|------|------|-------|-------|
| 2029 | 15.6 | 97.7 | solid | white |
| 2030 | 18.8 | 99.3 | solid | white |

The general procedure to prepare compounds 2032, 2033 and 2034 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (15 mL) was added to the vessel containing 2-CTC resin (0.20 mmol, 0.18 g, Sub = 1.10 mmol/g, 1.00 equiv.), and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-NMe<sub>2</sub> (79.3 mg, 0.20 mmol, 1.00 equiv.) in DCM (15 mL) was added to the resin, followed by addition of DIEA (0.13 mL, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.2 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (15 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (15 mL \* 5). The deprotection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.), and DIC (5.20 equiv.) in 50% NMP/DMF (15 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was judged to be complete. The reaction was then drained and washed with DMF (15 mL \* 5).
- 4) Steps 2~3 were repeated for the following amino acids elongation: Numbers # 3-11, in the Table below.
- 5) After coupling of the last amino acid, a solution of 2% DBU in DMF (15 mL) was added to the vessel, and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (30 mL \* 5), and MeOH (30 mL \* 3). The resin was drained and then dried under vacuum.

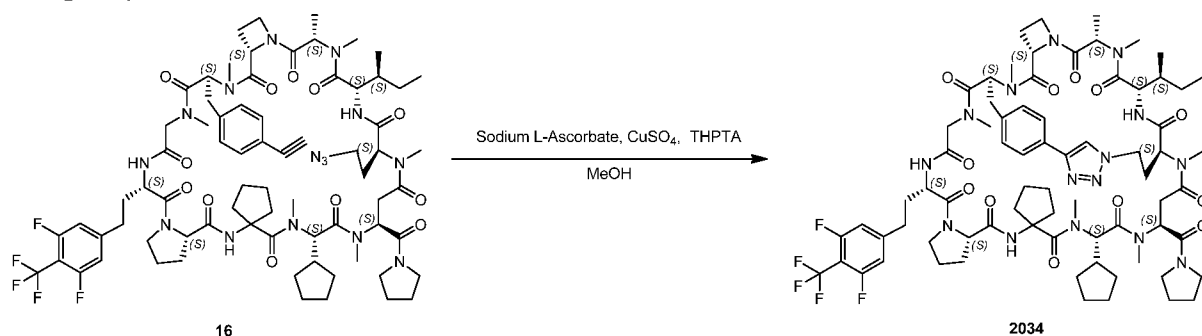
For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| # | Materials                                             | Coupling Reagents                        |
|---|-------------------------------------------------------|------------------------------------------|
| 1 | Fmoc- <i>N</i> -Me-Asp-NMe <sub>2</sub> (1.00 equiv.) | DIEA (4.00 equiv.)                       |
| 2 | Fmoc- <i>N</i> -Me-Gly(cPent)-OH (3.60 equiv.)        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3 | Fmoc-cLeu-OH (3.60 equiv.)                            | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |

|    |                                                                       |                                          |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc-N-Me-Phe(4-Acetylene)-OH (3.60 equiv.)                           | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 8  | Fmoc-Aze-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc-N-Me-Ala-OH (3.60 equiv.)                                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc-N-Me-Dab(N <sub>3</sub> )-OH (2.00 equiv.)                       | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |

Peptide cleavage and cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (20% HFIP/DCM, 15 mL) was added to the flask containing the side chain protected peptide at room temperature and the mixture was bubbled with N<sub>2</sub> for 10 min (15 mL\*3).
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give compound **15** (0.216 g, crude) as a white solid.
- 3) Compound **15** (0.216 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (150 mL). Next, HATU (110 mg, 0.290 mmol, 2.00 equiv.), HOBT (39.1 mg, 0.290 mmol, 2.00 equiv.), and DIEA (74.8 mg, 0.580 mmol, 4.00 equiv.) were added to the mixture. The mixture was then stirred at 25 °C for 2 h.
- 4) The mixture was concentrated under reduced pressure directly to give crude compound **16**.
- 5) The crude compound **16** was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **16** (103 mg, 48.3% yield, 94.7% purity) as a white solid. LCMS: R<sub>t</sub>= 0.404 min, MS cal.: 1471.74, MS observed: [M+H]<sup>+</sup> = 1473.0. HPLC: R<sub>t</sub>= 13.102 min, purity: 94.71%.



To complete the synthesis, a fresh mixture of sodium *L*-Ascorbate (0.40 M, 3.50 mL, 20.0 equiv.), CuSO<sub>4</sub> (0.40 M, 874 μL, 5.00 equiv.), and THPTA (152 mg, 350 μmol, 5.00 equiv.) were added to a solution of compound **16** (103 mg, 69.9 μmol, 1.00 equiv.) in MeOH (70 mL) under N<sub>2</sub> atmosphere at 0 °C. The mixture was then stirred at 0 °C for 1 h. LCMS indicated compound **16** was consumed and the desired MS (R<sub>t</sub> = 0.403 min, [M+H]<sup>+</sup> = 1473.5) was detected. The mixture was concentrated under reduced pressure. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) directly to afford **2034** (10.9 mg, 7.40 μmol, 10.6% yield, 96.7%

purity) as a white solid. LCMS:  $R_t = 1.208$  min, MS cal.: 1471.74, MS observed:  $[M+H]^+ = 1472.9$ . HPLC:  $R_t = 11.178$  min, purity: 96.74%.

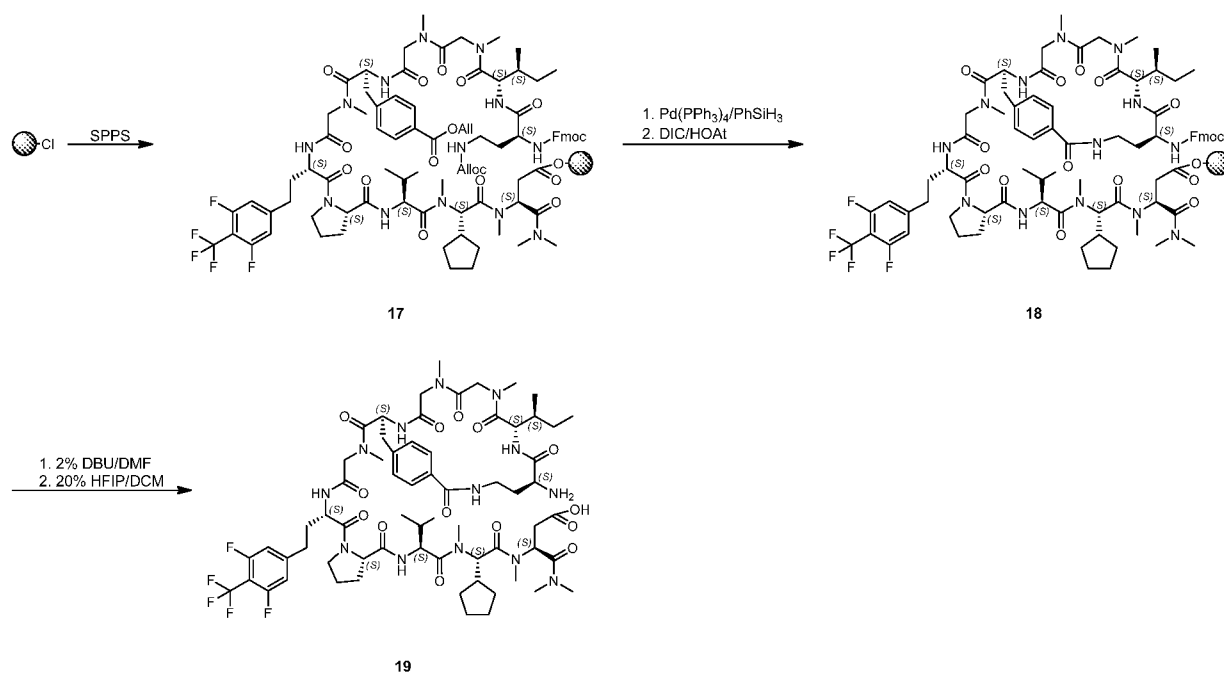
Purification Conditions:

|                         |                                            |    |    |
|-------------------------|--------------------------------------------|----|----|
| Instrument              | AUNO LC-2000                               |    |    |
| Column                  | YMC-Actus Triart C8, 30*150 mm, 10 $\mu$ m |    |    |
| Wavelength              | 220 nm & 254nm                             |    |    |
| Column Oven Temperature | Ambient                                    |    |    |
| Flow Rate               | 25.0 mL/min                                |    |    |
| Injector Volume         | 8 mL                                       |    |    |
| Diluent                 | ACN/Water=2/5                              |    |    |
| Mobile Phase A          | 0.1% TFA in water                          |    |    |
| Mobile Phase B          | ACN                                        |    |    |
| Gradient Program        | Time (min)                                 | A% | B% |
|                         | 0                                          | 65 | 35 |
|                         | 3                                          | 65 | 35 |
|                         | 43                                         | 35 | 65 |
|                         | 43.1                                       | 10 | 90 |
|                         | 48                                         | 10 | 90 |
|                         | 48.1                                       | 90 | 10 |
|                         | 53                                         | 90 | 10 |
| Retention Time          | 25.4 min                                   |    |    |

- 5 The same general procedure was used to prepare compounds 2032 and 2033 which were isolated in the following amount and purity:

| #           | Amount (mg) | Purity (%) | State | Color |
|-------------|-------------|------------|-------|-------|
| <b>2032</b> | 13.5        | 98.9       | solid | white |
| <b>2033</b> | 8.95        | 98.6       | solid | white |

The general procedure to prepare compounds 3001, 3002, and 3003 is as follows:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (20 mL) was added to the vessel containing 2-CTC Resin (0.20 mmol, 0.67 g, Sub = 1.10 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-NMe<sub>2</sub> (80.0 mg, 0.20 mmol, 1.00 equiv.) in DCM (20 mL) was added to the resin, followed by addition of DIEA (103 mg, 0.80 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.7 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (20 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (20 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was washed with DMF (20 mL \* 5). The deprotection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin showed blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.) and DIC (5.20 equiv.) in 50% NMP/DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 2 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin showed colorless, the coupling was judged to be complete. The reaction was then drained and washed with DMF (20 mL \* 5).
- 4) Steps 2~3 were repeated for the following amino acids elongation: Number # 3-11, in the Table below.
- 5) De-Alloc: A solution of PhSiH<sub>3</sub> (10.00 equiv.) and Pd(PPh<sub>3</sub>)<sub>4</sub> (0.10 equiv.) in DCM (10 mL) was added to the resin, and the mixture was bubbled with N<sub>2</sub> for 15 min \* 3 at 25 °C. The resin was then washed with DMF (20 mL \* 5). The de-Alloc reaction was monitored by ninhydrin test. If the resin showed blue or brownish red, the reaction was judged to be complete.

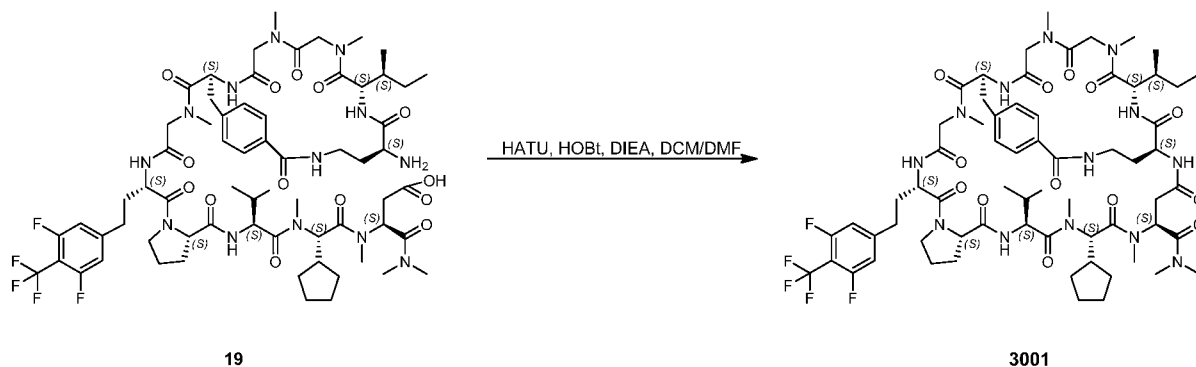
- 6) Cyclization: A solution of HOAt (0.60 mmol, 3.00 equiv.) and DIC (0.60 mmol, 3.00 equiv.) in DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 16 h at 25 °C. The coupling reaction was monitored by ninhydrin test. If the resin remained colorless, the coupling was judged to be complete. The resin was then washed with DMF (20 mL \* 5).
- 7) A solution of 2% DBU in DMF (20 mL) was added to the vessel, and the mixture was bubbled with N<sub>2</sub> flow for 10 min at 25 °C. The resin was then washed with DMF (20 mL \* 5), and MeOH (20 mL \* 3). The resin was drained and dried under vacuum.

For the solid phase synthesis, the amino acids were coupled in the following order and amounts:

| #  | Materials                                                             | Coupling reagents                        |
|----|-----------------------------------------------------------------------|------------------------------------------|
| 1  | Fmoc- <i>N</i> -Me-Asp-NMe <sub>2</sub> (1.00 equiv.)                 | DIEA (4.00 equiv.)                       |
| 2  | Fmoc- <i>N</i> -Me-Gly(cPent)-OH (3.60 equiv.)                        | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 3  | Fmoc-Val-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 4  | Fmoc-Pro-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (2.00 equiv.) | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |
| 6  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 7  | Fmoc-Phe(4-Carboxy-OAll)-OH (1.5 equiv.)                              | HOAt (0.96 equiv.) and DIC (2.17 equiv.) |
| 8  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 9  | Fmoc-Sar-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 10 | Fmoc-Ile-OH (3.60 equiv.)                                             | HOAt (2.30 equiv.) and DIC (5.20 equiv.) |
| 11 | Fmoc-Dab(Alloc)-OH (2.00 equiv.)                                      | HOAt (1.28 equiv.) and DIC (2.89 equiv.) |

Peptide cleavage and lactam cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (20% HFIP/DCM, 15 mL) was added to the flask containing the side chain protected peptide at room temperature and the mixture was bubbled with N<sub>2</sub> for 10 min (15 mL\*3).
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give compound **19** (0.19 g, crude) as an off-white solid. The structure was confirmed by LCMS (R<sub>t</sub> = 0.211 min, MS cal.: 1374.52, MS observed: [M+H]<sup>+</sup> = 1375.6).



Compound **19** (0.19 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (140 mL). Next, HATU (106 mg, 0.42 mmol, 2.00 equiv.), HOBt (37.8 mg, 0.28 mmol, 2.00 equiv.), and DIEA (72.4 mg, 0.56 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h. LCMS showed compound **19** was consumed and one main peak with the desired MS (R<sub>t</sub> = 0.323 min) was detected. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford **3001** (10.3 mg, 97.7% purity, 3.7% yield) as a white solid. LCMS: R<sub>t</sub> = 1.614 min, MS cal.: 1356.51, MS observed: [M+H]<sup>+</sup> = 1356.8. HPLC: R<sub>t</sub> = 10.082 min, purity: 97.7%.

#### Purification Conditions:

| Purification Conditions |                                         |
|-------------------------|-----------------------------------------|
| Sample Preparation      | Dissolve in ACN/Water=1/5               |
| Instrument              | AUNO LC-2000                            |
| Mobile Phase            | A: 0.1% TFA in water                    |
|                         | B: ACN                                  |
| Gradient                | 20-90%-60 min. Retention time: 33.6 min |
| Column                  | Boston Green-C18 8um 50*250mm           |
| Flow Rate               | 60 mL/min                               |
| Wavelength              | 220/254 nm                              |
| Oven Temperature        | Ambient                                 |

#### Analytical method:

| HPLC Method                |                                                    |                   |                   |
|----------------------------|----------------------------------------------------|-------------------|-------------------|
| Instrument                 | Thermo Scientific VANQUISH CORE                    |                   |                   |
| Mobile Phase               | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                            | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient<br>(35-65-20 min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                            | 0                                                  | 65                | 35                |
|                            | 20                                                 | 35                | 65                |
|                            | 20.1                                               | 10                | 90                |
|                            | 22                                                 | 10                | 90                |
| Column                     | YMC-Pack C4 150*4.6mm D.S-3um, 12nm                |                   |                   |
| Flow Rate                  | 1 mL/min                                           |                   |                   |
| Wavelength                 | 220 & 254 nm                                       |                   |                   |
| Oven Temp.                 | 25 °C                                              |                   |                   |

The same general procedure was used to prepare compounds 3002 and 3003 which were isolated in the following amount and purity:

| # | Amount (mg) | Purity(%) | State | Color |
|---|-------------|-----------|-------|-------|
|---|-------------|-----------|-------|-------|

|             |         |      |       |       |
|-------------|---------|------|-------|-------|
| <b>3002</b> | 15.4 mg | 97.0 | solid | white |
| <b>3003</b> | 13.1 mg | 95.9 | solid | white |

## HPLC Data and Methods

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method           | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                         |
|--------|------------|---------------|-----------------------|--------------------------|---------------------|-------------------------------------|
| 1001   | 97.6       | 9.88          | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1002   | 92.0       | 10.49         | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1003   | 98.8       | 9.19          | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1004   | 94.4       | 10.49         | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1005   | 90.3       | 11.48         | 40-70-20min-1mL-50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1006   | 97.9       | 9.43          | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1007   | 90.2       | 8.39          | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1008   | 93.7       | 9.03          | 60-90-20min-50        | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm       |
| 1010   | 98.8       | 11.95         | 40-70-20min-1mL-50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1011   | 87.8       | 12.43         | 50-80-20min-50        | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm       |
| 1012   | 94.4       | 8.75          | 45-75-20min-1mL-30-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1013   | 91.3       | 8.65          | 50-80-20min-1mL-50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1014   | 95.1       | 10.39         | 65-95-20min-50        | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm       |
| 1015   | 96.0       | 8.53          | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1016   | 92.9       | 10.82         | 40-70-20min-1mL-50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1017   | 90.1       | 8.92          | 40-70-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1018   | 96.4       | 8.63          | 45-75-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1019   | 95.5       | 11.25         | 35-65-20min-1mL-30    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4 150*4.6mmI.D.S-3um,12nm |
| 1020   | 91.8       | 9.19          | 55-85-20min-50        | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm       |
| 1021   | 96.4       | 9.23          | 50-80-20min-50        | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm       |

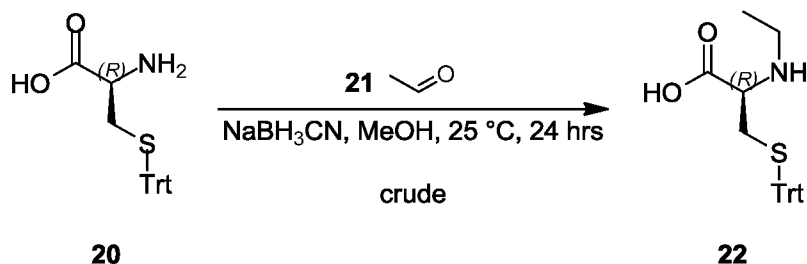
|      |      |       |                    |                          |               |                                            |
|------|------|-------|--------------------|--------------------------|---------------|--------------------------------------------|
| 1022 | 90.1 | 9.18  | 40-70-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 1023 | 95.2 | 10.50 | 50-80-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1024 | 93.7 | 9.48  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1025 | 94.0 | 11.09 | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1026 | 98.7 | 8.40  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1027 | 87.0 | 9.92  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1028 | 95.8 | 9.33  | 60-90-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1029 | 87.8 | 8.28  | 65-95-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1030 | 85.2 | 10.71 | 60-90-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1031 | 93.0 | 8.87  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1032 | 96.7 | 7.66  | 60-90-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1033 | 96.3 | 9.72  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1034 | 95.2 | 9.49  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 1035 | 97.5 | 8.72  | 60-90-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method        | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                                |
|--------|------------|---------------|--------------------|--------------------------|---------------------|--------------------------------------------|
| 2001   | 90.1       | 8.60          | 40-70-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2002   | 91.0       | 12.48         | 30-60-20min-1mL-30 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2003   | 96.5       | 8.59          | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2004   | 97.8       | 11.56         | 30-60-20min-1mL-30 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2005   | 95.2       | 11.42         | 30-60-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2006   | 98.6       | 8.87          | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2007   | 95.0       | 11.34         | 30-60-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2008   | 98.6       | 8.85          | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2009   | 98.2       | 10.10         | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2010   | 96.3       | 9.55          | 45-75-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18<br>110A,150*4.6mm           |
| 2011   | 99.4       | 9.79          | 45-75-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |

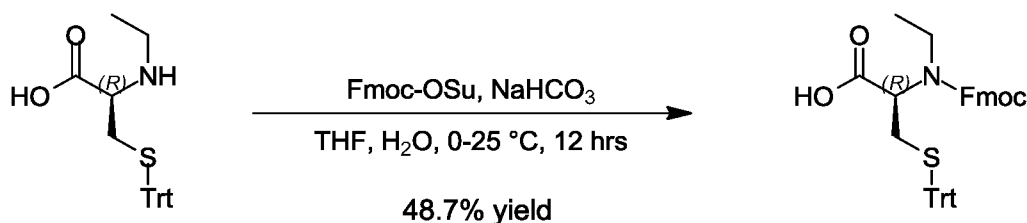
|      |      |       |                    |                          |               |                                            |
|------|------|-------|--------------------|--------------------------|---------------|--------------------------------------------|
| 2012 | 98.3 | 10.07 | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2013 | 99.4 | 10.56 | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2014 | 98.3 | 10.00 | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2015 | 99.6 | 10.62 | 40-70-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2016 | 97.1 | 11.48 | 30-60-20min-1mL-30 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2017 | 96.3 | 10.98 | 30-60-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2018 | 97.8 | 8.51  | 50-80-20min -50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2019 | 99.4 | 9.73  | 40-70-20min        | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2020 | 66.9 | 7.06  | 40-70-20min-1mL-30 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2021 | 96.4 | 8.83  | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 2022 | 97.9 | 10.11 | 45-75-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2023 | 97.0 | 9.82  | 45-75-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2024 | 97.7 | 8.43  | 50-80-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2025 | 99.2 | 10.08 | 45-75-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2026 | 96.4 | 7.93  | 50-80-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2027 | 96.3 | 7.95  | 55-85-20min -50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2028 | 98.6 | 10.26 | 50-80-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2029 | 97.7 | 9.24  | 50-80-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2030 | 99.3 | 8.97  | 50-80-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2031 | 96.9 | 8.37  | 50-80-20min -50    | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2032 | 98.9 | 12.39 | 45-75-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2033 | 98.6 | 9.36  | 55-85-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |
| 2034 | 96.7 | 11.18 | 45-75-20min-50     | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN | Gemini 5um C18<br>110A,150*4.6mm           |

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method        | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                                |
|--------|------------|---------------|--------------------|--------------------------|---------------------|--------------------------------------------|
| 3001   | 97.8       | 10.08         | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |
| 3002   | 97.1       | 10.03         | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18<br>110A,150*4.6mm           |
| 3003   | 96.0       | 8.95          | 35-65-20min-1mL-50 | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | YMC-Pack C4<br>150*4.6mmI.D.S-<br>3um,12nm |

## General procedure for the preparation of Fmoc-N-Et-Cys(Trt)-OH:



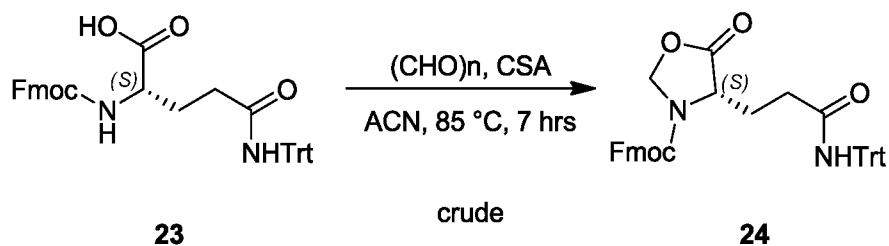
To a solution of compound **20** (8.00 g, 22.0 mmol, 1.00 equiv.) in MeOH (80.0 mL) was added NaBH<sub>3</sub>CN (1.38 g, 22.0 mmol, 1.00 equiv.). Next, compound **21** (1.55 g, 35.2 mmol, 1.98 mL, 1.60 equiv.) was added. The mixture was stirred at 25 °C for 24 h. LCMS showed compound **20** was consumed, and 79.4% of the desired mass was detected (*R*<sub>t</sub> = 1.42 min, MS cal.: 391.16, MS observed: [M+H]<sup>+</sup> = 392.1). The mixture was concentrated under reduced pressure to afford compound **22** (8.60 g, crude) as a white solid.



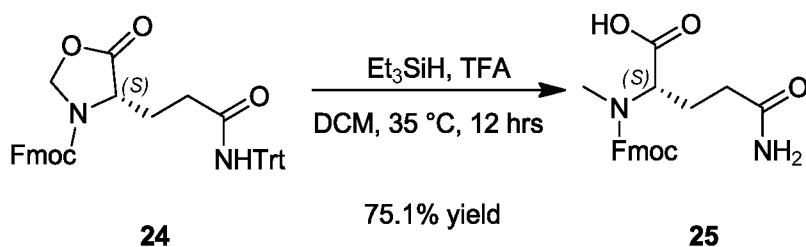
To a solution of compound **22** (8.60 g, 21.9 mmol, 1.00 equiv.) in THF (200 mL) was added NaHCO<sub>3</sub> (5.54 g, 65.9 mmol, 2.56 mL, 3.00 equiv.) in H<sub>2</sub>O (100 mL) at 0 °C. Next, Fmoc-OSu (7.48 g, 22.1 mmol, 1.01 equiv.) was added. The mixture was stirred at 25 °C for 12 h. LCMS showed compound **22** was consumed, and 71.5% of the desired mass was detected (*R*<sub>t</sub> = 0.64 min, MS cal.: 613.76, MS observed: [M+Na]<sup>+</sup> = 636.7). The mixture was adjusted to pH = 5 with 1 M HCl, and extracted with ethyl acetate (200 mL \* 2). The organic layers were combined and washed with water (300 mL \* 3) and brine (200 mL), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated in vacuum. Next, the crude was purified by prep-HPLC (neutral conditions) to afford **Fmoc-N-Et-Cys(Trt)-OH** (6.68 g, 10.7 mmol, 48.7% yield, 98.3% purity) as a white solid.

<sup>1</sup>H NMR: 400 MHz, DMSO-*d*<sub>6</sub>: δ: 7.97 - 7.84 (m, 2H), 7.79 - 7.54 (m, 2H), 7.51 - 7.07 (m, 19H), 4.50 - 4.23 (m, 2H), 4.21 - 3.70 (m, 2H), 3.21 (br dd, *J* = 7.2, 14.8 Hz, 1H), 3.15 - 2.92 (m, 1H), 2.92 - 2.63 (m, 2H), 1.07 - 0.69 (m, 3H).

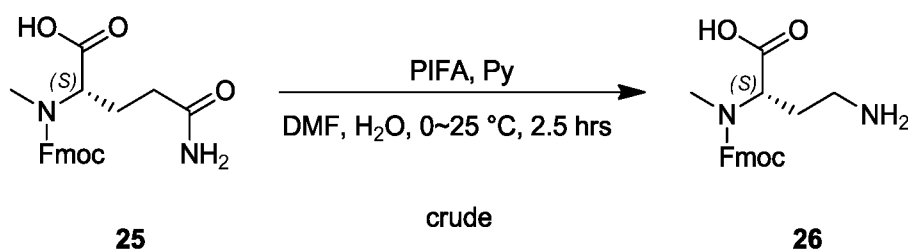
## General procedure for preparation of Fmoc-N-Me-Dab(Alloc)-OH



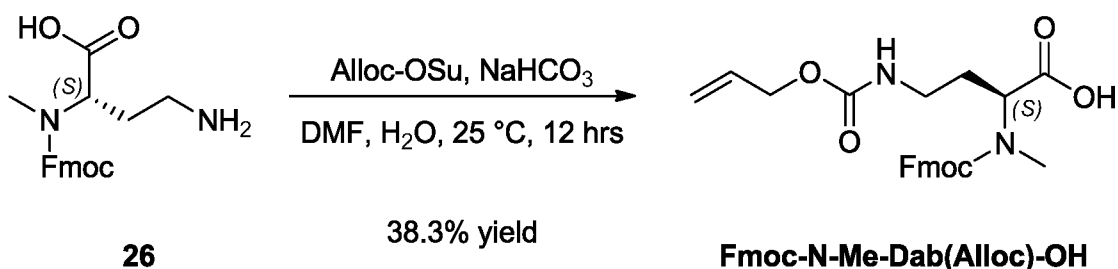
To a solution of compound **23** (100 g, 163 mmol, 1.00 equiv.) in ACN (1.00 L) was added CSA (3.23 g, 13.9 mmol, 0.08 equiv.) and (CHO)<sub>n</sub> (paraformaldehyde, 41.8 g, 818 mmol, 5.00 equiv.). The mixture was stirred at 85 °C for 7 h. LCMS showed compound **23** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 1.15$  min, MS cal.: 622.25, MS observed:  $[M+H]^+ = 623.7$ ). The mixture was filtered. The filtrate was diluted with water (1.00 L) and ethyl acetate (1.00 L), and the organic phase was washed with saturated NaHCO<sub>3</sub> solution (500 mL \* 3) and brine (500 mL \* 2), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure to afford compound **24** (99.0 g, crude) as a white solid. The structure was confirmed by LCMS.



To a solution of compound **24** (89.0 g, 113 mmol, 1.00 equiv.) in TFA (225 mL) and DCM (225 mL) was added Et<sub>3</sub>SiH (65.8 g, 565 mmol, 90.4 mL, 5.00 equiv.). The mixture was stirred at 35 °C for 12 h. LCMS showed that compound **24** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 0.39$  min, MS cal.: 382.15, MS observed:  $[M+H]^+ = 383.1$ ). The mixture was adjusted to pH = 8 with saturated NaHCO<sub>3</sub> solution, and then the mixture was extracted with MTBE (1.00 L \* 2). The aqueous phase was adjusted to pH = 5 with 1M HCl, and extracted with EtOAc (1.00 L \* 2). The organic layers were combined and washed with brine (1.00 L \* 2), dried over by Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure. The residue was triturated in ACN (200 mL) at 25 °C for 30 min. The mixture was filtered, and the residual solvent in the filter cake was removed under reduced pressure to yield compound **25** (33.1 g, 85.0 mmol, 75.1% yield, 98.3% purity) as a white solid. The structure was confirmed by <sup>1</sup>H NMR.



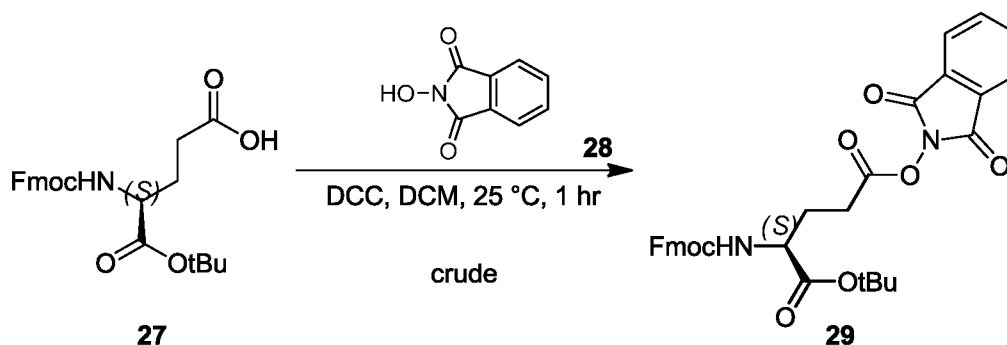
To a solution of compound **25** (12.6 g, 32.9 mmol, 1.00 equiv.) in DMF (60.0 mL) and H<sub>2</sub>O (60.0 mL) was added PIFA (21.2 g, 49.2 mmol, 1.50 equiv.). The mixture was stirred at 0 °C for 0.5 h. Pyridine (5.21 g, 65.8 mmol, 5.31 mL, 2.00 equiv.) was added to the mixture, and the reaction was stirred at 25 °C for 2 h. LCMS showed compound **25** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 0.32$  min, MS cal.: 354.16, MS observed:  $[M+H]^+ = 354.9$ ). The mixture was used in the next step directly without workup.



To the above solution of compound **26** was added NaHCO<sub>3</sub> (8.24 g, 98.1 mmol, 3.00 equiv.), and Alloc-OSu (9.70 g, 49.0 mmol, 1.50 equiv.). The mixture was stirred at 25 °C for 12 h. LCMS showed compound **26** was consumed completely, and the desired mass was detected ( $R_t = 1.45$  min, MS cal.: 438.18, MS observed:  $[M+H]^+ = 439.1$ ). The mixture was adjusted to pH = 5 with 1M HCl, and was extracted with ethyl acetate (200 mL \* 3). The organic layers were combined and washed with water (200 mL \* 3) and brine (200 mL \* 2), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure. The residue was purified by prep-HPLC (HCl condition) to yield **Fmoc-N-Me-Dab(Alloc)-OH** (5.50 g, 12.5 mmol, 38.3% yield, 98.6% purity) as a white solid. The structure was confirmed by <sup>1</sup>H NMR.

<sup>1</sup>H NMR: 400 MHz, DMSO-*d*<sub>6</sub>:  $\delta$ : 12.85 (br s, 1H), 7.89 (br d,  $J = 7.2$  Hz, 2H), 7.64 (br dd,  $J = 7.6, 14.2$  Hz, 2H), 7.41 - 7.39 (m, 2H), 7.33 - 7.31 (m, 2H), 7.25 - 7.19 (m, 1H), 5.90 - 5.84 (m, 1H), 5.28 - 5.12 (m, 2H), 4.50 - 4.27 (m, 6H), 2.97 - 2.92 (m, 2H), 2.78 (br d,  $J = 8.4$  Hz, 3H), 2.04 - 2.00 (m, 1H), 1.83 - 1.81 (m, 1H).

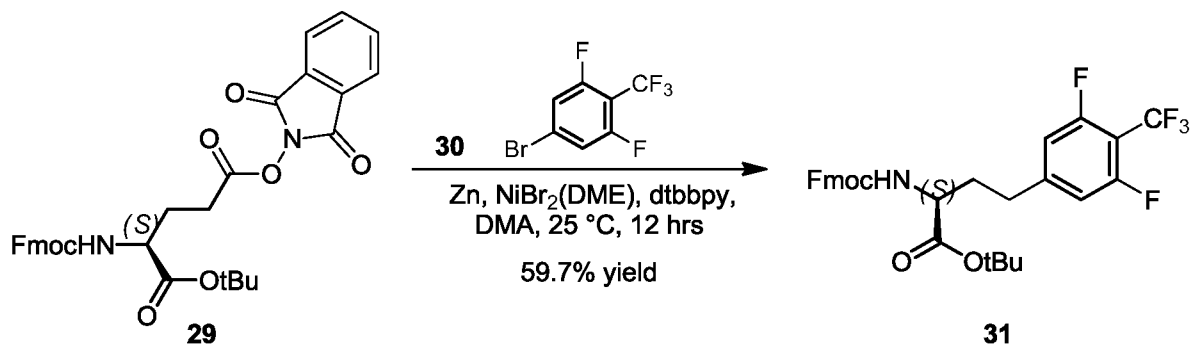
### General procedure for preparation of Intermediate 32



To a solution of compound **27** (50.0 g, 117 mmol, 1.00 equiv.) and compound **28** (21.0 g, 129 mmol, 1.10 equiv.) in DCM (250 mL) was added DCC (26.6 g, 26.1 mL, 129 mmol, 1.10

equiv.) under N<sub>2</sub>. The mixture was stirred at 25 °C for 1 h. TLC (petroleum ether:ethyl acetate = 2:1) indicated compound **27** ( $R_f$  = 0.24) was consumed and new spots ( $R_f$  = 0.15, 0.47, 0.66) formed. The mixture was filtered, and concentrated under reduced pressure to afford compound **29** (68.0 g, crude) as a white solid.

5

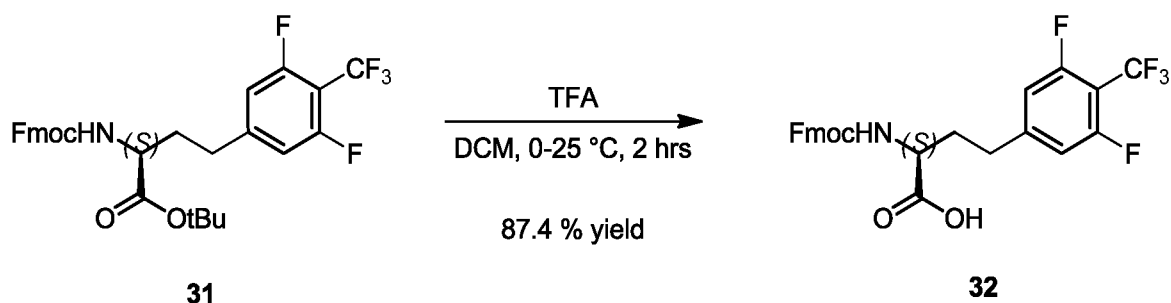


10

To a solution of dtbbpy (15.9 g, 59.5 mmol, 0.50 equiv.) in DMA (150 mL) was added NiBr<sub>2</sub>(DME) (18.3 g, 59.5 mmol, 0.50 equiv.). The mixture was stirred at 25 °C for 0.5 h, and then the mixture was added to a solution of compound **29** (68.0 g, 119 mmol, 1.00 equiv.), compound **30** (62.2 g, 238.3 mmol, 2.00 equiv.), and freshly activated Zn powder (34.6 g, 529 mmol, 4.44 equiv.) in DMA (500 mL). The mixture was stirred at 25 °C for 11.5 h. TLC (petroleum ether:ethyl acetate = 3:1) indicated compound **29** ( $R_f$  = 0.52) was consumed and new spots ( $R_f$  = 0.22, 0.45, 0.75) had formed. The mixture was filtered, quenched by the addition of 1 M HCl, and then diluted with water (1.9 L). Next, the mixture was extracted with ethyl acetate (650 mL \* 3). The organic phases was combined and washed with brine (650 mL \* 2), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (SiO<sub>2</sub>, petroleum ether:ethyl acetate = 1:0 to 1:1). Compound **31** (40.0 g, 71.2 mmol, 59.7% yield) was obtained as a white solid (petroleum ether:ethyl acetate = 3:1,  $R_f$  = 0.45).

15

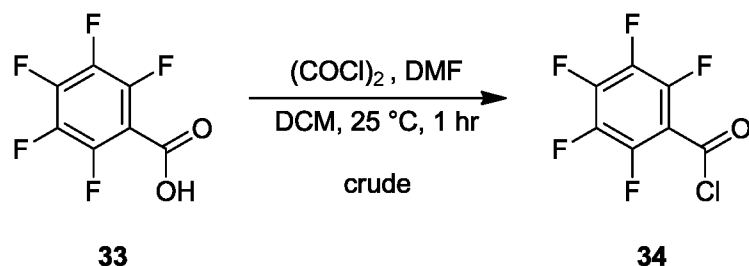
20



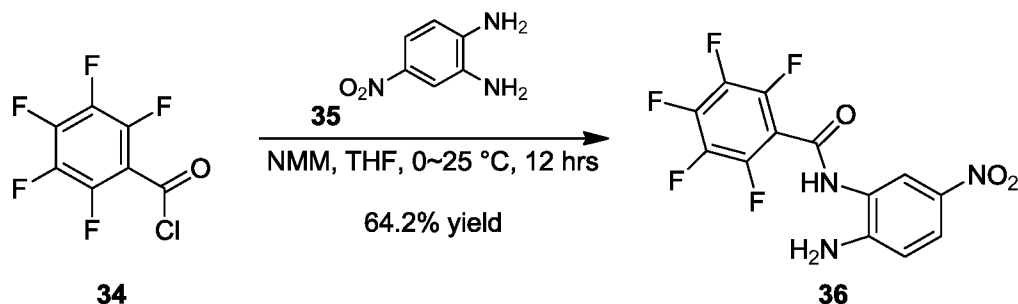
25

To a solution of compound **31** (39.5 g, 70.3 mmol, 1.00 equiv.) in DCM (200 mL) was added TFA (307 g, 2.69 mmol, 200 mL, 38.2 equiv.) at 0 °C. The mixture was stirred at 25 °C for 2 h. TLC (petroleum ether:ethyl acetate = 3:1) indicated compound **31** ( $R_f$  = 0.45) was consumed and a new spot ( $R_f$  = 0.27) had formed. The mixture was concentrated under reduced pressure followed by purification by prep-HPLC (HCl condition) to afford compound **32** (25.0 g, 49.1 mmol, 87.4% yield, 99.3% purity) as a white solid.

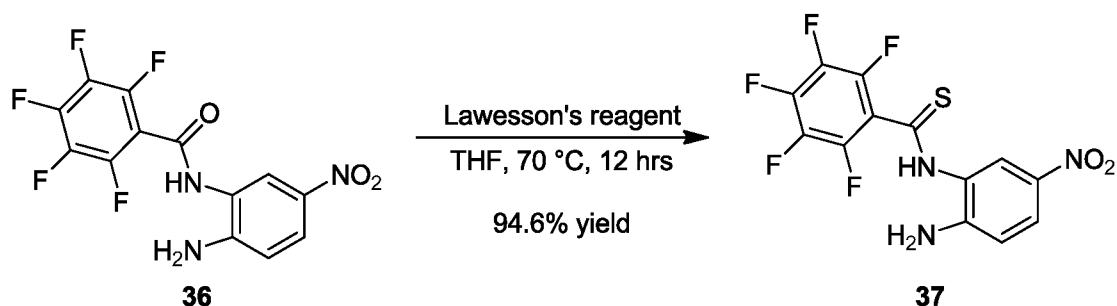
## General procedure for preparation of Intermediate 38



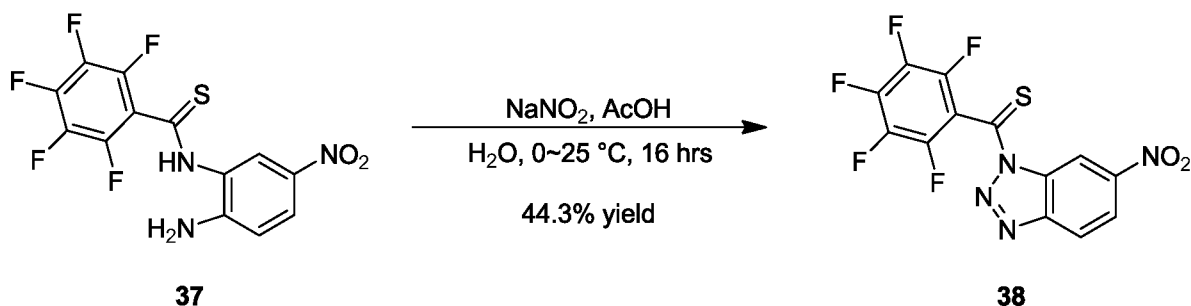
To a solution of compound **33** (20.0 g, 94.3 mmol, 1.00 equiv.) in DCM (200 mL) was added (COCl)<sub>2</sub> (17.9 g, 141 mmol, 12.4 mL, 1.50 equiv.) and DMF (14.0 mg, 0.18 mmol, 0.002 equiv.). The mixture was stirred at 25 °C for 1 h. TLC (petroleum ether:ethyl acetate = 1:1) indicated compound **33** (*R*<sub>f</sub> = 0.31) was consumed completely, and one new spot (*R*<sub>f</sub> = 0.98) had formed. The mixture was concentrated under reduced pressure to yield compound **44** (21.7 g, crude) as a white solid.



To a solution of compound **34** (21.7 g, 94.1 mmol, 1.00 equiv.) in THF (200 mL) was added NMM (12.3 g, 122 mmol, 13.4 mL, 1.30 equiv.) and compound **35** (14.4 g, 94.1 mmol, 1.00 equiv.) at 0 °C. The mixture was stirred at 25 °C for 12 h. LCMS showed compound **34** was consumed completely, and one main peak with the desired mass was detected (*R*<sub>t</sub> = 0.41 min, MS cal.: 347.13, MS observed: [M+H]<sup>+</sup> = 347.8). The mixture was concentrated under reduced pressure. The residue was triturated with MTBE (100 mL) at 25 °C for 30 min. Next, the mixture was filtered, and the residual solvent in the filter cake was removed under reduced pressure to afford compound **36** (21.0 g, 60.4 mmol, 64.2% yield, 94.6% purity) as a yellow solid. The structure was confirmed by LCMS.

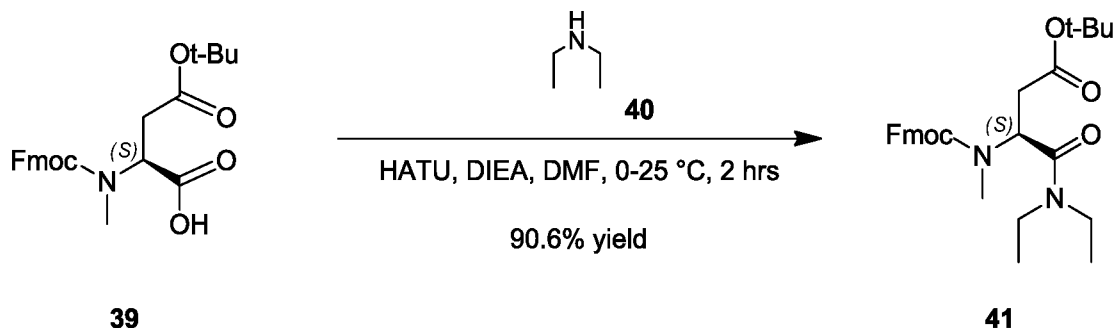


To a solution of compound **36** (21.0 g, 60.4 mmol, 1.00 equiv.) in THF (210 mL) was added Lawesson's reagent (22.0 g, 54.4 mmol, 0.90 equiv.). The mixture was stirred at 70 °C for 12 h. LCMS showed compound **36** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 0.45$  min, MS cal.: 363.01, MS observed:  $[M+H]^+ = 363.8$ ). The mixture was concentrated under reduced pressure. The residue was purified by column chromatography (SiO<sub>2</sub>, petroleum ether:ethyl acetate = 1:0 to 0:1) to afford compound **37** (20.8 g, 57.2 mmol, 94.6% yield, 91.4% purity) as a brown oil (TLC, petroleum ether: ethyl acetate = 2: 1,  $R_f = 0.38$ ). The structure was confirmed by LCMS.



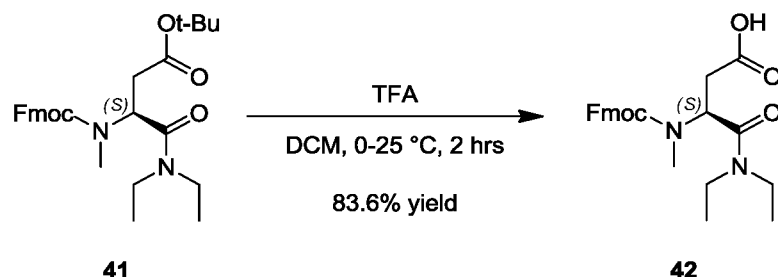
To a solution of compound **37** (20.8 g, 57.2 mmol, 1.00 equiv.) in AcOH (180 mL) and H<sub>2</sub>O (20.0 mL) was added NaNO<sub>2</sub> (5.92 g, 85.8 mmol, 1.50 equiv.) at 0 °C. The mixture was stirred at 25 °C for 16 h. LCMS showed compound **37** was consumed completely, and the desired mass was detected ( $R_t = 0.55$  min). The mixture was diluted with water (600 mL), and extracted with ethyl acetate (200 mL \* 3). The organic layers were combined and washed with water (500 mL \* 3) and brine (300 mL \* 2), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure. The residue was purified by prep-HPLC (HCl condition) to yield **38** (9.50 g, 21.4 mmol, 44.3% yield, 94.7% purity) as a red oil. The structure was confirmed by <sup>1</sup>H NMR. <sup>1</sup>H NMR: 400 MHz, CDCl<sub>3</sub>:  $\delta$ : 9.68 (s, 1H), 8.55 (dd,  $J = 1.6, 8.8$  Hz, 1H), 8.38 (d,  $J = 8.8$  Hz, 1H).

### General procedure for the preparation of Intermediate 42



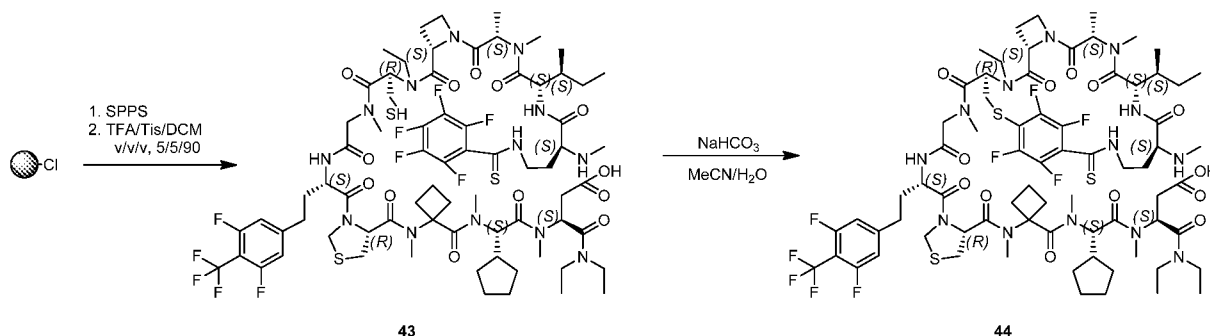
To a solution of compound **39** (10.0 g, 23.5 mmol, 1.00 equiv.) in DMF (100 mL) was added HATU (9.83 g, 25.8 mmol, 1.10 equiv.), DIEA (6.07 g, 7.76 mL, 47.0 mmol, 2.00 equiv.), and compound **40** (2.06 g, 2.90 mL, 28.2 mmol, 1.20 equiv.) at 0 °C. The mixture was stirred at 25 °C for 2 h. LCMS showed compound **39** was consumed completely, and one main peak with

the desired mass was detected ( $R_t = 2.16$  min, MS cal.: 480.26, MS observed:  $[M+H]^+ = 481.2$ ). The mixture was diluted with water (100 mL), and extracted with ethyl acetate (50.0 mL \* 3). The combined organic layers were washed with water (100 mL \* 3) and brine (100 mL \* 2), dried over  $\text{Na}_2\text{SO}_4$ , filtered and concentrated under reduced pressure to afford compound **41** (10.8 g, 21.3 mmol, 90.6% yield, 94.8% purity) as a brown oil.



To a solution of compound **41** (10.8 g, 21.3 mmol, 1.00 equiv.) in DCM (54.0 mL) was added TFA (82.8 g, 726 mmol, 54.0 mL, 34.1 equiv.) at 0 °C. The mixture was stirred at 25 °C for 2 h. LCMS showed compound **41** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 0.47$  min, MS cal.: 424.20, MS observed:  $[M+H]^+ = 425.3$ ). The mixture was concentrated under reduced pressure, and the residue was purified by reverse-phase HPLC (neutral conditions) to afford **42** (7.60 g, 17.8 mmol, 83.6% yield, 99.5% purity) as a brown solid.

### General procedure for the preparation of Compound 1118:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

- 1) Resin preparation: DCM (20.0 mL) was added to the vessel containing 2-CTC resin (0.30 mmol, 1.00 g, Sub = 0.30 mmol/g, 1.00 equiv.), and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-NEt<sub>2</sub> (128 mg, 0.30 mmol, 1.00 equiv.) in DCM (10.0 mL) was added to the resin, followed by the addition of DIEA (155 mg, 1.20 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (1.00 mL) was added to the mixture, and the solution was bubbled with N<sub>2</sub> for 30 min. The resin was drained, and washed with DMF (10 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 2% DBU in DMF (10.0 mL) was added to the resin, and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (10 mL \* 5).

The de-protection was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin displayed a blue or brownish red color.

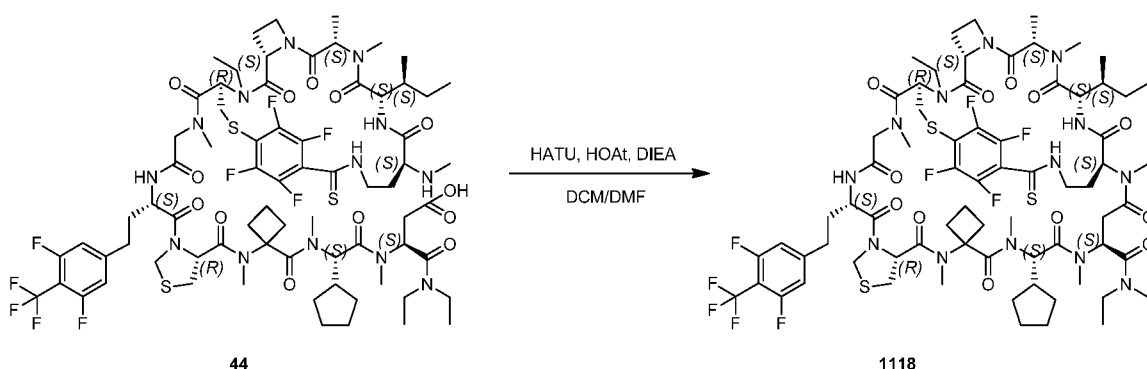
- 3) Coupling: A solution of Fmoc-N-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.), and DIC (5.20 equiv.) in 50% NMP/DMF (10.0 mL) was added to the resin with N<sub>2</sub> bubbling for 24 h at 40 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The mixture was then drained and washed with DMF (10 mL \* 5).
- 4) Steps 2~3 were repeated for the amino acids listed in the table below: Entries 3,5-7,9-11.
- 5) Coupling of **Fmoc-Thz-Cl**: A solution of DIEA (2.70 mmol, 6.00 equiv.) in DCM (10.0 mL) was added to the resin, followed by the addition of **Fmoc-Thz-Cl** (0.90 mmol, 3.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C. The coupling was monitored by ninhydrin test. If the resin remained colorless, the coupling was judged to be complete. The resin was then drained and washed with DMF (10 mL \* 5).
- 6) Step 5 was repeated for the coupling of **Fmoc-Aze-Cl** (Entry 8, Table Below).
- 7) De-Alloc: A solution of PhSiH<sub>3</sub> (10.0 equiv.) and Pd(PPh<sub>3</sub>)<sub>4</sub> (0.10 equiv.) in DCM (10.0 mL) was added to the resin, and the mixture was bubbled with N<sub>2</sub> for 15 min at 25 °C. The resin was drained and this process was repeated two more times. Next, the resin was washed with DMF (10 mL \* 5). The de-Alloc reaction was monitored by ninhydrin test. If the resin displayed a blue or brownish red color, the reaction was judged to be complete.
- 8) Coupling of Intermediate **38**: A solution of DIEA (0.90 mmol, 3.00 equiv.) in NMP (10.0 mL) was added to the resin, followed by the addition of Intermediate **38** (0.45 mmol, 1.50 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. The coupling was monitored by ninhydrin test. If the resin remained colorless, the coupling was judged to be complete. The resin was then washed with DMF (10 mL \* 5).
- 9) After coupling of Intermediate **38** was completed, a solution of 2% DBU in DMF (50.0 mL) was added to the vessel, and the mixture was bubbled with N<sub>2</sub> flow for 10 min at 25 °C. The resin was then washed with DMF (10 mL \* 5), and MeOH (10 mL \* 5). Next, the resin was drained and dried under vacuum.

| #  | Materials                                                         | Coupling Reagents                |
|----|-------------------------------------------------------------------|----------------------------------|
| 1  | Fmoc-N-Me-Asp-NEt <sub>2</sub> (1.00 eq)                          | DIEA (4.00 eq)                   |
| 2  | Fmoc-N-Me-Gly(cPent)-OH (3.60 eq)                                 | HOAt (2.30 eq) and DIC (5.20 eq) |
| 3  | Fmoc-N-Me-cVal-OH (3.60 eq)                                       | HOAt (2.30 eq) and DIC (5.20 eq) |
| 4  | Fmoc-Thz-Cl (3.00 eq)                                             | DIEA (6.00 eq)                   |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (1.50 eq) | HOAt (0.96 eq) and DIC (2.17 eq) |
| 6  | Fmoc-Sar-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |
| 7  | Fmoc-NEt-Cys(Trt)-OH (1.50 eq)                                    | HOAt (0.96 eq) and DIC (2.17 eq) |
| 8  | Fmoc-Aze-Cl (3.00 eq)                                             | DIEA (6.00 eq)                   |
| 9  | Fmoc-N-Me-Ala-OH (3.60 eq)                                        | HOAt (2.30 eq) and DIC (5.20 eq) |
| 10 | Fmoc-Ile-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |

|    |                                   |                                  |
|----|-----------------------------------|----------------------------------|
| 11 | Fmoc-N-Me-Dab(Alloc)-OH (1.50 eq) | HOAt (0.96 eq) and DIC (2.17 eq) |
| 12 | Intermediate 38 (1.50 eq)         | DIEA (3.00 eq)                   |

Peptide cleavage and thioether cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (TFA/TIS/DCM, v/v/v, 5/5/90, 30 mL) was added to the flask containing the side chain protected peptide, and the mixture was stirred at 25 °C for 10 min.
- 2) The mixture was then filtered. The filtrate was collected, and concentrated under reduced pressure.
- 3) The solid was precipitated in cold isopropyl ether (50 mL) and centrifuged (3 mins at 3000 rpm). The solid was dried under reduced pressure to afford the linear peptide intermediate **43** (0.50 g, crude) as a light-yellow solid.
- 4) To a solution of compound **43** (0.50 g, 0.3 mmol, 1.00 equiv.) in 50% ACN/H<sub>2</sub>O (300 mL) was added NaHCO<sub>3</sub> until the pH reached ~8. Next, the mixture was stirred at 25 °C for 16 h. LCMS indicated compound **43** was consumed, and one main peak with the desired mass ( $R_t = 0.520$  min, MS cal.: 1620.8, MS observed:  $[M+H]^+ = 1621.4$ ) was detected. 1M HCl was added until the pH reached ~6. The mixture was then concentrated and dried by lyophilization.
- 5) The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **44** (77.0 mg, 47.5 μmol, 15.3% yield, 96.7% purity) as a light-yellow solid.



Compound **44** (77.0 mg, 47.5 μmol, 1.00 equiv.) was dissolved in 50% DCM/DMF (94 mL). Next, HATU (36 mg, 0.094 mmol, 2.00 equiv.), HOAt (13 mg, 0.094 mmol, 2.00 equiv.) and DIEA (24 mg, 0.188 mmol, 4.00 equiv.) were added to the mixture. The reaction was stirred at 25 °C for 12 h. LCMS indicated compound **44** was consumed, and one main peak with the desired MS ( $R_t = 0.475$  min, MS cal.: 1602.8, MS observed:  $[M+H]^+ = 1603.0$ ) was detected. The mixture was concentrated under reduced pressure to give a residue. The residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford **1118** (14.7 mg, 90.0% purity, 2.75% yield) as a light-yellow solid.

Purification Conditions:

|                   |                                                    |
|-------------------|----------------------------------------------------|
| <b>Instrument</b> | AUNO LC-2000                                       |
| <b>Column</b>     | YMC-Actus Triart C8-S 150*30.0 mmI.D.S-10 μm,20 nm |

|                                |                   |    |    |
|--------------------------------|-------------------|----|----|
| <b>Wavelength</b>              | 220 nm & 254 nm   |    |    |
| <b>Column Oven Temperature</b> | Ambient           |    |    |
| <b>Flow Rate</b>               | 25.0 mL/min       |    |    |
| <b>Injector Volume</b>         | 7 mL              |    |    |
| <b>Diluent</b>                 | ACN/Water = 1/5   |    |    |
| <b>Mobile Phase A</b>          | 0.1% TFA in water |    |    |
| <b>Mobile Phase B</b>          | ACN               |    |    |
| <b>Gradient Program</b>        | Time (min)        | A% | B% |
|                                | 0                 | 85 | 15 |
|                                | 3                 | 85 | 15 |
|                                | 3.1               | 44 | 56 |
|                                | 43                | 14 | 86 |
|                                | 43.1              | 10 | 90 |
|                                | 48                | 10 | 90 |
|                                | 48.1              | 90 | 10 |
|                                | 58                | 90 | 10 |
| <b>Retention Time</b>          | 29.2 min          |    |    |

Analytical HPLC Conditions:

|                               |                                                    |                   |                   |
|-------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>             | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                               | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (60-90-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                               | 0                                                  | 40                | 60                |
|                               | 20                                                 | 10                | 90                |
|                               | 20.1                                               | 10                | 90                |
|                               | 22                                                 | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1103</b> | 10.4        | 93.2      | off-white solid    |
| <b>1116</b> | 23.7        | 88.7      | yellow solid       |
| <b>1117</b> | 20.0        | 93.2      | yellow solid       |
| <b>1118</b> | 14.7        | 90.1      | off-white solid    |
| <b>1122</b> | 7.7         | 88.7      | off-white solid    |
| <b>1123</b> | 12.8        | 95.2      | light yellow solid |
| <b>1124</b> | 10.0        | 91.6      | light yellow solid |
| <b>1131</b> | 6.7         | 84.4      | light yellow solid |

|             |      |      |                    |
|-------------|------|------|--------------------|
| <b>1132</b> | 4.0  | 81.9 | light yellow solid |
| <b>1170</b> | 37.2 | 84.8 | light yellow solid |
| <b>1171</b> | 18.7 | 86.6 | light yellow solid |
| <b>1172</b> | 35.1 | 84.8 | yellow solid       |
| <b>1173</b> | 29.6 | 91.3 | light yellow solid |
| <b>1175</b> | 22.6 | 91.9 | light yellow solid |
| <b>1176</b> | 30.5 | 91.6 | light yellow solid |
| <b>1177</b> | 21.7 | 90.2 | light yellow solid |
| <b>1178</b> | 24.5 | 92.1 | light yellow solid |
| <b>1180</b> | 3.6  | 87.2 | off-white solid    |
| <b>1182</b> | 22.2 | 93.4 | light yellow solid |
| <b>1184</b> | 25.6 | 90.7 | light yellow solid |
| <b>1185</b> | 19.8 | 88.6 | light yellow solid |
| <b>1186</b> | 24.8 | 92.1 | light yellow solid |
| <b>1187</b> | 23.1 | 93.5 | light yellow solid |
| <b>1188</b> | 24.0 | 93.3 | light yellow solid |
| <b>1189</b> | 27.4 | 87.4 | light yellow solid |
| <b>1190</b> | 47.6 | 16.7 | light yellow solid |
| <b>1200</b> | 27.8 | 95.5 | light yellow solid |
| <b>1201</b> | 23.5 | 88.8 | light yellow solid |
| <b>1202</b> | 35.5 | 88.7 | light yellow solid |
| <b>1206</b> | 20.1 | 95.2 | white solid        |
| <b>1207</b> | 9.5  | 87.4 | light yellow solid |
| <b>1208</b> | 7.7  | 77.0 | light yellow solid |
| <b>1211</b> | 11.9 | 80.2 | light yellow solid |
| <b>1212</b> | 14.2 | 83.9 | light yellow solid |
| <b>1215</b> | 12.2 | 83.1 | light yellow solid |
| <b>1220</b> | 36.4 | 92.2 | light yellow solid |
| <b>1223</b> | 13.6 | 75.4 | light yellow solid |
| <b>1224</b> | 5.5  | 88.7 | light yellow solid |
| <b>1225</b> | 5.4  | 78.2 | light yellow solid |
| <b>1227</b> | 39.8 | 94.6 | light yellow solid |
| <b>1228</b> | 20.4 | 93.3 | light yellow solid |
| <b>1229</b> | 39.0 | 96.7 | light yellow solid |
| <b>1231</b> | 8.0  | 91.9 | light yellow solid |

| <b>Cmpd #</b> | <b>Purity (%)</b> | <b>HPLC RT (min)</b> | <b>HPLC Method</b> | <b>HPLC Mobile Phase A</b>   | <b>HPLC Mobile Phase B</b> | <b>HPLC Column</b>            |
|---------------|-------------------|----------------------|--------------------|------------------------------|----------------------------|-------------------------------|
| <b>1103</b>   | 93.2              | 10.92                | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1116</b>   | 88.7              | 12.01                | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1117</b>   | 93.2              | 13.10                | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1118</b>   | 90.1              | 11.67                | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1122</b>   | 88.7              | 13.15                | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1123</b>   | 95.2              | 10.47                | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |

|      |      |       |                |                              |                   |                                  |
|------|------|-------|----------------|------------------------------|-------------------|----------------------------------|
| 1124 | 91.6 | 12.93 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1131 | 84.4 | 11.03 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1132 | 81.9 | 11.23 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1170 | 84.8 | 11.83 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1171 | 86.6 | 11.77 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1172 | 84.8 | 11.65 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1173 | 91.3 | 11.48 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1175 | 91.9 | 9.88  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1176 | 91.6 | 12.28 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1177 | 90.2 | 10.04 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1178 | 92.1 | 8.98  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1180 | 87.2 | 11.44 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1182 | 93.4 | 10.39 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1184 | 90.7 | 11.50 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1185 | 88.6 | 11.15 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1186 | 92.1 | 11.30 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1187 | 93.5 | 10.61 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1188 | 93.3 | 10.99 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1189 | 87.4 | 10.90 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1190 | 16.7 | 17.37 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1200 | 95.5 | 10.65 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1201 | 88.8 | 10.71 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1202 | 88.7 | 11.79 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1206 | 95.2 | 10.28 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1207 | 87.4 | 12.08 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1208 | 77.0 | 13.24 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1211 | 80.2 | 12.90 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1212 | 83.9 | 11.98 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1215 | 83.1 | 11.39 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1220 | 92.2 | 11.71 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1223 | 75.4 | 11.09 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1224 | 88.7 | 12.24 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
| 1225 | 78.2 | 12.86 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |

|             |      |       |                |                              |                   |                               |
|-------------|------|-------|----------------|------------------------------|-------------------|-------------------------------|
| <b>1227</b> | 94.6 | 12.96 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1228</b> | 93.3 | 13.00 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1229</b> | 96.7 | 11.34 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1231</b> | 91.9 | 13.52 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

|                               |                                                    |                   |                   |
|-------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>             | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                               | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (55-85-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                               | 0                                                  | 45                | 55                |
|                               | 20                                                 | 15                | 85                |
|                               | 20.1                                               | 10                | 90                |
|                               | 22                                                 | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

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| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1095</b> | 36.9        | 84.1      | yellow solid       |
| <b>1101</b> | 13.5        | 92.7      | yellow solid       |
| <b>1197</b> | 36.8        | 95.2      | light yellow solid |
| <b>1198</b> | 36.6        | 97.5      | light yellow solid |
| <b>1199</b> | 39.6        | 96.3      | light yellow solid |
| <b>1203</b> | 27.1        | 94.7      | light yellow solid |
| <b>1204</b> | 14.9        | 90.0      | white solid        |
| <b>1205</b> | 10.3        | 91.1      | white solid        |
| <b>1209</b> | 34.7        | 89.4      | light yellow solid |
| <b>1210</b> | 37.4        | 92.4      | light yellow solid |
| <b>1219</b> | 22.2        | 92.2      | light yellow solid |
| <b>1226</b> | 20.2        | 91.5      | light yellow solid |
| <b>1230</b> | 34.5        | 91.7      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1095</b> | 84.1       | 11.28         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1101</b> | 92.7       | 12.50         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1197</b> | 95.2       | 12.65         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |       |                |                              |                   |                               |
|-------------|------|-------|----------------|------------------------------|-------------------|-------------------------------|
| <b>1198</b> | 97.5 | 11.80 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1199</b> | 96.3 | 12.18 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1203</b> | 94.7 | 13.03 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1204</b> | 90.0 | 12.73 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1205</b> | 91.1 | 12.56 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1209</b> | 89.4 | 12.14 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1210</b> | 92.4 | 13.40 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1219</b> | 92.2 | 13.36 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1226</b> | 91.5 | 13.79 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1230</b> | 91.7 | 12.93 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                               |                                                    |                   |                   |
|-------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>             | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                               | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (65-95-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                               | 0                                                  | 35                | 65                |
|                               | 20                                                 | 5                 | 95                |
|                               | 20.1                                               | 10                | 90                |
|                               | 22                                                 | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                                       |                   |                   |
| <b>Oven Tem.</b>              | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

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| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1091</b> | 19.9        | 92.8      | off-white solid    |
| <b>1121</b> | 11.4        | 87.2      | light green solid  |
| <b>1125</b> | 15.9        | 93.2      | light yellow solid |
| <b>1127</b> | 6.3         | 93.8      | yellow solid       |
| <b>1128</b> | 3.1         | 92.2      | light yellow solid |
| <b>1129</b> | 17.6        | 91.0      | yellow solid       |

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|--------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| 1091   | 92.8       | 8.94          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| 1121   | 87.2       | 10.77         | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|      |      |       |                |                              |                   |                               |
|------|------|-------|----------------|------------------------------|-------------------|-------------------------------|
| 1125 | 93.2 | 12.31 | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1127 | 93.8 | 9.23  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1128 | 92.2 | 9.21  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1129 | 91.0 | 9.58  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

|                                  |                                                    |                   |                   |
|----------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>                | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>              | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                  | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (50-80-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                  | 0                                                  | 50                | 50                |
|                                  | 20                                                 | 20                | 80                |
|                                  | 20.1                                               | 10                | 90                |
|                                  | 22                                                 | 10                | 90                |
| <b>Column</b>                    | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                 | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>                | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>                | 50 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1080</b> | 2.2         | 37.9      | light yellow solid |
| <b>1094</b> | 10.5        | 96.6      | white solid        |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1080</b> | 37.9       | 14.11         | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1094</b> | 96.6       | 12.45         | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

|                                  |                                                    |                   |                   |
|----------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>                | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>              | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                  | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (55-85-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                  | 0                                                  | 45                | 55                |
|                                  | 20                                                 | 15                | 85                |
|                                  | 20.1                                               | 10                | 90                |
|                                  | 22                                                 | 10                | 90                |
| <b>Column</b>                    | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                 | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>                | 220 & 254 nm                                       |                   |                   |

|           |       |
|-----------|-------|
| Oven Tem. | 50 °C |
|-----------|-------|

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1040</b> | 10.1        | 88.6      | white solid        |
| <b>1054</b> | 13.3        | 95.3      | off-white solid    |
| <b>1058</b> | 11.0        | 91.4      | white solid        |
| <b>1059</b> | 19.3        | 93.7      | light yellow solid |
| <b>1077</b> | 10.4        | 93.1      | off-white solid    |
| <b>1088</b> | 11.0        | 96.2      | off-white solid    |
| <b>1089</b> | 14.7        | 95.0      | off-white solid    |
| <b>1090</b> | 15.6        | 90.5      | off-white solid    |
| <b>1096</b> | 10.5        | 90.0      | off-white solid    |
| <b>1097</b> | 13.0        | 83.6      | off-white solid    |
| <b>1098</b> | 33.8        | 95.1      | off-white solid    |
| <b>1217</b> | 32.1        | 93.1      | light yellow solid |
| <b>1218</b> | 31.4        | 93.6      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1040</b> | 88.6       | 9.21          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1054</b> | 95.3       | 9.87          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1058</b> | 91.4       | 11.19         | 10-80-6MIN-50  | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1059</b> | 93.7       | 10.21         | 10-80-6MIN-50  | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1077</b> | 93.1       | 11.01         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1088</b> | 96.2       | 10.53         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1089</b> | 95.0       | 11.83         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1090</b> | 90.5       | 12.50         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1096</b> | 90.0       | 10.14         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1097</b> | 83.6       | 11.51         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1098</b> | 95.1       | 10.31         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1217</b> | 93.1       | 12.55         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1218</b> | 93.6       | 12.54         | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

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Analytical HPLC Conditions:

|                                  |                                                    |                   |                   |
|----------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>                | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>              | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                  | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (60-90-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |

|                   |                               |    |    |
|-------------------|-------------------------------|----|----|
|                   | 0                             | 40 | 60 |
|                   | 20                            | 10 | 90 |
|                   | 20.1                          | 10 | 90 |
|                   | 22                            | 10 | 90 |
| <b>Column</b>     | Gemini C18 5um 110A 150*4.6mm |    |    |
| <b>Flow Rate</b>  | 1 mL/min                      |    |    |
| <b>Wavelength</b> | 220 & 254 nm                  |    |    |
| <b>Oven Temp.</b> | 50 °C                         |    |    |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1041</b> | 12.8        | 90.0      | off-white solid    |
| <b>1042</b> | 8.3         | 92.0      | white solid        |
| <b>1043</b> | 11.0        | 90.3      | light yellow solid |
| <b>1044</b> | 4.6         | 83.4      | white solid        |
| <b>1046</b> | 12.1        | 67.2      | off-white solid    |
| <b>1049</b> | 7.1         | 83.8      | light yellow solid |
| <b>1050</b> | 1.1         | 64.7      | purple solid       |
| <b>1051</b> | 19.9        | 87.6      | white solid        |
| <b>1053</b> | 1.6         | 91.6      | white solid        |
| <b>1055</b> | 13.3        | 92.4      | white solid        |
| <b>1056</b> | 17.0        | 95.9      | light yellow solid |
| <b>1057</b> | 4.6         | 68.5      | white solid        |
| <b>1060</b> | 11.8        | 96.0      | off-white solid    |
| <b>1061</b> | 4.8         | 76.5      | off-white solid    |
| <b>1062</b> | 3.2         | 78.6      | off-white solid    |
| <b>1063</b> | 5.9         | 72.9      | off-white solid    |
| <b>1064</b> | 6.9         | 82.5      | off-white solid    |
| <b>1065</b> | 2.9         | 86.8      | off-white solid    |
| <b>1066</b> | 9.3         | 83.0      | off-white solid    |
| <b>1068</b> | 4.5         | 82.7      | off-white solid    |
| <b>1069</b> | 2.9         | 79.4      | off-white solid    |
| <b>1070</b> | 19.6        | 90.1      | off-white solid    |
| <b>1071</b> | 7.2         | 90.9      | white solid        |
| <b>1073</b> | 14.2        | 94.4      | white solid        |
| <b>1074</b> | 16.4        | 90.5      | light yellow solid |
| <b>1076</b> | 16.6        | 93.5      | off-white solid    |
| <b>1079</b> | 6.1         | 94.3      | white solid        |
| <b>1083</b> | 5.8         | 91.8      | off-white solid    |
| <b>1085</b> | 7.8         | 80.5      | off-white solid    |
| <b>1093</b> | 13.3        | 97.5      | light yellow solid |
| <b>1099</b> | 6.3         | 95.7      | yellow solid       |
| <b>1100</b> | 34.2        | 98.0      | off-white solid    |
| <b>1102</b> | 30.4        | 96.8      | off-white solid    |
| <b>1104</b> | 11.2        | 95.6      | off-white solid    |

|             |      |      |                    |
|-------------|------|------|--------------------|
| <b>1108</b> | 11.6 | 97.0 | off-white solid    |
| <b>1112</b> | 7.7  | 85.0 | off-white solid    |
| <b>1113</b> | 23.3 | 95.8 | light yellow solid |
| <b>1114</b> | 34.7 | 93.2 | light yellow solid |
| <b>1115</b> | 22.9 | 97.0 | light yellow solid |
| <b>1119</b> | 30.1 | 83.2 | yellow solid       |
| <b>1120</b> | 3.6  | 90.3 | off-white solid    |
| <b>1126</b> | 27.5 | 90.5 | yellow solid       |
| <b>1213</b> | 10.2 | 82.1 | light yellow solid |
| <b>1214</b> | 31.5 | 95.4 | light yellow solid |
| <b>1222</b> | 2.8  | 74.7 | light yellow solid |

| <b>Cmpd #</b> | <b>Purity (%)</b> | <b>HPLC RT (min)</b> | <b>HPLC Method</b> | <b>HPLC Mobile Phase A</b>   | <b>HPLC Mobile Phase B</b> | <b>HPLC Column</b>            |
|---------------|-------------------|----------------------|--------------------|------------------------------|----------------------------|-------------------------------|
| <b>1041</b>   | 90.0              | 9.2                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1042</b>   | 92.0              | 10.0                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1043</b>   | 90.3              | 10.6                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1044</b>   | 83.4              | 11.2                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1046</b>   | 67.2              | 9.6                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1049</b>   | 83.8              | 10.7                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1050</b>   | 64.7              | 10.2                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1051</b>   | 87.6              | 9.9                  | 10-80-6MIN-50      | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1053</b>   | 91.6              | 9.8                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1055</b>   | 92.4              | 10.7                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1056</b>   | 95.9              | 9.8                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1057</b>   | 68.5              | 10.7                 | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1060</b>   | 96.0              | 8.8                  | 10-80-6MIN-50      | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1061</b>   | 76.5              | 8.6                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1062</b>   | 78.6              | 7.6                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1063</b>   | 72.9              | 8.9                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1064</b>   | 82.5              | 9.2                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1065</b>   | 86.8              | 9.9                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1066</b>   | 83.0              | 8.8                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1068</b>   | 82.7              | 8.1                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1069</b>   | 79.4              | 9.0                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1070</b>   | 90.1              | 8.9                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>1071</b>   | 90.9              | 7.8                  | 60-90-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |

|      |      |      |                |                              |                   |                               |
|------|------|------|----------------|------------------------------|-------------------|-------------------------------|
| 1073 | 94.4 | 8.4  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1074 | 90.5 | 7.4  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1076 | 93.5 | 9.5  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1079 | 94.3 | 8.0  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1083 | 91.8 | 9.8  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1085 | 80.5 | 10.9 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1093 | 97.5 | 10.7 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1099 | 95.7 | 8.2  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1100 | 98.0 | 9.1  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1102 | 96.8 | 10.1 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1104 | 95.6 | 9.3  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1108 | 97.0 | 11.4 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1112 | 85.0 | 12.2 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1113 | 95.8 | 11.7 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1114 | 93.2 | 10.3 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1115 | 97.0 | 11.3 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1119 | 83.2 | 12.1 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1120 | 90.3 | 11.3 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1126 | 90.5 | 13.2 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1213 | 82.1 | 10.4 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1214 | 95.4 | 9.9  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 1222 | 74.7 | 9.8  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                                  |                                                    |                   |                   |
|----------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>                | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>              | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                  | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (65-95-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                  | 0                                                  | 35                | 65                |
|                                  | 20                                                 | 5                 | 95                |
|                                  | 20.1                                               | 10                | 90                |
|                                  | 22                                                 | 10                | 90                |
| <b>Column</b>                    | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                 | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>                | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>                | 50 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1036</b> | 1.9         | 67.0      | white solid        |
| <b>1037</b> | 2.9         | 91.3      | off-white solid    |
| <b>1038</b> | 2.4         | 95.0      | white solid        |
| <b>1039</b> | 13.6        | 95.9      | light yellow solid |
| <b>1045</b> | 5.4         | 81.6      | light yellow solid |
| <b>1047</b> | 19.3        | 92.1      | off-white solid    |
| <b>1048</b> | 12.5        | 90.2      | light yellow solid |
| <b>1052</b> | 5.1         | 78.2      | off-white solid    |
| <b>1072</b> | 15.4        | 90.1      | light yellow solid |
| <b>1075</b> | 15.5        | 90.3      | light yellow solid |
| <b>1078</b> | 9.6         | 97.0      | off-white solid    |
| <b>1081</b> | 18.3        | 94.3      | off-white solid    |
| <b>1082</b> | 1.5         | 85.5      | white solid        |
| <b>1084</b> | 15.1        | 82.4      | light yellow solid |
| <b>1086</b> | 16.1        | 95.4      | yellow solid       |
| <b>1087</b> | 16.0        | 94.2      | light yellow solid |
| <b>1092</b> | 13.8        | 95.9      | light yellow solid |
| <b>1105</b> | 6.3         | 85.5      | light yellow solid |
| <b>1106</b> | 10.4        | 84.1      | light yellow solid |
| <b>1107</b> | 14.4        | 96.1      | off-white solid    |
| <b>1109</b> | 18.6        | 95.4      | off-white solid    |
| <b>1110</b> | 17.2        | 87.7      | light yellow solid |
| <b>1111</b> | 20.0        | 85.2      | light yellow solid |
| <b>1179</b> | 16.7        | 81.2      | light yellow solid |
| <b>1181</b> | 21.1        | 80.1      | light yellow solid |
| <b>1216</b> | 33.8        | 95.1      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1036</b> | 67.0       | 9.5           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1037</b> | 91.3       | 8.7           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1038</b> | 95.0       | 9.2           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1039</b> | 95.9       | 9.8           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1045</b> | 81.6       | 9.3           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1047</b> | 92.1       | 10.0          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1048</b> | 90.2       | 8.0           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1052</b> | 78.2       | 12.0          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1072</b> | 90.1       | 10.0          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |      |                |                              |                   |                               |
|-------------|------|------|----------------|------------------------------|-------------------|-------------------------------|
| <b>1075</b> | 90.3 | 9.6  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1078</b> | 97.0 | 10.8 | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1081</b> | 94.3 | 7.5  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1082</b> | 85.5 | 8.0  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1084</b> | 82.4 | 7.8  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1086</b> | 95.4 | 8.8  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1087</b> | 94.2 | 9.2  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1092</b> | 95.9 | 9.0  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1105</b> | 85.5 | 7.7  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1106</b> | 84.1 | 9.9  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1107</b> | 96.1 | 9.1  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1109</b> | 95.4 | 9.5  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1110</b> | 87.7 | 9.4  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1111</b> | 85.2 | 9.3  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1179</b> | 81.2 | 9.9  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1181</b> | 80.1 | 8.4  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1216</b> | 95.1 | 9.1  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

| <b>Instrument</b>                 | Thermo Scientific VANQUISH CORE                    |                   |                   |
|-----------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Mobile Phase</b>               | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                   | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (70-100-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                   | 0                                                  | 30                | 70                |
|                                   | 20                                                 | 0                 | 100               |
|                                   | 20.1                                               | 10                | 90                |
|                                   | 22                                                 | 10                | 90                |
| <b>Column</b>                     | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                  | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>                 | 220 & 254 nm                                       |                   |                   |
| <b>Oven Tem.</b>                  | 50 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1067</b> | 9.4         | 90.2      | light yellow solid |

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method     | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|--------|------------|---------------|-----------------|------------------------------|---------------------|-------------------------------|
| 1067   | 90.2       | 8.5           | 70-100-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

| Instrument             | Thermo Scientific VANQUISH CORE                    |                   |                   |
|------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                        | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (60-90-20min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                        | 0                                                  | 40                | 60                |
|                        | 20                                                 | 10                | 90                |
|                        | 20.1                                               | 10                | 90                |
|                        | 22                                                 | 10                | 90                |
| Column                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate              | 1 mL/min                                           |                   |                   |
| Wavelength             | 220 & 254 nm                                       |                   |                   |
| Oven Temp.             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

5

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1133</b> | 12.1        | 90.1      | light yellow       |
| <b>1134</b> | 24.6        | 92.9      | light yellow       |
| <b>1135</b> | 10.6        | 90.5      | light yellow       |
| <b>1142</b> | 23.0        | 90.7      | light yellow solid |
| <b>1143</b> | 16.9        | 92.8      | off-white solid    |
| <b>1145</b> | 15.6        | 93.8      | light yellow solid |
| <b>1148</b> | 21.5        | 87.5      | light yellow       |
| <b>1149</b> | 3.9         | 89.3      | off-white          |
| <b>1150</b> | 13.8        | 92.9      | off-white          |
| <b>1151</b> | 31.6        | 91.6      | light yellow       |
| <b>1152</b> | 3.3         | 91.0      | off-white          |
| <b>1156</b> | 32.9        | 94.2      | light yellow solid |
| <b>1159</b> | 11.2        | 83.2      | off-white solid    |
| <b>1160</b> | 12.3        | 88.1      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1133</b> | 90.1       | 10.2          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1134</b> | 92.9       | 11.3          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1135</b> | 90.5       | 11.2          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1142</b> | 90.7       | 11.6          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1143</b> | 92.8       | 10.4          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |      |                |                              |                   |                               |
|-------------|------|------|----------------|------------------------------|-------------------|-------------------------------|
| <b>1145</b> | 93.8 | 11.9 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1148</b> | 87.5 | 9.1  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1149</b> | 89.3 | 9.1  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1150</b> | 92.9 | 11.0 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1151</b> | 91.6 | 11.0 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1152</b> | 91.0 | 11.0 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1156</b> | 94.2 | 9.9  | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1159</b> | 83.2 | 11.8 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1160</b> | 88.1 | 12.2 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                               |                                                    |                   |                   |
|-------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>             | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                               | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (55-85-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                               | 0                                                  | 45                | 55                |
|                               | 20                                                 | 15                | 85                |
|                               | 20.1                                               | 10                | 90                |
|                               | 22                                                 | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1136</b> | 18.9        | 91.1      | light yellow       |
| <b>1137</b> | 1.9         | 95.1      | white              |
| <b>1157</b> | 20.3        | 90.0      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1136</b> | 91.1       | 11.8          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1137</b> | 95.1       | 11.7          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1157</b> | 90.0       | 14.0          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                     |                                                    |
|---------------------|----------------------------------------------------|
| <b>Instrument</b>   | Thermo Scientific VANQUISH CORE                    |
| <b>Mobile Phase</b> | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |

|                               | B: ACN (0.075% TFA in ACN)    |                   |                   |
|-------------------------------|-------------------------------|-------------------|-------------------|
| <b>Gradient (65-95-20min)</b> | Time (min)                    | Percent Solvent A | Percent Solvent B |
|                               | 0                             | 35                | 65                |
|                               | 20                            | 5                 | 95                |
|                               | 20.1                          | 10                | 90                |
|                               | 22                            | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                      |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                  |                   |                   |
| <b>Oven Temp.</b>             | 30 °C                         |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1130</b> | 32.6        | 93.0      | light yellow       |
| <b>1138</b> | 31.7        | 94.7      | light yellow solid |
| <b>1139</b> | 14.2        | 93.2      | light yellow solid |
| <b>1141</b> | 19.5        | 86.9      | light yellow solid |
| <b>1146</b> | 40.4        | 90.1      | light yellow       |
| <b>1147</b> | 14.1        | 80.9      | light yellow       |
| <b>1154</b> | 27.7        | 93.5      | white solid        |
| <b>1169</b> | 35.9        | 90.4      | light yellow solid |
| <b>1174</b> | 33.5        | 90.5      | light yellow solid |
| <b>1183</b> | 21.2        | 93.7      | light yellow solid |
| <b>1191</b> | 23.4        | 92.1      | off-white solid    |
| <b>1192</b> | 58.9        | 46.4      | white solid        |
| <b>1196</b> | 19.0        | 89.7      | off-white solid    |
| <b>1221</b> | 22.9        | 84.7      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>1130</b> | 93.0       | 10.4          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1138</b> | 94.7       | 11.0          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1139</b> | 93.2       | 9.9           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1141</b> | 86.9       | 11.2          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1146</b> | 90.1       | 8.5           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1147</b> | 80.9       | 8.5           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1154</b> | 93.5       | 9.2           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1169</b> | 90.4       | 8.2           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1174</b> | 90.5       | 9.8           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>1183</b> | 93.7       | 10.5          | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |      |                |                              |                   |                               |
|-------------|------|------|----------------|------------------------------|-------------------|-------------------------------|
| <b>1191</b> | 92.1 | 8.5  | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1192</b> | 46.4 | 15.4 | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1196</b> | 89.7 | 11.7 | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>1221</b> | 84.7 | 10.3 | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                               |                                                    |                   |                   |
|-------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>             | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                               | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (60-90-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                               | 0                                                  | 40                | 60                |
|                               | 20                                                 | 10                | 90                |
|                               | 20.1                                               | 10                | 90                |
|                               | 22                                                 | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

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| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1153</b> | 23.4        | 95.0      | light yellow solid |
| <b>1161</b> | 18.5        | 83.8      | light yellow solid |
| <b>1163</b> | 26.1        | 93.9      | light yellow solid |
| <b>1165</b> | 29.5        | 91.9      | light yellow solid |
| <b>1167</b> | 24.6        | 92.4      | light yellow solid |
| <b>1168</b> | 20.1        | 86.0      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                    |
|-------------|------------|---------------|-------------|--------------------------|---------------------|--------------------------------|
| <b>1153</b> | 95.0       | 9.14          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/CAN       | Gemini 5um C18 110A, 150*4.6mm |
| <b>1161</b> | 83.8       | 11.56         | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A, 150*4.6mm |
| <b>1163</b> | 93.9       | 9.90          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A, 150*4.6mm |
| <b>1165</b> | 91.9       | 11.03         | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A, 150*4.6mm |
| <b>1167</b> | 92.4       | 12.19         | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A, 150*4.6mm |
| <b>1168</b> | 86.0       | 10.79         | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A, 150*4.6mm |

## Analytical HPLC Conditions:

|                   |                                 |
|-------------------|---------------------------------|
| <b>Instrument</b> | Thermo Scientific VANQUISH CORE |
|-------------------|---------------------------------|

| Mobile Phase           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|------------------------|----------------------------------------------------|-------------------|-------------------|
|                        | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (55-85-20min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                        | 0                                                  | 45                | 55                |
|                        | 20                                                 | 15                | 85                |
|                        | 20.1                                               | 10                | 90                |
|                        | 22                                                 | 10                | 90                |
| Column                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate              | 1 mL/min                                           |                   |                   |
| Wavelength             | 220 & 254 nm                                       |                   |                   |
| Oven Temp.             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1155</b> | 20.4        | 92.1      | light yellow solid |
| <b>1158</b> | 8.0         | 78.9      | light yellow solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|-------------|--------------------------|---------------------|-------------------------------|
| <b>1155</b> | 92.1       | 11.8          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1158</b> | 78.9       | 13.2          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |

5

Analytical HPLC Conditions:

| Instrument             | Thermo Scientific VANQUISH CORE                    |                   |                   |
|------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                        | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (65-95-20min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                        | 0                                                  | 35                | 65                |
|                        | 20                                                 | 5                 | 95                |
|                        | 20.1                                               | 10                | 90                |
|                        | 22                                                 | 10                | 90                |
| Column                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate              | 1 mL/min                                           |                   |                   |
| Wavelength             | 220 & 254 nm                                       |                   |                   |
| Oven Temp.             | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **1118** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1144</b> | 20.4        | 90.3      | light yellow solid |
| <b>1162</b> | 16.0        | 80.0      | off-white solid    |
| <b>1164</b> | 28.2        | 96.0      | light yellow solid |

|             |     |      |                    |
|-------------|-----|------|--------------------|
| <b>1166</b> | 4.9 | 94.1 | light yellow solid |
|-------------|-----|------|--------------------|

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|-------------|--------------------------|---------------------|-------------------------------|
| <b>1144</b> | 90.3       | 10.8          | 65-95-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1162</b> | 80.0       | 10.4          | 65-95-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1164</b> | 96.0       | 9.1           | 65-95-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1166</b> | 94.1       | 10.1          | 65-95-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |

## Analytical HPLC Conditions:

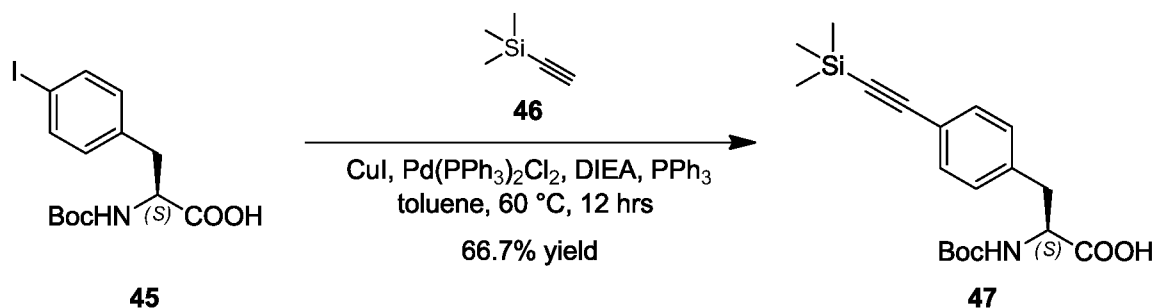
| Instrument              | Thermo Scientific VANQUISH CORE                    |                   |                   |
|-------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase            | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                         | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (70-100-20min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                         | 0                                                  | 30                | 70                |
|                         | 20                                                 | 0                 | 100               |
|                         | 20.1                                               | 10                | 90                |
|                         | 22                                                 | 10                | 90                |
| Column                  | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate               | 1 mL/min                                           |                   |                   |
| Wavelength              | 220 & 254 nm                                       |                   |                   |
| Oven Temp.              | 30 °C                                              |                   |                   |

- 5 The same general procedure that was used to prepare **1118** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>1140</b> | 35.1        | 86.8      | light yellow solid |
| <b>1193</b> | 23.2        | 92.2      | light yellow solid |
| <b>1194</b> | 39.2        | 22.7      | off-white solid    |
| <b>1195</b> | 35.4        | 88.3      | light yellow solid |

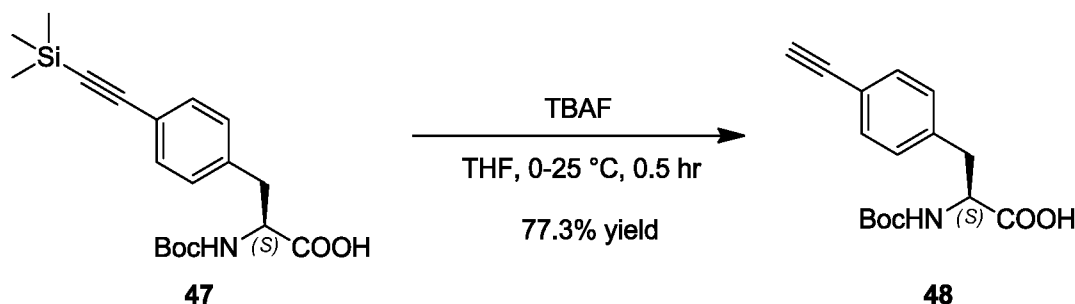
| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method  | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|--------------|--------------------------|---------------------|-------------------------------|
| <b>1140</b> | 86.8       | 10.1          | 70-100-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1193</b> | 92.2       | 7.7           | 70-100-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1194</b> | 22.7       | 13.1          | 70-100-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>1195</b> | 88.3       | 9.0           | 70-100-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |

General procedure for the preparation of Intermediate **47**:



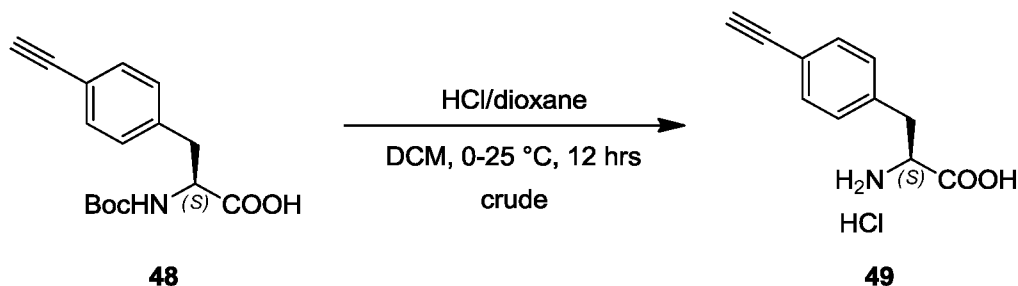
To a solution of compound **45** (25.0 g, 63.9 mmol, 1.00 equiv.) and compound **46** (12.5 g, 127 mmol, 17.7 mL, 2.00 equiv.) in toluene (400 mL) was added DIEA (16.5 g, 127 mmol, 22.2 mL, 2.00 equiv.), Pd(PPh<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> (2.24 g, 3.20 mmol, 0.05 equiv.), PPh<sub>3</sub> (1.68 g, 6.39 mmol, 0.10 equiv.) and CuI (1.22 g, 6.39 mmol, 0.10 equiv.). The mixture was stirred at 60 °C for 12 hrs under N<sub>2</sub> atmosphere. LCMS showed compound **45** was consumed, and the desired mass was detected by LCMS (R<sub>t</sub> = 0.50 min, MS cal.: 361.17, MS observed: [M+Na]<sup>+</sup> = 384.1). The reaction mixture was filtered, and the filtrate was adjusted to pH = 4 with 0.5 M HCl aqueous solution. The mixture was extracted with ethyl acetate (300 mL \* 3). Next, the combined organic layers were washed with water (400 mL \* 2) and brine 300 mL, dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated in vacuum to provide crude product which was purified by column chromatography (SiO<sub>2</sub>, petroleum ether: ethyl acetate = 100: 0 to 3: 1, R<sub>f</sub> = 0.21, petroleum ether: ethyl acetate=1: 1). Compound **47** (19.0 g, 42.6 mmol, 66.7% yield, 81.2% purity) was obtained as a brown solid.

General procedure for the preparation of Intermediate **48**:



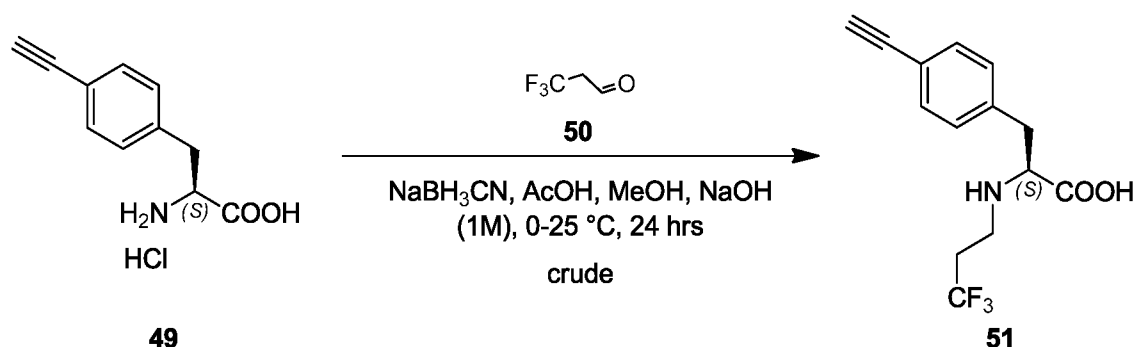
To a solution of compound **47** (9.00 g, 20.2 mmol, 1.00 equiv.) in THF (108 mL) was added TBAF (1.00 M, 30.3 mL, 1.50 equiv.) at 0 °C. The mixture was stirred at 25 °C for 0.5 hr. LCMS indicated compound **47** was consumed, and the desired mass was detected by LCMS (R<sub>t</sub> = 0.41 min, MS cal.: 289.13, MS observed: [2M+H]<sup>+</sup> = 579.3). The reaction mixture was adjusted to pH = 4 with 0.5 M HCl aqueous solution. The mixture was extracted with ethyl acetate (300 mL \* 3), and the combined organic layers were washed with water (400 mL \* 3) and brine 300 mL, dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated in vacuum to provide crude product which was purified by reverse-phase HPLC (HCl condition). Compound **48** (4.60 g, 15.6 mmol, 77.3% yield, 98.3% purity) was obtained as a brown solid.

General procedure for the preparation of Intermediate **49**:



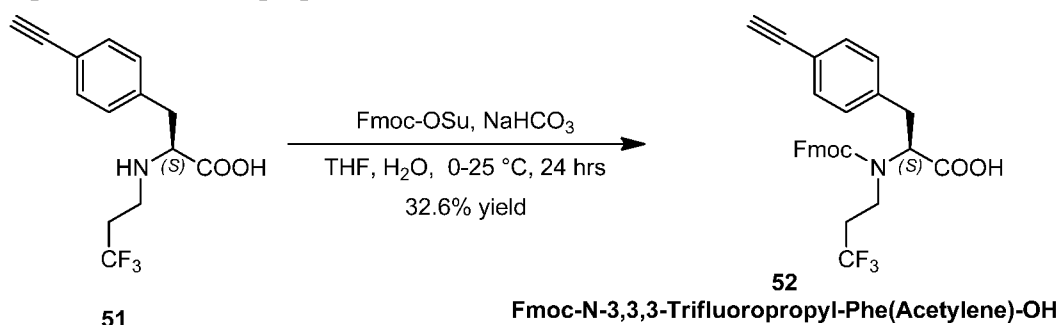
To a solution of compound **48** (4.60 g, 15.6 mmol, 1.00 equiv.) in DCM (20.0 mL) was added HCl/dioxane (2.00 M, 45.2 mL, 5.79 equiv.) at 0 °C. Next, the mixture was stirred at 25 °C for 12 hrs. LCMS indicated compound **48** was consumed, and the desired mass was detected by LCMS ( $R_t = 0.14$  min, MS cal.: 189.08, MS observed:  $[M+H]^+ = 190.0$ ). Next, the reaction mixture was concentrated under reduced pressure to obtain compound **49** (3.12 g, crude, HCl salt) as a white solid which was used in the next step without further purification.

General procedure for the preparation of Intermediate **51**:



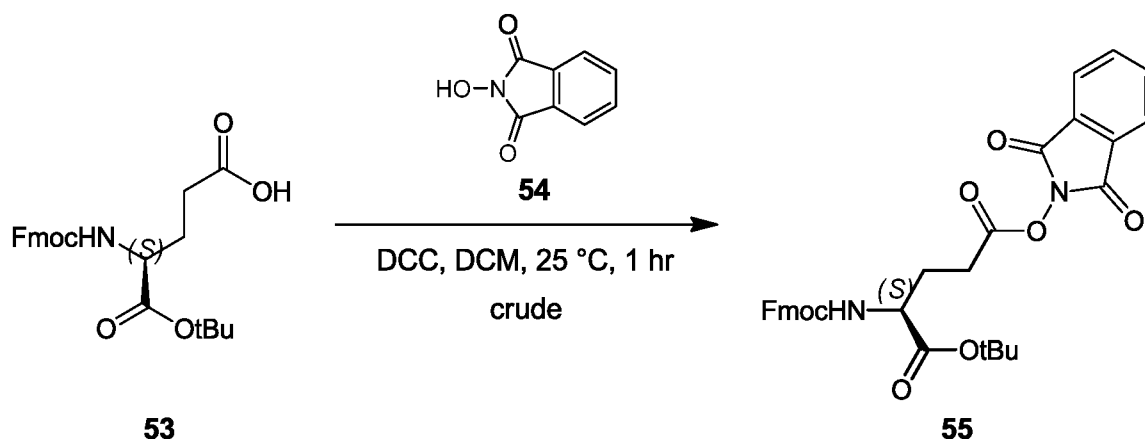
To a solution of compound **49** (3.12 g, 13.8 mmol, 1.00 equiv., HCl salt) in MeOH (32.0 mL) was added NaOH (1 M, 691  $\mu\text{L}$ , 0.05 equiv.), and AcOH (830 mg, 13.8 mmol, 791  $\mu\text{L}$ , 1.00 equiv.) at 0 °C. Next, compound **50** and NaBH<sub>3</sub>CN (2.61 g, 41.4 mmol, 3.00 equiv.) were added, and the mixture solution was stirred at 25 °C at 24 hrs. LCMS indicated compound **49** was consumed, and the desired mass was detected by LCMS ( $R_t = 0.24$  min, MS cal.: 285.1, MS observed:  $[M+H]^+ = 286.0$ ). The reaction mixture was concentrated under reduced pressure to provide compound **51** (3.94 g, crude) as a white solid which was used in the next step without further purification.

General procedure for the preparation of Intermediate **52**:



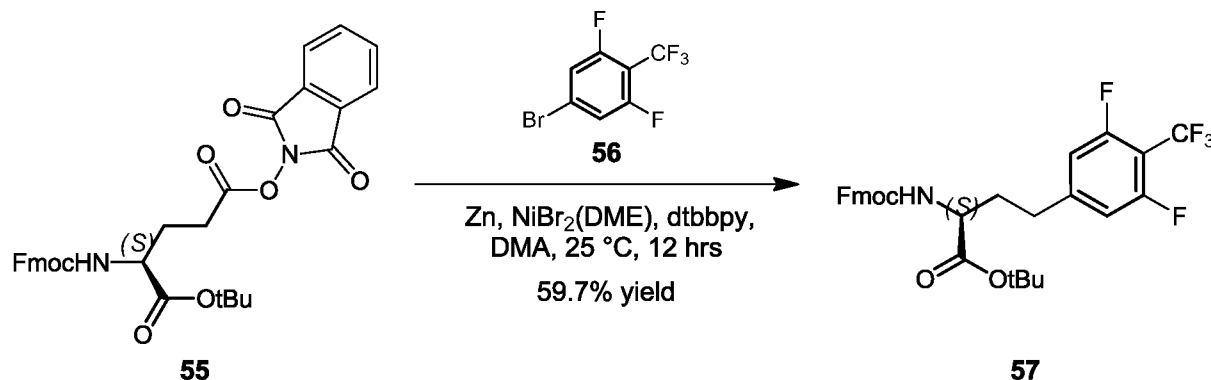
To a solution of compound **51** (3.94 g, 13.8 mmol, 1.00 equiv.) in THF (400 mL) was added a solution of NaHCO<sub>3</sub> (4.64 g, 55.2 mmol, 2.15 mL, 4.00 equiv.) in H<sub>2</sub>O (200 mL) at 0 °C. Next, Fmoc-OSu (4.66 g, 13.8 mmol, 1.00 equiv.) was added. The mixture was stirred at 25 °C for 24 hrs. LCMS indicated compound **51** was consumed and the desired mass was detected by LCMS (*R*<sub>t</sub> = 0.50 min, MS cal.: 507.17, MS observed: [M+H]<sup>+</sup> = 508.1). The mixture was adjusted to pH = 3 with 1M HCl aqueous solution and extracted with ethyl acetate (300 mL \* 2). The combined organic layers were washed with water (200 mL \* 3) and brine 300 mL, dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated in vacuum to provide crude product which was purified by prep-HPLC (HCl condition) to furnish compound **52**, **Fmoc-N-3,3,3-Trifluoropropyl-Phe(Acetylene)-OH** (2.30 g, 4.51 mmol, 32.6% yield, 99.6% purity) which was obtained as a white solid.

General procedure for the preparation of Intermediate **55**:



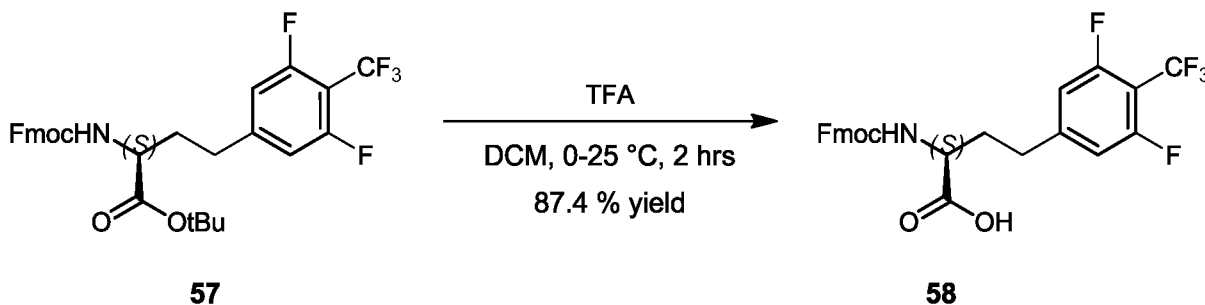
To a solution of compound **53** (50.0 g, 117 mmol, 1.00 equiv.) and compound **54** (21.0 g, 129 mmol, 1.10 equiv.) in DCM (250 mL) was added DCC (26.6 g, 129 mmol, 26.1 mL, 1.10 equiv.) under N<sub>2</sub>. The mixture was stirred at 25 °C for 1 h. TLC (petroleum ether:ethyl acetate = 2:1) indicated compound **53** (*R*<sub>f</sub> = 0.24) was consumed, and new spots (*R*<sub>f</sub> = 0.15, 0.47, 0.66) had formed. The reaction mixture was filtered and concentrated under reduced pressure to afford compound **55** (68.0 g, crude) as a white solid.

General procedure for the preparation of Intermediate **57**:



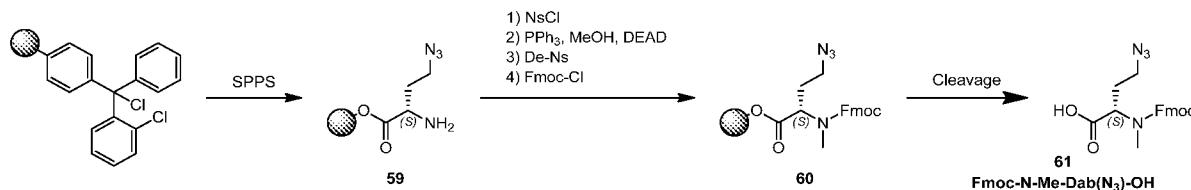
To a solution of dtbbpy (15.9 g, 59.5 mmol, 0.50 equiv.) in DMA (150 mL) was added NiBr<sub>2</sub>(DME) (18.3 g, 59.5 mmol, 0.50 equiv.), and the mixture was stirred at 25 °C for 0.5 h. Next, the mixture was added to a solution of compound **55** (68.0 g, 119 mmol, 1.00 equiv.), compound **56** (62.2 g, 238.3 mmol, 2.00 equiv.), and freshly activated Zn powder (34.6 g, 529 mmol, 4.44 equiv.) in DMA (500 mL). The mixture was stirred at 25 °C for 11.5 h. TLC (petroleum ether:ethyl acetate = 3:1) indicated compound **55** ( $R_f$  = 0.52) was consumed, and new spots ( $R_f$  = 0.22, 0.45, 0.75) had formed. The reaction mixture was filtered and quenched by addition of 1 M HCl, and then diluted with water (1.9 L), and extracted with ethyl acetate (650 mL \* 3). The organic phase was combined and washed with brine (650 mL \* 2), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (SiO<sub>2</sub>, petroleum ether:ethyl acetate = 1:0 to 1:1). Compound **57** (40.0 g, 71.2 mmol, 59.7% yield) was obtained as a white solid (petroleum ether:ethyl acetate = 3:1,  $R_f$  = 0.45).

General procedure for the preparation of Intermediate **58**:



To a solution of compound **57** (39.5 g, 70.3 mmol, 1.00 equiv.) in DCM (200 mL) was added TFA (307 g, 2.69 mmol, 200 mL, 38.2 equiv.) at 0 °C. The mixture was stirred at 25 °C for 2 h. TLC (petroleum ether:ethyl acetate = 3:1) indicated compound **57** ( $R_f$  = 0.45) was consumed, and a new spot ( $R_f$  = 0.27) had formed. The reaction mixture was concentrated under reduced pressure followed by purification by prep-HPLC (HCl condition). Compound **58** (25.0 g, 49.1 mmol, 87.4% yield, 99.3% purity) was obtained as a white solid.

General procedure for the preparation of Intermediate **61**:



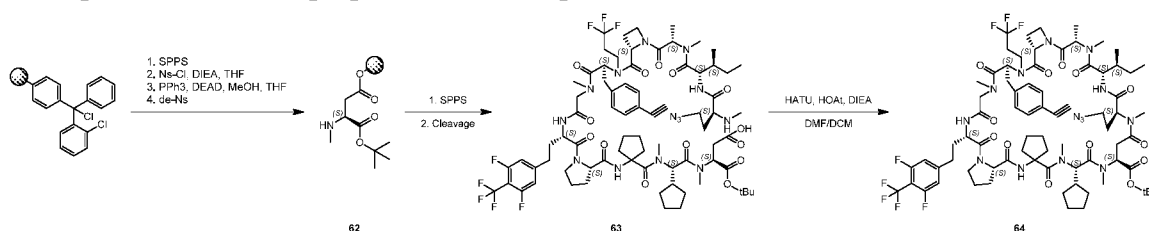
Intermediate **61** was synthesized using standard Fmoc chemistry as follows:

- 1) Resin preparation: DCM (50 mL) was added to the vessel containing 2-CTC Resin (10.0 mmol, 10.2 g, Sub = 0.980 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained, and a solution of Fmoc-Dab(N<sub>3</sub>)-OH (3.67 g, 10.0 mmol, 1.00 equiv.) in DCM (50 mL) was added to the resin, followed by addition of DIEA (7.5 mL, 40.0 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH

(11.0 mL) was added to the mixture, and bubbled with N<sub>2</sub> for 30 min. The resin was then drained and washed with DMF (100 mL \* 5).

- 2) Deprotection: 20% piperidine in DMF (50 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (100 mL \* 5) and THF (100 mL \* 2). The de-protection was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin was blue or brownish red.
- 3) A solution of 2-nitrobenzenesulfonyl chloride (6.00 equiv.) and DIEA (12.0 equiv.) in THF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 1 h at 25 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The reaction was then drained and washed with DMF (100 mL \* 5) and THF (100 mL \* 2).
- 4) A solution of triphenylphosphine (10.0 equiv.) and MeOH (10.0 equiv.) in THF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 5 min at 25 °C. Next, DEAD (10.0 equiv.) was added to the resin with N<sub>2</sub> bubbling at 0 °C. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for a total of 2 h. The reaction was then drained, and washed with DMF (100 mL \* 5) and THF (100 mL \* 2). This process was repeated one more time. The mixture was then drained, and washed with DMF (100 mL \* 5).
- 5) A solution of sodium benzenethiolate (2.00 equiv.) in DMF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 1 h at 25 °C. The reaction was then drained, and washed with DMF (100 mL \* 5). This process was repeated one more time. The de-protection was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin remained blue or brownish red.
- 6) A solution of Fmoc-Cl (3.00 equiv.) and DIEA (6.00 equiv.) in DMF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 1 h at 25 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The reaction was then drained and washed with DMF (100 mL \* 5), and MeOH (100 mL \* 3). Next, the resin was drained, and then dried under vacuum.
- 7) Cleavage cocktail (20% HFIP/DCM, 100 mL) was added to the flask containing the side chain protected peptide at room temperature, and the mixture was bubbled with N<sub>2</sub> for 10 min (100 mL\*3).
- 8) The mixture was filtered. The filtrate was collected, and concentrated under reduced pressure to give compound **61**, **Fmoc-N-Me-Dab(N<sub>3</sub>)-OH** (2.8 g, crude) as a white solid.

General procedure for the preparation of Compound **2114**:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

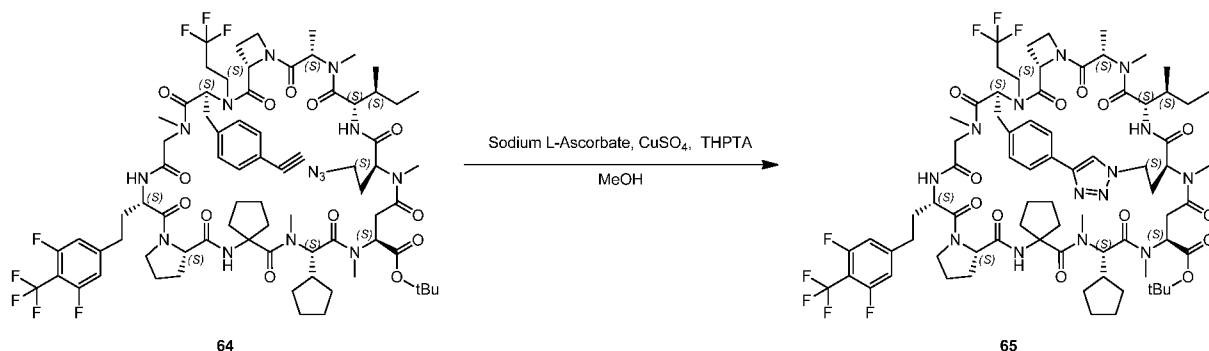
- 1) Resin preparation: DCM (15 mL) was added to the vessel containing 2-CTC Resin (1.00 mmol, 3.33 g, Sub = 0.300 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained, and a solution of Fmoc-Asp-O $t$ Bu (412 mg, 1.00 mmol, 1.00 equiv.) in DCM (15 mL) was added to the resin, followed by addition of DIEA (0.75 mL, 4.00 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (3.50 mL) was added to the mixture and bubbled with N<sub>2</sub> for 30 min. The resin was then drained and washed with DMF (100 mL \* 5).
- 2) Deprotection: 20% piperidine in DMF (40 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (100 mL \* 5). The de-protection was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin turned blue or brownish red.
- 3) A solution of 2-nitrobenzenesulfonyl chloride (6.00 equiv.) and DIEA (12.0 equiv.) in THF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 1 h at 25 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The reaction was then drained, and washed with DMF (100 mL \* 5) and THF (100 mL \* 2).
- 4) A solution of triphenylphosphine (10.0 equiv.) and MeOH (10.0 equiv.) in THF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 5 min at 25 °C. Next, DEAD (10.0 equiv.) was added to the resin with N<sub>2</sub> bubbling at 0 °C. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. The reaction was then drained, and washed with DMF (100 mL \* 5) and THF (100 mL \* 2). This process was repeated one more time. The mixture was then drained and washed with DMF (100 mL \* 5).
- 5) A solution of sodium benzenethiolate (2.00 equiv.) in DMF (50 mL) was added to the resin with N<sub>2</sub> bubbling for 1 h at 25 °C. The reaction was then drained and washed with DMF (100 mL \* 5). This process was repeated one more time. The de-protection was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin turned blue or brownish red.
- 6) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.) and DIC (5.20 equiv.) in 50% NMP/DMF (15 mL) was added to the resin with N<sub>2</sub> bubbling for 16 h at 25 °C. The coupling reaction was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The mixture was then drained and washed with DMF (100 mL \* 5).
- 7) Deprotection: 2% DBU in DMF (40 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (100 mL \* 5). The de-protection reaction was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin turned blue or brownish red.
- 8) Steps 6~7 were repeated for the amino acids (Entries 3-7, 9-11) in the Table below.

- 9) Coupling of **Fmoc-Aze-Cl**: A solution of DIEA (6.00 mmol, 6.00 equiv.) in DCM (40.0 mL) was added to the resin, followed by the addition of **Fmoc-Aze-Cl** (3.00 mmol, 3.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C. The coupling reaction was monitored by ninhydrin test. If the resin remained colorless, the coupling was judged to be complete. The resin was then washed with DMF (100 mL \* 5).
- 10) After coupling of the last amino acid, a solution of 2% DBU in DMF (40 mL) was added to the vessel, and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (100 mL \* 5) and MeOH (100 mL \* 3). The resin was then drained and dried under vacuum.

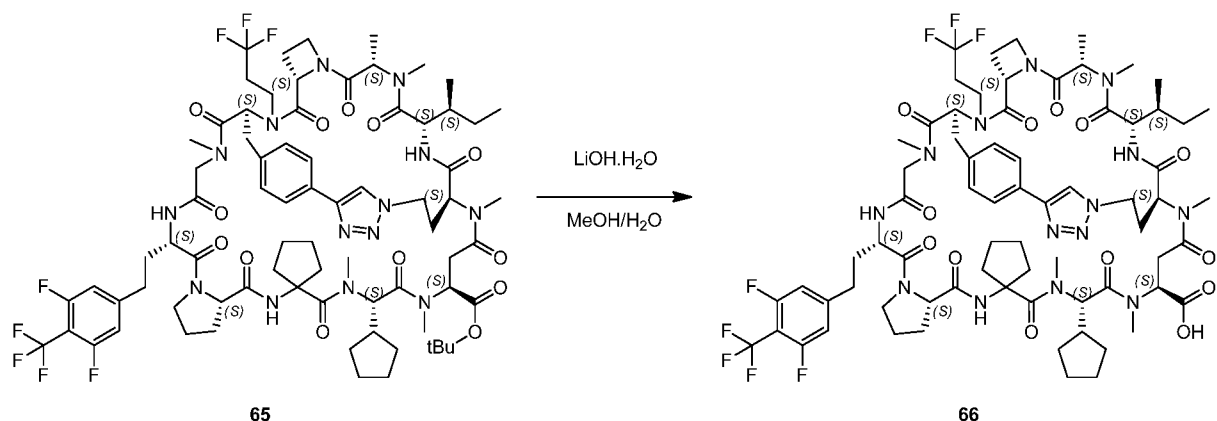
| #  | Materials                                                         | Coupling reagents                |
|----|-------------------------------------------------------------------|----------------------------------|
| 1  | Fmoc-Asp-OtBu (1.00 eq)                                           | DIEA (4.00 eq)                   |
| 2  | Fmoc-N-Me-Gly(cPent)-OH (3.60 eq)                                 | HOAt (2.30 eq) and DIC (5.20 eq) |
| 3  | Fmoc-cLeu-OH (3.60 eq)                                            | HOAt (2.30 eq) and DIC (5.20 eq) |
| 4  | Fmoc-Pro-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (1.50 eq) | HOAt (0.96 eq) and DIC (2.17 eq) |
| 6  | Fmoc-Sar-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |
| 7  | Fmoc-N-3,3,3-Trifluoropropyl-Dab(N <sub>3</sub> )-OH (1.50 eq)    | HOAt (0.96 eq) and DIC (2.17 eq) |
| 8  | Fmoc-Aze-Cl (3.00 eq)                                             | DIEA (6.00 eq)                   |
| 9  | Fmoc-N-Me-Ala-OH (3.60 eq)                                        | HOAt (2.30 eq) and DIC (5.20 eq) |
| 10 | Fmoc-Ile-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |
| 11 | Fmoc-N-Me-Dab(N <sub>3</sub> )-OH (1.50 eq)                       | HOAt (0.96 eq) and DIC (2.17 eq) |

Peptide cleavage and cyclization were conducted according to the following steps:

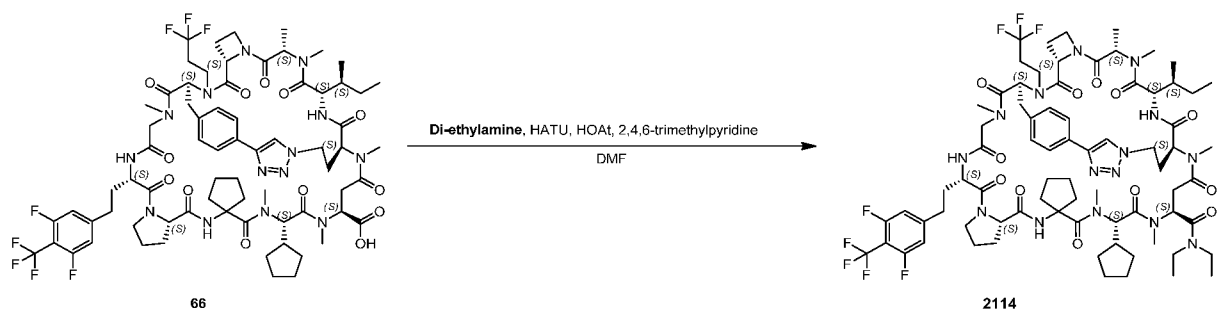
- 1) Cleavage cocktail (20% HFIP/DCM, 50 mL) was added to the flask containing the side chain protected peptide at room temperature, and the mixture was bubbled with N<sub>2</sub> for 10 min.
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give compound **63** (1.27 g, crude) as a white solid.
- 3) Compound **63** (1.27 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (970 mL). Next, HATU (613 mg, 1.61 mmol, 2.00 equiv.), HOAt (218 mg, 1.61 mmol, 2.00 equiv.), and DIEA (416 mg, 3.22 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 2 h. LCMS indicated compound **63** was consumed, and desired mass (R<sub>t</sub> = 0.481 min, MS cal.: 1556.75, MS observed: [M+H]<sup>+</sup> = 1557.9) was detected.
- 4) The mixture was concentrated under reduced pressure, and the crude was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **64** (350 mg, 19.2% yield, 85.6% purity) as a white solid.



A fresh mixture of sodium *L*-Ascorbate (0.40 M, 11.2 mL, 20.0 equiv.), CuSO<sub>4</sub> (0.40 M, 2.80 mL, 5.00 equiv.), and THPTA (488 mg, 1.12 mmol, 5.00 equiv.) was added to a solution of compound **64** (350 mg, 224 μmol, 1.00 equiv.) in MeOH (230 mL) under N<sub>2</sub> atmosphere at 0 °C. Next, the mixture was stirred at 0 °C for 1 h. LCMS indicated compound **64** was consumed and the desired mass ( $R_t = 0.422$  min, MS cal.: 1557.67, MS observed:  $[M+H]^+ = 1558.5$ ) was detected. Next, the mixture was concentrated under reduced pressure, and the residue was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **65** (160 mg, 102 μmol, 45.3% yield, 92.7% purity) as a white solid.



To a solution of compound **65** (160 mg, 102 μmol, 1.00 equiv.) in MeOH (1.0 mL) and H<sub>2</sub>O (0.5 mL) was added LiOH·H<sub>2</sub>O (68.9 mg, 1.64 mmol, 16.0 equiv.) at 20 °C. The mixture was then stirred at 20 °C for 24 h. LCMS indicated compound **65** was consumed, and the desired mass ( $R_t = 0.356$  min, MS cal.: 1500.68, MS observed:  $[M+H]^+ = 1501.7$ ) was detected. The mixture was then purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **66** (126 mg, 83.9 μmol, 80.6% yield, 98.54% purity) as a white solid.



To a solution of compound **66** (5.00 mg, 3.33  $\mu\text{mol}$ , 1.00 equiv.) and di-ethylamine (974  $\mu\text{g}$ , 13.3  $\mu\text{mol}$ , 4.00 equiv.) in DMF (0.05 mL) was added HATU (2.53 mg, 6.66  $\mu\text{mol}$ , 2.00 equiv.), HOAt (906  $\mu\text{g}$ , 6.66  $\mu\text{mol}$ , 2.00 equiv.), and 2,4,6-trimethylpyridine (3.23 mg, 26.6  $\mu\text{mol}$ , 3.60  $\mu\text{L}$ , 8.00 equiv.) at 20 °C. The mixture was then stirred at 20 °C for 2 h. LCMS indicated compound **66** was consumed, and the desired mass ( $R_t = 0.400$  min, MS cal.: 1555.76, MS observed:  $[\text{M}+\text{H}]^+ = 1557.9$ ) was detected. Next, the mixture was purified by prep-HPLC (A: 0.1% TFA in  $\text{H}_2\text{O}$ , B: ACN) to afford compound **2114** (3.39 mg, 2.17  $\mu\text{mol}$ , 59.1% yield, 90.5% purity) as a white solid.

Purification Conditions:

|                                |                                                                 |    |    |
|--------------------------------|-----------------------------------------------------------------|----|----|
| <b>Instrument</b>              | AUNO LC-2000                                                    |    |    |
| <b>Column</b>                  | YMC-Actus Triart Prep C8-S, 30 $\times$ 150 mm, 5 $\mu\text{m}$ |    |    |
| <b>Wavelength</b>              | 220 nm & 254nm                                                  |    |    |
| <b>Column Oven Temperature</b> | Ambient                                                         |    |    |
| <b>Flow Rate</b>               | 25.0 mL/min                                                     |    |    |
| <b>Injector Volume</b>         | 8 mL                                                            |    |    |
| <b>Diluent</b>                 | ACN/Water=1/4                                                   |    |    |
| <b>Mobile Phase A</b>          | 0.1% TFA in water                                               |    |    |
| <b>Mobile Phase B</b>          | ACN                                                             |    |    |
| <b>Gradient Program</b>        | Time (min)                                                      | A% | B% |
|                                | 0                                                               | 50 | 50 |
|                                | 40                                                              | 20 | 80 |
|                                | 40.1                                                            | 10 | 90 |
|                                | 45                                                              | 10 | 90 |
|                                | 45.1                                                            | 90 | 10 |
|                                | 50                                                              | 90 | 10 |
| <b>Retention Time</b>          | 28 min                                                          |    |    |

Analytical HPLC Conditions:

|                                 |                                                             |                   |                   |
|---------------------------------|-------------------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>               | Thermo Scientific VANQUISH CORE                             |                   |                   |
| <b>Mobile Phase</b>             | A: $\text{H}_2\text{O}$ (0.1% TFA in $\text{H}_2\text{O}$ ) |                   |                   |
|                                 | B: ACN (0.075% TFA in ACN)                                  |                   |                   |
| <b>Gradient(55-85-20min-50)</b> | Time (min)                                                  | Percent Solvent A | Percent Solvent B |
|                                 | 0                                                           | 45                | 55                |
|                                 | 20                                                          | 15                | 85                |
|                                 | 20.1                                                        | 10                | 90                |
|                                 | 22                                                          | 10                | 90                |
| <b>Column</b>                   | Gemini C18 5 $\mu\text{m}$ 110A 150*4.6mm                   |                   |                   |
| <b>Flow Rate</b>                | 1 mL/min                                                    |                   |                   |
| <b>Wavelength</b>               | 220 & 254 nm                                                |                   |                   |
| <b>Oven Tem.</b>                | 50°C                                                        |                   |                   |

The same general procedure that was used to prepare **2114** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance      |
|-------------|-------------|-----------|-----------------|
| <b>2047</b> | 4.4         | 99.3      | white solid     |
| <b>2056</b> | 3.5         | 96.8      | white solid     |
| <b>2058</b> | 4.3         | 95.3      | white solid     |
| <b>2062</b> | 4.4         | 91.3      | white solid     |
| <b>2069</b> | 4.1         | 95.8      | white solid     |
| <b>2072</b> | 6.4         | 91.8      | colorless oil   |
| <b>2075</b> | 3.2         | 97.4      | white solid     |
| <b>2078</b> | 2.4         | 83.3      | white solid     |
| <b>2085</b> | 2.4         | 97.1      | white solid     |
| <b>2086</b> | 3.1         | 96.6      | white solid     |
| <b>2087</b> | 3.9         | 90.0      | white solid     |
| <b>2104</b> | 4.5         | 90.4      | white solid     |
| <b>2105</b> | 4.1         | 96.5      | white solid     |
| <b>2106</b> | 2.8         | 94.3      | white solid     |
| <b>2107</b> | 2.8         | 96.3      | white solid     |
| <b>2113</b> | 3.1         | 91.8      | off-white solid |
| <b>2114</b> | 3.4         | 90.5      | white solid     |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2047</b> | 99.3       | 11.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2056</b> | 96.8       | 9.6           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2058</b> | 95.3       | 10.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2062</b> | 91.3       | 11.1          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2069</b> | 95.8       | 11.7          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2072</b> | 91.8       | 14.3          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2075</b> | 97.4       | 9.0           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2078</b> | 83.3       | 9.5           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2085</b> | 97.1       | 9.2           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2086</b> | 96.6       | 10.2          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2087</b> | 90.0       | 8.8           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2104</b> | 90.4       | 10.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2105</b> | 96.5       | 11.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2106</b> | 94.3       | 10.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2107</b> | 96.3       | 9.7           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2113</b> | 91.8       | 9.8           | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |      |                |                              |                   |                                  |
|-------------|------|------|----------------|------------------------------|-------------------|----------------------------------|
| <b>2114</b> | 90.5 | 11.0 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
|-------------|------|------|----------------|------------------------------|-------------------|----------------------------------|

Analytical HPLC Conditions:

|                                 |                                                    |                   |                   |
|---------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>               | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                 | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(40-70-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                 | 0                                                  | 60                | 40                |
|                                 | 20                                                 | 30                | 70                |
|                                 | 20.1                                               | 10                | 90                |
|                                 | 22                                                 | 10                | 90                |
| <b>Column</b>                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>               | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>               | 50°C                                               |                   |                   |

The same general procedure that was used to prepare **2114** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2111</b> | 4.1         | 99.2      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                      |
|-------------|------------|---------------|----------------|------------------------------|---------------------|----------------------------------|
| <b>2111</b> | 99.2       | 8.4           | 40-70-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A<br>150*4.6mm |

Analytical HPLC Conditions:

|                                  |                                                    |                   |                   |
|----------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>                | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>              | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                  | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (50-80-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                  | 0                                                  | 50                | 50                |
|                                  | 20                                                 | 20                | 80                |
|                                  | 20.1                                               | 10                | 90                |
|                                  | 22                                                 | 10                | 90                |
| <b>Column</b>                    | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                 | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>                | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>                | 50°C                                               |                   |                   |

The same general procedure that was used to prepare **2114** was also used to prepare the peptides listed below which were analyzed by the method above.

| # | Amount (mg) | Purity(%) | Appearance |
|---|-------------|-----------|------------|
|---|-------------|-----------|------------|

|             |     |      |             |
|-------------|-----|------|-------------|
| <b>2046</b> | 2.9 | 87.4 | white solid |
| <b>2048</b> | 3.9 | 99.5 | white solid |
| <b>2066</b> | 4.3 | 95.9 | white solid |
| <b>2067</b> | 4.3 | 93.8 | white solid |
| <b>2068</b> | 3.4 | 99.3 | white solid |
| <b>2077</b> | 3.2 | 97.7 | white solid |
| <b>2080</b> | 2.8 | 95.5 | white solid |
| <b>2090</b> | 5.6 | 95.7 | white solid |
| <b>2108</b> | 3.9 | 96.2 | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2046</b> | 87.4       | 10.4          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2048</b> | 99.5       | 10.8          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2066</b> | 95.9       | 10.1          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2067</b> | 93.8       | 11.1          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2068</b> | 99.3       | 11.3          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2077</b> | 97.7       | 12.6          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2080</b> | 95.5       | 9.3           | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2090</b> | 95.7       | 10.7          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2108</b> | 96.2       | 11.8          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                                 |                                                    |                   |                   |
|---------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>               | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                 | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(60-90-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                 | 0                                                  | 40                | 60                |
|                                 | 20                                                 | 10                | 90                |
|                                 | 20.1                                               | 10                | 90                |
|                                 | 22                                                 | 10                | 90                |
| <b>Column</b>                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>               | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>               | 50°C                                               |                   |                   |

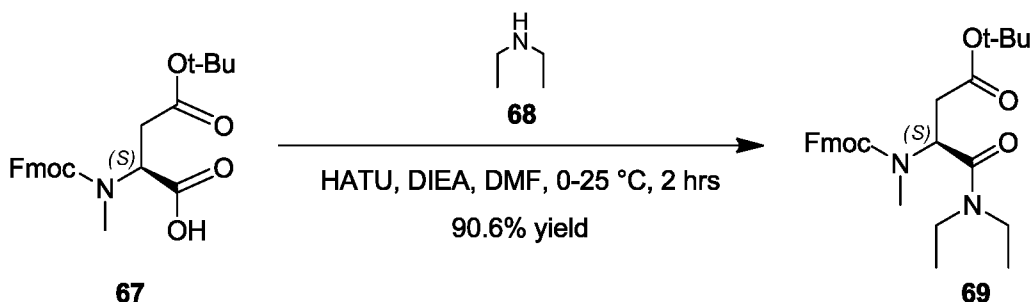
- 5 The same general procedure that was used to prepare **2114** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2035</b> | 9.0         | 99.2      | white solid |
| <b>2057</b> | 3.8         | 93.8      | white solid |

|             |     |      |                 |
|-------------|-----|------|-----------------|
| <b>2059</b> | 4.5 | 95.2 | white solid     |
| <b>2063</b> | 5.3 | 95.8 | white solid     |
| <b>2073</b> | 3.3 | 96.6 | white solid     |
| <b>2076</b> | 3.4 | 97.6 | white solid     |
| <b>2084</b> | 4.2 | 94.4 | white solid     |
| <b>2088</b> | 3.5 | 95.6 | white solid     |
| <b>2089</b> | 3.4 | 93.8 | off-white solid |
| <b>2103</b> | 2.7 | 94.0 | white solid     |

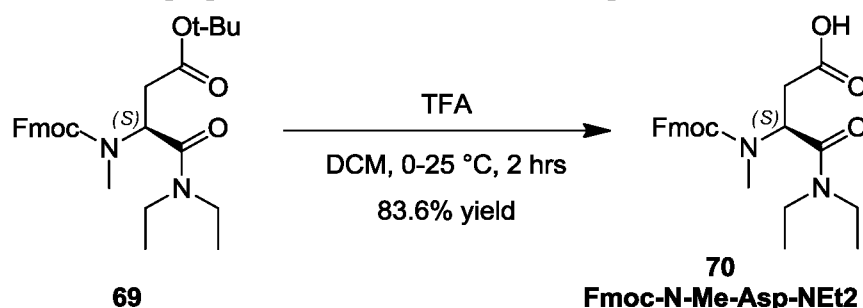
| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2035</b> | 99.2       | 10.8          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2057</b> | 93.8       | 8.2           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2059</b> | 95.2       | 10.9          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2063</b> | 95.8       | 10.7          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2073</b> | 96.6       | 11.7          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2076</b> | 97.6       | 12.3          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2084</b> | 94.4       | 8.8           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2088</b> | 95.6       | 11.3          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2089</b> | 93.8       | 9.8           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2103</b> | 94.0       | 9.6           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

General procedure for the preparation of intermediate **69**:



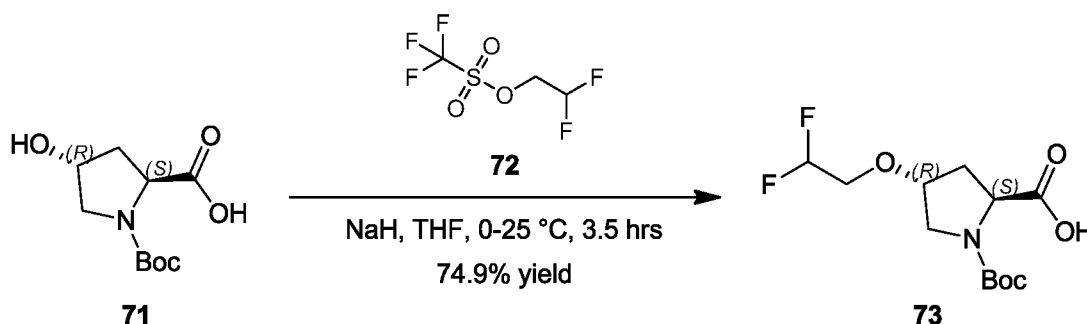
To a solution of compound **67** (10.0 g, 23.5 mmol, 1.00 equiv.) in DMF (100 mL) was added HATU (9.83 g, 25.8 mmol, 1.10 equiv.), DIEA (6.07 g, 47.0 mmol, 7.76 mL, 2.00 equiv.) and compound **68** (2.06 g, 28.2 mmol, 2.90 mL, 1.20 equiv.) at 0 °C. The mixture was stirred at 25 °C for 2 hrs. LCMS showed compound **67** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 2.16$  min, MS cal.: 480.26, MS observed:  $[M+H]^+ = 481.2$ ). The reaction mixture was diluted with water (100 mL), and extracted with ethyl acetate (50.0 mL \* 3). The combined organic layers were washed with water (100 mL \* 3) and brine (100 mL \* 2), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure to afford compound **69** (10.8 g, 21.3 mmol, 90.6% yield, 94.8% purity) as a brown oil.

General procedure for the preparation of **70**, Fmoc-N-Me-Asp-NEt<sub>2</sub>:



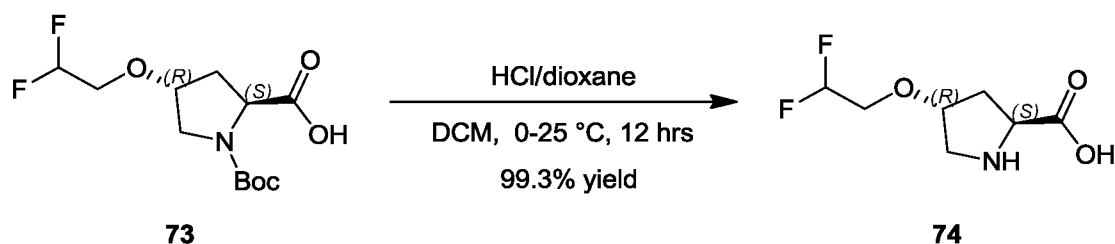
To a solution of compound **69** (10.8 g, 21.3 mmol, 1.00 equiv.) in DCM (54.0 mL) was added TFA (82.8 g, 726 mmol, 54.0 mL, 34.1 equiv.) at 0 °C. The mixture was stirred at 25 °C for 2 hrs. LCMS showed compound **69** was consumed completely, and one main peak with the desired mass was detected ( $R_t = 0.47$  min, MS cal.: 424.20, MS observed:  $[M+H]^+ = 425.3$ ). The mixture was concentrated under reduced pressure. Next, the mixture was purified by reversed-phase HPLC (neutral conditions) to afford **70**, Fmoc-N-Me-Asp-NEt<sub>2</sub> (7.60 g, 17.8 mmol, 83.6% yield, 99.5% purity) as a brown solid.

General procedure for the preparation of Intermediate **73**:



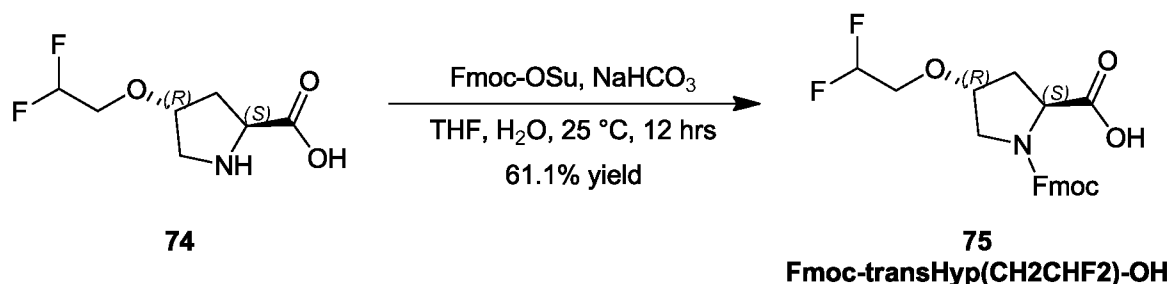
To a solution of compound **71** (23.0 g, 99.4 mmol, 1.00 equiv.) in THF (450 mL) was added NaH (8.35 g, 208 mmol, 60% purity, 2.10 equiv.) at 0 °C under N<sub>2</sub> atmosphere. Next, the mixture was stirred at 25 °C for 30 min. A solution of compound **72** (23.4 g, 109 mmol, 1.10 equiv.) in THF (15.0 mL) was then added dropwise, and the mixture was stirred at 25 °C for 3 hrs under N<sub>2</sub> atmosphere. LCMS showed compound **71** was consumed and the desired mass was detected by LCMS ( $R_t = 0.33$  min, MS cal.: 295.12, MS observed:  $[M+Na]^+ = 318.2$ ). The reaction mixture was quenched by addition of water (200 mL), and the mixture was stirred at 25 °C for 1 hr. Next, the mixture was extracted with MTBE (150 mL \* 3). The aqueous phase was adjusted to pH = 3 with 1M HCl aqueous solution, and extracted with ethyl acetate (200 mL \* 3). The combined organic layers were washed with brine (200 mL), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated under reduced pressure to afford compound **73** (22.0 g, 74.5 mmol, 74.9% yield, 100% purity) as a colorless oil which was used in the next step without further purification.

General procedure for the preparation of Intermediate **74**:



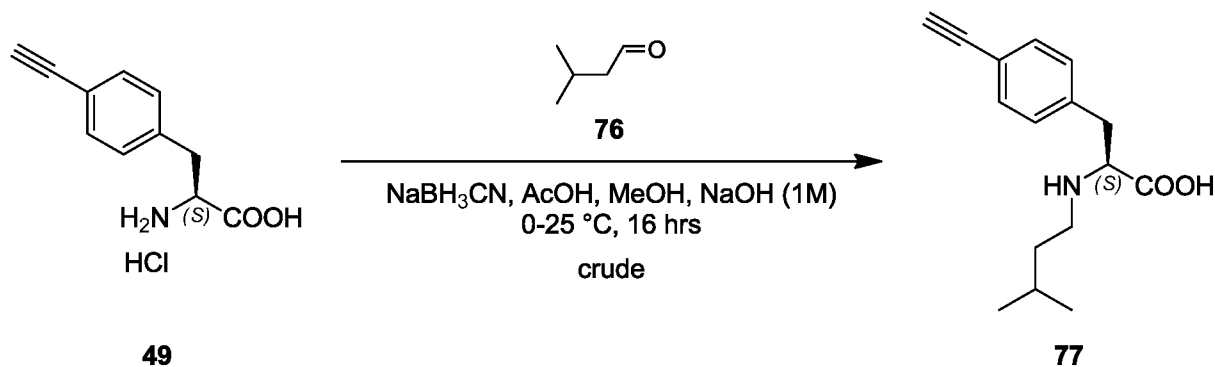
To a solution of compound **73** (10.0 g, 33.8 mmol, 1.00 equiv.) in DCM (30.0 mL) was added HCl/dioxane (2.00 M, 72.4 mL, 4.28 equiv.) at 0 °C. The mixture was stirred at 25 °C for 12 hrs. LCMS showed compound **73** was consumed, and the desired mass was detected by LCMS ( $R_t = 0.28$  min, MS cal.: 195.07, MS observed:  $[M+H]^+ = 196.0$ ). The reaction mixture was concentrated under reduced pressure to afford compound **74** (7.30 g, 33.6 mmol, 99.3% yield, HCl salt) as a white solid which was used in the next step without further purification.

General procedure for preparation of **75**, Fmoc-transHyp(CH<sub>2</sub>CHF<sub>2</sub>)-OH:



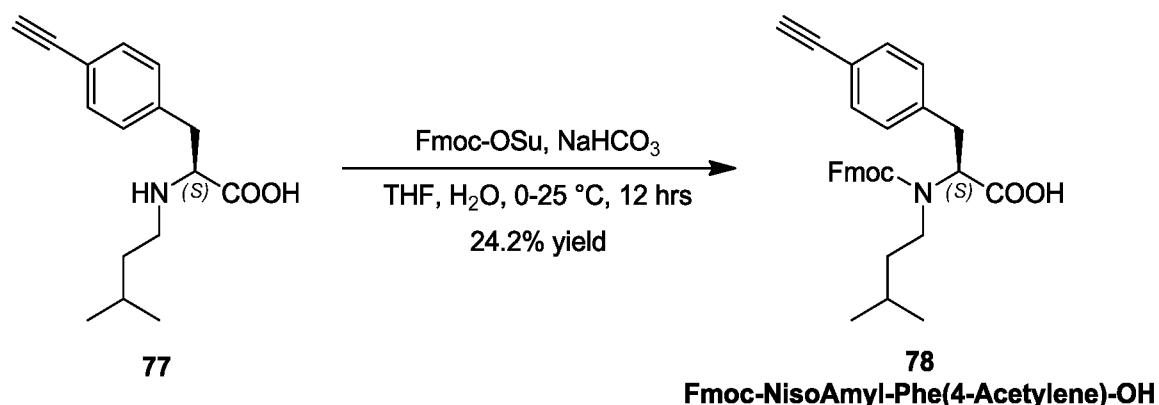
To a solution of compound **74** (7.30 g, 33.6 mmol, 1.00 equiv., HCl salt) in THF (120 mL) was added NaHCO<sub>3</sub> (8.48 g, 100 mmol, 3.93 mL, 3.00 equiv.) in H<sub>2</sub>O (40.0 mL), and then Fmoc-OSu (10.2 g, 30.2 mmol, 0.90 equiv.) was added. The mixture was stirred at 25 °C for 12 hrs. LCMS showed compound **74** was consumed, and the desired mass was detected by LCMS ( $R_t = 1.36$  min, MS cal.: 417.14, MS observed:  $[M+H]^+ = 418.3$ ). The mixture was adjusted to pH = 3 with 1M HCl aqueous solution, and extracted with ethyl acetate (300 mL \* 3). The combined organic layers were washed with water (200 mL \* 3) and brine (100 mL), dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated in vacuum. The residue was triturated in MTBE (90.0 mL) at 25 °C for 3 hrs. **Compound 75, Fmoc-transHyp(CH<sub>2</sub>CHF<sub>2</sub>)-OH** (8.69 g, 20.5 mmol, 61.1% yield, 98.8% purity) was obtained as a white solid.

General procedure for the preparation of Intermediate **77**:



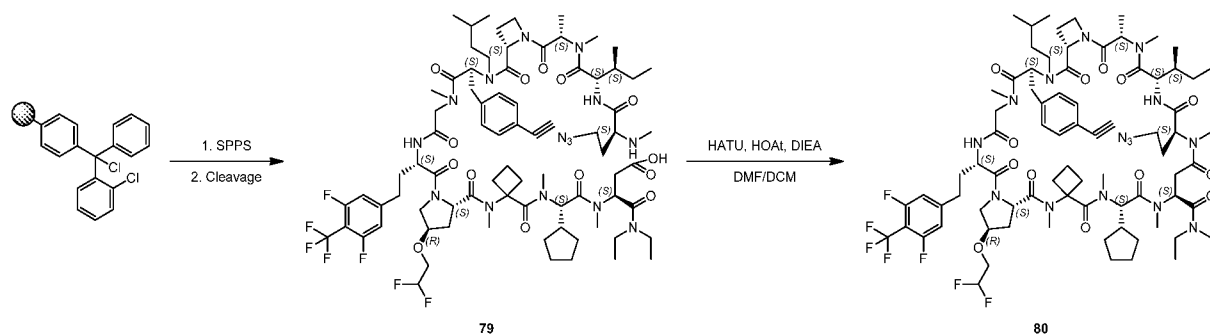
To a solution of compound **49** (1.70 g, 7.53 mmol, 1.00 equiv., HCl salt) in MeOH (30.0 mL) was added NaOH (1 M, 691  $\mu$ L, 0.05 equiv.), AcOH (22.6 mg, 376  $\mu$ mol, 21.5  $\mu$ L, 0.05 equiv.) at 0 °C, compound **76** (583 mg, 6.78 mmol, 743  $\mu$ L, 0.90 equiv.) and NaBH<sub>3</sub>CN (1.28 g, 20.3 mmol, 2.70 equiv.). Next, the mixture was stirred at 25 °C for 16 hrs. LCMS showed compound **49** was consumed, and the desired mass was detected by LCMS ( $R_t$  = 0.28 min, MS cal.: 259.16, MS observed:  $[M+H]^+$  = 260.1). The reaction mixture was concentrated under reduced pressure to obtain compound **77** (1.95 g, crude) as a white solid which was used in the next step without further purification.

General procedure for preparation of **78**, **Fmoc-NisoAmyl-Phe(4-Acetylene)-OH**:



To a solution of compound **77** (1.95 g, 7.52 mmol, 1.00 equiv.) in THF (60.0 mL) was added NaHCO<sub>3</sub> (1.89 g, 22.5 mmol, 877  $\mu$ L, 3.00 equiv.) in H<sub>2</sub>O (20.0 mL) at 0 °C, followed by the addition of Fmoc-OSu (2.28 g, 6.77 mmol, 0.90 equiv.). The mixture was then stirred at 25 °C for 12 hrs. LCMS showed compound **77** was consumed, and the desired mass was detected by LCMS ( $R_t$  = 0.52 min, MS cal.: 481.23, MS observed:  $[M+H]^+$  = 482.2). The mixture was adjusted to pH = 3 with 1M HCl aqueous solution, and extracted with ethyl acetate (100 mL \*2). The combined organic layers were washed with water (90.0 mL \* 3) and brine 90.0 mL, dried over Na<sub>2</sub>SO<sub>4</sub>, filtered and concentrated in vacuum. The residue was purified by prep-HPLC (HCl conditions) to afford **78**, **Fmoc-NisoAmyl-Phe(4-Acetylene)-OH** (900 mg, 1.82 mmol, 24.2% yield, 97.4% purity) as a white solid.

General procedure for the preparation of Compound **2152**:



The peptide was synthesized using standard Fmoc chemistry according to the following steps.

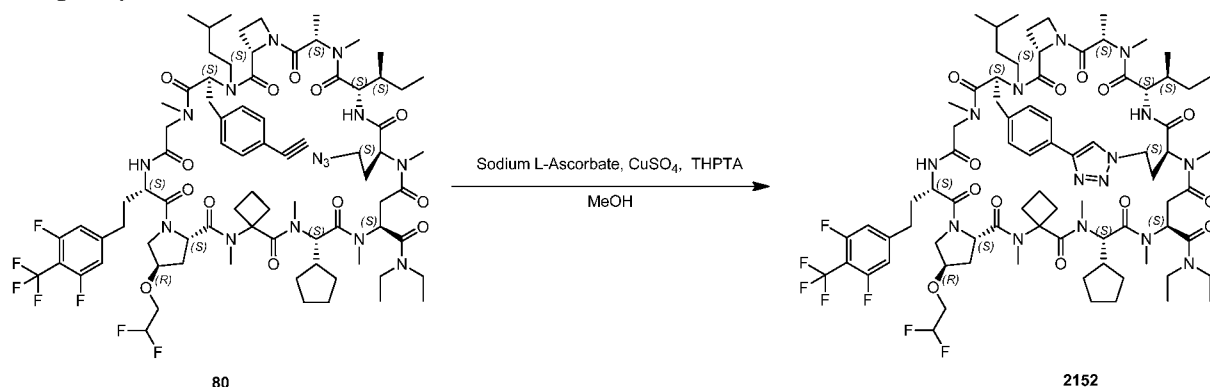
- 1) Resin preparation: DCM (5 mL) was added to the vessel containing 2-CTC Resin (0.200 mmol, 0.667 g, Sub = 0.300 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-*N*-Me-Asp-NEt<sub>2</sub> (84.9 mg, 0.200 mmol, 1.00 equiv.) in DCM (5 mL) was added to the resin, followed by the addition of DIEA (0.15 mL, 0.800 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (0.7 mL) was added to the mixture, and the solution was bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (15 mL \* 5).
- 2) Deprotection: 2% DBU in DMF (10 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (15 mL \* 5). The deprotection was monitored by ninhydrin test or chloranil test. Completion of the reaction was determined if the resin turned blue or brownish red.
- 3) Coupling: A solution of Fmoc-*N*-Me-Gly(cPent)-OH (3.60 equiv.), HOAt (2.30 equiv.), and DIC (5.20 equiv.) in 50% NMP/DMF (5 mL) was added to the resin with N<sub>2</sub> bubbling for 16 h at 25 °C. The coupling was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The mixture was then drained and washed with DMF (15 mL \* 5).
- 4) Steps 2~3 were repeated for the amino acids (Entries 3,5-7,9-11) in the Table below.
- 5) Coupling of Fmoc-transHyp(CH<sub>2</sub>CHF<sub>2</sub>)-Cl: A solution of DIEA (1.20 mmol, 6.00 equiv.) in DCM (2.0 mL) was added to the resin, followed by the dropwise addition of Fmoc-transHyp(CH<sub>2</sub>CHF<sub>2</sub>)-Cl (0.60 mmol, 3.00 equiv.). The mixture was then bubbled with N<sub>2</sub> flow at 25 °C. The coupling was monitored by ninhydrin test. If the resin remained colorless, the coupling was judged to be complete. The resin was then washed with DMF (100 mL \* 5).
- 6) Step 5 was repeated for the coupling of **Fmoc-Aze-Cl** (Entry 8, Table below).
- 7) After coupling of the last amino acid, a solution of 2% DBU in DMF (15 mL) was added to the vessel and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (30 mL \* 5), and MeOH (30 mL \* 3). Finally, the resin was drained and dried under vacuum.

| # | Materials                                | Coupling Reagents |
|---|------------------------------------------|-------------------|
| 1 | Fmoc-N-Me-Asp-NEt <sub>2</sub> (1.00 eq) | DIEA (4.00 eq)    |

|    |                                                                   |                                  |
|----|-------------------------------------------------------------------|----------------------------------|
| 2  | Fmoc-N-Me-Gly(cPent)-OH (3.60 eq)                                 | HOAt (2.30 eq) and DIC (5.20 eq) |
| 3  | Fmoc-N-Me-cVal-OH (3.60 eq)                                       | HOAt (2.30 eq) and DIC (5.20 eq) |
| 4  | Fmoc-transHyp(CH <sub>2</sub> CHF <sub>2</sub> )-Cl (3.00 eq)     | DIEA (6.00 eq)                   |
| 5  | Fmoc-homoPhe(3,5-F <sub>2</sub> -4-CF <sub>3</sub> )-OH (1.50 eq) | HOAt (0.96 eq) and DIC (2.17 eq) |
| 6  | Fmoc-Sar-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |
| 7  | Fmoc-NisoAmyl-Phe(4-Acetylene)-OH (1.50 eq)                       | HOAt (0.96 eq) and DIC (2.17 eq) |
| 8  | Fmoc-Aze-Cl (3.00 eq)                                             | DIEA (6.00 eq)                   |
| 9  | Fmoc-N-Me-Ala-OH (3.60 eq)                                        | HOAt (2.30 eq) and DIC (5.20 eq) |
| 10 | Fmoc-Ile-OH (3.60 eq)                                             | HOAt (2.30 eq) and DIC (5.20 eq) |
| 11 | Fmoc-N-Me-Dab(N <sub>3</sub> )-OH (1.50 eq)                       | HOAt (0.96 eq) and DIC (2.17 eq) |

Peptide cleavage and cyclization were conducted according to the following steps:

- 1) Cleavage cocktail (20% HFIP/DCM, 10 mL) was added to the flask containing the side chain protected peptide at room temperature, and the mixture was bubbled with N<sub>2</sub> for 10 min (10 mL\*3).
- 2) The mixture was filtered. The filtrate was collected and concentrated under reduced pressure to give compound **79** (0.180 g, crude) as a white solid.
- 3) Compound **79** (0.180 g, crude, 1.00 equiv.) was dissolved in 50% DCM/DMF (140 mL). Next, HATU (83.9 mg, 0.221 mmol, 2.00 equiv.), HOAt (30.1 mg, 0.221 mmol, 2.00 equiv.), and DIEA (57.0 mg, 0.442 mmol, 4.00 equiv.) were added to the mixture. The mixture was stirred at 25 °C for 4 h. LCMS indicated compound **79** was consumed, and the desired MS (R<sub>t</sub> = 0.533 min, MS cal.: 1610.8, MS observed: [M+H]<sup>+</sup> = 1611.7) was detected.
- 4) The mixture was concentrated under reduced pressure, and the crude was purified by prep-HPLC (A: 0.1% TFA in H<sub>2</sub>O, B: ACN) to afford compound **80** (80.0 mg, 41.4% yield, 92.2% purity) as a white solid.



A fresh mixture of sodium *L*-Ascorbate (0.40 M, 0.48 mL, 20.0 equiv.), CuSO<sub>4</sub> (0.40 M, 621 μL, 5.00 eq.), and THPTA (108 mg, 248 μmol, 5.00 equiv.) was added to a solution of compound **80** (80.0 mg, 49.7 μmol, 1.00 equiv.) in MeOH (50 mL) under N<sub>2</sub> atmosphere at 0 °C. Next, the mixture was stirred at 0 °C for 1 h. LCMS indicated compound **80** was consumed, and the desired MS (R<sub>t</sub> = 0.487 min, MS cal.: 1610.80, MS observed: [M+H]<sup>+</sup> = 1612.2) was detected. The mixture was concentrated under reduced pressure. The residue was purified by prep-HPLC

(A: 0.1% TFA in H<sub>2</sub>O, B: ACN) directly to afford **2152** (39.1 mg, 22.5  $\mu$ mol, 45.3% yield, 92.7% purity) as a white solid.

Purification Conditions:

|                                |                                          |    |    |
|--------------------------------|------------------------------------------|----|----|
| <b>Instrument</b>              | AUNO LC-2000                             |    |    |
| <b>Column</b>                  | Luna C8, 21.2 $\times$ 250 mm, 5 $\mu$ m |    |    |
| <b>Wavelength</b>              | 220 nm & 254 nm                          |    |    |
| <b>Column Oven Temperature</b> | Ambient                                  |    |    |
| <b>Flow Rate</b>               | 20.0 mL/min                              |    |    |
| <b>Injector Volume</b>         | 8 mL                                     |    |    |
| <b>Diluent</b>                 | ACN/Water=2/5                            |    |    |
| <b>Mobile Phase A</b>          | 0.1% TFA in water                        |    |    |
| <b>Mobile Phase B</b>          | ACN                                      |    |    |
| <b>Gradient Program</b>        | Time (min)                               | A% | B% |
|                                | 0                                        | 80 | 20 |
|                                | 5                                        | 44 | 56 |
|                                | 45                                       | 14 | 86 |
|                                | 45.1                                     | 10 | 90 |
|                                | 50                                       | 10 | 90 |
|                                | 50.1                                     | 90 | 10 |
|                                | 56                                       | 90 | 10 |
| <b>Retention Time</b>          | 27.2 min                                 |    |    |

5 Analytical HPLC Conditions:

|                              |                                                    |                   |                   |
|------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>            | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>          | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                              | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(55-85-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                              | 0                                                  | 45                | 55                |
|                              | 20                                                 | 15                | 85                |
|                              | 20.1                                               | 10                | 90                |
|                              | 22                                                 | 10                | 90                |
| <b>Column</b>                | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>             | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>            | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>            | 30°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance      |
|-------------|-------------|-----------|-----------------|
| <b>2137</b> | 9.1         | 96.4      | off-white solid |
| <b>2146</b> | 36.3        | 97.9      | white solid     |
| <b>2147</b> | 31.8        | 99.4      | white solid     |

|             |      |      |             |
|-------------|------|------|-------------|
| <b>2150</b> | 47.5 | 98.4 | white solid |
| <b>2151</b> | 36.5 | 97.7 | white solid |
| <b>2152</b> | 39.1 | 92.8 | White solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2137</b> | 96.4       | 11.0          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2146</b> | 97.9       | 10.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2147</b> | 99.4       | 11.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2150</b> | 98.4       | 12.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2151</b> | 97.7       | 13.0          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2152</b> | 92.8       | 13.3          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

| Instrument               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|--------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                          | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(45-75-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                          | 0                                                  | 55                | 45                |
|                          | 20                                                 | 25                | 75                |
|                          | 20.1                                               | 10                | 90                |
|                          | 22                                                 | 10                | 90                |
| Column                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate                | 1 mL/min                                           |                   |                   |
| Wavelength               | 220 & 254 nm                                       |                   |                   |
| Oven Temp.               | 50°C                                               |                   |                   |

- 5 The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>2036</b> | 15.9        | 94.4      | light yellow solid |
| <b>2037</b> | 20.7        | 94.2      | white solid        |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2036</b> | 94.4       | 9.4           | 45-75-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2037</b> | 94.2       | 11.2          | 45-75-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

| Instrument   | Thermo Scientific VANQUISH CORE                    |  |  |
|--------------|----------------------------------------------------|--|--|
| Mobile Phase | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |  |  |

|                           |                               |                   |                   |
|---------------------------|-------------------------------|-------------------|-------------------|
|                           | B: ACN (0.075% TFA in ACN)    |                   |                   |
| Gradient (50-80-20min-50) | Time (min)                    | Percent Solvent A | Percent Solvent B |
|                           | 0                             | 50                | 50                |
|                           | 20                            | 20                | 80                |
|                           | 20.1                          | 10                | 90                |
|                           | 22                            | 10                | 90                |
| Column                    | Gemini C18 5um 110A 150*4.6mm |                   |                   |
| Flow Rate                 | 1 mL/min                      |                   |                   |
| Wavelength                | 220 & 254 nm                  |                   |                   |
| Oven Temp.                | 50°C                          |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2042</b> | 10.4        | 99.3      | white solid |
| <b>2045</b> | 17.7        | 96.9      | white solid |
| <b>2050</b> | 13.0        | 99.7      | white solid |
| <b>2051</b> | 17.9        | 99.1      | white solid |
| <b>2052</b> | 17.1        | 95.2      | white solid |
| <b>2060</b> | 12.2        | 92.4      | white solid |
| <b>2079</b> | 16.6        | 94.6      | white solid |
| <b>2082</b> | 11.4        | 95.8      | white solid |
| <b>2099</b> | 20.0        | 98.9      | white solid |
| <b>2115</b> | 12.3        | 98.6      | white solid |
| <b>2116</b> | 14.2        | 99.3      | white solid |
| <b>2120</b> | 7.8         | 96.7      | white solid |
| <b>2121</b> | 33.1        | 98.6      | white solid |
| <b>2127</b> | 18.8        | 97.4      | white solid |
| <b>2128</b> | 10.7        | 98.2      | white solid |
| <b>2130</b> | 15.7        | 95.8      | white solid |
| <b>2132</b> | 20.9        | 98.9      | white solid |
| <b>2133</b> | 29.9        | 99.1      | white solid |
| <b>2142</b> | 32.2        | 98.5      | white solid |
| <b>2148</b> | 13.9        | 99.2      | white solid |
| <b>2149</b> | 8.6         | 97.8      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2042</b> | 99.3       | 10.8          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2045</b> | 96.9       | 11.1          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2050</b> | 99.7       | 11.0          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2051</b> | 99.1       | 10.4          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2052</b> | 95.2       | 10.8          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |      |                |                              |                   |                               |
|-------------|------|------|----------------|------------------------------|-------------------|-------------------------------|
| <b>2060</b> | 92.4 | 10.6 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2079</b> | 94.6 | 9.7  | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2082</b> | 95.8 | 12.1 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2099</b> | 98.9 | 10.4 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2115</b> | 98.6 | 10.8 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2116</b> | 99.3 | 11.7 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2120</b> | 96.7 | 10.8 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2121</b> | 98.6 | 11.1 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2127</b> | 97.4 | 11.4 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2128</b> | 98.2 | 11.4 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2130</b> | 95.8 | 12.2 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2132</b> | 98.9 | 13.2 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2133</b> | 99.1 | 12.6 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2142</b> | 98.5 | 9.8  | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2148</b> | 99.2 | 10.5 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| <b>2149</b> | 97.8 | 10.9 | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

| <b>Instrument</b>               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|---------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Mobile Phase</b>             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                 | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(55-85-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                 | 0                                                  | 45                | 55                |
|                                 | 20                                                 | 15                | 85                |
|                                 | 20.1                                               | 10                | 90                |
|                                 | 22                                                 | 10                | 90                |
| <b>Column</b>                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                | 1 mL/Min                                           |                   |                   |
| <b>Wavelength</b>               | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>               | 50°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

5

| <b>#</b>    | <b>Amount (mg)</b> | <b>Purity(%)</b> | <b>Appearance</b> |
|-------------|--------------------|------------------|-------------------|
| <b>2043</b> | 8.3                | 97.7             | white solid       |
| <b>2044</b> | 19.3               | 98.3             | white solid       |
| <b>2049</b> | 10.6               | 95.6             | white solid       |
| <b>2053</b> | 0.5                | 70.6             | white solid       |

|             |      |      |             |
|-------------|------|------|-------------|
| <b>2054</b> | 1.1  | 96.5 | white solid |
| <b>2055</b> | 3.6  | 93.9 | white solid |
| <b>2061</b> | 12.2 | 96.2 | white solid |
| <b>2070</b> | 10.9 | 98.3 | white solid |
| <b>2071</b> | 10.6 | 96.1 | white solid |
| <b>2074</b> | 5.4  | 86.9 | white solid |
| <b>2081</b> | 4.0  | 93.1 | white solid |
| <b>2083</b> | 10.4 | 95.1 | white solid |
| <b>2091</b> | 8.5  | 98.1 | white solid |
| <b>2092</b> | 11.0 | 96.8 | white solid |
| <b>2095</b> | 4.4  | 93.6 | white solid |
| <b>2096</b> | 2.1  | 91.0 | white solid |
| <b>2097</b> | 17.6 | 95.6 | white solid |
| <b>2098</b> | 16.6 | 95.8 | white solid |
| <b>2101</b> | 6.2  | 99.2 | white solid |
| <b>2112</b> | 14.6 | 94.7 | white solid |
| <b>2118</b> | 12.2 | 99.2 | white solid |
| <b>2119</b> | 16.4 | 98.4 | white solid |
| <b>2122</b> | 8.4  | 97.3 | white solid |
| <b>2123</b> | 21.5 | 97.7 | white solid |
| <b>2124</b> | 20.5 | 97.2 | white solid |
| <b>2125</b> | 15.9 | 96.2 | white solid |
| <b>2126</b> | 11.8 | 96.1 | white solid |
| <b>2129</b> | 18.5 | 98.1 | white solid |
| <b>2131</b> | 15.5 | 99.4 | white solid |
| <b>2134</b> | 20.5 | 97.3 | white solid |
| <b>2138</b> | 21.1 | 95.9 | white solid |
| <b>2140</b> | 11.2 | 97.2 | white solid |

| <b>Cmpd #</b> | <b>Purity (%)</b> | <b>HPLC RT (min)</b> | <b>HPLC Method</b> | <b>HPLC Mobile Phase A</b>   | <b>HPLC Mobile Phase B</b> | <b>HPLC Column</b>            |
|---------------|-------------------|----------------------|--------------------|------------------------------|----------------------------|-------------------------------|
| <b>2043</b>   | 97.7              | 10.1                 | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2044</b>   | 98.3              | 11.7                 | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2049</b>   | 95.6              | 10.8                 | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2053</b>   | 70.6              | 10.1                 | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2054</b>   | 96.5              | 12.2                 | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2055</b>   | 93.9              | 11.4                 | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2061</b>   | 96.2              | 9.1                  | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2070</b>   | 98.3              | 9.1                  | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2071</b>   | 96.1              | 9.9                  | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2074</b>   | 86.9              | 9.4                  | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |
| <b>2081</b>   | 93.1              | 8.4                  | 55-85-20MIN-50     | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN          | Gemini 5um C18 110A 150*4.6mm |

|      |      |      |                |                              |                   |                               |
|------|------|------|----------------|------------------------------|-------------------|-------------------------------|
| 2083 | 95.1 | 10.1 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2091 | 98.1 | 8.3  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2092 | 96.8 | 9.3  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2095 | 93.6 | 9.0  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2096 | 91.0 | 8.6  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2097 | 95.6 | 8.2  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2098 | 95.8 | 9.0  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2101 | 99.2 | 8.9  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2112 | 94.7 | 8.5  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2118 | 99.2 | 8.6  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2119 | 98.4 | 9.4  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2122 | 97.3 | 9.3  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2123 | 97.7 | 10.1 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2124 | 97.2 | 8.9  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2125 | 96.2 | 9.0  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2126 | 96.1 | 9.8  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2129 | 98.1 | 9.8  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2131 | 99.4 | 9.7  | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2134 | 97.3 | 10.6 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2138 | 95.9 | 10.0 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |
| 2140 | 97.2 | 10.1 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A 150*4.6mm |

## Analytical HPLC Conditions:

|                                 |                                                    |                   |                   |
|---------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>               | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                                 | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(60-90-20min-50)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                                 | 0                                                  | 40                | 60                |
|                                 | 20                                                 | 10                | 90                |
|                                 | 20.1                                               | 10                | 90                |
|                                 | 22                                                 | 10                | 90                |
| <b>Column</b>                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>                | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>               | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>               | 50°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2093</b> | 10.0        | 97.9      | white solid |
| <b>2094</b> | 3.4         | 93.4      | white solid |
| <b>2109</b> | 1.5         | 98.7      | white solid |
| <b>2135</b> | 34.1        | 98.0      | white solid |
| <b>2136</b> | 24.3        | 98.1      | white solid |
| <b>2139</b> | 10.3        | 98.4      | white solid |
| <b>2141</b> | 3.8         | 98.2      | white solid |
| <b>2166</b> | 5.1         | 95.8      | white solid |
| <b>2173</b> | 16.4        | 98.1      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2093</b> | 97.9       | 7.1           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2094</b> | 93.4       | 6.3           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2109</b> | 98.7       | 9.2           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2135</b> | 98.0       | 8.2           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2136</b> | 98.1       | 8.8           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2139</b> | 98.4       | 8.2           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2141</b> | 98.2       | 8.9           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2166</b> | 95.8       | 11.2          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2173</b> | 98.1       | 10.5          | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

#### 5 Analytical HPLC Conditions:

| Instrument            | Thermo Scientific VANQUISH CORE                    |                   |                   |
|-----------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase          | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                       | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(60-90-20min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                       | 0                                                  | 40                | 60                |
|                       | 20                                                 | 10                | 90                |
|                       | 20.1                                               | 10                | 90                |
|                       | 22                                                 | 10                | 90                |
| Column                | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate             | 1 mL/min                                           |                   |                   |
| Wavelength            | 220 & 254 nm                                       |                   |                   |
| Oven Temp.            | 30°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2110</b> | 14.7        | 98.6      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2110</b> | 98.6       | 9.5           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

| Instrument               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|--------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                          | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(65-95-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                          | 0                                                  | 35                | 65                |
|                          | 20                                                 | 5                 | 95                |
|                          | 20.1                                               | 10                | 90                |
|                          | 22                                                 | 10                | 90                |
| Column                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate                | 1 mL/min                                           |                   |                   |
| Wavelength               | 220 & 254 nm                                       |                   |                   |
| Oven Temp.               | 50°C                                               |                   |                   |

- 5 The same general procedure that was used to prepare **2152** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2100</b> | 13.3        | 96.9      | white solid |
| <b>2102</b> | 15.7        | 98.4      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2100</b> | 96.9       | 7.8           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2102</b> | 98.4       | 8.7           | 65-95-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

| Instrument               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|--------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                          | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(45-75-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                          | 0                                                  | 55                | 45                |
|                          | 20                                                 | 25                | 75                |
|                          | 20.1                                               | 10                | 90                |
|                          | 22                                                 | 10                | 90                |
| Column                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |

|                   |              |
|-------------------|--------------|
| <b>Flow Rate</b>  | 1 mL/min     |
| <b>Wavelength</b> | 220 & 254 nm |
| <b>Oven Temp.</b> | 50°C         |

The same general procedure that was used to prepare **2152** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2117</b> | 14.4        | 99.4      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2117</b> | 99.4       | 13.5          | 45-75-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

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Analytical HPLC Conditions:

|                              |                                                    |                   |                   |
|------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>            | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>          | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                              | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(55-85-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                              | 0                                                  | 45                | 55                |
|                              | 20                                                 | 15                | 85                |
|                              | 20.1                                               | 10                | 90                |
|                              | 22                                                 | 10                | 90                |
| <b>Column</b>                | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>             | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>            | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>            | 30°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2145</b> | 26.6        | 98.2      | white solid |
| <b>2153</b> | 26.3        | 99.4      | white solid |
| <b>2154</b> | 28.1        | 99.1      | white solid |
| <b>2155</b> | 31.8        | 96.0      | white solid |
| <b>2156</b> | 28.4        | 95.5      | white solid |

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| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2145</b> | 98.2       | 10.2          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2153</b> | 99.4       | 10.4          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2154</b> | 99.1       | 10.1          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| <b>2155</b> | 96.0       | 10.2          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

|             |      |     |                |                              |                   |                                  |
|-------------|------|-----|----------------|------------------------------|-------------------|----------------------------------|
| <b>2156</b> | 95.5 | 9.4 | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN | Gemini 5um C18 110A<br>150*4.6mm |
|-------------|------|-----|----------------|------------------------------|-------------------|----------------------------------|

Analytical HPLC Conditions:

|                               |                                                    |                   |                   |
|-------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>             | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>           | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                               | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient (50-80-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                               | 0                                                  | 50                | 50                |
|                               | 20                                                 | 20                | 80                |
|                               | 20.1                                               | 10                | 90                |
|                               | 22                                                 | 10                | 90                |
| <b>Column</b>                 | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>              | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>             | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>             | 30°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

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| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2143</b> | 3.1         | 96.8      | white solid |
| <b>2157</b> | 22.2        | 96.9      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                      |
|-------------|------------|---------------|----------------|------------------------------|---------------------|----------------------------------|
| <b>2143</b> | 96.8       | 10.0          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A<br>150*4.6mm |
| <b>2157</b> | 96.9       | 12.0          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A<br>150*4.6mm |

Analytical HPLC Conditions:

|                              |                                                    |                   |                   |
|------------------------------|----------------------------------------------------|-------------------|-------------------|
| <b>Instrument</b>            | Thermo Scientific VANQUISH CORE                    |                   |                   |
| <b>Mobile Phase</b>          | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                              | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| <b>Gradient(60-90-20min)</b> | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                              | 0                                                  | 40                | 60                |
|                              | 20                                                 | 10                | 90                |
|                              | 20.1                                               | 10                | 90                |
|                              | 22                                                 | 10                | 90                |
| <b>Column</b>                | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| <b>Flow Rate</b>             | 1 mL/min                                           |                   |                   |
| <b>Wavelength</b>            | 220 & 254 nm                                       |                   |                   |
| <b>Oven Temp.</b>            | 30°C                                               |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2144</b> | 31.2        | 95.8      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2144</b> | 95.8       | 8.6           | 60-90-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

5 Analytical HPLC Conditions:

| Instrument            | Thermo Scientific VANQUISH CORE                    |                   |                   |
|-----------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase          | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                       | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(55-85-20min) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                       | 0                                                  | 45                | 55                |
|                       | 20                                                 | 15                | 85                |
|                       | 20.1                                               | 10                | 90                |
|                       | 22                                                 | 10                | 90                |
| Column                | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate             | 1 mL/min                                           |                   |                   |
| Wavelength            | 220 & 254 nm                                       |                   |                   |
| Oven Temp.            | 30 °C                                              |                   |                   |

The same general procedure that was used to prepare **2152** was also used to prepare the peptide listed below which was analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance  |
|-------------|-------------|-----------|-------------|
| <b>2158</b> | 15.4        | 92.1      | white solid |
| <b>2168</b> | 21.7        | 96.4      | white solid |
| <b>2169</b> | 25.3        | 97.6      | white solid |
| <b>2170</b> | 33.8        | 91.2      | white solid |
| <b>2172</b> | 5.1         | 99.4      | white solid |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|-------------|--------------------------|---------------------|-------------------------------|
| <b>2158</b> | 92.1       | 10.9          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2168</b> | 96.4       | 9.90          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2169</b> | 97.6       | 10.1          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2170</b> | 91.2       | 12.2          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2172</b> | 99.4       | 12.9          | 55-85-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |

10

Analytical HPLC Conditions:

| Instrument | Thermo Scientific VANQUISH CORE |
|------------|---------------------------------|
|------------|---------------------------------|

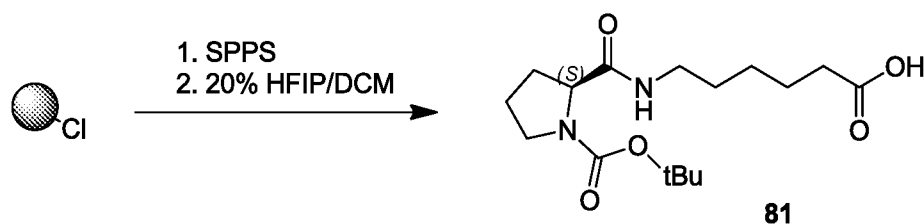
|                              |                                                    |                          |                          |
|------------------------------|----------------------------------------------------|--------------------------|--------------------------|
| <b>Mobile Phase</b>          | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                          |                          |
|                              | B: ACN (0.075% TFA in ACN)                         |                          |                          |
| <b>Gradient(60-90-20min)</b> | <b>Time (min)</b>                                  | <b>Percent Solvent A</b> | <b>Percent Solvent B</b> |
|                              | 0                                                  | 40                       | 60                       |
|                              | 20                                                 | 10                       | 90                       |
|                              | 20.1                                               | 10                       | 90                       |
|                              | 22                                                 | 10                       | 90                       |
| <b>Column</b>                | Gemini C18 5um 110A 150*4.6mm                      |                          |                          |
| <b>Flow Rate</b>             | 1 mL/min                                           |                          |                          |
| <b>Wavelength</b>            | 220 & 254 nm                                       |                          |                          |
| <b>Oven Temp.</b>            | 30 °C                                              |                          |                          |

The same general procedure that was used to prepare **2152** was also used to prepare the peptides listed below which were analyzed by the method above.

| #           | Amount (mg) | Purity(%) | Appearance      |
|-------------|-------------|-----------|-----------------|
| <b>2159</b> | 4.5         | 95.2      | white solid     |
| <b>2160</b> | 12.2        | 92.4      | white solid     |
| <b>2161</b> | 12.2        | 96.2      | white solid     |
| <b>2162</b> | 4.4         | 91.3      | white solid     |
| <b>2163</b> | 18.3        | 97.9      | white solid     |
| <b>2164</b> | 13.5        | 97.0      | off-white solid |
| <b>2165</b> | 6.4         | 98.2      | white solid     |
| <b>2167</b> | 35.5        | 95.8      | white solid     |
| <b>2171</b> | 2.0         | 96.7      | white solid     |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method | HPLC Mobile Phase A      | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|-------------|--------------------------|---------------------|-------------------------------|
| <b>2159</b> | 97.7       | 8.95          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2160</b> | 93.9       | 9.19          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2161</b> | 93.3       | 9.64          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2162</b> | 97.2       | 8.53          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2163</b> | 97.9       | 8.89          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2164</b> | 97.0       | 11.62         | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2165</b> | 98.2       | 9.97          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2167</b> | 95.8       | 8.62          | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |
| <b>2171</b> | 96.7       | 10.79         | 60-90-20MIN | 0.1%TFA/H <sub>2</sub> O | 0.075%TFA/ACN       | Gemini 5um C18 110A,150*4.6mm |

General procedure for the preparation of Intermediate **81**:



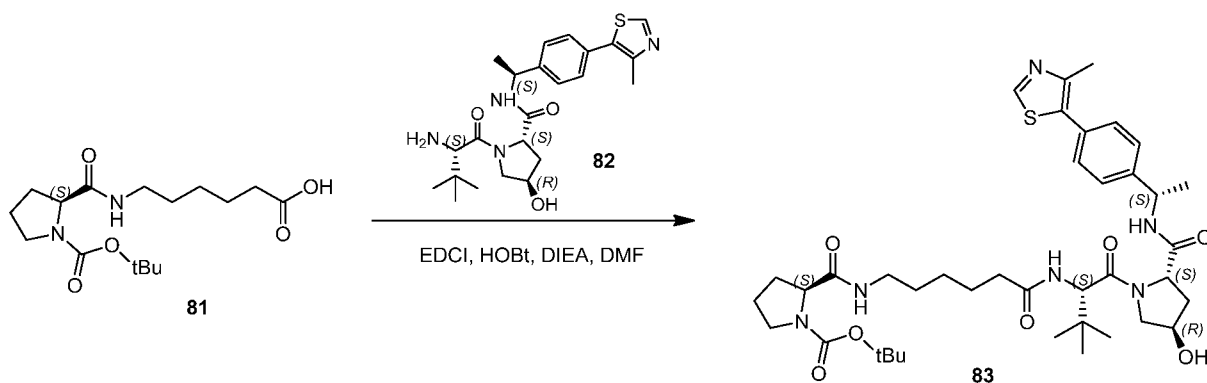
The intermediate **81** was synthesized using standard Fmoc chemistry.

- 1) Resin preparation: DCM (15 mL) was added to the vessel containing 2-CTC Resin (1.00 mmol, 1.02 g, Sub = 0.980 mmol/g, 1.00 equiv.) and the resin was swelled for 10 min. Next, the resin was drained. A solution of Fmoc-6-Ahx-OH (354 mg, 1.00 mmol, 1.00 equiv.) in DCM (15 mL) was added to the resin, followed by the addition of DIEA (0.75 mL, 4.00 mmol, 4.00 equiv.) dropwise. The mixture was bubbled with N<sub>2</sub> flow at 25 °C for 2 h. Next, MeOH (1.10 mL) was added to the mixture, and the mixture was bubbled with N<sub>2</sub> for 30 min. The resin was drained and washed with DMF (100 mL \* 5) before proceeding to the next step.
- 2) Deprotection: 20% piperidine in DMF (40 mL) was added to the resin and the mixture was bubbled with N<sub>2</sub> for 10 min at 25 °C. The resin was then washed with DMF (100 mL \* 5). The de-protection was monitored by ninhydrin test or chloranil test. The reaction was judged to be complete if the resin turned blue or brownish red.
- 3) Coupling: A solution of Boc-Pro-OH (3.00 equiv.), HATU (2.85 equiv.), and DIEA (6.00 equiv.) in DMF (15 mL) was added to the resin with N<sub>2</sub> bubbling for 1 h at 25 °C. The coupling was monitored by ninhydrin test or chloranil test. If the resin remained colorless, the coupling was judged to be complete. The mixture was then drained, and washed with DMF (100 mL \* 5) and MeOH (100 mL \* 3). Finally, the resin was drained, and then dried under vacuum.

#### Peptide Cleavage:

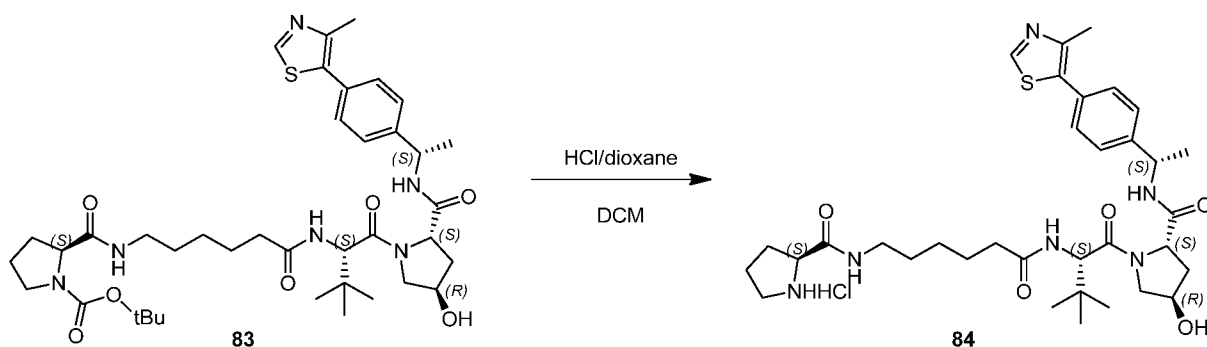
- 1) Cleavage cocktail (20% HFIP/DCM, 50 mL) was added to the flask containing the side chain protected peptide at room temperature, and the mixture was bubbled with N<sub>2</sub> for 10 min (50 mL\*3).
- 2) The mixture was then filtered. The filtrate was collected and concentrated under reduced pressure to give intermediate **81** (317 mg, crude) as a white oil.

General procedure for the preparation of intermediate **83**:



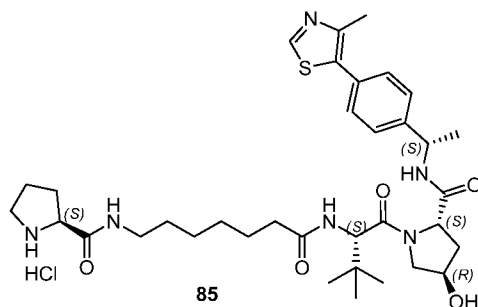
To a solution of compound **81** (50.0 mg, 112  $\mu\text{mol}$ , 1.00 equiv.) and compound **82** (44.3 mg, 134  $\mu\text{mol}$ , 1.20 equiv.) in DMF (0.5 mL) was added EDCI (43.1 mg, 224  $\mu\text{mol}$ , 2.00 equiv.), HOBt (30.3 mg, 224  $\mu\text{mol}$ , 2.00 equiv.), and DIEA (58.1 mg, 449  $\mu\text{mol}$ , 78.3  $\mu\text{L}$ , 4.00 equiv.) at 20  $^{\circ}\text{C}$ . Next, the mixture was stirred at 20  $^{\circ}\text{C}$  for 6 h. LCMS indicated compound **81** was consumed and the desired MS ( $R_t = 0.430$  min, MS cal.: 754.41, MS observed:  $[\text{M}+\text{H}]^+ = 755.6$ ) was detected. The mixture was then purified by prep-HPLC (A: 0.1% TFA in  $\text{H}_2\text{O}$ , B: ACN) to afford compound **83** (99.0 mg, 131  $\mu\text{mol}$ , 86.1% yield, 98.4% purity) as a white solid.

General procedure for preparation of intermediate **84**.

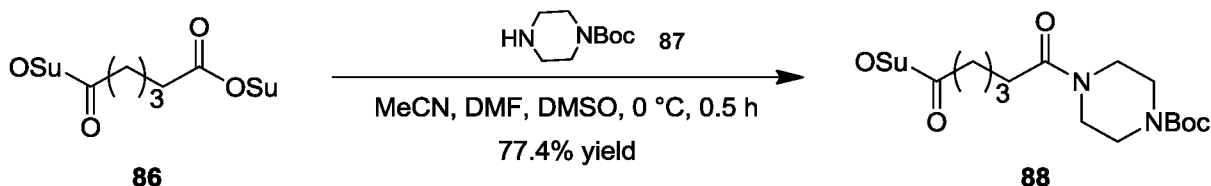


To a solution of compound **83** (99.0 mg, 131  $\mu\text{mol}$ , 1.00 equiv.) in DCM (3.0 mL) was added HCl/dioxane (2 M, 20 mL, 389 equiv.) at 20  $^{\circ}\text{C}$ . Next, the mixture was stirred at 20  $^{\circ}\text{C}$  for 1 h. LCMS indicated compound **83** was consumed, and the desired MS ( $R_t = 0.313$  min, MS cal.: 654.36, MS observed:  $[\text{M}+\text{H}]^+ = 655.4$ ) was detected. The reaction was concentrated under pressure. The residue was dissolved in MeCN/ $\text{H}_2\text{O}$  (5.00 mL), and lyophilized to afford intermediate **84** (80.0 mg, crude) as a white solid.

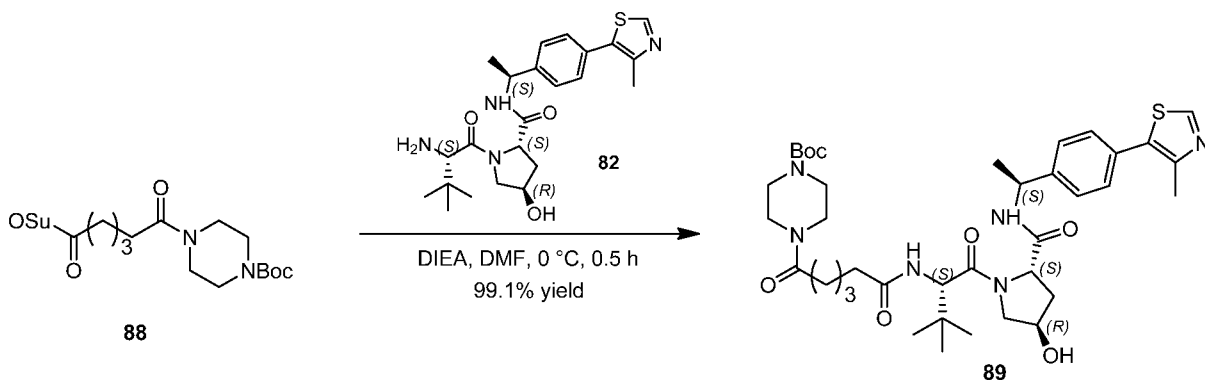
Intermediate **85** was synthesized in the same manner as intermediate **84**.



General procedure for preparation of intermediate **88**.



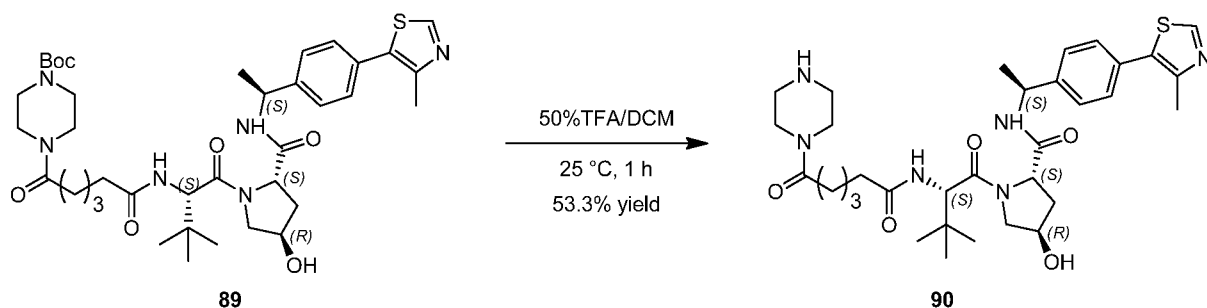
- 5 A solution of compound **87** (456 mg, 2.45 mmol, 1.00 equiv.) in DMF (5.00 mL) was added dropwise to a mixture of compound **86** (2.50 g, 7.35 mmol, 3.00 equiv.) in DMF (10.0 mL), DMSO (10.0 mL), and ACN (5.00 mL) at 0 °C. Next, the mixture was stirred at 0 °C for 0.5 h. LCMS showed compound **87** was consumed, and the desired mass ( $R_t = 1.19$  min, MS cal.: 411.2, MS observed:  $[M+H]^+ = 412.2$ ) was detected. The mixture was purified by prep-HPLC (TFA conditions) to give compound **88** (780 mg, 1.90 mmol, 77.4% yield) as a white solid.
- 10



- 15 To a solution of compound **88** (100 mg, 243  $\mu$ mol, 1.05 equiv.) and compound **82** (103 mg, 231  $\mu$ mol, 1.00 equiv.) in DMF (1.00 mL) was added DIEA (60.0 mg, 462  $\mu$ mol, 2.00 equiv.) at 0 °C under  $N_2$ . The mixture was then stirred at 0 °C for 0.5 h. LCMS showed compound **88** was consumed, and the desired mass ( $R_t = 1.463$  min, MS cal.: 740.3, MS observed:  $[M+H]^+ = 741.5$ ) was detected. The mixture was diluted with  $H_2O$  (5.00 mL), and extracted with ethyl acetate (2.00 mL \* 4), dried over  $Na_2SO_4$ , filtered and concentrated to give compound **89** (170 mg, crude) as yellow oil.

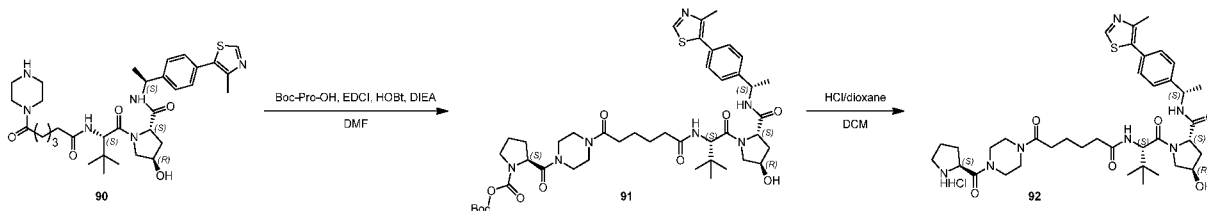
20

General procedure for the preparation of intermediate **90**.



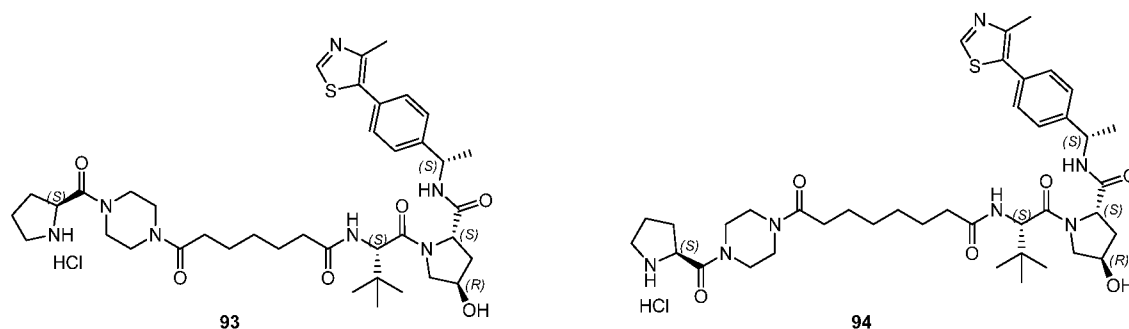
To a solution of compound **89** (170 mg, 229  $\mu$ mol, 1.00 equiv.) in DCM (2.00 mL) was added TFA (2.00 mL, 27.0 mmol) at 25  $^{\circ}$ C. Next, the reaction was stirred at 25  $^{\circ}$ C for 1 h. LCMS showed compound **89** was consumed, and the desired mass ( $R_t$  = 1.132 min, MS cal.: 640.34, MS observed:  $[M+H]^+ = 640.8$ ) was detected. The mixture was concentrated under reduced pressure. The residue was purified by prep-HPLC (TFA conditions) to provide intermediate **90** (80.0 mg, 122  $\mu$ mol, 53.3% yield, 97.9% purity, TFA salt) as a white solid.

### General procedure for the preparation of intermediate **92**.

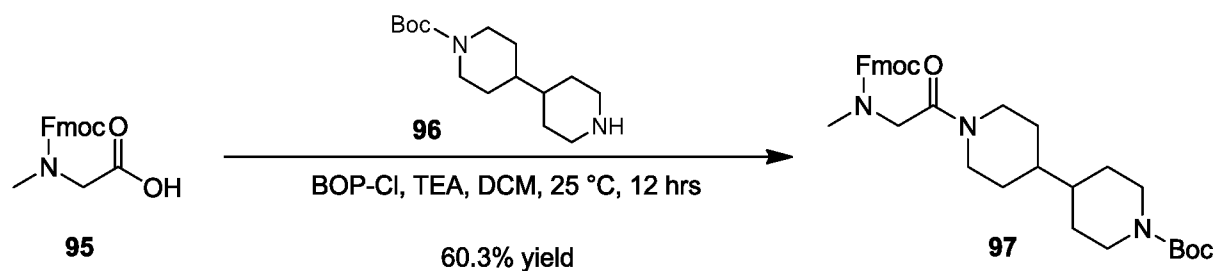


To a solution of intermediate **90** (20.0 mg, 31.2  $\mu\text{mol}$ , 1.00 equiv.) and Boc-Pro-OH (8.06 mg, 37.4  $\mu\text{mol}$ , 1.20 equiv.) in DMF (0.2 mL) was added EDCI (11.9 mg, 62.4  $\mu\text{mol}$ , 2.00 equiv.), HOBt (8.43 mg, 62.4  $\mu\text{mol}$ , 2.00 equiv.), and DIEA (16.1 mg, 124  $\mu\text{mol}$ , 21.7  $\mu\text{L}$ , 4.00 equiv.) at 20  $^{\circ}\text{C}$ . Next, the mixture was stirred at 20  $^{\circ}\text{C}$  for 12 h. LCMS indicated **90** was consumed, and the desired mass ( $R_t$  = 0.420 min, MS cal.: 837.45, MS observed:  $[\text{M}+\text{H}]^+ = 838.6$ ) was detected. The mixture was then concentrated under reduced pressure. The residue was dissolved in DCM (1.0 mL), and HCl/dioxane (2 M, 5 mL, 320 equiv) was added at 20  $^{\circ}\text{C}$ . Next, the mixture was stirred at 20  $^{\circ}\text{C}$  for 1 h. LCMS indicated compound **91** was consumed, and the desired mass ( $R_t$  = 0.313 min,  $[\text{M}+\text{H}]^+ = 655.4$ ) was detected. The reaction was concentrated under reduced pressure. The residue was purified by prep-HPLC (A: 0.1% TFA in  $\text{H}_2\text{O}$ , B: ACN) to afford **92** (19.0 mg, 35.7  $\mu\text{mol}$ , 94.4% purity, 82.5% yield) as a white solid.

Intermediates **93** and **94** were synthesized in a similar manner.



General procedure for the preparation of intermediate **97**:

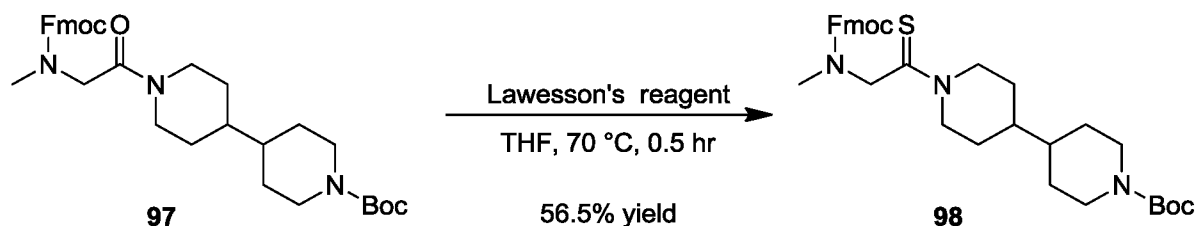


To a solution of compound **95** (1.00 g, 3.21 mmol, 1.00 equiv.) in DCM (24.0 mL) was added compound **96** (1.29 g, 4.82 mmol, 1.50 equiv.), TEA (487 mg, 4.82 mmol, 670  $\mu$ L, 1.50 equiv.), and BOP-Cl (1.23 g, 4.82 mmol, 1.50 equiv.). The mixture was stirred at 25°C for 12 hrs. LCMS showed compound **96** was consumed, and the desired mass was detected by LCMS ( $R_t$  = 0.52 min, MS cal.: 561.32, MS observed:  $[M+H]^+ = 562.3$ ). The mixture was diluted with water 100 mL, and extracted with ethyl acetate (150 mL \*3). The combined organic layers were washed with citric acid aqueous solution (100 mL \* 2) and brine (90.0 mL), dried over  $Na_2SO_4$ , filtered and concentrated in vacuum. The residue was purified by column chromatography ( $SiO_2$ , petroleum ether: ethyl acetate=100: 0 to 2: 1, petroleum ether: ethyl acetate = 1: 1,  $R_f$  = 0.16). Compound **97** (1.10 g, 1.94 mmol, 60.3% yield, 98.9% purity) was obtained as a white solid.

$^1H$  NMR: 400 MHz,  $DMSO-d_6$ .

$\delta$ : 7.89 (t,  $J$  = 8.0 Hz, 2H), 7.67 (d,  $J$  = 7.2 Hz, 1H), 7.57 (br d,  $J$  = 3.2 Hz, 1H), 7.41 (q,  $J$  = 7.2 Hz, 2H), 7.37 - 7.21 (m, 2H), 4.45 - 3.50 (m, 10H), 2.82 (d,  $J$  = 16.4 Hz, 4H), 2.72 - 2.55 (m, 2H), 2.48 - 2.40 (m, 1H), 1.78 - 1.52 (m, 4H), 1.38 (s, 9H), 1.30 (br d,  $J$  = 2.0 Hz, 1H), 1.11 - 0.85 (m, 4H).

General procedure for the preparation of intermediate **98**:



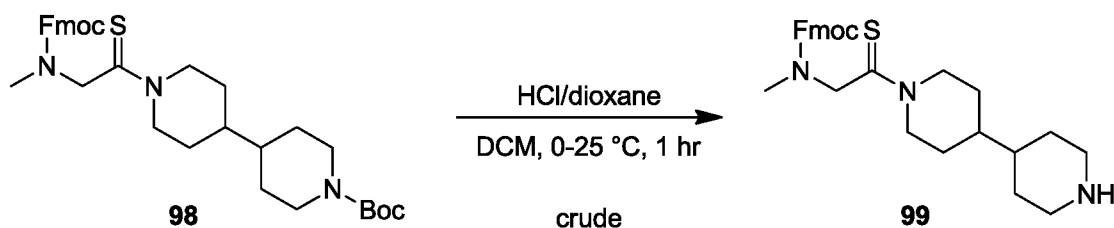
To a solution of compound **97** (633 mg, 1.11 mmol, 1.00 equiv.) in THF (6.30 mL) was added Lawesson's reagent (676 mg, 1.67 mmol, 1.50 equiv.). The mixture was stirred at 70 °C

for 0.5 hr. LCMS indicated that 19.5% of compound **97** remained, and 60.7% of the desired mass was detected by LCMS ( $R_t = 0.56$  min, MS cal.: 577.3, MS observed:  $[M+H]^+ = 578.2$ ). The mixture was concentrated under reduced pressure, and the residue was purified by prep-HPLC (HCl conditions). Compound **98** (367 mg, 630  $\mu$ mol, 56.5% yield, 99.2% purity) was obtained as a white solid.

$^1\text{H}$  NMR: 400 MHz, DMSO- $d_6$

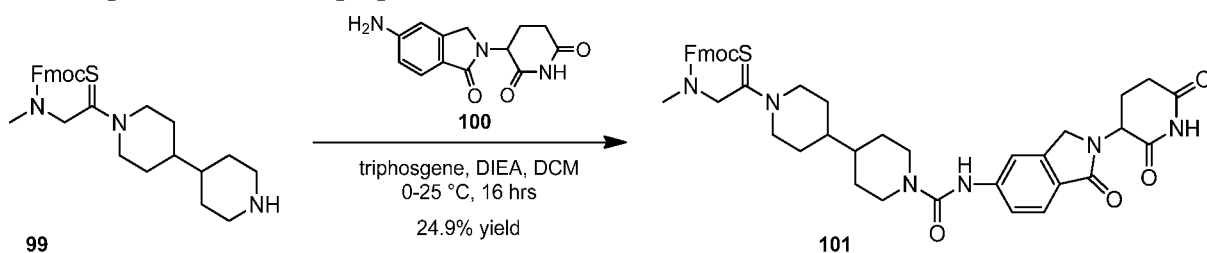
$\delta$ : 7.89 (br t,  $J = 8.4$  Hz, 2H), 7.70 (br d,  $J = 6.8$  Hz, 1H), 7.59 (br t,  $J = 7.2$  Hz, 1H), 7.46 - 7.25 (m, 4H), 5.34 (br d,  $J = 12.0$  Hz, 1H), 4.42 - 3.69 (m, 9H), 3.18 - 3.04 (m, 1H), 3.02 - 2.84 (m, 4H), 2.67 - 2.59 (m, 1H), 1.80 - 1.54 (m, 4H), 1.38 (s, 9H), 1.28 - 0.88 (m, 6H).

General procedure for the preparation of intermediate **99**:



To a solution of compound **98** (200 mg, 343  $\mu$ mol, 1.00 equiv.) in DCM (1.00 mL) was added HCl/dioxane (2.0 M, 2.0 mL, 11.6 equiv.) at 0  $^\circ\text{C}$ . The mixture was stirred at 25  $^\circ\text{C}$  for 1 hr. LCMS showed compound **98** was consumed, and the desired mass was detected by LCMS ( $R_t = 0.37$  min, MS cal.: 477.24, MS observed:  $[M+H]^+ = 478.2$ ). The mixture was concentrated under reduced pressure to afford compound **99** (200 mg, crude, HCl salt) as a white solid which was used in the next step without further purification.

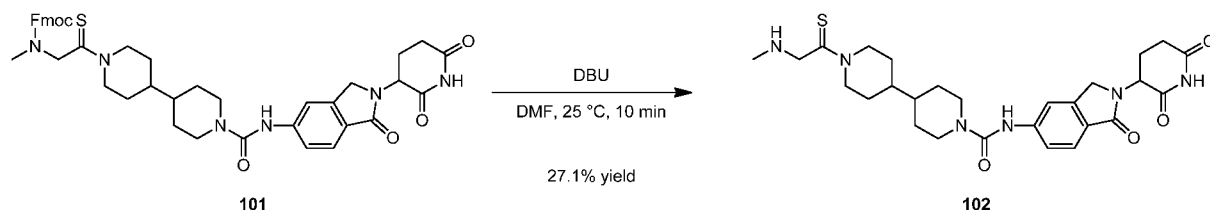
General procedure for the preparation of intermediate **101**:



To a solution of compound **99** (37.0 mg, 142  $\mu$ mol, 1.00 equiv.) in DCM (4.00 mL) was added compound **100** (36.8 mg, 142  $\mu$ mol, 1.00 equiv.), DIEA (36.8 mg, 49.7  $\mu$ L, 285  $\mu$ mol, 2.00 equiv.), and triphosgene (63.5 mg, 214  $\mu$ mol, 1.50 equiv.) under  $\text{N}_2$  atmosphere. Next, the mixture was stirred at 0  $^\circ\text{C}$  for 1 hr, and then a solution of compound **100** (100 mg, 194  $\mu$ mol, 1.36 equiv., HCl) in DCM (1.00 mL) was added. The mixture was stirred at 25  $^\circ\text{C}$  for 15 hrs. LCMS showed compound **24** was consumed, and 39.7% of the desired mass was detected by LCMS ( $R_t = 0.47$  min, MS cal.: 762.32, MS observed:  $[M+H]^+ = 763.2$ ). The mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was then purified

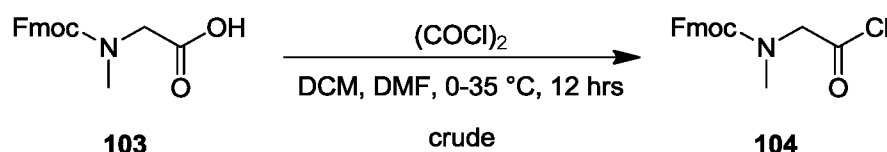
by prep-HPLC (TFA conditions). Compound **101** (55.0 mg, 71.2  $\mu$ mol, 98.8% purity, 24.9% yield) was obtained as a white solid.

General procedure for the preparation of intermediate **102**:



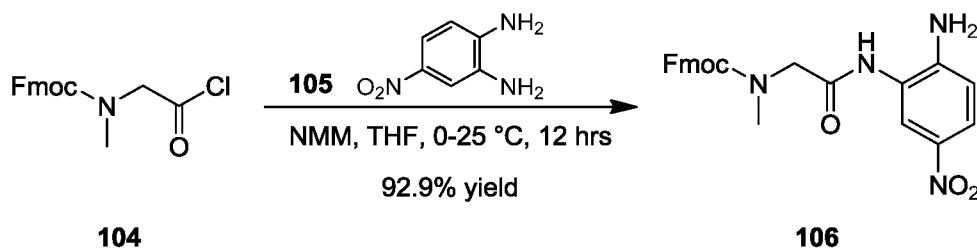
To a solution of compound **101** (20.0 mg, 25.9  $\mu$ mol, 1.00 equiv.) in DMF (0.30 mL) was added DBU (3.94 mg, 3.90  $\mu$ L, 25.9  $\mu$ mol, 1.00 equiv.). The mixture was stirred at 25  $^\circ$ C for 10 min. LCMS showed compound **101** was consumed, and 31.2% of the desired mass was detected by LCMS ( $R_f$  = 0.79 min, MS cal.: 540.25, MS observed:  $[M+H]^+$  = 541.5). The mixture was concentrated under reduced pressure, and the residue was purified by prep-HPLC (HCl conditions). Compound **102** (4.00 mg, 7.04  $\mu$ mol, 95.2% purity, 27.1% yield) was obtained as a white solid.

General procedure for the preparation of intermediate **104**:



To a solution of compound **103** (30.0 g, 96.3 mmol, 1.00 equiv.) in DCM (300 mL) was added DMF (14.0 mg, 14.8  $\mu$ L, 192  $\mu$ mol, 0.002 equiv.) and  $(\text{COCl})_2$  (24.4 g, 192 mmol, 16.8 mL, 2.00 equiv.) at 0  $^\circ$ C. The mixture was stirred at 35 $^\circ$ C for 12 hrs. A small aliquot of the reaction was quenched with a drop of MeOH. TLC (petroleum ether: ethyl acetate = 1: 1) indicated compound **103** ( $R_f$  = 0.09) was consumed, and one new spot ( $R_f$  = 0.70) was detected. The mixture was then concentrated under reduced pressure to afford compound **104** (30.9 g, crude) as brown oil, and was used in the next step without further purification.

General procedure for the preparation of intermediate **106**:



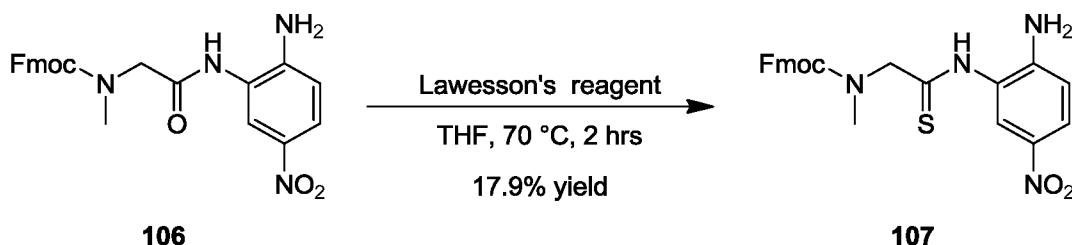
To a solution of compound **104** (14.3 g, 93.7 mmol, 1.00 equiv.) in THF (150 mL) was added NMM (12.3 g, 13.3 mL, 121 mmol, 1.3 equiv.). Next, a solution of compound **105** (30.9 g, 93.7 mmol, 1.00 equiv.) in THF (400 mL) was added at 0  $^\circ$ C, and the mixture was stirred for

1 hr. The mixture was then stirred at 25 °C for 11 hrs. LCMS showed compound **104** was consumed, and 95.2% of desired mass was detected by LCMS ( $R_t = 0.45$  min, MS cal.: 446.16, MS observed:  $[M+H]^+ = 447.1$ ). The residue was diluted with water (600 mL), and extracted with ethyl acetate (400 mL \* 3). The combined organic layers were combined and washed with water (400 mL \* 3) and brine (300 mL \* 2), dried over  $Na_2SO_4$ , filtered and concentrated under reduced pressure. The crude was triturated in EtOH (200 mL) at 50 °C for 30 min. Compound **106** (40.0 g, 87.0 mmol, 97.2% purity, 92.9% yield) was obtained as a yellow solid.

$^1H$  NMR: 400 MHz,  $DMSO-d_6$

$\delta$ : 9.34 (br d,  $J = 8.8$  Hz, 1H), 8.54 - 8.11 (m, 1H), 8.07 - 7.78 (m, 3H), 7.75 - 7.56 (m, 2H), 7.51 - 7.32 (m, 3H), 7.30 - 7.07 (m, 1H), 6.77 (dd,  $J = 9.2, 12.4$  Hz, 1H), 6.53 (br d,  $J = 11.2$  Hz, 2H), 4.51 - 4.27 (m, 3H), 3.65 - 3.36 (m, 2H), 2.94 (d,  $J = 2.4$  Hz, 3H).

General procedure for the preparation of intermediate **107**:

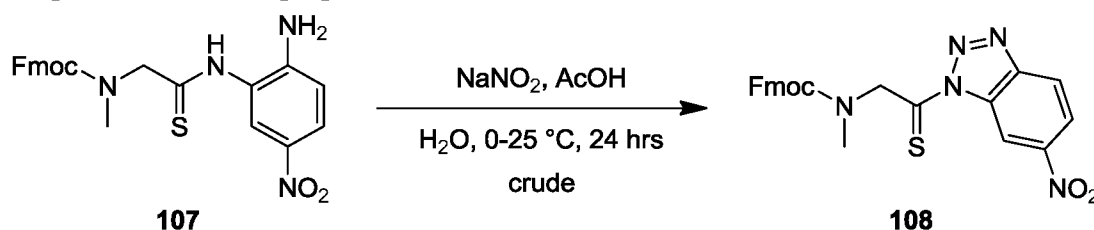


To a solution of compound **106** (20.0 g, 43.5 mmol, 1.00 equiv.) in THF (500 mL) was added Lawesson's reagent (15.8 g, 39.1 mmol, 0.90 equiv.). The mixture was stirred at 70 °C for 2 hrs. LCMS showed compound **106** was consumed and the desired mass was detected by LCMS ( $R_t = 0.47$  min, MS cal.: 462.14, MS observed:  $[M+Na]^+ = 485.1$ ). The mixture was concentrated under reduced pressure. The crude was first purified by column chromatography ( $SiO_2$ , petroleum ether: ethyl acetate = 80: 1 to 0: 1,  $R_f = 0.40$ , petroleum ether: ethyl acetate = 1:1), and then the material was further purified by reverse-phase HPLC (AcOH conditions) to afford compound **107** (4.00 g, 7.79 mmol, 17.9% yield, 90.1% purity) as a yellow solid.

$^1H$  NMR: 400 MHz,  $DMSO-d_6$ .

$\delta$ : 11.07 (s, 1H), 8.10 - 7.80 (m, 4H), 7.69 (br dd,  $J = 7.6, 12.0$  Hz, 2H), 7.55 - 7.19 (m, 4H), 7.02 - 6.72 (m, 1H), 6.60 (br d,  $J = 16.4$  Hz, 2H), 4.66 - 4.01 (m, 5H), 3.11 - 2.86 (m, 3H).

General procedure for the preparation of intermediate **108**:



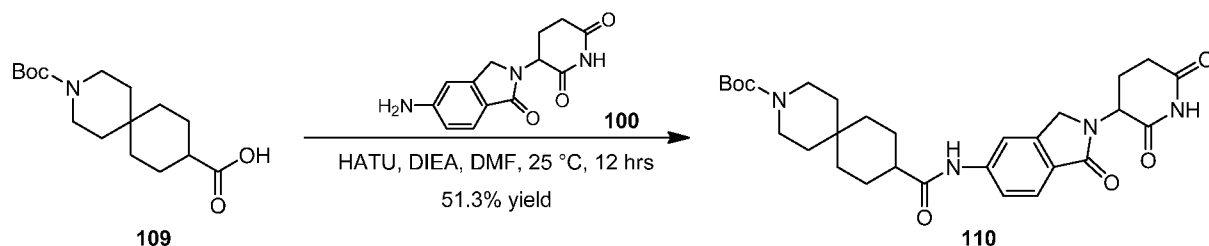
To a solution of compound **107** (1.00 g, 1.95 mmol, 1.00 equiv.) in AcOH (12.0 mL) and  $H_2O$  (1.30 mL) was added  $NaNO_2$  (201 mg, 2.92 mmol, 1.50 equiv.) at 0 °C. The mixture was stirred

at 25 °C for 24 hrs. LCMS showed compound **107** was consumed and 46.4% of the desired mass was detected by LCMS ( $R_t = 0.54$  min, MS cal.: 473.12, MS observed:  $[M+Na]^+ = 495.9$ ). Ice water (50 mL) was added to the reaction, and the mixture was filtered. The filter cake was concentrated under reduced pressure to afford intermediate **108** (700 mg, crude) as a yellow solid.

$^1H$  NMR: 400 MHz,  $CDCl_3$

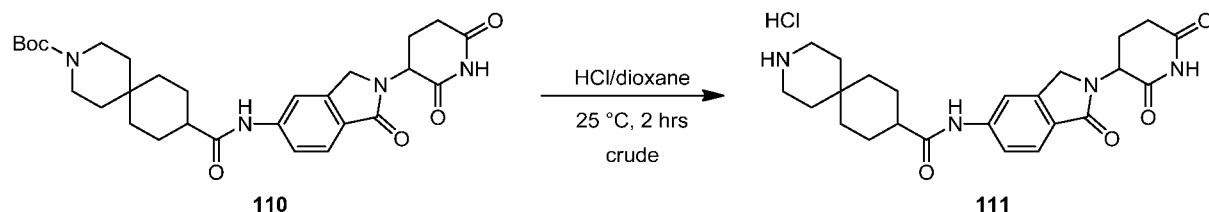
$\delta$ : 9.95 - 8.87 (m, 1H), 8.77 - 8.06 (m, 2H), 7.83 - 7.64 (m, 1H), 7.61 - 7.26 (m, 5H), 7.21 - 6.93 (m, 2H), 5.33 - 4.70 (m, 1H), 4.68 - 4.40 (m, 2H), 4.39 - 4.00 (m, 1H), 3.93 - 3.39 (m, 1H), 3.33 - 2.79 (m, 1H).

General procedure for the preparation of intermediate **110**:



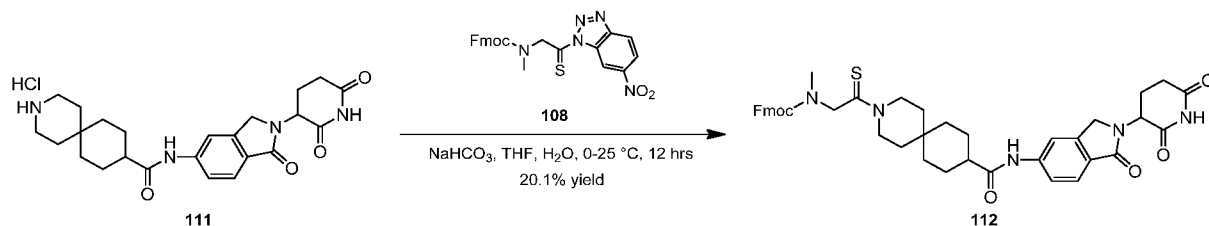
To a solution of compound **109** (500 mg, 1.68 mmol, 1.00 equiv.) in DMF (5.00 mL) was added DIEA (434 mg, 585  $\mu$ L, 3.36 mmol, 2.00 equiv.), HATU (767 mg, 2.02 mmol, 1.20 equiv.), and compound **100** (435 mg, 1.68 mmol, 1.00 equiv.). The mixture was stirred at 25 °C for 12 hrs. LCMS showed compound **109** was consumed and the desired mass ( $R_t = 0.44$  min, MS cal.: 538.28, MS observed:  $[M-t-Bu+H]^+ = 483.0$ ) was detected. The mixture was adjusted to pH = 6 with acetic acid, and purified by prep-HPLC (formic acid conditions) to afford compound **110** (510 mg, 863  $\mu$ mol, 91.2% purity, 51.3% yield) as a brown solid.

General procedure for the preparation of intermediate **111**:



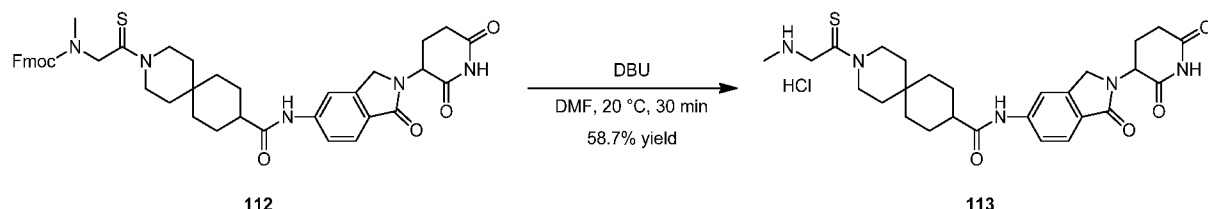
HCl/dioxane (2.00 M, 25.0 mL, 50.0 mmol, 57.9 equiv.) was added to compound **110** (510 mg, 863  $\mu$ mol, 1.00 equiv.), and the mixture was stirred at 25 °C for 2 hrs. LCMS showed compound **110** was consumed and the desired mass ( $R_t = 0.24$  min, MS cal.: 438.23, MS observed:  $[M+H]^+ = 439.0$ ) was detected. The mixture was concentrated under vacuum to afford compound **111** (462 mg, crude, HCl salt) as a brown solid.

General procedure for the preparation of intermediate **112**:



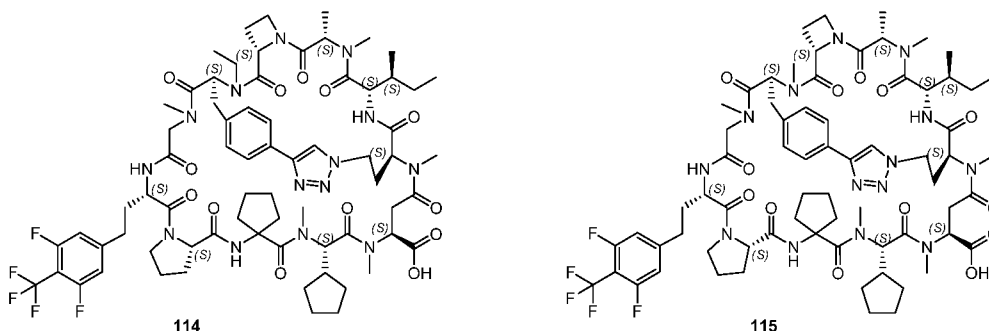
To a solution of compound **111** (50.0 mg, 105  $\mu\text{mol}$ , 1.00 equiv., HCl salt) in THF (2.00 mL) was added  $\text{NaHCO}_3$  (17.6 mg, 210  $\mu\text{mol}$ , 2.00 equiv.) in  $\text{H}_2\text{O}$  (0.50 mL) and intermediate **108** (149 mg, 315  $\mu\text{mol}$ , 3.00 equiv.) at 0  $^\circ\text{C}$ . The mixture was stirred at 25  $^\circ\text{C}$  for 12 hrs. LCMS showed compound **111** was consumed and the desired mass ( $R_t = 1.83$  min, MS cal.: 747.31, MS observed:  $[\text{M}+\text{H}]^+ = 748.3$ ) was detected. The mixture was filtered and purified by prep-HPLC (formic acid conditions) to afford compound **112** (17.0 mg, 21.1  $\mu\text{mol}$ , 20.1% yield, 93.1% purity) as a brown solid.

General procedure for the preparation of intermediate **113**:



To a solution of compound **112** (27.0 mg, 36.1  $\mu\text{mol}$ , 1.00 equiv.) in DMF (0.50 mL) was added DBU (138 mg, 137  $\mu\text{L}$ , 912  $\mu\text{mol}$ , 25.2 equiv.). The mixture was stirred at 20  $^\circ\text{C}$  for 30 min. LCMS showed compound **112** was consumed and the desired mass ( $R_t = 0.30$  min, MS cal.: 525.24, MS observed:  $[\text{M}+\text{H}]^+ = 526.5$ ) was detected. The mixture was purified by prep-HPLC (HCl condition) to afford **113** (12.0 mg, 21.2  $\mu\text{mol}$ , 99.4% purity, 58.7% yield, HCl salt) as a yellow solid.

Intermediates **114** and **115** were synthesized following the same strategy used to prepare compound **66**.



Using peptide intermediate **115** and intermediates **84**, **85**, **93** and **94**, respectively, compounds **2038**, **2039**, **2040** and **2041** were prepared using the same general procedure that was used to prepare **2114**.

| #    | Amount (mg) | Purity(%) | Appearance  |
|------|-------------|-----------|-------------|
| 2038 | 10.4        | 99.2      | white solid |
| 2039 | 9.2         | 97.4      | white solid |
| 2040 | 10.0        | 97.4      | white solid |
| 2041 | 10.1        | 98.6      | white solid |

Analytical HPLC Conditions:

| Instrument               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|--------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                          | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(45-75-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                          | 0                                                  | 55                | 45                |
|                          | 20                                                 | 25                | 75                |
|                          | 20.1                                               | 10                | 90                |
|                          | 22                                                 | 10                | 90                |
| Column                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate                | 1 mL/min                                           |                   |                   |
| Wavelength               | 220 & 254 nm                                       |                   |                   |
| Oven Temp.               | 50°C                                               |                   |                   |

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|--------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| 2038   | 99.2       | 9.6           | 45-75-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| 2039   | 97.4       | 10.5          | 45-75-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |
| 2041   | 98.6       | 9.5           | 45-75-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

5 Analytical HPLC Conditions:

| Instrument               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|--------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                          | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(40-70-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                          | 0                                                  | 60                | 40                |
|                          | 20                                                 | 30                | 70                |
|                          | 20.1                                               | 10                | 90                |
|                          | 22                                                 | 10                | 90                |
| Column                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate                | 1 mL/min                                           |                   |                   |
| Wavelength               | 220 & 254 nm                                       |                   |                   |
| Oven Temp.               | 50°C                                               |                   |                   |

| Cmpd # | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|--------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| 2040   | 97.4       | 11.8          | 40-70-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Similarly, using peptide intermediate **66** and intermediates **102** and **113**, respectively, compounds **2064** and **2065** were prepared using the same general procedure that was used to prepare **2114**.

| #           | Amount (mg) | Purity(%) | Appearance         |
|-------------|-------------|-----------|--------------------|
| <b>2064</b> | 2.9         | 90.0      | light yellow solid |
| <b>2065</b> | 2.5         | 87.6      | light yellow solid |

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Analytical HPLC Conditions:

| Instrument                | Thermo Scientific VANQUISH CORE                    |                   |                   |
|---------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase              | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                           | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient (50-80-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                           | 0                                                  | 50                | 50                |
|                           | 20                                                 | 20                | 80                |
|                           | 20.1                                               | 10                | 90                |
|                           | 22                                                 | 10                | 90                |
| Column                    | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate                 | 1 mL/min                                           |                   |                   |
| Wavelength                | 220 & 254 nm                                       |                   |                   |
| Oven Temp.                | 50°C                                               |                   |                   |

| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2064</b> | 90.0       | 9.95          | 50-80-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

Analytical HPLC Conditions:

| Instrument               | Thermo Scientific VANQUISH CORE                    |                   |                   |
|--------------------------|----------------------------------------------------|-------------------|-------------------|
| Mobile Phase             | A: H <sub>2</sub> O (0.1% TFA in H <sub>2</sub> O) |                   |                   |
|                          | B: ACN (0.075% TFA in ACN)                         |                   |                   |
| Gradient(55-85-20min-50) | Time (min)                                         | Percent Solvent A | Percent Solvent B |
|                          | 0                                                  | 45                | 55                |
|                          | 20                                                 | 15                | 85                |
|                          | 20.1                                               | 10                | 90                |
|                          | 22                                                 | 10                | 90                |
| Column                   | Gemini C18 5um 110A 150*4.6mm                      |                   |                   |
| Flow Rate                | 1 mL/min                                           |                   |                   |
| Wavelength               | 220 & 254 nm                                       |                   |                   |
| Oven Tem.                | 50°C                                               |                   |                   |

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| Cmpd #      | Purity (%) | HPLC RT (min) | HPLC Method    | HPLC Mobile Phase A          | HPLC Mobile Phase B | HPLC Column                   |
|-------------|------------|---------------|----------------|------------------------------|---------------------|-------------------------------|
| <b>2065</b> | 87.6       | 7.75          | 55-85-20MIN-50 | 0.1% TFA in H <sub>2</sub> O | 0.075% TFA in ACN   | Gemini 5um C18 110A 150*4.6mm |

**SPR Procedure to Determine Binding Affinity:**

The binding affinity of each peptide was tested by Surface Plasmon Resonance (SPR) using a Biacore 8K instrument (Cytiva). The target KRAS (1-185) proteins carrying different KRAS mutations were expressed in *E. coli* BL21 (DE3) strain with an N-terminal His tag and a C-terminal AVI tag. All proteins were purified using a Ni-NTA affinity purification system followed by size-exclusion chromatography (SEC). Purified proteins were biotinylated using a Biotin-protein ligase (Genecopoeia). For the preparation of GDP-loaded protein, GDP was added at a molar ratio of 40 to the KRAS protein together with 5 mM MgCl<sub>2</sub>. When preparing GppNHp-loaded proteins, 1 mM EDTA was added to biotinylated KRAS protein and incubated at 4 °C overnight. GppNHp was then added at a molar ratio of 40 to the KRAS protein together with 5 mM MgCl<sub>2</sub>. Incubation with GDP or GppNHp was carried out at room temperature for 2 hours on a rotator before the protein was aliquoted and stored at -80 °C. Next, the proteins were purified using Zeba™ Micro Spin Desalting Columns (Thermo).

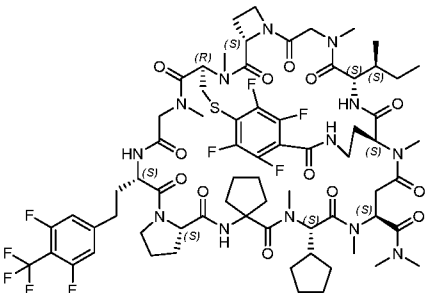
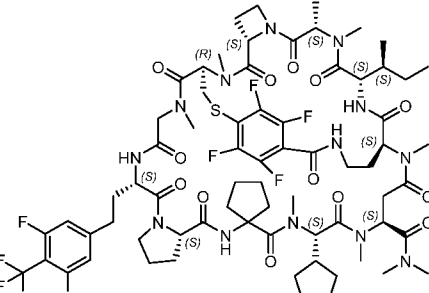
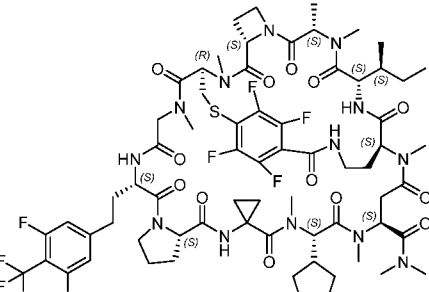
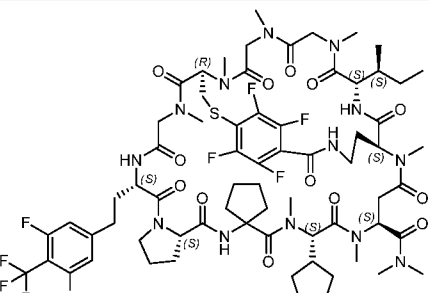
To test the binding affinity of the peptide compounds, GDP-loaded or GppNHp-loaded KRAS protein was immobilized on the surface of a Sensor Chip CAP (Cytiva) coated with Biotin CAPture Reagent to reach about 500 RU. The binding evaluation of each compound was performed by the Single Cycle Kinetics method, with compound in running buffer being added to the chip with or without immobilized KRAS protein to obtain the binding response. HBS buffer supplemented with 10 mM MgCl<sub>2</sub>, 0.05% Tween-20 and 5% DMSO together with 5 μM GDP or GppNHp was used as running buffer for GDP-loaded KRAS proteins and GppNHp-loaded KRAS proteins, respectively. The association time was set as 75 seconds and the dissociation phase was set as 3500 seconds. The Biotin CAPture Reagent and KRAS protein were immobilized for each cycle, and at the end of each cycle, the sensor chip was regenerated with the Regeneration solution. The resulting sensorgrams were subjected to curve fitting based on a 1:1 binding model using the Biacore Insight Evaluation Software to determine the dissociation constant, K<sub>D</sub>, of each compound for the respective KRAS proteins.

**Cell Proliferation Assay:**

Cancer cell lines GP2d (KRAS-G12D), NCI-H441 (KRAS-G12V), A375 (KRAS WT, BRAF V600E) and KRAS WT cell line HEK293 were grown in DMEM/F12 supplemented with 10% FBS. The day before the assay, the cell lines were seeded in 96-well plates in 100 μL of complete culture medium with GP2d and NCI-H441 at a concentration of 3000 cells/well and 4000 cells/well, respectively, and HEK293 and A375 at a concentration of 1500 cells/well. All cells were incubated overnight at 37 °C in a 5% carbon dioxide gas incubator, followed by the addition of 20 μL of serially diluted compound solutions. After 2 days of incubation, a new series of compound solutions with concentrations 7-fold to the final concentrations in cells were prepared and added to the cells. After incubating for 2 more days, 50 μL of CellTiter-Glo® (Promega) was added to each well and luminescence was measured by an EnVision 2105

multimode plate reader (PerkinElmer). The cell growth inhibitory activity of each test compound was calculated as the 50% growth inhibitory concentration (IC<sub>50</sub> value) relative to no drug treatment.

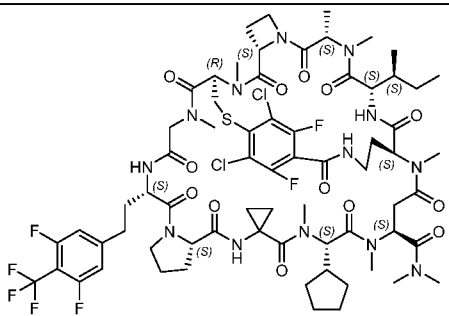
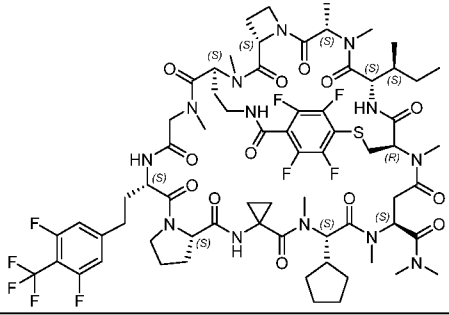
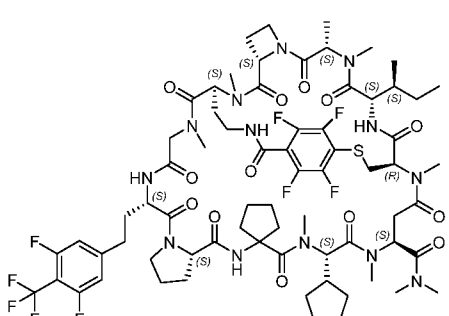
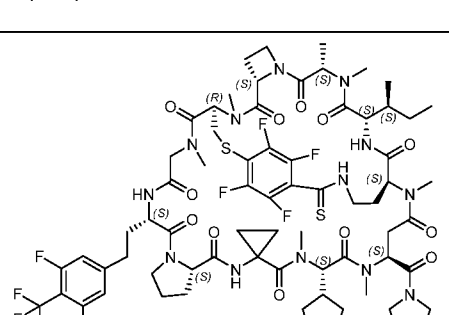
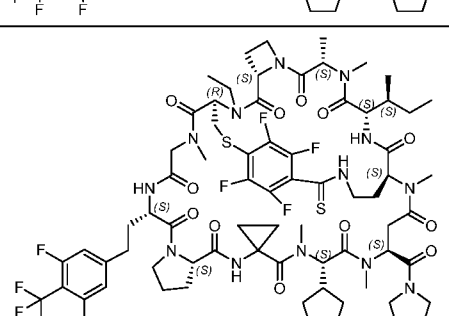
## 5 LCMS and SPR Binding Affinity for Compounds 1001-1035:

| Cmpd # | Structure                                                                           | LCMS RT (min) | MS (m/z) | LCMS Method | MS Polarity        | G12V GDP KD nM |
|--------|-------------------------------------------------------------------------------------|---------------|----------|-------------|--------------------|----------------|
| 1001   |    | 1.70          | 1,512.8  | 10-80-2min  | [M+H] <sup>+</sup> | 0.062          |
| 1002   |   | 1.72          | 1,526.9  | 10-80-2min  | [M+H] <sup>+</sup> | 0.064          |
| 1003   |  | 1.68          | 1,498.8  | 10-80-2min  | [M+H] <sup>+</sup> | 0.114          |
| 1004   |  | 1.73          | 1,500.8  | 10-80-2min  | [M+H] <sup>+</sup> | 0.146          |

|      |  |      |         |           |                    |       |
|------|--|------|---------|-----------|--------------------|-------|
| 1005 |  | 1.85 | 1,514.9 | 0-60-2min | [M+H] <sup>+</sup> | 0.301 |
| 1006 |  | 1.83 | 1,512.8 | 0-60-2min | [M+H] <sup>+</sup> | 0.304 |
| 1007 |  | 1.78 | 1,484.8 | 0-60-2min | [M+H] <sup>+</sup> | 0.854 |
| 1008 |  | 1.90 | 1,543.3 | 0-60-2min | [M+H] <sup>+</sup> | 0.440 |
| 1010 |  | 1.82 | 1,540.9 | 5-95-2min | [M+H] <sup>+</sup> | 0.944 |

|             |  |             |                |                         |                          |               |
|-------------|--|-------------|----------------|-------------------------|--------------------------|---------------|
| <b>1011</b> |  | <b>1.85</b> | <b>1,516.9</b> | <b>0-60-2min</b>        | <b>[M+H]<sup>+</sup></b> | <b>0.979</b>  |
| <b>1012</b> |  | <b>1.85</b> | <b>1,540.0</b> | <b>0-60-2min</b>        | <b>[M+H]<sup>+</sup></b> | <b>1.436</b>  |
| <b>1013</b> |  | <b>1.52</b> | <b>1,581.0</b> | <b>40-100-<br/>2min</b> | <b>[M+H]<sup>+</sup></b> | <b>4.673</b>  |
| <b>1014</b> |  | <b>1.60</b> | <b>1,609.5</b> | <b>40-100-<br/>2min</b> | <b>[M+H]<sup>+</sup></b> | <b>6.244</b>  |
| <b>1015</b> |  | <b>1.80</b> | <b>1,460.7</b> | <b>0-60-2min</b>        | <b>[M+H]<sup>+</sup></b> | <b>11.205</b> |

|             |  |             |                |                  |                          |              |
|-------------|--|-------------|----------------|------------------|--------------------------|--------------|
| <b>1016</b> |  | <b>1.85</b> | <b>1,540.9</b> | <b>0-60-2min</b> | <b>[M+H]<sup>+</sup></b> | <b>58.45</b> |
| <b>1017</b> |  | <b>1.80</b> | <b>1,488.9</b> | <b>0-60-2min</b> | <b>[M+H]<sup>+</sup></b> | <b>67.22</b> |
| <b>1018</b> |  | <b>1.85</b> | <b>1,540.8</b> | <b>0-60-2min</b> | <b>[M+H]<sup>+</sup></b> | <b>92.00</b> |
| <b>1019</b> |  | <b>1.80</b> | <b>1,474.8</b> | <b>0-60-2min</b> | <b>[M+H]<sup>+</sup></b> | <b>978</b>   |
| <b>1020</b> |  | <b>1.85</b> | <b>1,560.2</b> | <b>0-60-2min</b> | <b>[M+H]<sup>+</sup></b> | <b>0.387</b> |

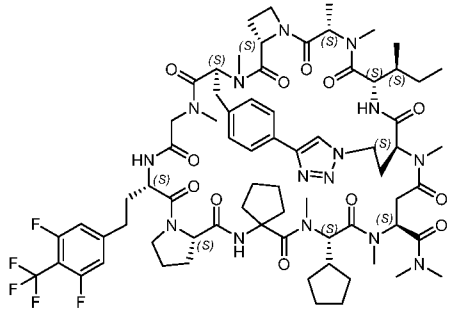
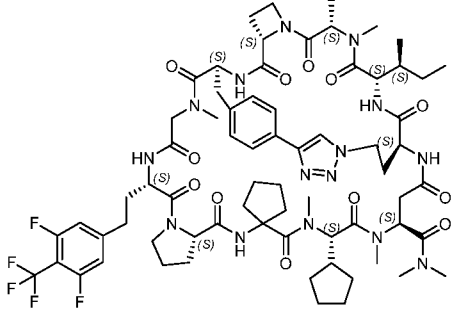
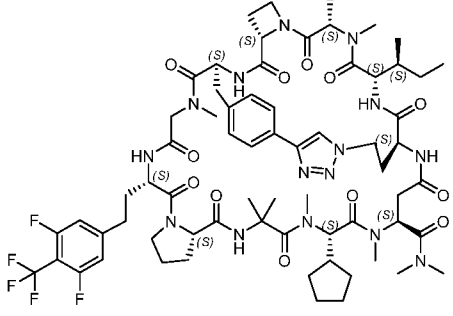
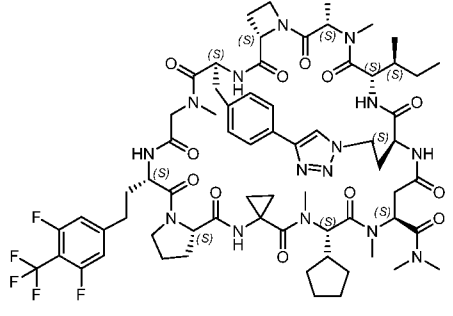
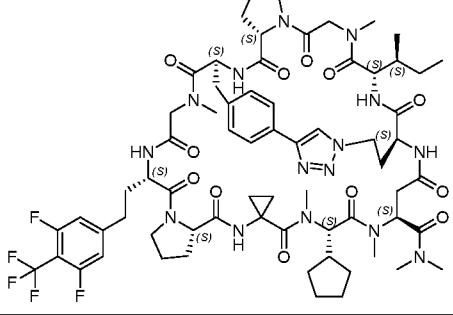
|      |                                                                                     |      |         |             |                    |        |
|------|-------------------------------------------------------------------------------------|------|---------|-------------|--------------------|--------|
| 1021 |    | 1.83 | 1,532.3 | 0-60-2min   | [M+H] <sup>+</sup> | 0.989  |
| 1022 |    | 1.42 | 1,499.3 | 40-100-2min | [M+H] <sup>+</sup> | 31.232 |
| 1023 |   | 1.84 | 1,526.9 | 0-60-2min   | [M+H] <sup>+</sup> | 19.413 |
| 1024 |  | 1.37 | 1,541.6 | 40-100-2min | [M+H] <sup>+</sup> | 0.212  |
| 1025 |  | 1.44 | 1,555.6 | 40-100-2min | [M+H] <sup>+</sup> | 0.766  |

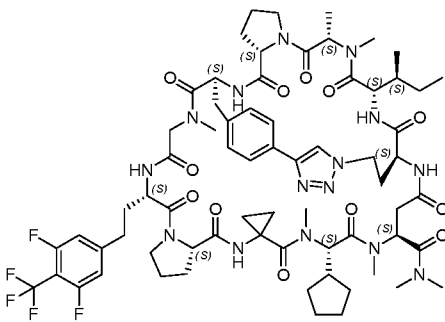
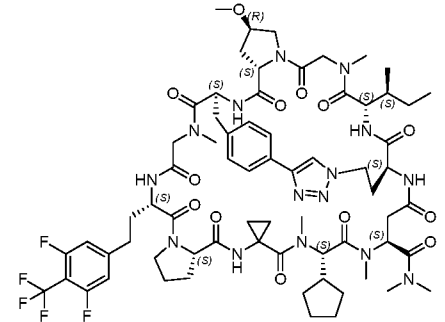
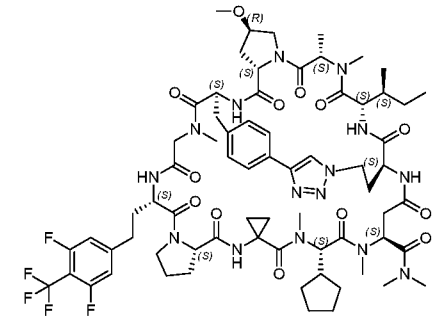
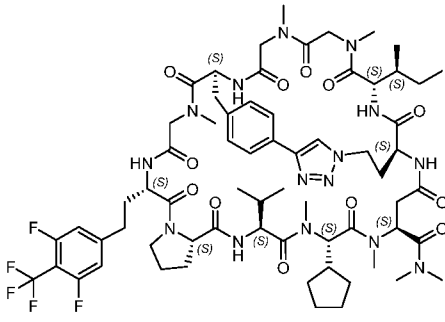
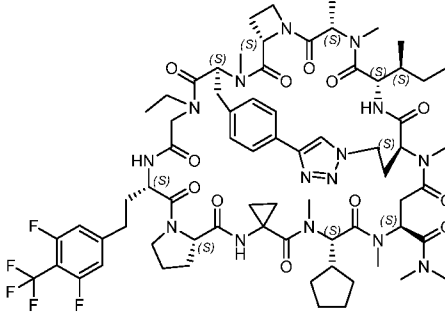
|             |  |             |                |                         |                    |              |
|-------------|--|-------------|----------------|-------------------------|--------------------|--------------|
| <b>1026</b> |  | <b>1.38</b> | <b>1,524.6</b> | <b>40-100-<br/>2min</b> | [M+H] <sup>+</sup> | <b>456</b>   |
| <b>1027</b> |  | <b>1.41</b> | <b>1,586.5</b> | <b>40-100-<br/>2min</b> | [M+H] <sup>+</sup> | <b>0.620</b> |
| <b>1028</b> |  | <b>1.46</b> | <b>1,569.6</b> | <b>40-100-<br/>2min</b> | [M+H] <sup>+</sup> | <b>0.315</b> |
| <b>1029</b> |  | <b>1.53</b> | <b>1,582.8</b> | <b>40-100-<br/>2min</b> | [M+H] <sup>+</sup> | <b>1.623</b> |
| <b>1030</b> |  | <b>1.52</b> | <b>1,554.9</b> | <b>40-100-<br/>2min</b> | [M+H] <sup>+</sup> | <b>48.08</b> |

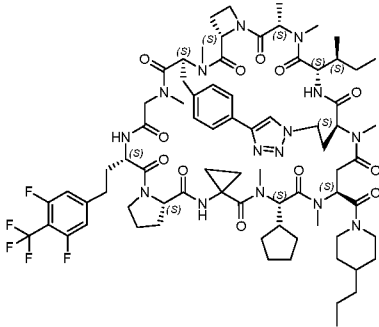
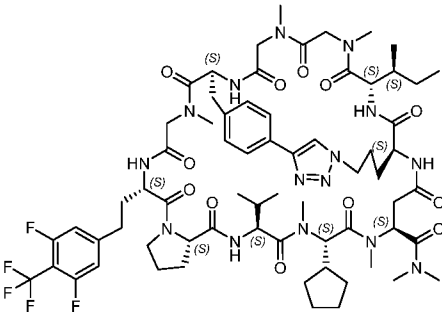
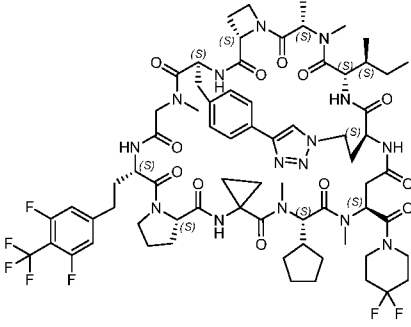
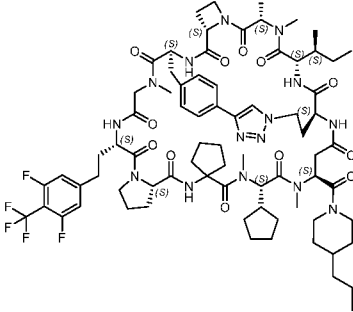
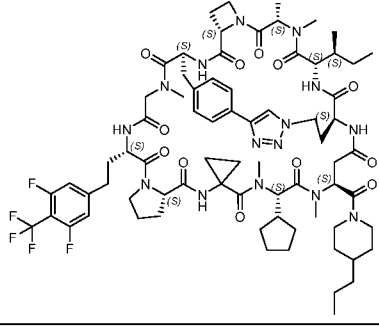
|      |  |      |         |             |                    |       |
|------|--|------|---------|-------------|--------------------|-------|
| 1031 |  | 1.36 | 1,552.8 | 40-100-2min | [M+H] <sup>+</sup> | 0.721 |
| 1032 |  | 1.43 | 1,566.8 | 40-100-2min | [M+H] <sup>+</sup> | 0.426 |
| 1033 |  | 1.43 | 1,566.9 | 40-100-2min | [M+H] <sup>+</sup> | 4.876 |
| 1034 |  | 1.41 | 1,564.8 | 40-100-2min | [M+H] <sup>+</sup> | 0.328 |
| 1035 |  | 1.48 | 1,614.9 | 40-100-2min | [M+H] <sup>+</sup> | 1.564 |

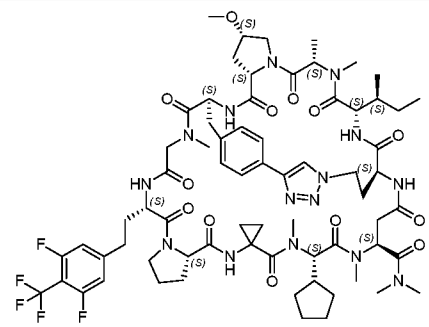
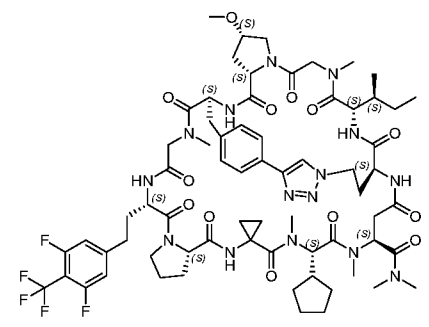
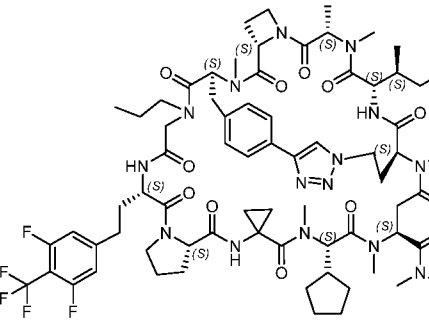
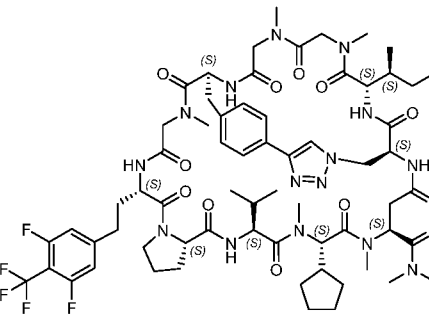
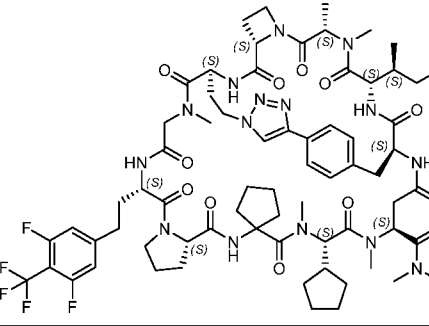
### LCMS and SPR Binding Affinity for Compounds 2001-2034:

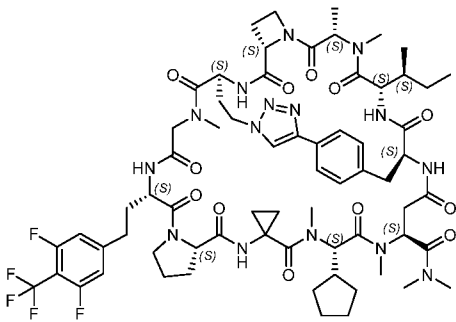
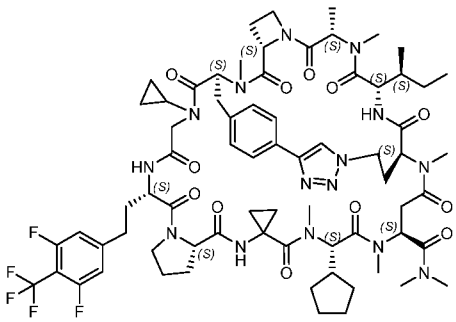
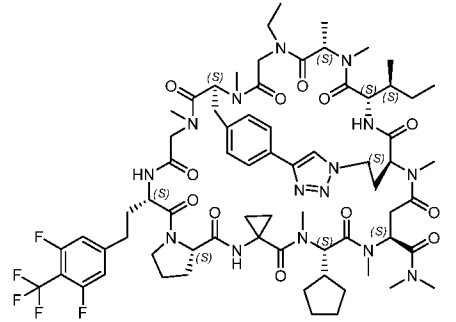
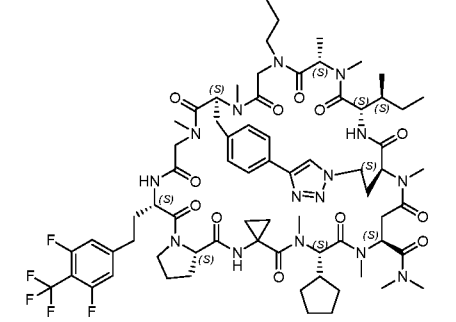
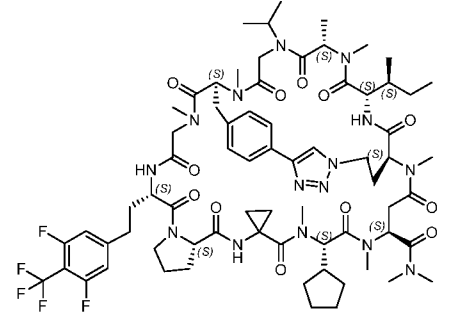
| Cmpd # | Structure | LCMS RT (min) | MS (m/z) | LCMS Method | MS Polarity | G12V GDP KD nM |
|--------|-----------|---------------|----------|-------------|-------------|----------------|
|--------|-----------|---------------|----------|-------------|-------------|----------------|

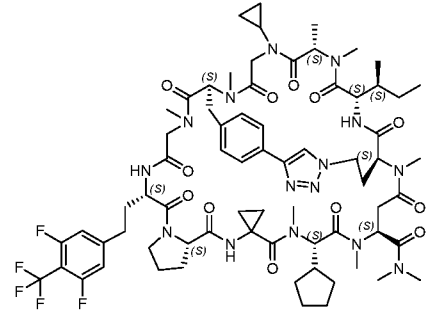
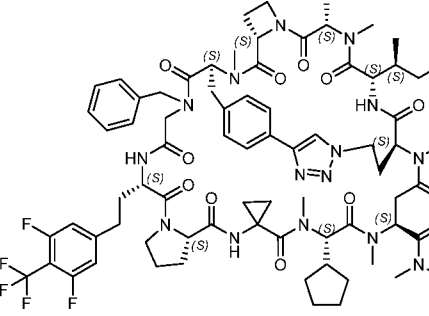
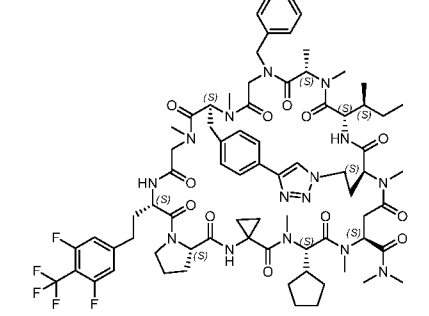
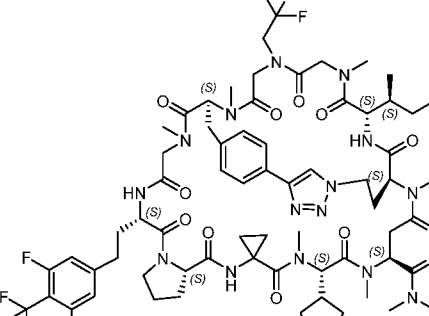
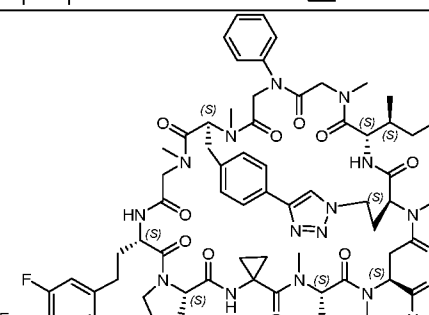
|      |                                                                                     |      |         |            |                    |       |
|------|-------------------------------------------------------------------------------------|------|---------|------------|--------------------|-------|
| 2001 |    | 1.69 | 1,446.9 | 5-95-2min  | [M+H] <sup>+</sup> | 0.016 |
| 2002 |    | 1.63 | 1,418.8 | 10-80-2min | [M+H] <sup>+</sup> | 0.082 |
| 2003 |   | 1.75 | 1,392.9 | 0-60-2min  | [M+H] <sup>+</sup> | 0.254 |
| 2004 |  | 1.73 | 1,390.9 | 0-60-2min  | [M+H] <sup>+</sup> | 0.305 |
| 2005 |  | 1.62 | 1,390.8 | 5-95-2min  | [M+H] <sup>+</sup> | 0.464 |

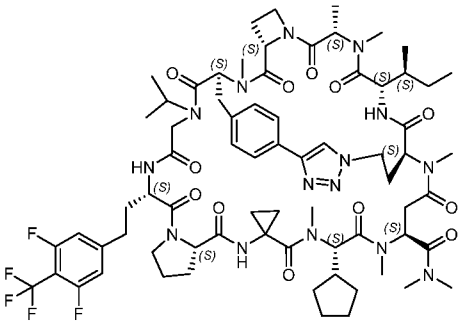
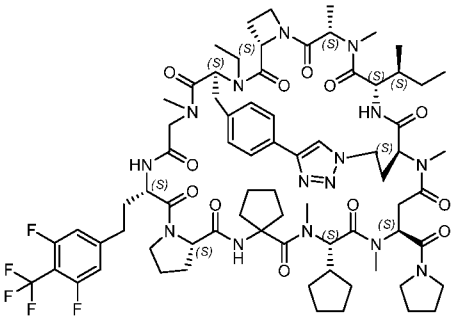
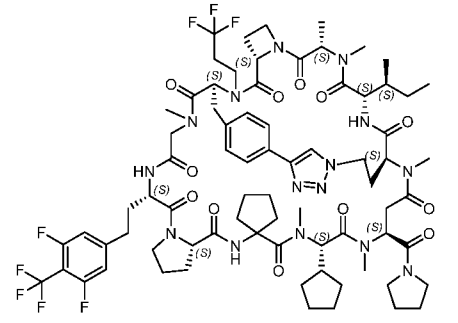
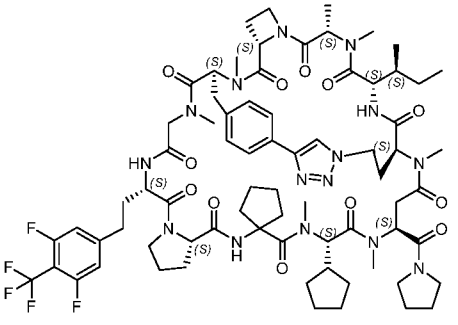
|      |                                                                                     |      |         |            |                    |       |
|------|-------------------------------------------------------------------------------------|------|---------|------------|--------------------|-------|
| 2006 |    | 1.75 | 1,405.9 | 0-60-2min  | [M+H] <sup>+</sup> | 0.649 |
| 2007 |    | 1.73 | 1,420.8 | 0-60-2min  | [M+H] <sup>+</sup> | 0.860 |
| 2008 |   | 1.60 | 1,434.7 | 10-80-2min | [M+H] <sup>+</sup> | 1.188 |
| 2009 |  | 1.75 | 1,380.8 | 0-60-2min  | [M+H] <sup>+</sup> | 1.281 |
| 2010 |  | 1.78 | 1,433.4 | 0-60-2min  | [M+H] <sup>+</sup> | 1.871 |

|      |                                                                                     |      |         |             |                    |       |
|------|-------------------------------------------------------------------------------------|------|---------|-------------|--------------------|-------|
| 2011 |    | 1.88 | 1,500.9 | 0-60-2min   | [M+H] <sup>+</sup> | 2.659 |
| 2012 |    | 1.09 | 1,394.8 | 40-100-2min | [M+H] <sup>+</sup> | 2.962 |
| 2013 |   | 1.78 | 1,466.8 | 0-60-2min   | [M+H] <sup>+</sup> | 3.141 |
| 2014 |  | 1.88 | 1,501.1 | 0-60-2min   | [M+H] <sup>+</sup> | 3.310 |
| 2015 |  | 1.85 | 1,472.9 | 0-60-2min   | [M+H] <sup>+</sup> | 4.462 |

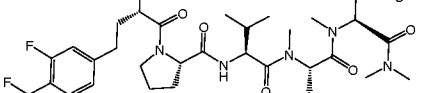
|      |                                                                                     |      |         |           |                    |        |
|------|-------------------------------------------------------------------------------------|------|---------|-----------|--------------------|--------|
| 2016 |    | 1.75 | 1,434.9 | 0-60-2min | [M+H] <sup>+</sup> | 16.51  |
| 2017 |    | 1.73 | 1,420.9 | 0-60-2min | [M+H] <sup>+</sup> | 24.65  |
| 2018 |   | 1.80 | 1,447.4 | 0-60-2min | [M+H] <sup>+</sup> | 25.15  |
| 2019 |  | 1.78 | 1,366.8 | 0-60-2min | [M+H] <sup>+</sup> | 43.82  |
| 2020 |  | 1.78 | 1,418.9 | 0-60-2min | [M+H] <sup>+</sup> | 474.93 |

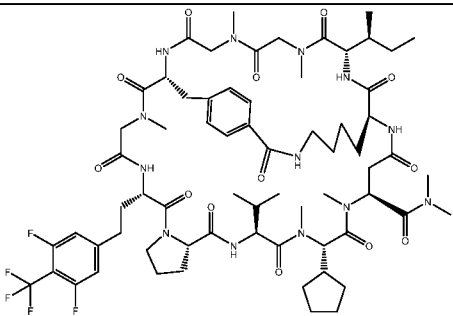
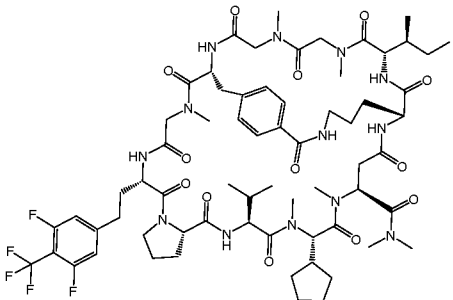
|      |                                                                                     |      |         |           |                    |       |
|------|-------------------------------------------------------------------------------------|------|---------|-----------|--------------------|-------|
| 2021 |    | 1.75 | 1,390.8 | 0-60-2min | [M+H] <sup>+</sup> | 3,961 |
| 2022 |    | 1.78 | 1,445.4 | 0-60-2min | [M+H] <sup>+</sup> | 3,768 |
| 2023 |   | 1.78 | 1,420.9 | 0-60-2min | [M+H] <sup>+</sup> | 1,783 |
| 2024 |  | 1.80 | 1,435.0 | 0-60-2min | [M+H] <sup>+</sup> | 2,952 |
| 2025 |  | 1.78 | 1,434.9 | 0-60-2min | [M+H] <sup>+</sup> | 23.55 |

|      |                                                                                     |      |         |           |                    |        |
|------|-------------------------------------------------------------------------------------|------|---------|-----------|--------------------|--------|
| 2026 |    | 1.80 | 1,432.9 | 0-60-2min | [M+H] <sup>+</sup> | 7.014  |
| 2027 |    | 1.83 | 1,495.4 | 0-60-2min | [M+H] <sup>+</sup> | 1,658  |
| 2028 |   | 1.83 | 1,482.9 | 0-60-2min | [M+H] <sup>+</sup> | 1.716  |
| 2029 |  | 1.83 | 1,460.6 | 0-60-2min | [M+H] <sup>+</sup> | 0.378  |
| 2030 |  | 1.83 | 1,454.6 | 0-60-2min | [M+H] <sup>+</sup> | 31.348 |

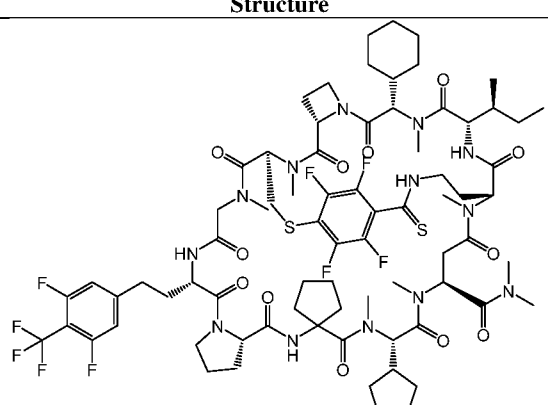
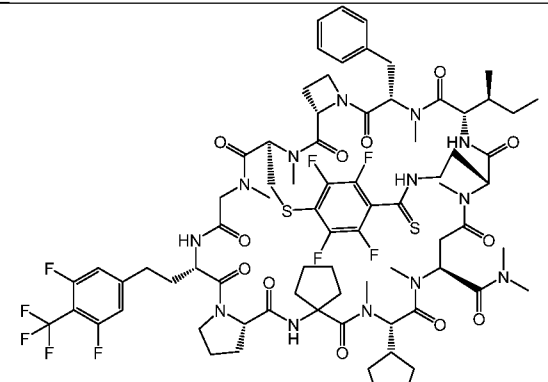
|      |                                                                                     |      |         |             |                    |       |
|------|-------------------------------------------------------------------------------------|------|---------|-------------|--------------------|-------|
| 2031 |    | 1.80 | 1,447.4 | 0-60-2min   | [M+H] <sup>+</sup> | 4,514 |
| 2032 |    | 1.28 | 1,486.9 | 40-100-2min | [M+H] <sup>+</sup> | 0.052 |
| 2033 |   | 1.48 | 1,555.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.033 |
| 2034 |  | 1.23 | 1,472.9 | 40-100-2min | [M+H] <sup>+</sup> | 0.072 |

### LCMS and SPR Binding Affinity for Compounds 3001-3003:

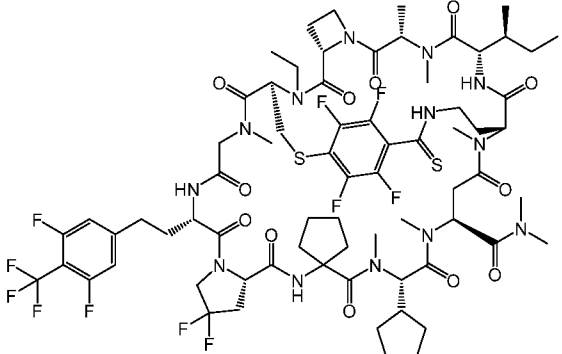
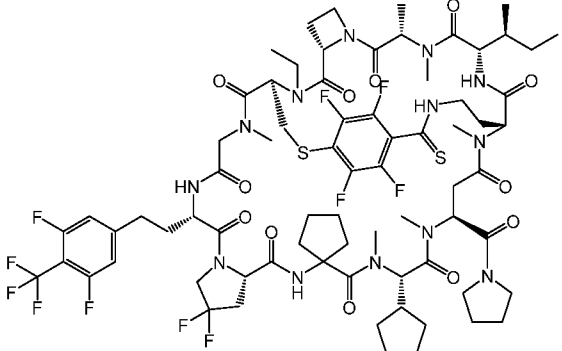
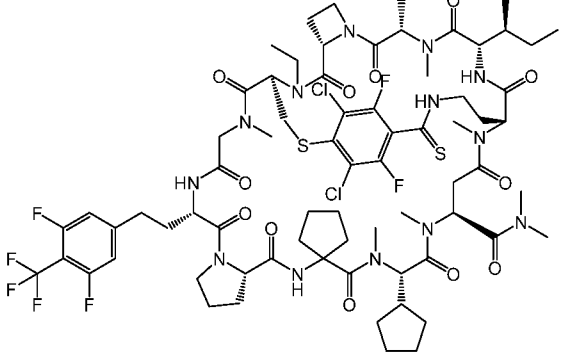
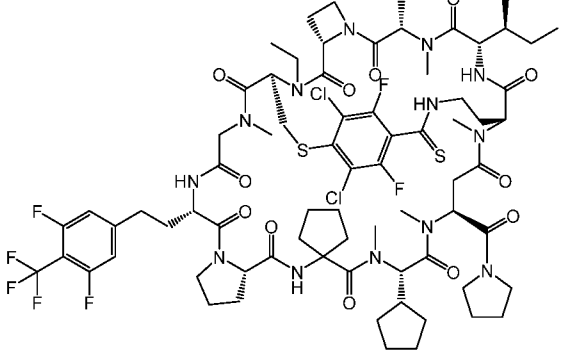
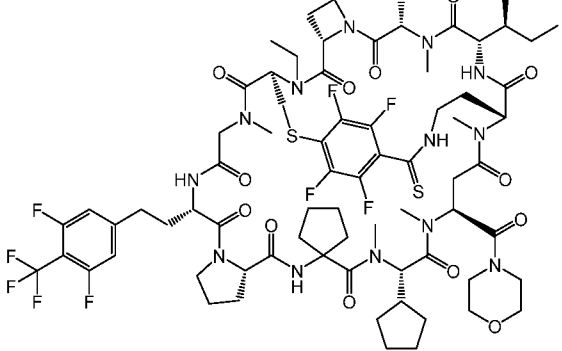
| Cmpd # | Structure                                                                           | LCMS RT (min) | MS (m/z) | LCMS Method | MS Polarity        | G12V GDP KD nM |
|--------|-------------------------------------------------------------------------------------|---------------|----------|-------------|--------------------|----------------|
| 3001   |  | 1.63          | 1,356.8  | 10-80-2min  | [M+H] <sup>+</sup> | 10.2           |

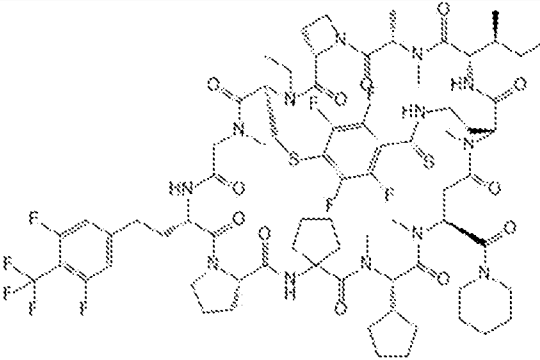
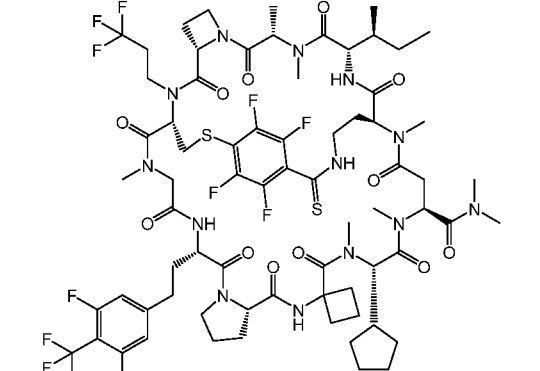
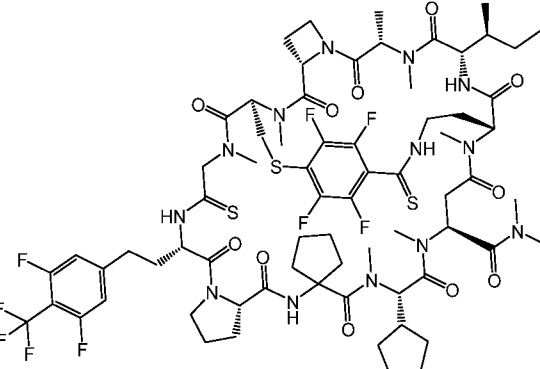
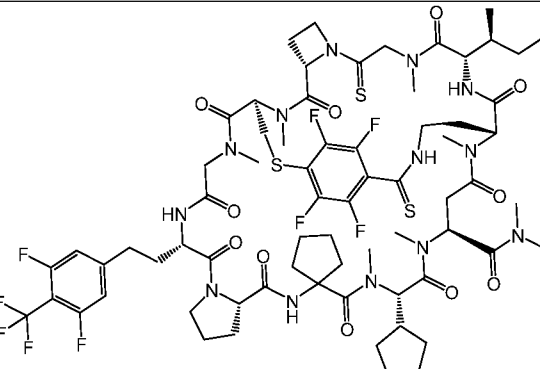
|      |                                                                                   |      |         |            |                    |       |
|------|-----------------------------------------------------------------------------------|------|---------|------------|--------------------|-------|
| 3002 |  | 1.78 | 1,384.8 | 0-60-2min  | [M+H] <sup>+</sup> | 52.1  |
| 3003 |  | 1.63 | 1,370.8 | 10-80-2min | [M+H] <sup>+</sup> | 145.0 |

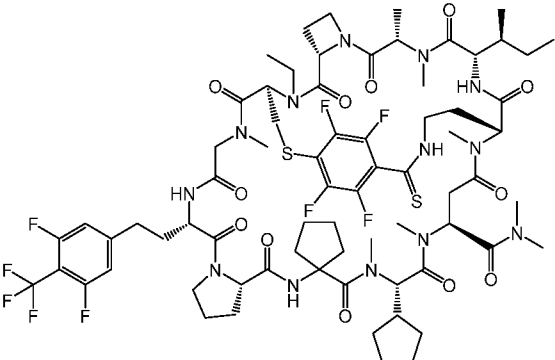
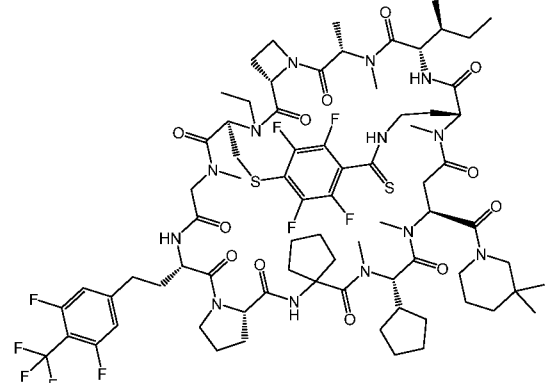
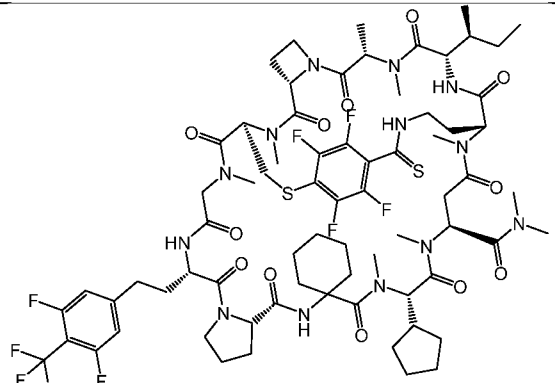
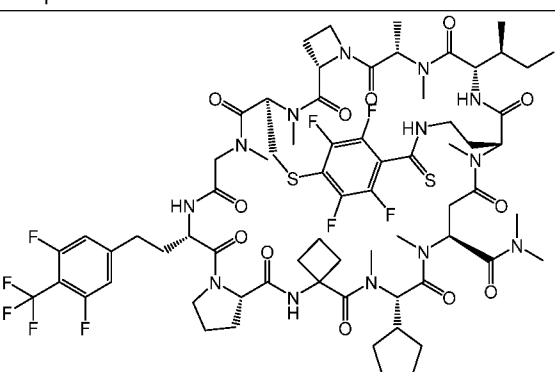
### LCMS and SPR Binding Affinity for Compounds 1036-1168 and 2035-2162:

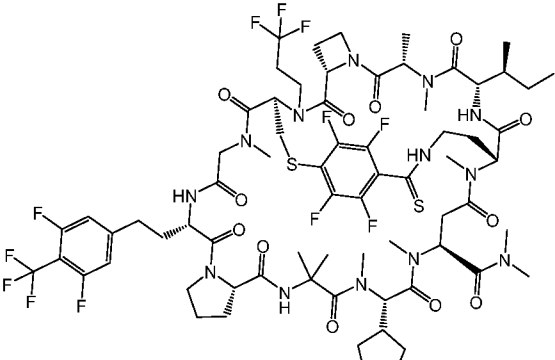
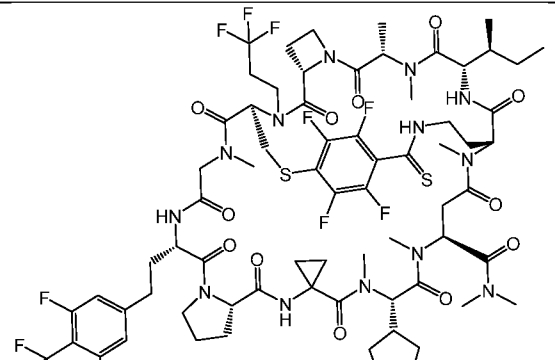
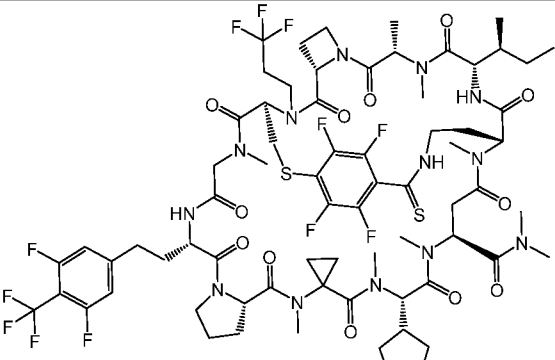
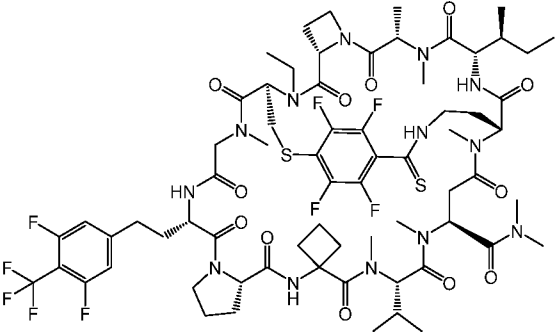
| Cmpd # | Structure                                                                           | SMILES                                                                                                                                                                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1036   |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C2CCCCC2)C(=O)N3[C@@H](CC3)C(=O)N(C)[C@@H]4C(=O)N(C)CC(=O)N[C@@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)C(=O)N6[C@@H](CCC6)C(=O)NC7(CC7)C(=O)N(C)[C@@H](C8CCCC8)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C9=C(F)C(F)=C(SC4)C(F)=C9F)C(=O)N1)C(=O)N(C)C</chem>       |
| 1037   |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](CC2=CC=CC=C2)C(=O)N3[C@@H](CC3)C(=O)N(C)[C@@H]4C(=O)N(C)CC(=O)N[C@@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)C(=O)N6[C@@H](CCC6)C(=O)NC7(CCCC7)C(=O)N(C)[C@@H](C8CCCC8)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C9=C(F)C(F)=C(SC4)C(F)=C9F)C(=O)N1)C(=O)N(C)C</chem> |

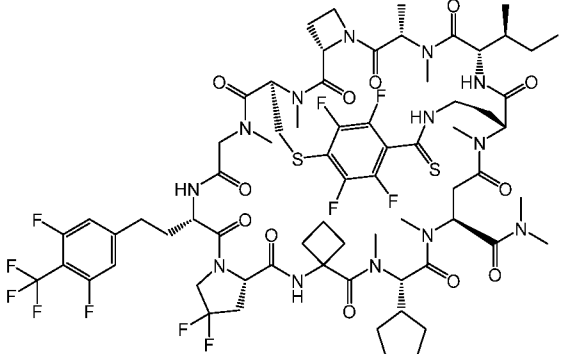
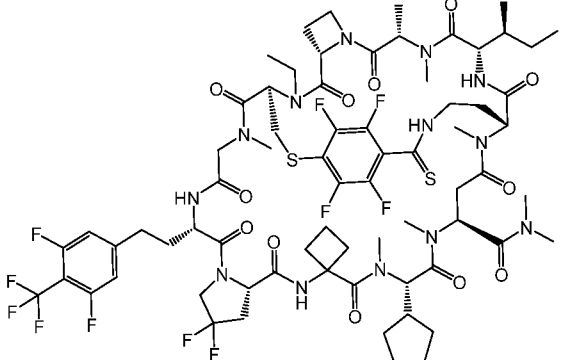
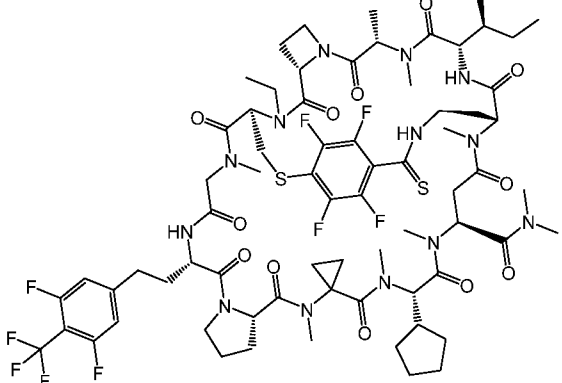
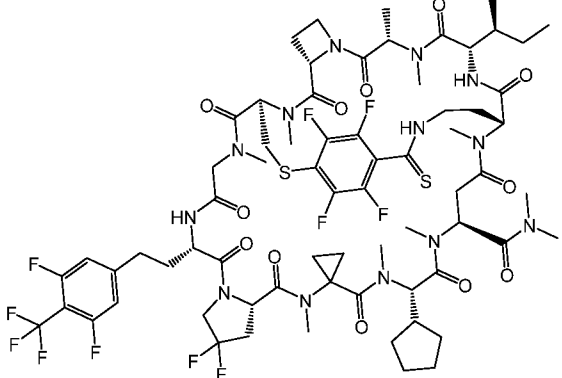
|      |  |                                                                                                                                                                                                                                                                                                 |
|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1038 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>   |
| 1039 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem> |
| 1040 |  | <chem>CCO[C@H]1CN2[C@@H](C1)C(=O)NC3(CC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(C)[C@@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C2=O)C(=O)N(C)C</chem>    |
| 1041 |  | <chem>CCO[C@H]1CN2[C@@H](C1)C(=O)NC3(CCCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(C)[C@@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C2=O)C(=O)N(C)C</chem> |

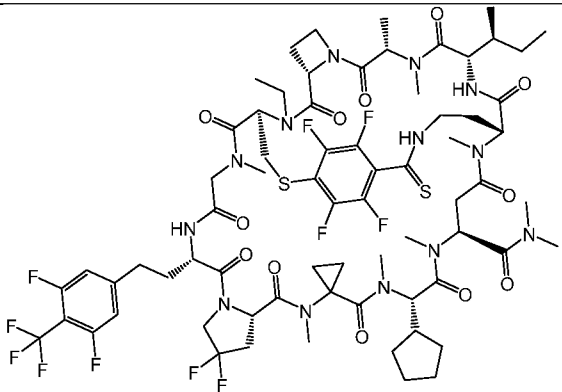
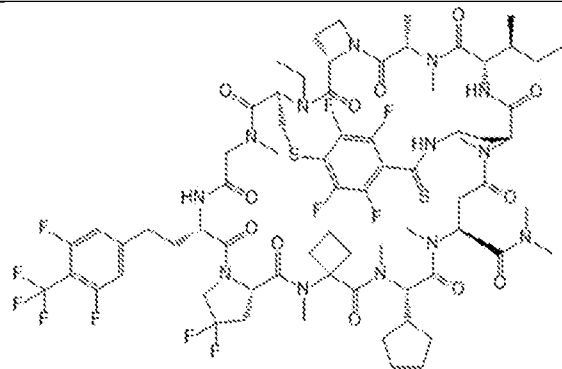
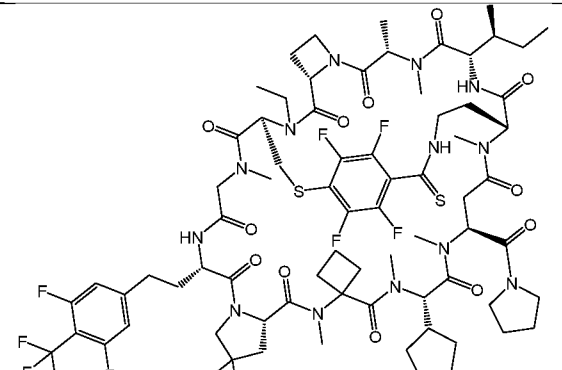
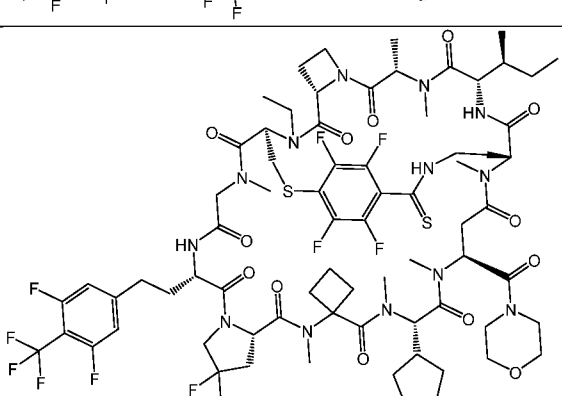
|      |                                                                                     |                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1042 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>    |
| 1043 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(CS3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem> |
| 1044 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(Cl)=C(SC3)C(Cl)=C8F)C(=O)N1)C(=O)N(C)C</chem>       |
| 1045 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(Cl)=C(SC3)C(Cl)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>     |
| 1046 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem>      |

|      |                                                                                     |                                                                                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1047 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCCC9</chem>    |
| 1048 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem> |
| 1049 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=S)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>        |
| 1050 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=S)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>                |

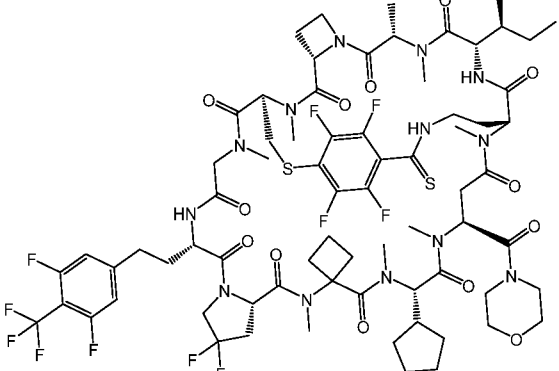
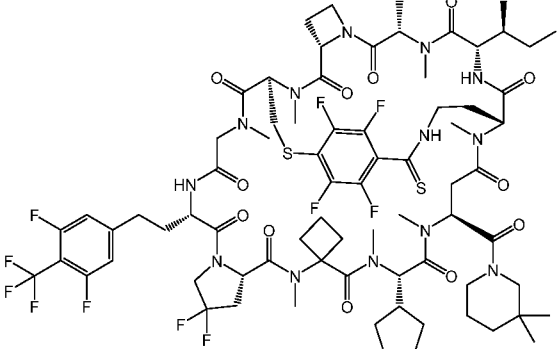
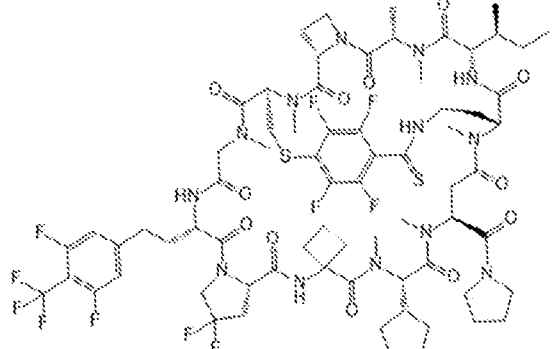
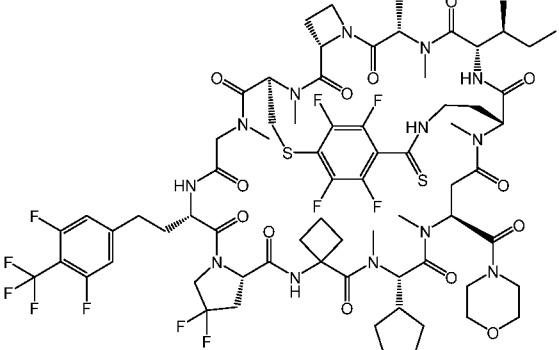
|      |                                                                                     |                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1051 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>          |
| 1052 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CC(C)(C)CCC9</chem> |
| 1053 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5C(CCC5)C(=O)NC6(CCCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>                 |
| 1054 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>            |

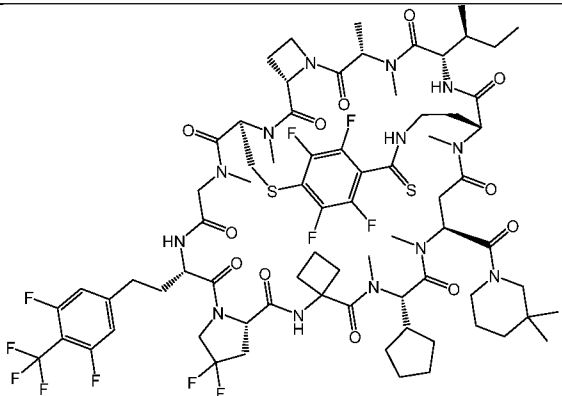
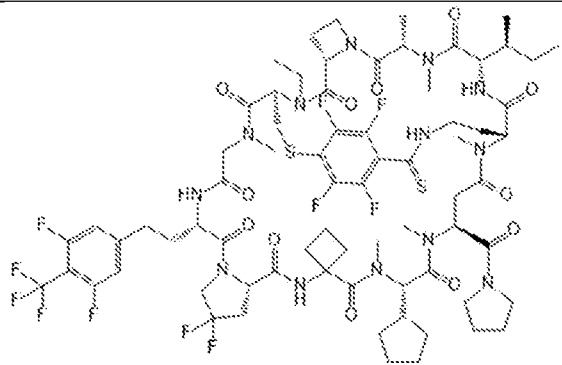
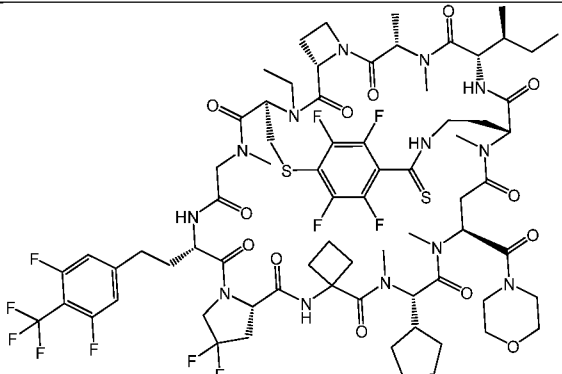
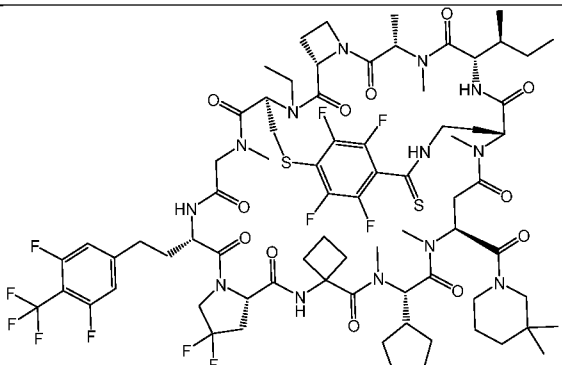
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|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1055 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC(C)(C)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC3)C(F)=C7F)C(=O)N1)C(=O)N(C)C</chem>    |
| 1056 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>    |
| 1057 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem> |
| 1058 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>          |

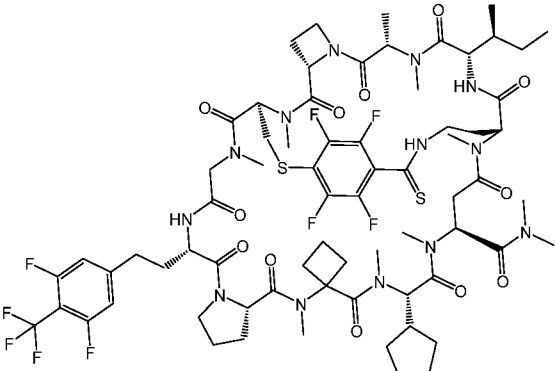
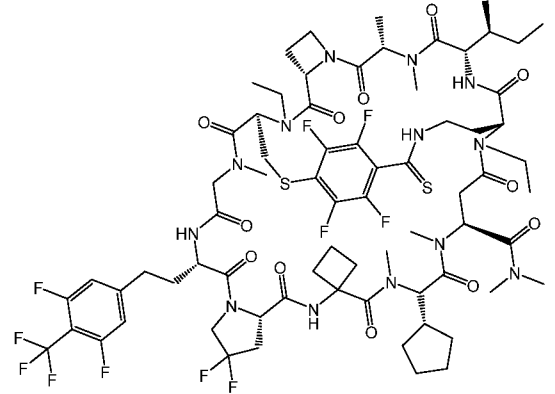
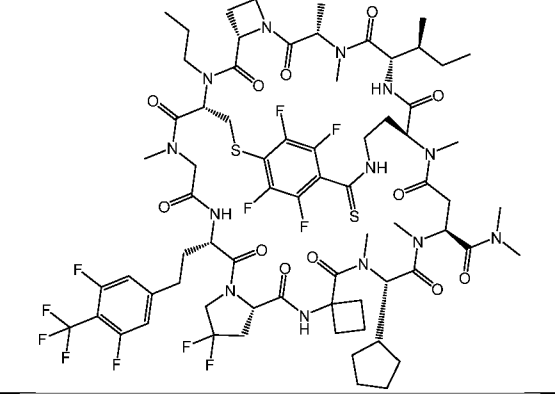
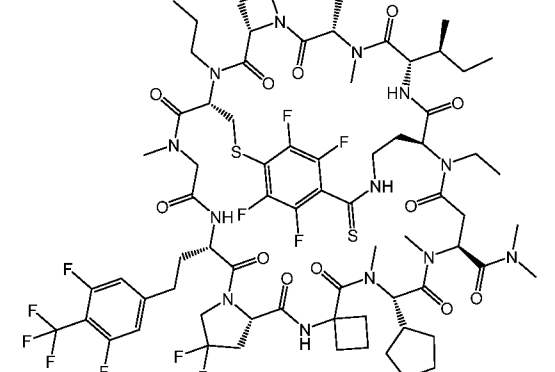
|      |                                                                                     |                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1059 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>    |
| 1060 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem> |
| 1061 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>     |
| 1062 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1063 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>    |
| 1064 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>    |
| 1065 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>  |
| 1066 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem> |

|      |  |                                                                                                                                                                                                                                                                                                       |
|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1067 |  | <chem>CC[C@@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CC(C)(C)CCC9</chem> |
| 1068 |  | <chem>CC[C@@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>          |
| 1069 |  | <chem>CC[C@@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>               |
| 1070 |  | <chem>CC[C@@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>        |

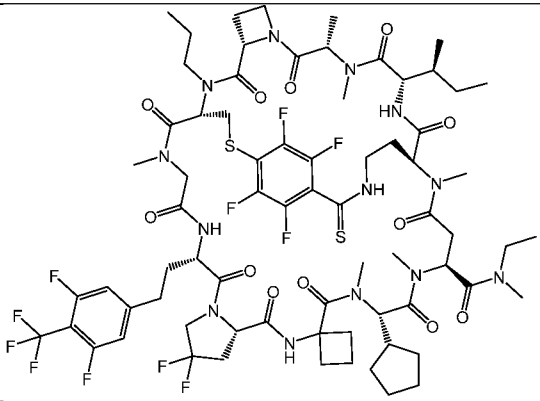
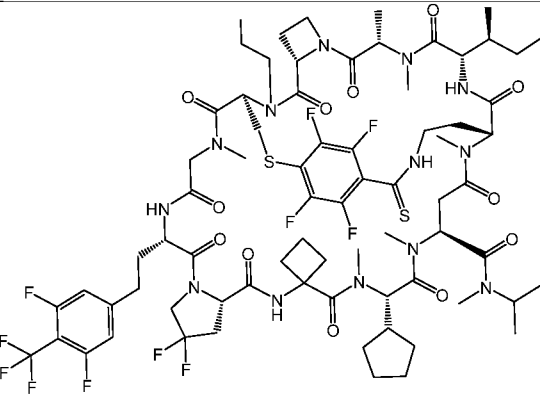
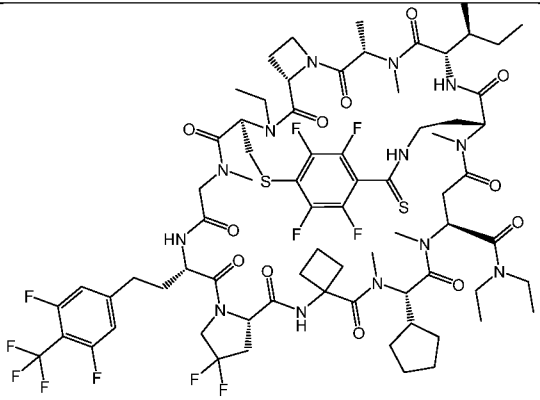
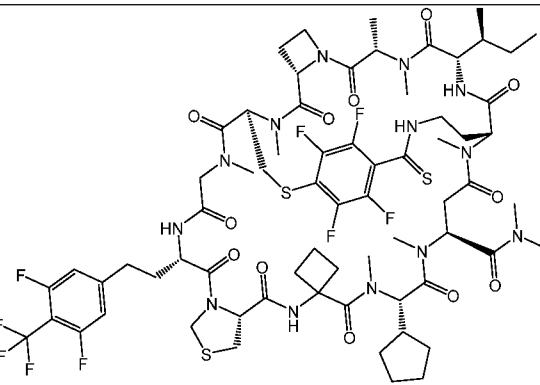
|      |                                                                                     |                                                                                                                                                                                                                                                                                                        |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1071 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC(C6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem>       |
| 1072 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC(C6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CC(C)(C)CCC9</chem> |
| 1073 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>            |
| 1074 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem>           |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1075 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CC(C)(C)CCC9</chem>  |
| 1076 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>        |
| 1077 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem>       |
| 1078 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CC(C)(C)CCC9</chem> |

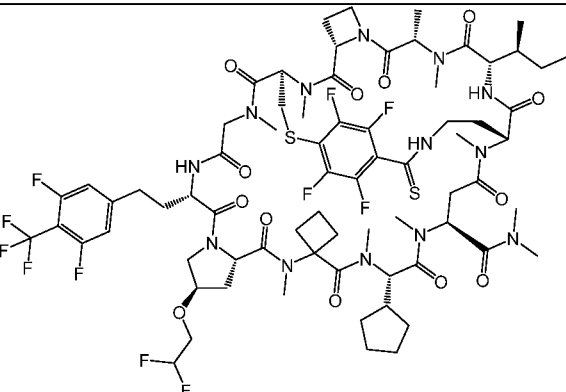
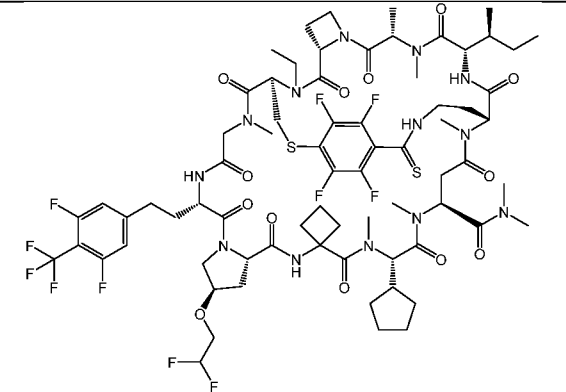
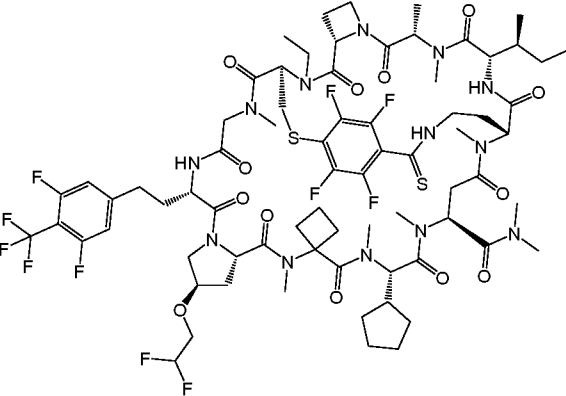
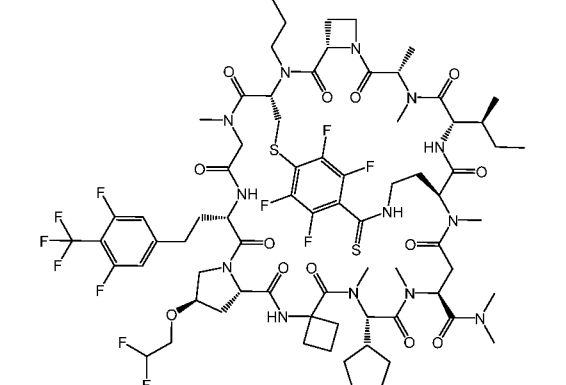
|      |                                                                                     |                                                                                                                                                                                                                                                                                        |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1079 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>  |
| 1080 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)C(C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(CC)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem> |
| 1081 |   | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>  |
| 1082 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem> |

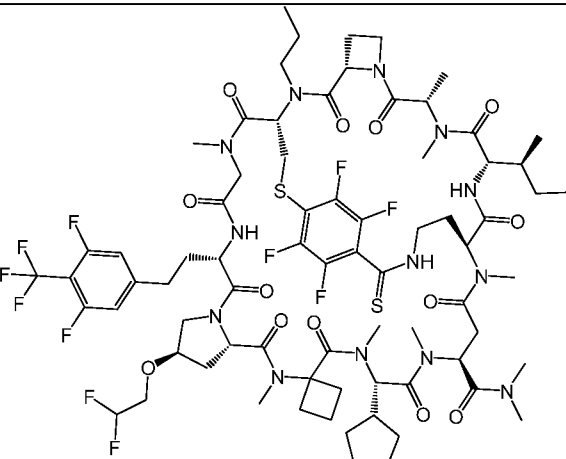
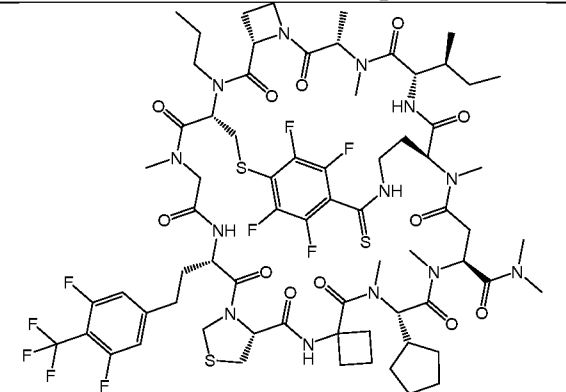
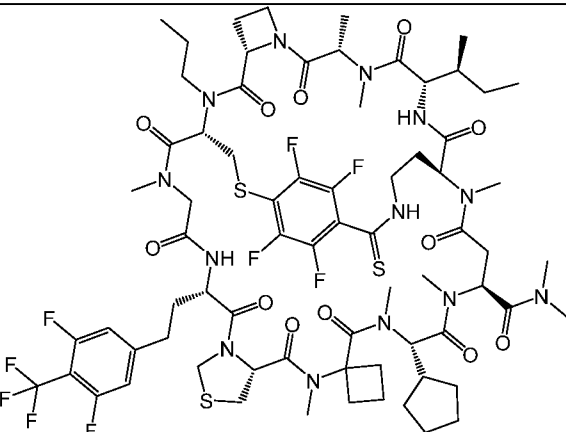
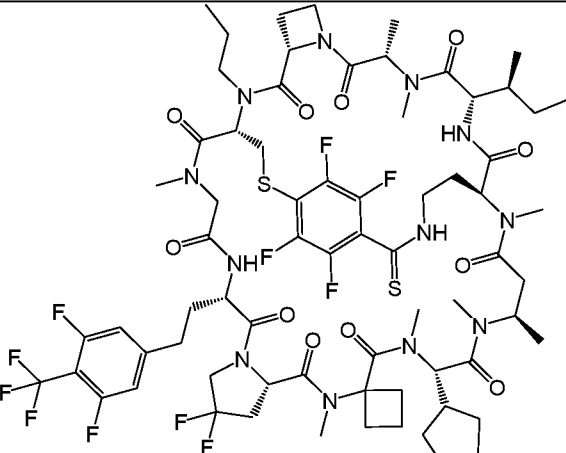
|      |  |                                                                                                                                                                                                                                                                                                 |
|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1083 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC(C6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(CC)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem> |
| 1084 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)C(F)(F)C6)C(=O)[C@H](CC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>         |
| 1085 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)C(F)(F)C6)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@H]1CSC7=C(F)C(F)=C(C(=S)NCC3)C(F)=C7F)CC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>        |
| 1086 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C)CC(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(S(C3)C(F)=C8F)C(=O)N1</chem>              |

|      |  |                                                                                                                                                                                                                                                                                      |
|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1087 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@H]3C(=O)N(C)C(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>    |
| 1088 |  | <chem>CN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CC)C(=O)[C@H]7N(C7)C(=O)[C@H](C)N(C)C(=O)[C@H](C(C)C)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>      |
| 1089 |  | <chem>CN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@@H](NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)C8CCCC8)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 1090 |  | <chem>CN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CC)C(=O)[C@H]7N(C7)C(=O)[C@H](C)N(C)C(=O)[C@@H](NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1)C9CCCC9</chem>   |

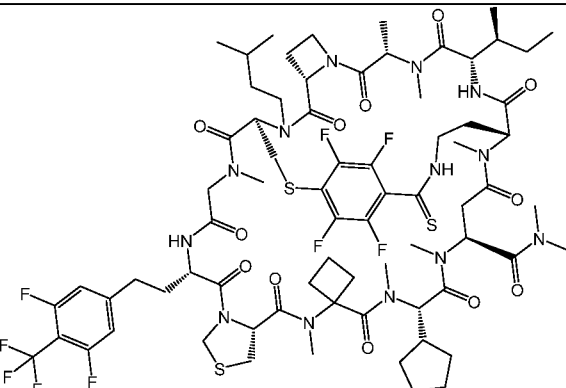
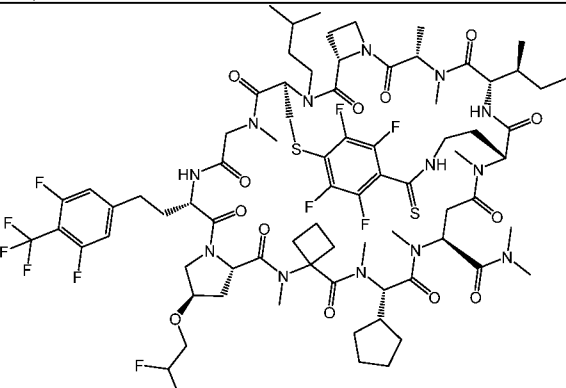
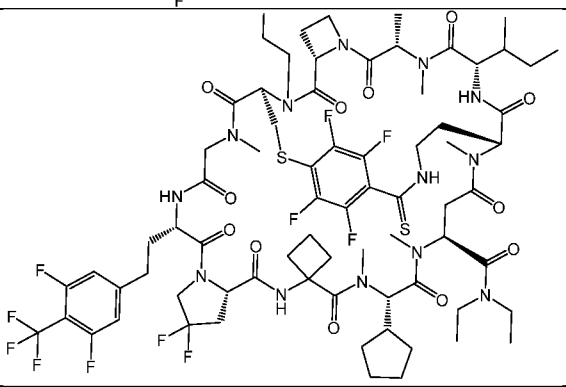
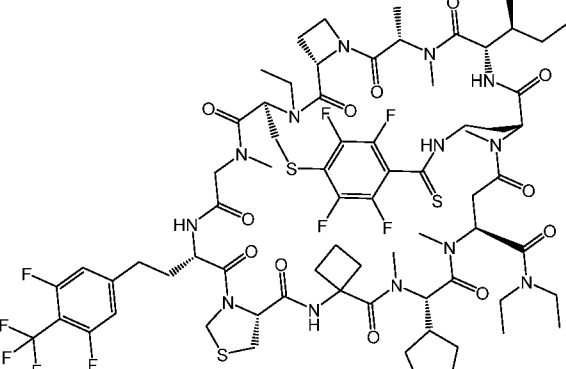
|      |                                                                                     |                                                                                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1091 |    | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=C(F)C(F)=C(C(F)=C8)C(F)(F)F</chem>   |
| 1092 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>  |
| 1093 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(CC)CC</chem> |
| 1094 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>          |

|      |  |                                                                                                                                                                                                                                                                                                     |
|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1095 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem>               |
| 1096 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>             |
| 1097 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>             |
| 1098 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H]([N](C(=O)CN(C)C(=O)[C@H]3N(C)C(=O)[C@H]4N(C)CC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)NC2=O)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1099 |    | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@H](C)C(=O)N6[C@@H](CC6)C(=O)N(C)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1</chem>   |
| 1100 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]4N(CC)C(=O)[C@H]5N(CC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O</chem>      |
| 1101 |   | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(C)C</chem> |
| 1102 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]4N(CCC)C(=O)[C@H]5N(CC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O</chem>      |

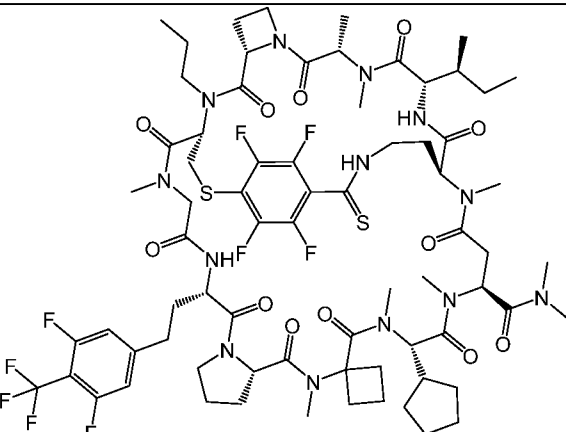
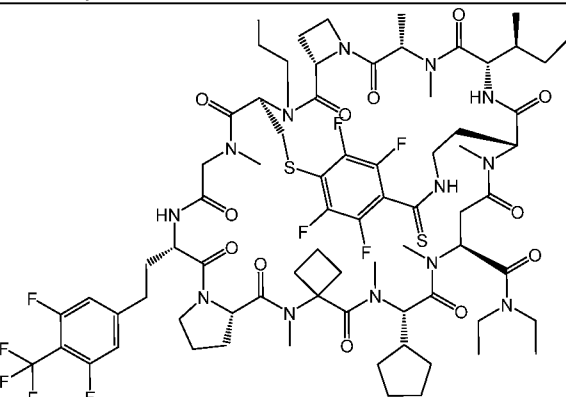
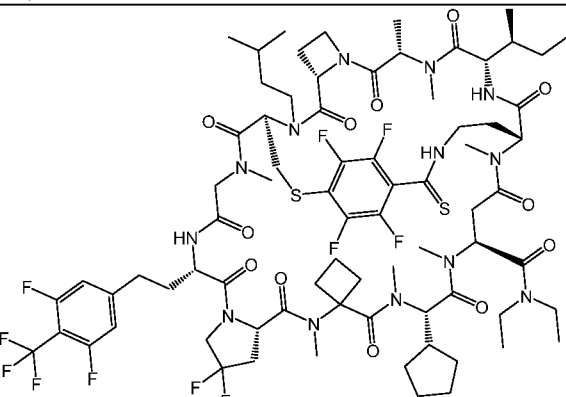
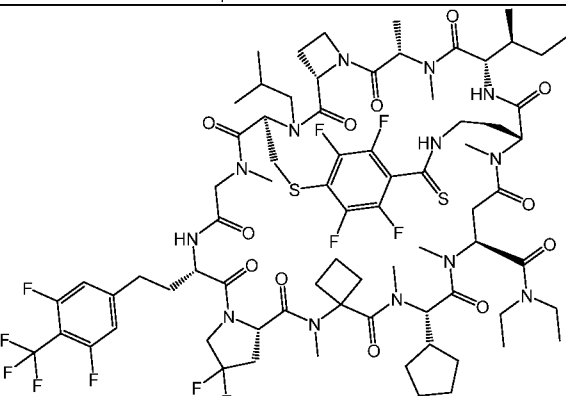
|      |                                                                                     |                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1103 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)[C@H](OCC(F)F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem> |
| 1104 |   | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(C)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>                     |
| 1105 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>                  |
| 1106 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>                  |

|      |  |                                                                                                                                                                                                                                                                                                    |
|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1107 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>        |
| 1108 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>              |
| 1109 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@H]3N(CCC(C)C)C(=O)[C@H]4N(CC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)NC2=O)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem> |
| 1110 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>     |

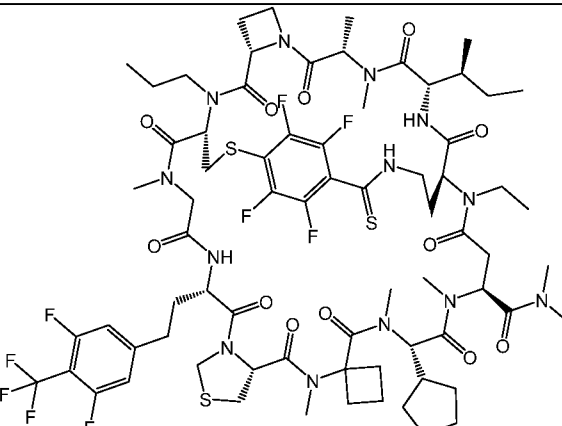
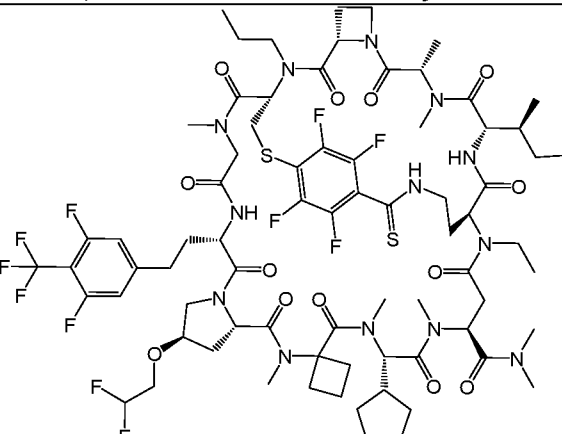
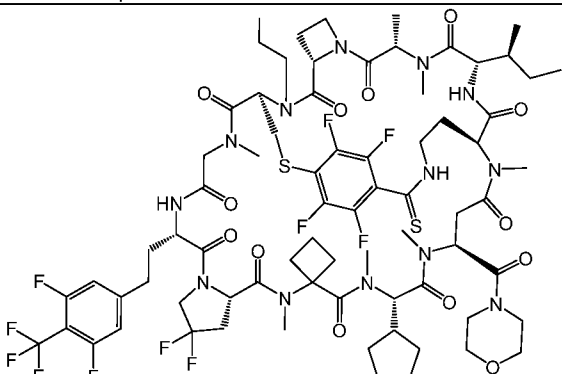
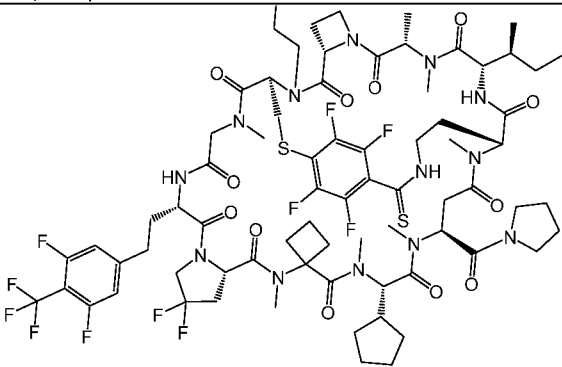
|      |                                                                                     |                                                                                                                                                                                                                                                                                                        |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1111 |    | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>               |
| 1112 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]4N(CCC(C)C)C(=O)[C@H]5N(CC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)N(C)C2=O</chem> |
| 1113 |   | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H](C(C)CC)NC(=O)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>                   |
| 1114 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CSC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CC)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>                     |

|      |  |                                                                                                                                                                                                                                                                                              |
|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1115 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CSC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>          |
| 1116 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)C(C8=CC(F)=C(C(F)=C8)C(F)(F)F</chem> |
| 1117 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)C([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C(C(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>    |
| 1118 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CSC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>       |

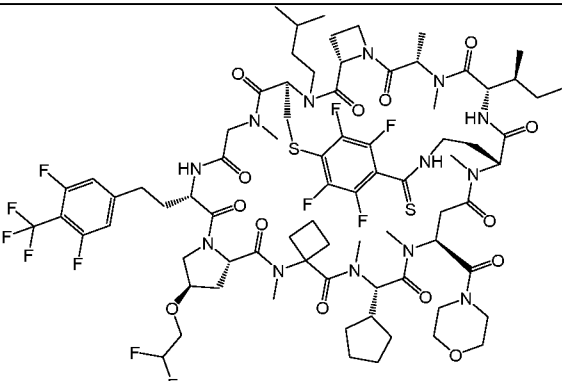
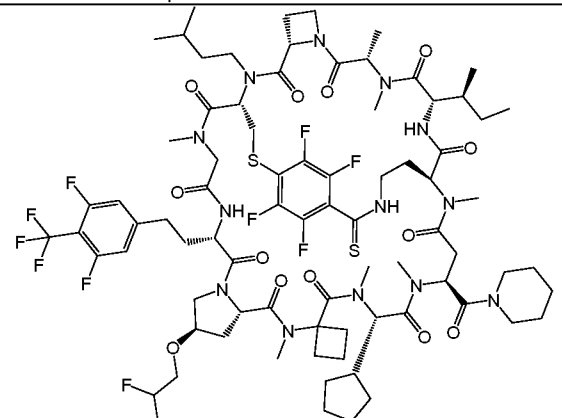
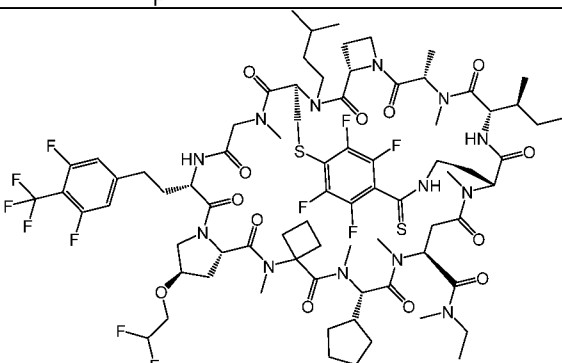
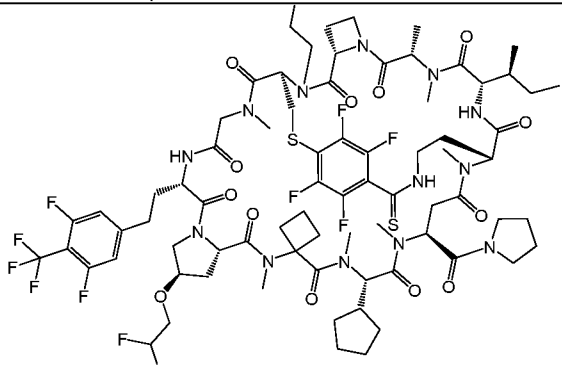
|      |  |                                                                                                                                                                                                                                                                                                                                    |
|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1119 |  | <chem>CCCN1[C@@H]2C(=O)N(C)CC(=O)N[C@@H](CC<br/>C3=CC(F)=C(C(F)=C3)C(F)(F)C(=O)N4[C@@H](<br/>CSC4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@H](C6<br/>CCCC6)C(=O)N(C)[C@H](CC(=O)N(C)[C@H](C<br/>CNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C<br/>@@H])([C@@H](C)CC)C(=O)N(C)[C@H](C)C(=O)<br/>N)N8[C@@H](CC8)C1=O)C(=O)N(CC)CC</chem>            |
| 1120 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC<br/>2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C<br/>4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CC<br/>C)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[<br/>C@H])([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)<br/>C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CC<br/>C8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>         |
| 1121 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC<br/>2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(<br/>F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F<br/>F)NC(=O)CN(C)C(=O)[C@H]6N(CCC)C(=O)[C@H]<br/>7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H](C@@<br/>H)(C)CC)NC(=O)[C@@H](CCNC(=S)C8=C(F)C(F)=<br/>C(SC6)C(F)=C8F)N(CC)C(=O)C1</chem>          |
| 1122 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C<br/>(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)<br/>C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H]<br/>(C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6<br/>N(C)[C@H](OCC(F)F)C6)C(=O)[C@H](CCC7=CC(F)<br/>=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]<br/>1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1123 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)CC6C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>                   |
| 1124 |   | <chem>CCCN1[C@@H]2C(=O)N(C)CC(=O)N[C@@H](CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CCC4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](C)CNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N8[C@@H](CC8)C1=O)C(=O)N(CC)CC</chem>          |
| 1125 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]1CSC7=C(F)C(F)=C(C(=S)NCC3)C(F)=C7F)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>   |
| 1126 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC(C)C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(CC)CC</chem> |

|      |  |                                                                                                                                                                                                                                                                                                    |
|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1127 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC(C)C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(C)C</chem> |
| 1128 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>          |
| 1129 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>       |
| 1130 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>      |

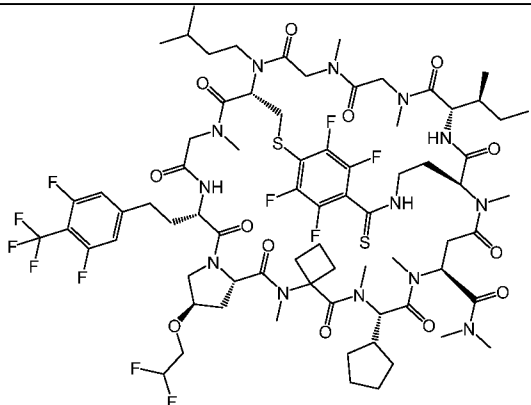
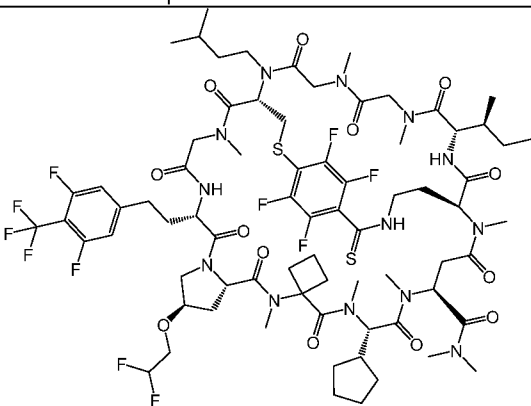
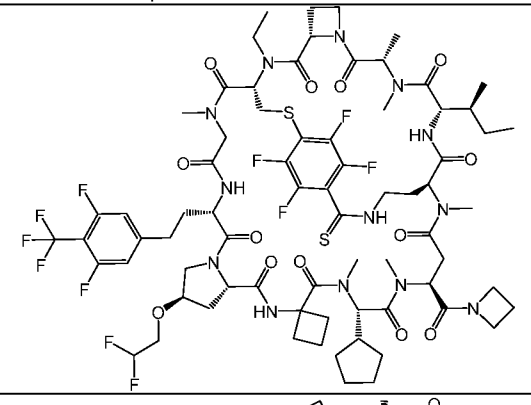
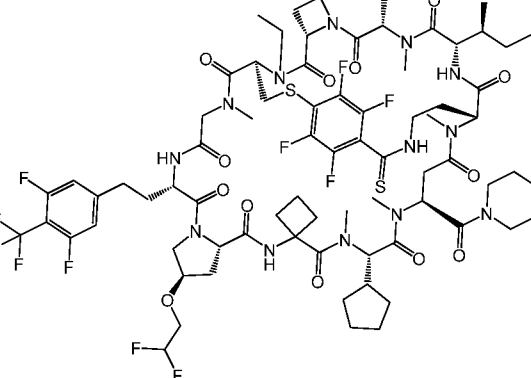
|      |                                                                                     |                                                                                                                                                                                                                                                                                                        |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1131 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(CC)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>              |
| 1132 |   | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(CC)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1</chem> |
| 1133 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N4CCOCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>          |
| 1134 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>           |

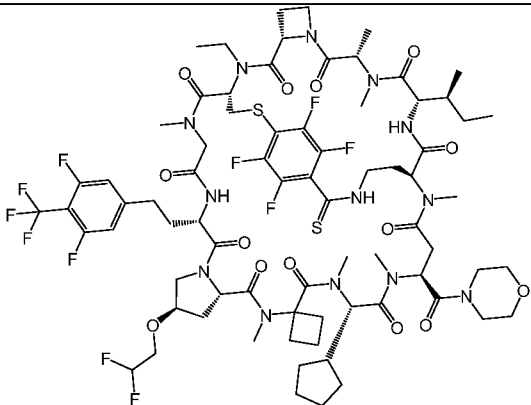
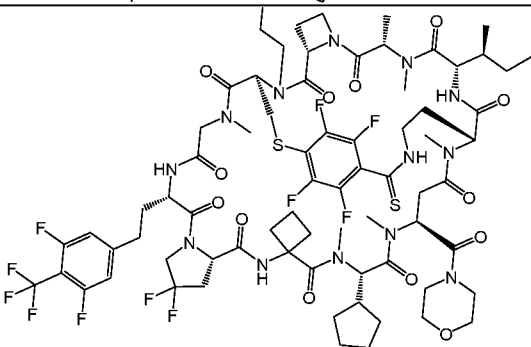
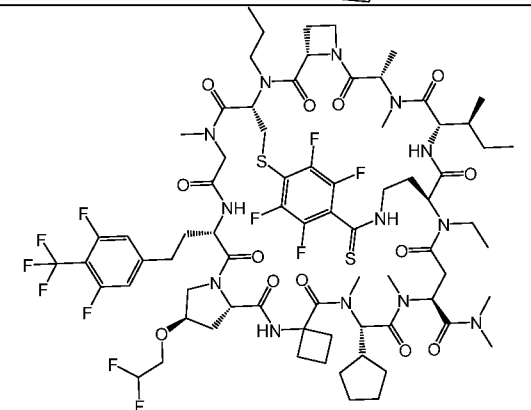
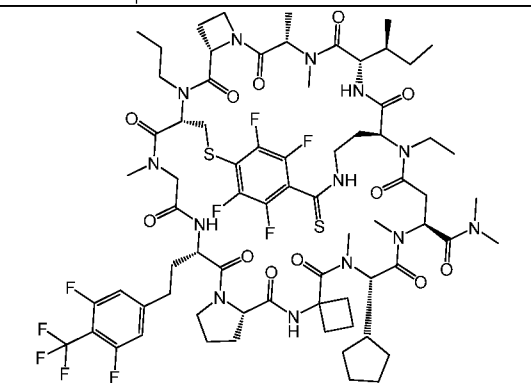
|      |  |                                                                                                                                                                                                                                                                                                             |
|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1135 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)C(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CSC7=C(F)C(F)=C(C(=S)NCC3)C(F)=C7F)CCC8C=C(Cl)C(=CC=8)C(F)(F)F</chem>                     |
| 1136 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(C)C</chem>                             |
| 1137 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(C)C</chem>                             |
| 1138 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCCC9</chem> |

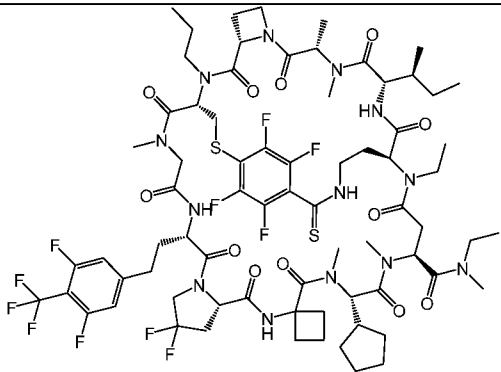
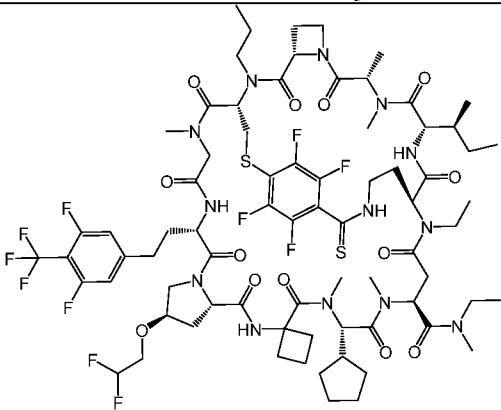
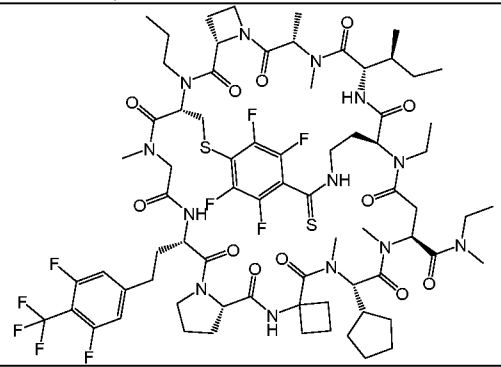
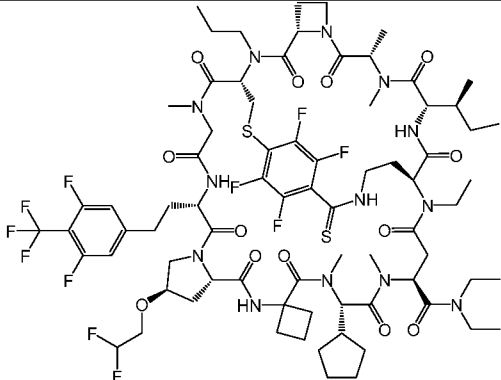
|      |                                                                                     |                                                                                                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1139 |    | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCOCC9</chem> |
| 1140 |    | <chem>CCSC1=C(F)C(F)=C(C(=S)NCC[C@@H]2N(C)C(=O)C[C@@H](C(=O)N3CCCCC3)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)[C@H](OCC(F)F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)CN(CCC(C)C)C(=O)[C@H]8N(CC8)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC2=O)C(F)=C1F</chem>          |
| 1141 |   | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem>   |
| 1142 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=C(C(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCOCC9</chem>    |

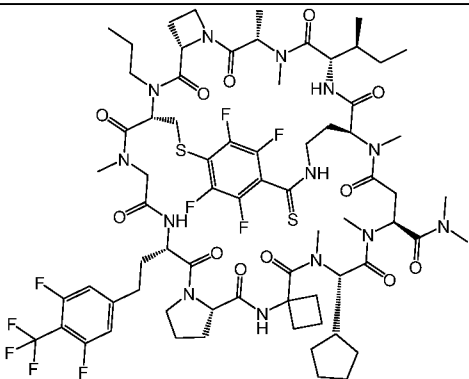
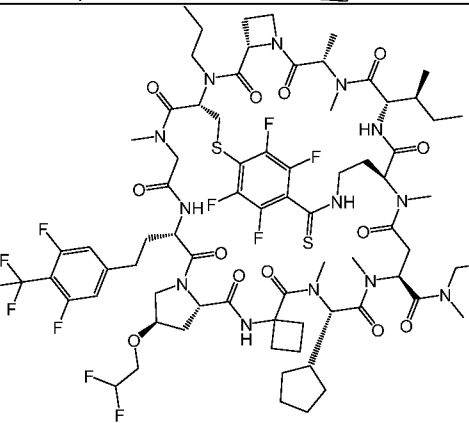
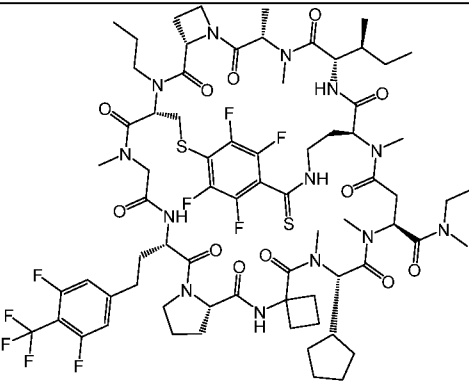
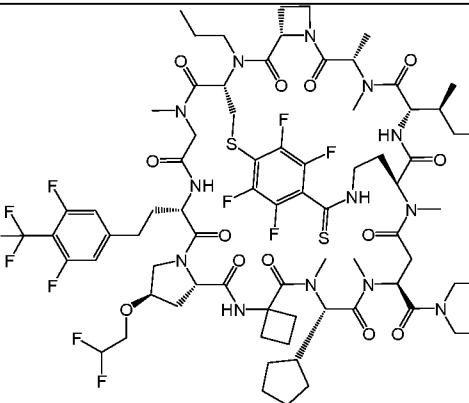
|      |  |                                                                                                                                                                                                                                                                                                       |
|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1143 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N4CCOCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(C[C@H](OCC(F)F)C7)C(=O)[C@H](CCC8=C(C(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem> |
| 1144 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(C[C@H](OCC(F)F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>     |
| 1145 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C[C@H](OCC(F)F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F</chem>      |
| 1146 |  | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(C)C</chem>                 |

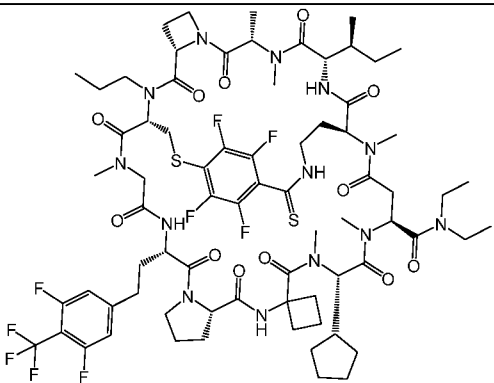
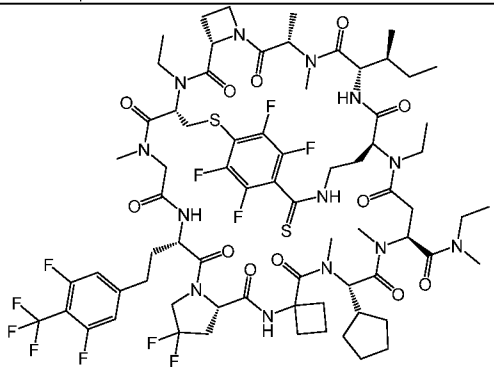
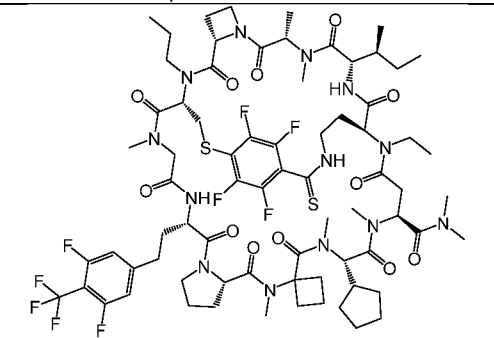
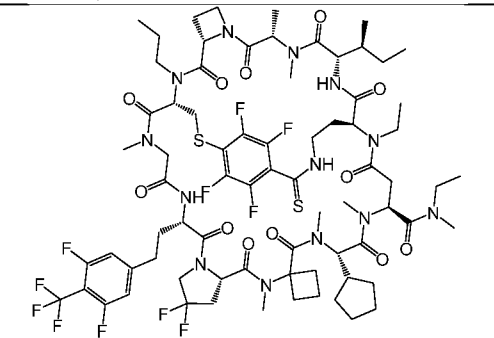
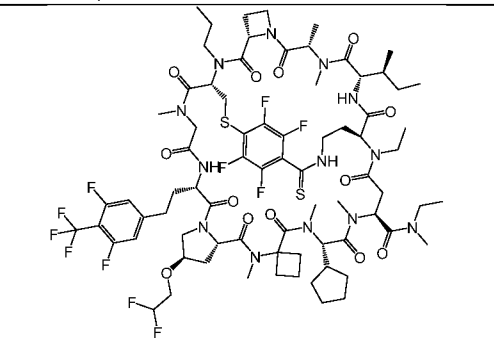
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|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1147 |  | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)C(=O)N4[C@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(C)C</chem>         |
| 1148 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC(=O)N(CC)C[C@H](CSC6=C(F)C(F)=C(C(=S)NCC5)C(F)=C6F)C(=O)N(C)CC(=O)N[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)C(=O)N2C1</chem>    |
| 1149 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC(=O)N(CC)C[C@H](CSC6=C(F)C(F)=C(C(=S)NCC5)C(F)=C6F)C(=O)N(C)CC(=O)N[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)C(=O)N2C1</chem>    |
| 1150 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CCC(C)C)C(=O)CN(C)C(=O)CN(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC4)C(F)=C5F)N(C)C(=O)[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)N(C)C2=O</chem> |

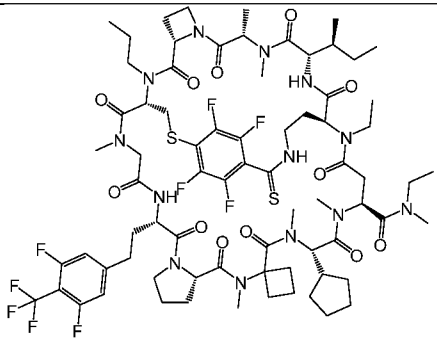
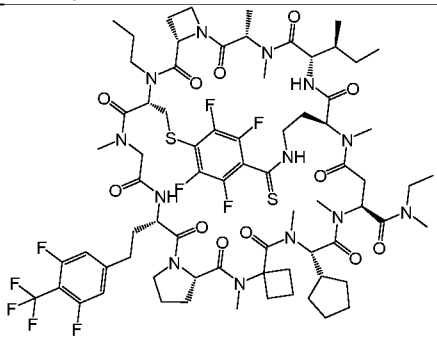
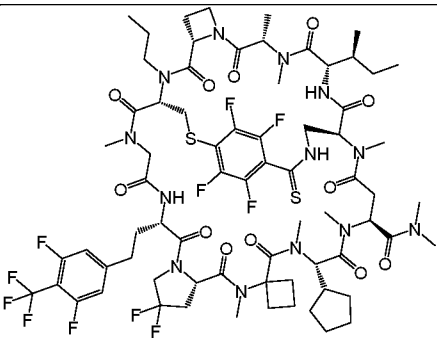
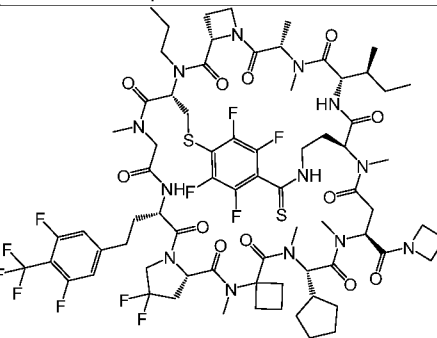
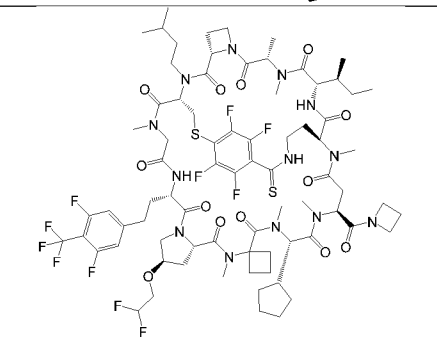
|      |                                                                                     |                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1151 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CCC(C)C)C(=O)CN(C)C(=O)CN(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC4)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)N(C)C2=O</chem>          |
| 1152 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CCC(C)C)C(=O)CN(C)C(=O)CN(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC4)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)N(C)C2=O</chem>          |
| 1153 |   | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]3N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N6CCCC6)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 1154 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]3N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N6CCOCC6)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CC8)NC2=O)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>  |

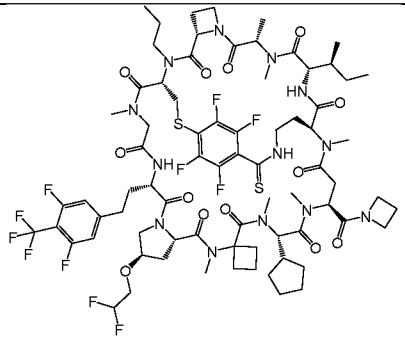
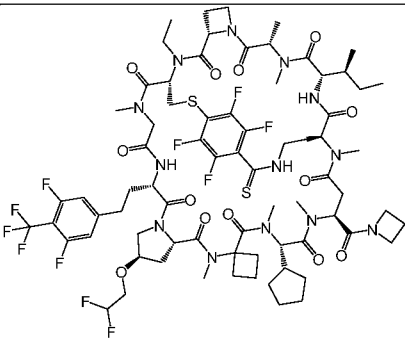
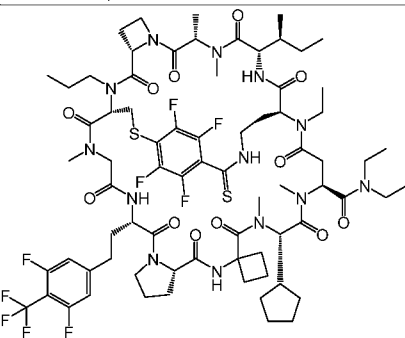
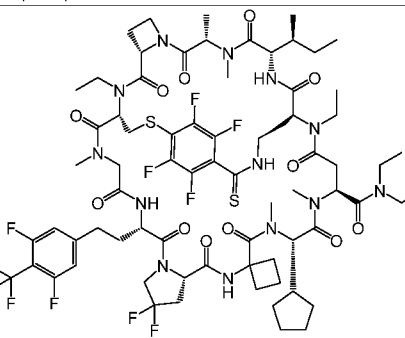
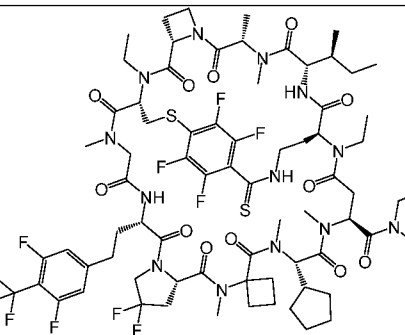
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|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1155 |    | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCOC9</chem> |
| 1156 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem>         |
| 1157 |   | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]3N(CCC)C(=O)[C@@H]4N(CC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(CC)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)NC2=O)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>      |
| 1158 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(CC)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>                  |

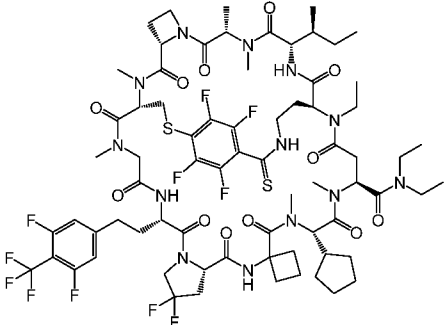
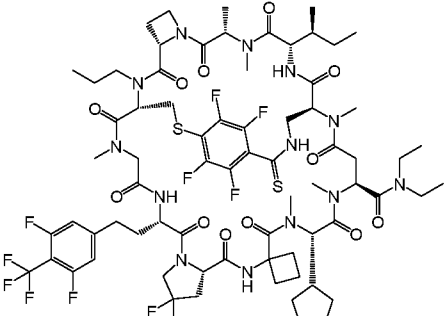
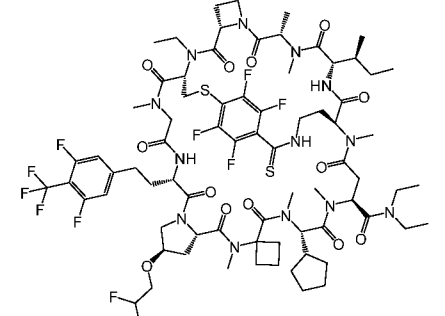
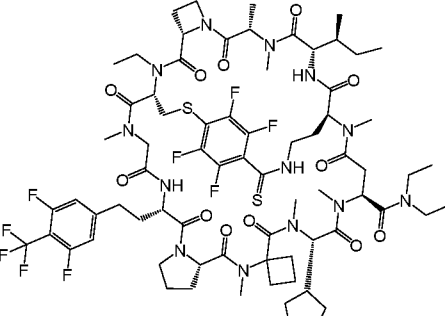
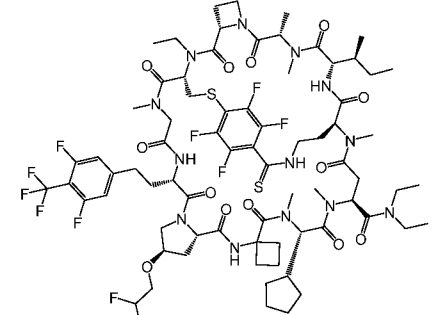
|      |                                                                                     |                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1159 |    | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)N(C(=O)CN(C)C(=O)[C@H]6N(CCC)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(CC)C(=O)C1</chem>            |
| 1160 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]3N(CCC)C(=O)[C@H]4N(CC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(CC)C(=O)C[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CC7)NC2=O)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>     |
| 1161 |   | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(CCC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>                 |
| 1162 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CCC)C(=O)[C@H]5N(CC5)C(=O)[C@@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(CC)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O</chem> |

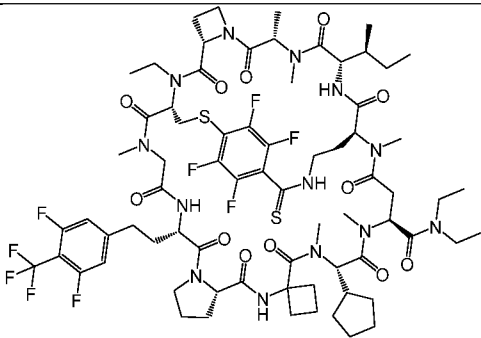
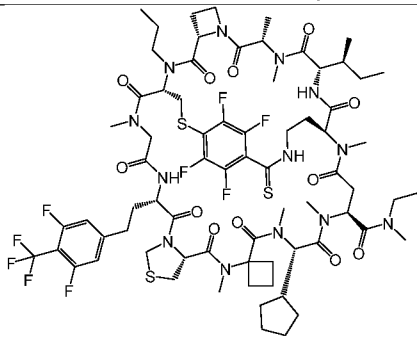
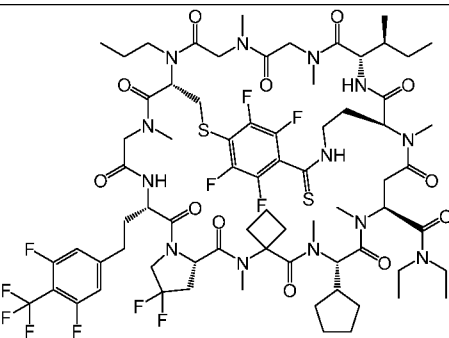
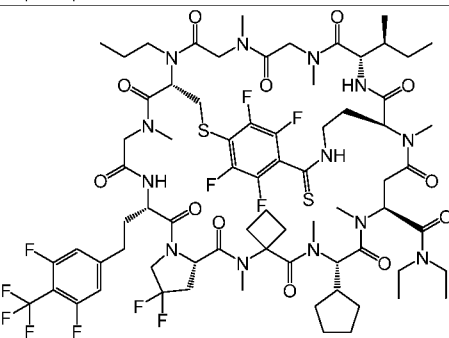
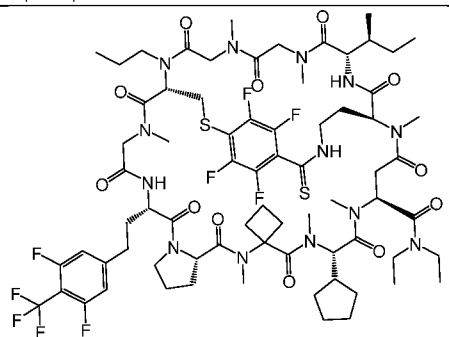
|      |                                                                                     |                                                                                                                                                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1163 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H]([C@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H]([C@H](C=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CCC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCCC3)C(F)=C8F</chem>           |
| 1164 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CCC)C(=O)[C@H]5N(CCC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)C[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O</chem>     |
| 1165 |   | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CCC)C(=O)[C@H]7N(CCC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>                   |
| 1166 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]3N(CCC)C(=O)[C@H]4N(CCC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N(C)CC)CC)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)NC2=O)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem> |

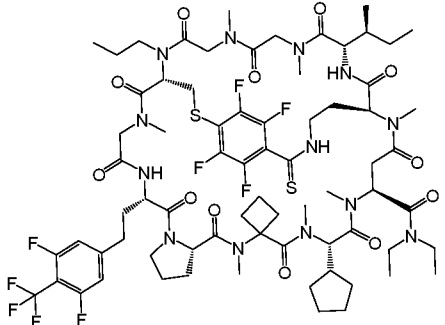
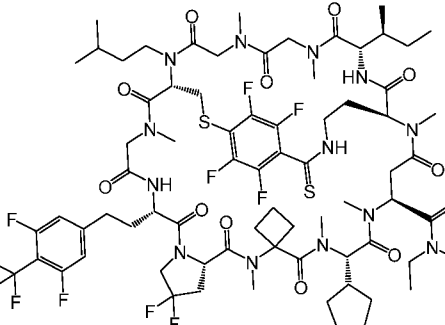
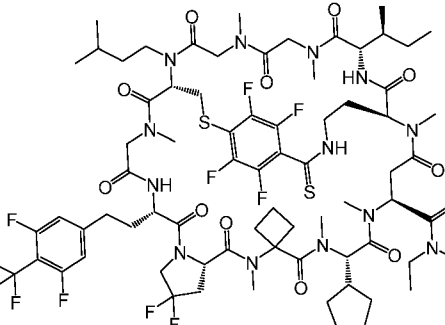
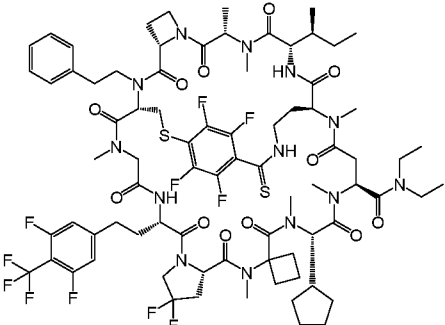
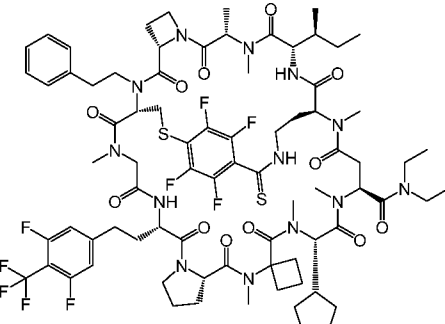
|      |                                                                                     |                                                                                                                                                                                                                                                                                                          |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1167 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC)C(=O)[C@H]7N(C)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)C)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>                          |
| 1168 |    | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(C(C(F)(F)C4)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@@H]5N(CCC)C(=O)[C@H]6N(CCC)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>               |
| 1169 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)CC(=O)N(CC)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1</chem>                  |
| 1170 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CSC7=C(F)C(F)=C(C(=S)NCC3)C(F)=C7F)C(C8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>               |
| 1171 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem> |

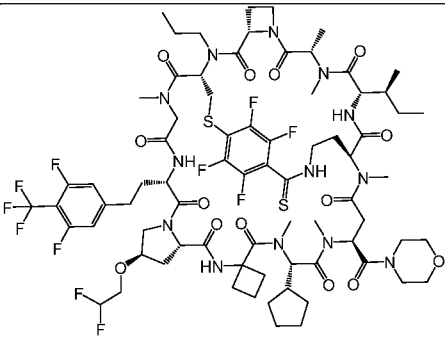
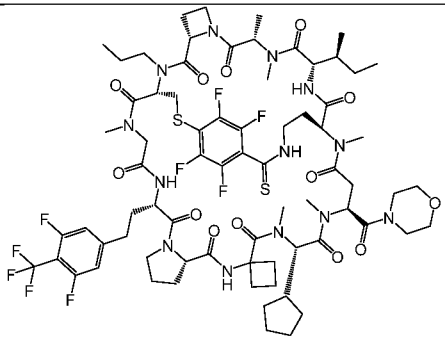
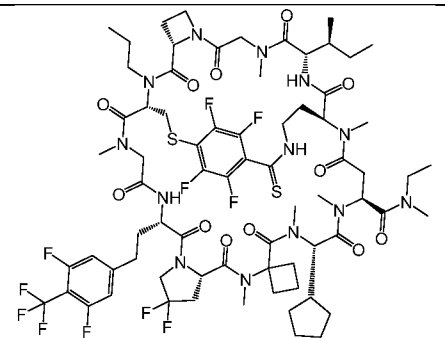
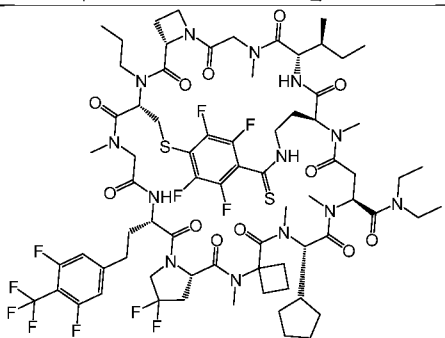
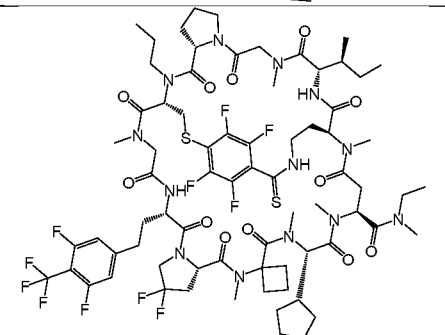
|      |                                                                                     |                                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1172 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CCC6)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@H]1CSC7=C(F)C(F)=C(C(=S)NCC3)C(F)=C7F)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>                        |
| 1173 |    | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CCC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=C(F)=C(C(F)=C8)C(F)(F)F</chem>                        |
| 1174 |   | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(C)C(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NC3)C(F)=C8F</chem>                    |
| 1175 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(C)C(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>                 |
| 1176 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCC9</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                                            |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1177 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(C)[C@H](OCC(F)F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CS(C9=C(F)C(F)=C(C(=S)NCCC3)C(F)=C9F</chem>                       |
| 1178 |    | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@H]([C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@H](C)CC)C(=O)N(C)[C@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCC9</chem>                        |
| 1179 |   | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC)C(=O)[C@H]7N(C)C7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)C)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(CC)C(=O)C1</chem>                                        |
| 1180 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CC)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(CC)C(=O)C1</chem>                                |
| 1181 |  | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CC)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CC(C8=CC(F)=C(C(F)=C8)C(F)(F)F</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                   |
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| 1182 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@@H]5N(C)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CCC8=C(F)=C(C(F)=C8)C(F)(F)F</chem>           |
| 1183 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CCC)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>          |
| 1184 |   | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CC)C(=O)[C@H]5N(CC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)[C@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)N(C)C2=O</chem> |
| 1185 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=C(F)=C(C(F)=C8)C(F)(F)F</chem>              |
| 1186 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]3N(CC)C(=O)[C@H]4N(CC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)[C@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C6CCCC6)N(C)C(=O)C7(CCC7)NC2=O)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>   |

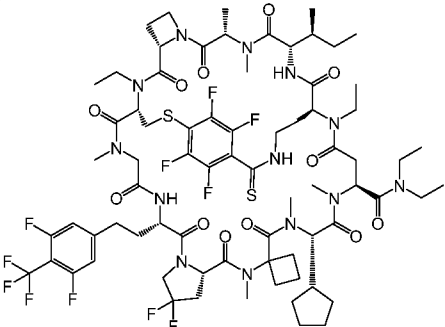
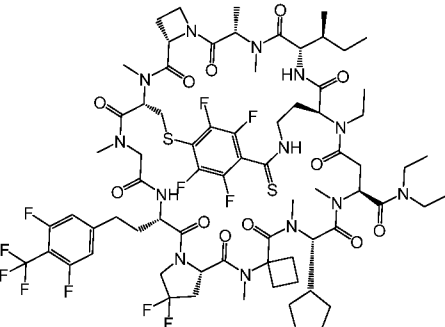
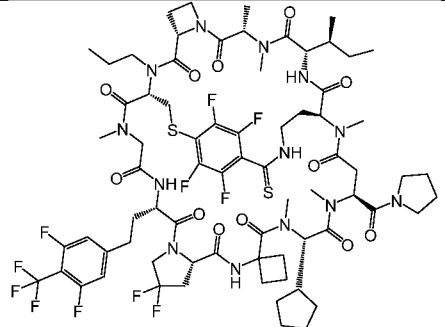
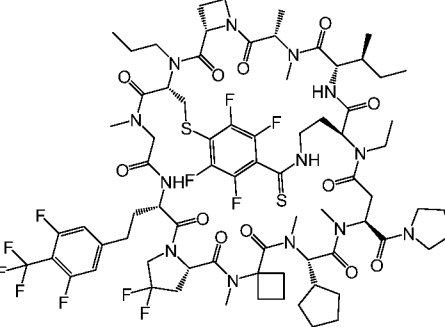
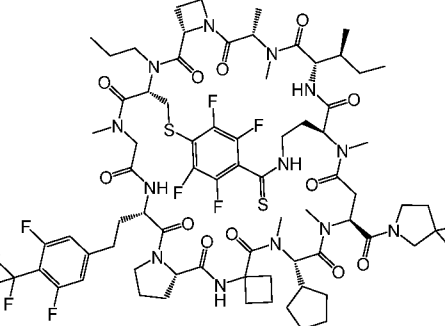
|      |                                                                                     |                                                                                                                                                                                                                                                                                    |
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| 1187 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CC)C(=O)[C@H]7N(CC)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@H](C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>    |
| 1188 |    | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CSC4)C(=O)[C@H](NC(=O)CN(C)C(=O)[C@H]5N(CCC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=C(F)=C(C(F)=C8)C(F)(F)F</chem> |
| 1189 |   | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>  |
| 1190 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>  |
| 1191 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>       |

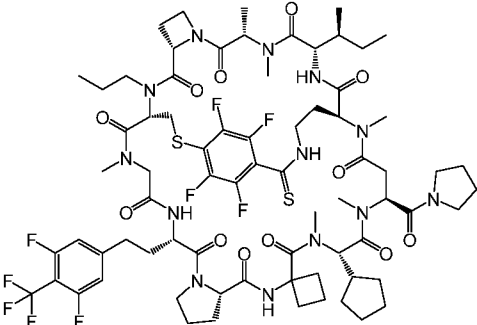
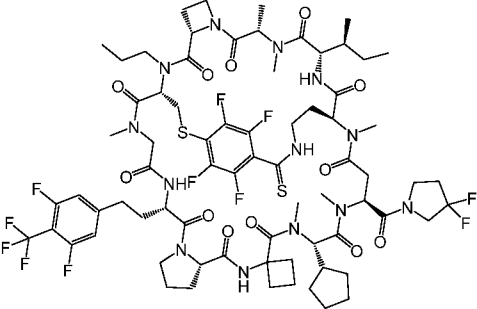
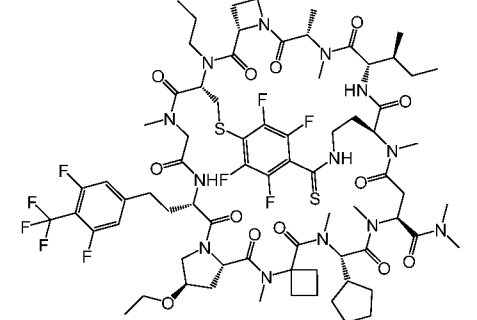
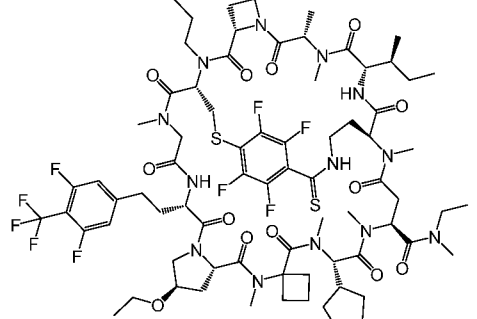
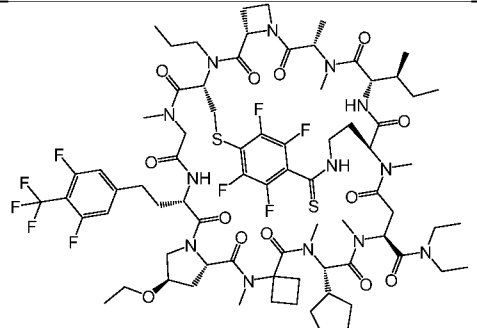
|      |                                                                                     |                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1192 |    | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>                             |
| 1193 |    | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>                   |
| 1194 |   | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>                   |
| 1195 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CCC4)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(CCC6=CC=CC=C6)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC5)C(F)=C8F)N(C)C(=O)C1)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 1196 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(CCC6=CC=CC=C6)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC5)C(F)=C8F)N(C)C(=O)C1)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>       |

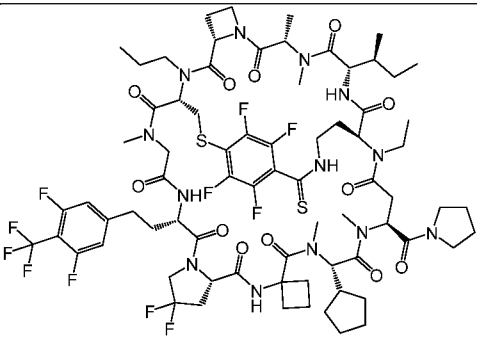
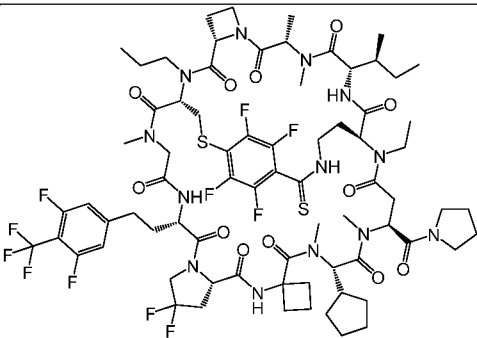
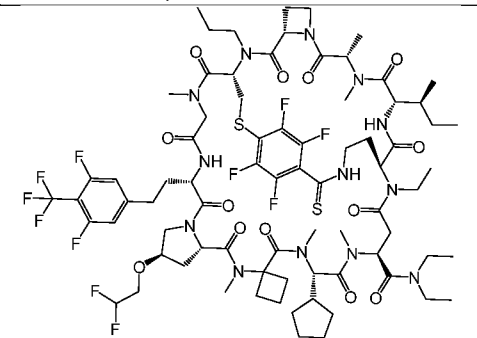
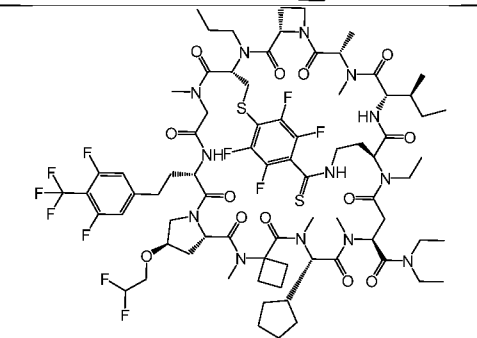
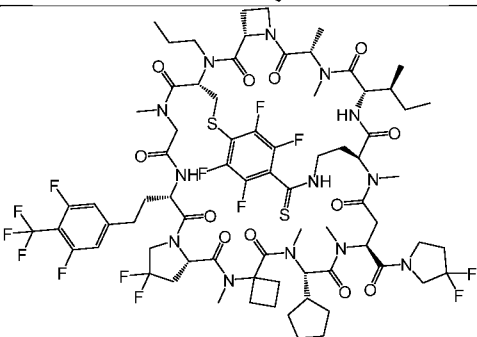
|      |                                                                                     |                                                                                                                                                                                                                                                                                                          |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1197 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]3N(CCC)C(=O)[C@@H]4N(CC4)C(=O)[C@@H](C)N(C)C(=O)[C@@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C5=C(F)C(F)=C(SC3)C(F)=C5F)N(C)C(=O)C[C@@H](C(=O)N6CCOCC6)N(C)C(=O)[C@@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 1198 |    | <chem>CCC1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC4)C(=O)NC5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N8[C@@H](CC8)C1=O)C(=O)N9CCOCC9</chem>                  |
| 1199 |   | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(CCC)C(=O)[C@@H]6N(CC6)C(=O)CN(C)C(=O)[C@@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=CC(F)=C(C(F)=C8)C(F)(F)F</chem>                 |
| 1200 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(CC)CC</chem>                |
| 1201 |  | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]6N(CCC)C(=O)[C@@H]7N(CCC7)C(=O)CN(C)C(=O)[C@@H]([C@@H](C)C)NC(=O)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(C)C(=O)C1</chem>                 |

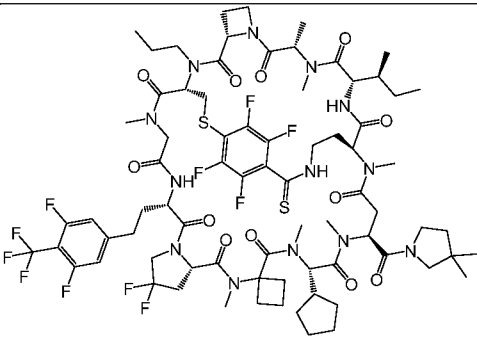
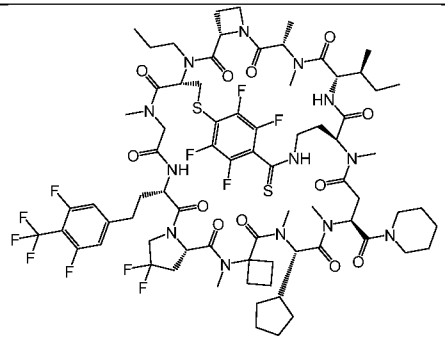
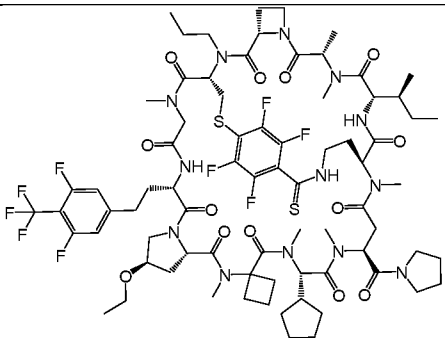
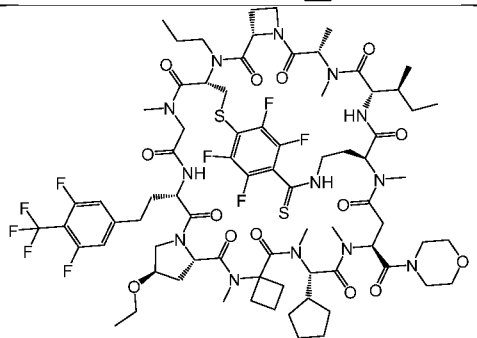
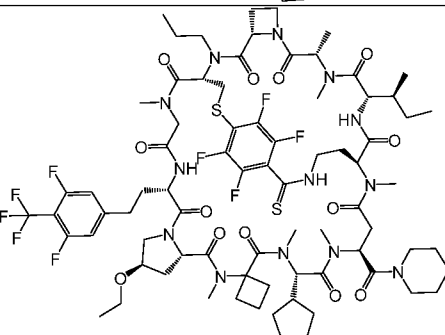
|      |  |                                                                                                                                                                                                                                                                                                    |
|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1202 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=O)N2[C@@H](CCC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(CC)CC</chem>         |
| 1203 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N(CC)CC</chem>    |
| 1204 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(C)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=C(C(F)=C(C(F)=C8)C(F)(F)F</chem>            |
| 1205 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CSC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(C)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(C)C(=O)C1)CCC8=C(C(F)=C(C(F)=C8)C(F)(F)F</chem>            |
| 1206 |  | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(C)C(=O)[C@H]5N(CC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)N(C)C2=O</chem> |

|      |  |                                                                                                                                                                                                                                                                                               |
|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1207 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>            |
| 1208 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O)C(=O)N(CC)CC</chem>            |
| 1209 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C)C(=O)N(C)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCOCC9</chem> |
| 1210 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N8[C@@H](CC8)C1=O)C(=O)N9CCCC9</chem> |
| 1211 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)N(C(=O)CN(C)C(=O)[C@@H]6N(CCC)C(=O)[C@H]7N(CCC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C8=C(F)C(F)=C(SC6)C(F)=C8F)N(CC)C(=O)C1</chem>     |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1212 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CC8=CC(F)=C(C(F)=C8)C(F)F</chem>      |
| 1213 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]5N(C)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C7=C(F)C(F)=C(SC5)C(F)=C7F)N(CC)C(=O)C1)CC8=CC(F)=C(C(F)=C8)C(F)F</chem>       |
| 1214 |   | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)NC(=O)[C@H]7N(C)C(F)(F)C7)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CSC8=C(F)C(F)=C(C(=S)NCC3)C(F)=C8F)CCC9=CC(F)=C(C(F)=C9)C(F)F</chem>          |
| 1215 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem> |
| 1216 |  | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)F)C(=O)N4[C@@H](CC4)C(=O)NC5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N8[C@@H](CC8)C1=O)C(=O)N9CC(C)C)CC9</chem>      |

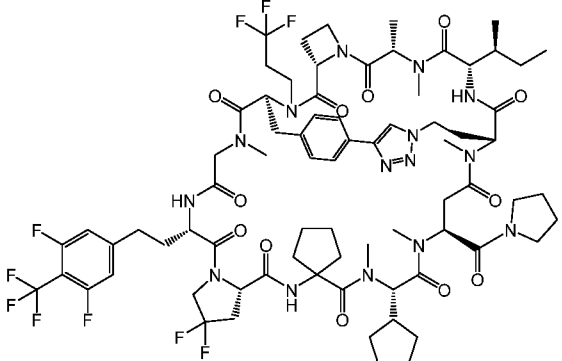
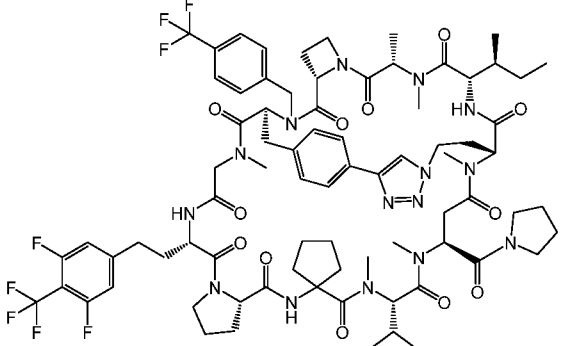
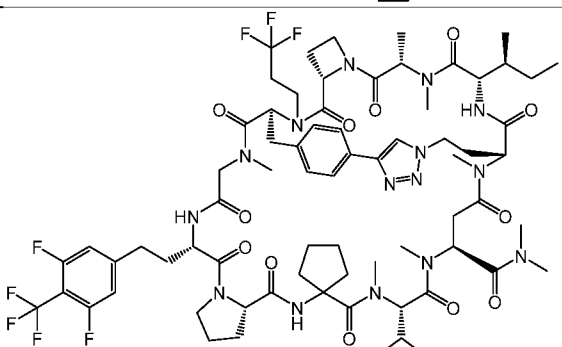
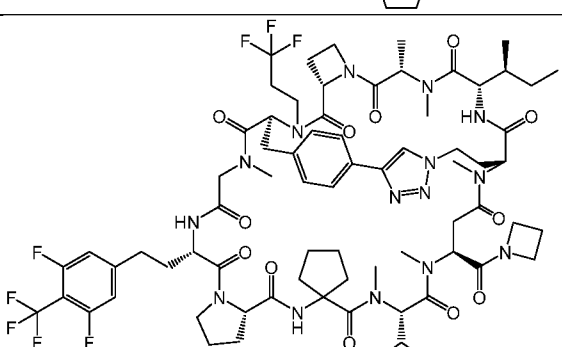
|      |                                                                                     |                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1217 |    | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC4)C(=O)NC5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N8[C@@H](CC8)C1=O)C(=O)N9CCCC9</chem>              |
| 1218 |    | <chem>CCCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC4)C(=O)NC5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCNC(=S)C7=C(F)C(F)=C(SC2)C(F)=C7F)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N8[C@@H](CC8)C1=O)C(=O)N9CC(F)(F)CC9</chem>        |
| 1219 |   | <chem>CCO[C@@H]1C[C@@H]2N(C1)C(=O)[C@H](CCC3=CC(F)=C(C(F)=C3)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]4N(CCC)C(=O)[C@H]5N(CCC5)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCNC(=S)C6=C(F)C(F)=C(SC4)C(F)=C6F)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)N(C)C2=O</chem>      |
| 1220 |  | <chem>CCO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem>  |
| 1221 |  | <chem>CCO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(CC)CC</chem> |

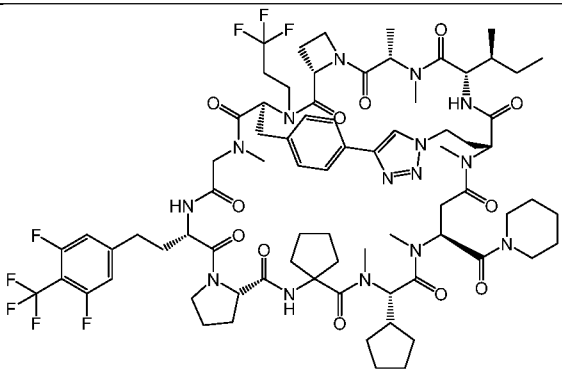
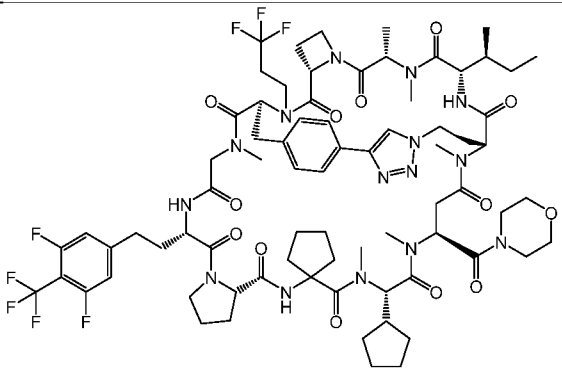
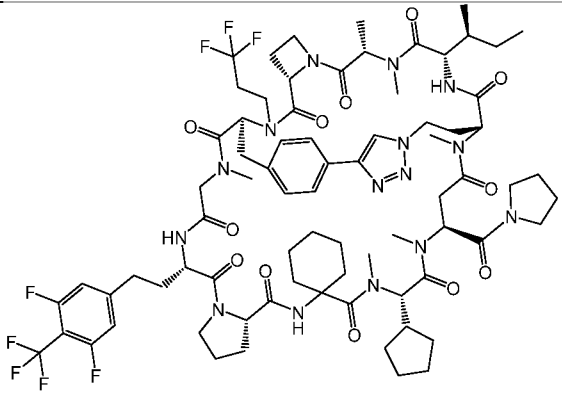
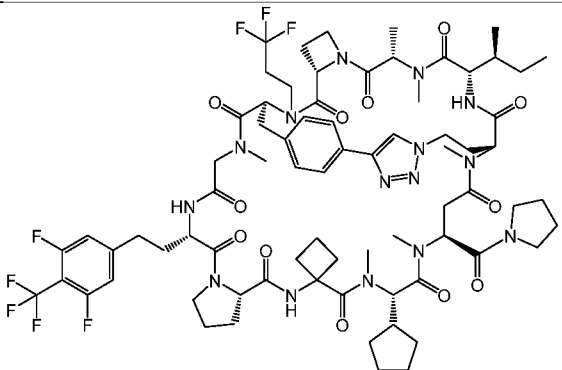
|      |                                                                                     |                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1222 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>         |
| 1223 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCNC(=S)C8=C(F)C(F)=C(SC3)C(F)=C8F)C(=O)N1)C(=O)N9CCCC9</chem>         |
| 1224 |   | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem> |
| 1225 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem> |
| 1226 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@H](C(=O)N4CC(F)(F)CC4N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>          |

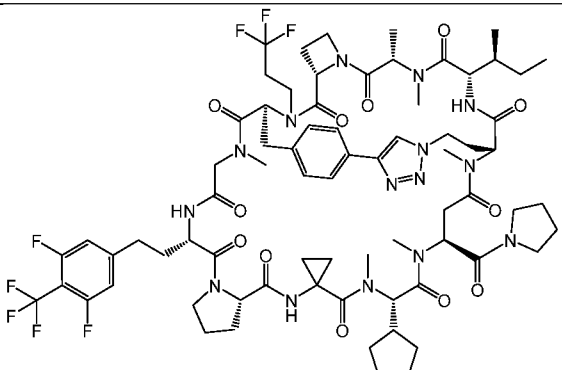
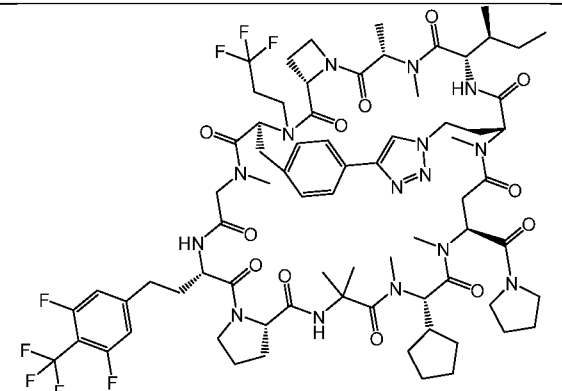
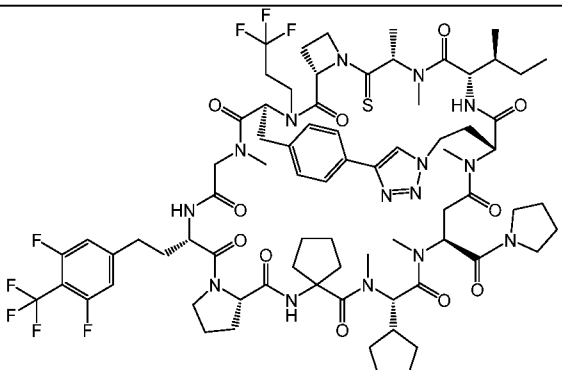
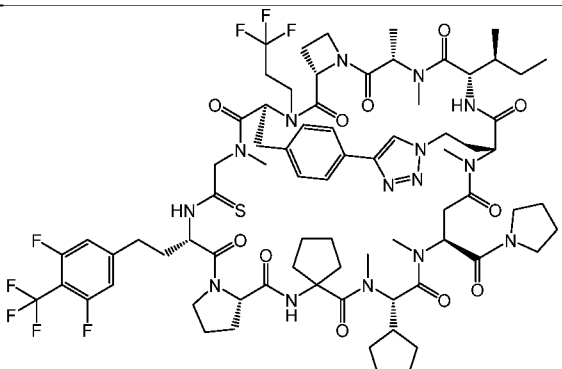
|      |                                                                                     |                                                                                                                                                                                                                                                                                                    |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1227 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N4CC(C)(C)CC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>   |
| 1228 |    | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CSC9=C(F)C(F)=C(C(=S)NCC3)C(F)=C9F</chem>         |
| 1229 |   | <chem>CCO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@H](C4CCCC4)C(=O)N(C)[C@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCCC9</chem> |
| 1230 |  | <chem>CCO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@H](C4CCCC4)C(=O)N(C)[C@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCOC9</chem> |
| 1231 |  | <chem>CCO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@H](C4CCCC4)C(=O)N(C)[C@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CSC7=C(F)C(F)=C(C(=S)NCC5)C(F)=C7F)C(=O)N(C)CC(=O)N[C@@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)C(=O)N2C1)C(=O)N9CCCC9</chem> |

|      |  |                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2035 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)CC%10CCCCC%10</chem>                                                                                                              |
| 2036 |  | <chem>CCO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)N(C)C(=O)[C@H]3N(C)C(=O)[C@H]4N(CC4)C(=O)[C@H](C)N(C)C(=O)[C@H](C)C(=O)[C@H](C)CC)NC(=O)[C@@H](CCN5C=C(N=N5)C6=CC=C(C=C6)C3)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(NC2=O)CC8)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>                                                                                                                   |
| 2037 |  | <chem>CCO[C@H]1CN2[C@@H](C1)C(=O)NC3(CCCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(C)[C@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C2=O)C(=O)N(C)C</chem>                                                                                                                          |
| 2038 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10[C@@H](CCC%10)C(=O)NCCCCC(=O)N[C@@H](C(C)C)C(=O)N%11[C@@H](C)[C@@H](O)C%11)C(=O)N[C@@H](C)C%12=CC=C(C=C%12)C%13=C(C)N=CS%13</chem> |

|      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2039 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10[C@@H](CCC%10)C(=O)NCCCCCCC(=O)N[C@@H](C(C)(C)C)C(=O)N%11[C@@H](C[C@@H](O)C%11)C(=O)N[C@@H](C)C%12=CC=C(C=C%12)C%13=C(C)N=CS%13</chem>                  |
| 2040 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10[C@@H](CCC%10)C(=O)N%11CCN(C(=O)CCCCC(=O)N[C@@H](C(C)(C)C)C(=O)N%12[C@@H](C[C@@H](O)C%12)C(=O)N[C@@H](C)C%13=CC=C(C=C%13)C%14=C(C)N=CS%14)CC%11</chem>  |
| 2041 |  | <chem>C(CCCC(=O)N1CCN(CC1)C(=O)[C@H]2N(CCC2)C(=O)[C@@H]3N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)NC(=O)[C@H]6N(CCC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]8N(C)C(=O)[C@H]9N(CCC9)C(=O)[C@@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCN%10C=C(N=N%10)C%11=CC=C(C=C%11)C8)N(C)C(=O)C3)CCC(=O)N[C@@H](C(C)(C)C)C(=O)N%12[C@@H](C[C@@H](O)C%12)C(=O)N[C@@H](C)C%13=CC=C(C=C%13)C%14=C(C)N=CS%14</chem> |
| 2042 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>                                                                                                                                   |

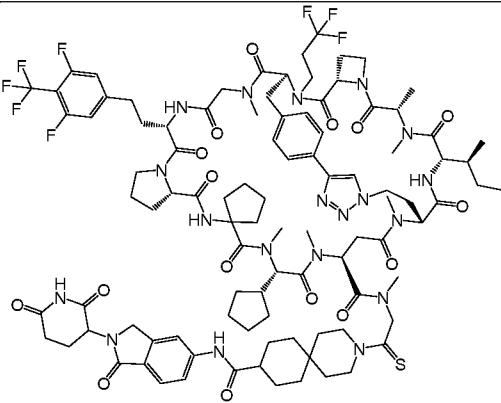
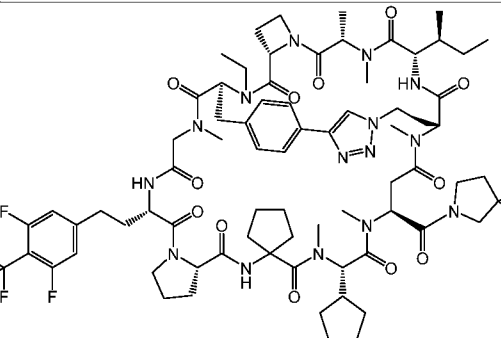
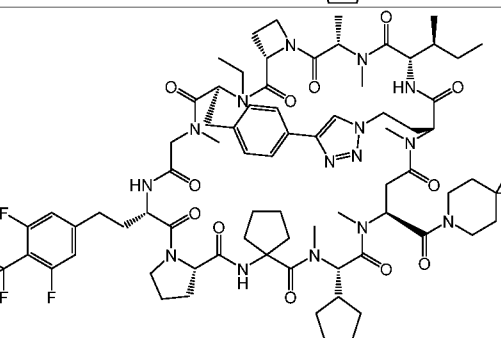
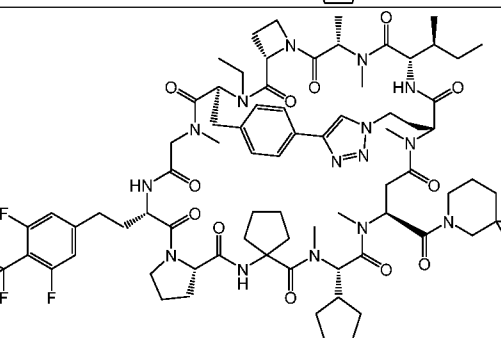
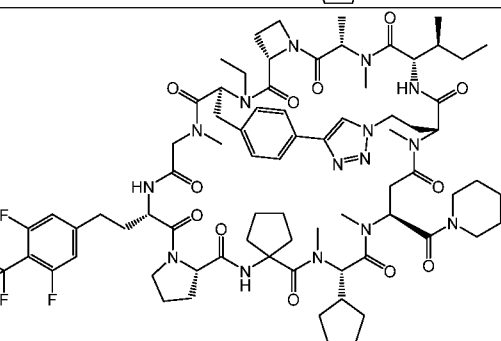
|      |                                                                                     |                                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2043 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CC(F)F)C5C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>               |
| 2044 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC3=CC=C(C=C3)C(F)F)F[C@@H]4C(=O)N(C)CC(=O)N[C@@H](CCC5=C(C(F)=C(C(F)=C5)C(F)F)C(=O)N6[C@@H](CCC6)C(=O)NC7(CCCC7)C(=O)N(C)[C@@H](C8CCCC8)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN9C=C(N=N9)C%10=CC=C(C=C%10)C4)C(=O)N1)C(=O)N%11CCCC%11</chem> |
| 2045 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>                         |
| 2046 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>                   |

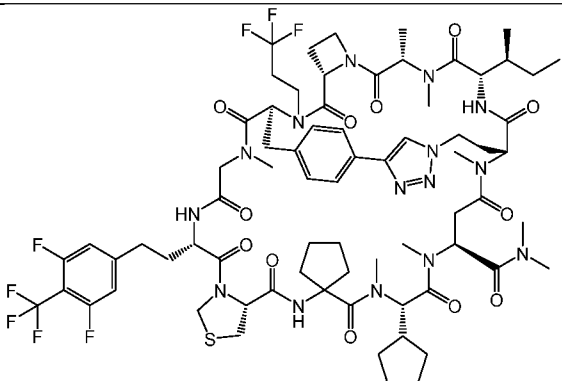
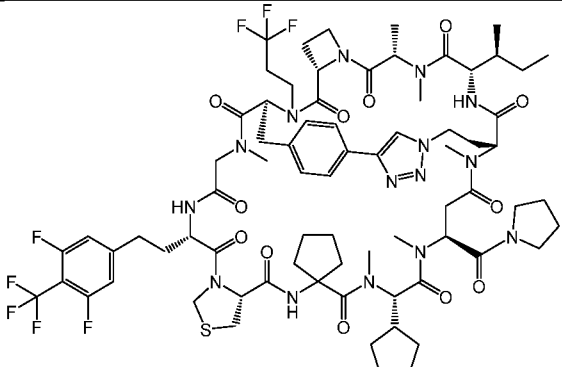
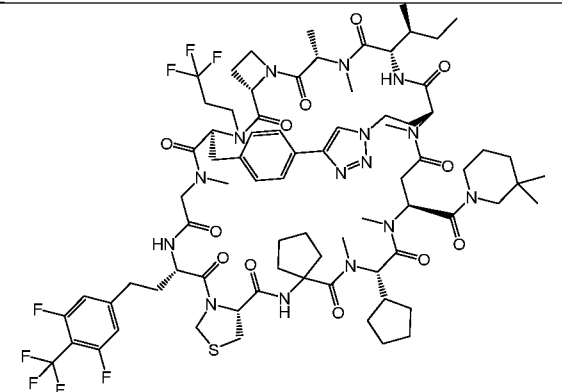
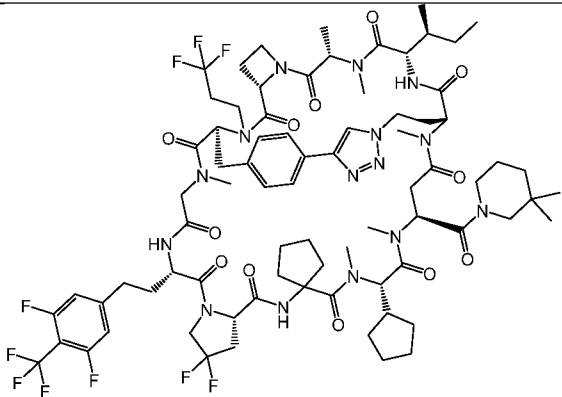
|      |                                                                                     |                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2047 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5C(CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C)C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCCC%10</chem>    |
| 2048 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5C(CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C)C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>    |
| 2049 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C)C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>  |
| 2050 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C)C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem> |

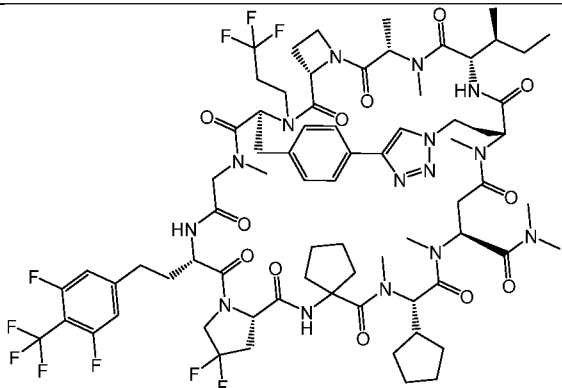
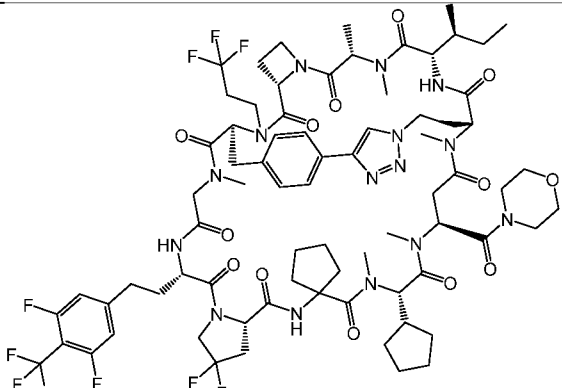
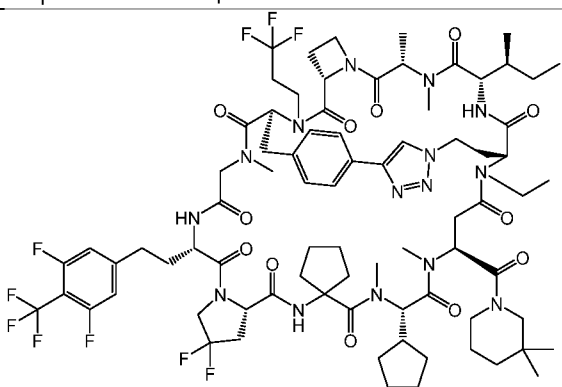
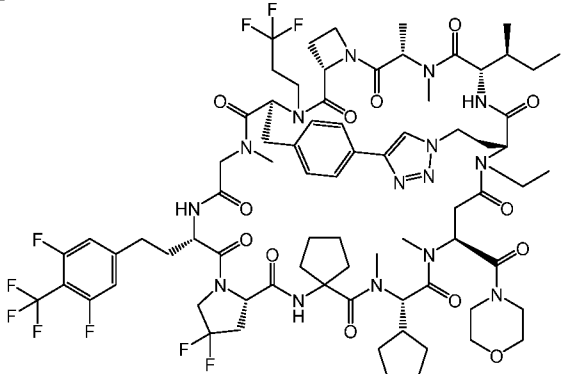
|      |                                                                                     |                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2051 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>  |
| 2052 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC(C)(C)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C3)C(=O)N1)C(=O)N9CCCC9</chem>      |
| 2053 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=S)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem> |
| 2054 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=S)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>  |

|      |  |                                                                                                                                                                                                                                                                                                        |
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| 2055 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=S)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>                 |
| 2056 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(F)CC%10</chem>   |
| 2057 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(F)CCC%10</chem>  |
| 2058 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCC(F)(F)CC%10</chem>  |
| 2059 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCC(C)(C)CCC%10</chem> |

|      |  |                                                                                                                                                                                                                                                                                                                                                                                   |
|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2060 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>                                                                                  |
| 2061 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>                                                                                  |
| 2062 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5C(CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)CC(F)(F)F</chem>                                                                                       |
| 2063 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)CC%10=CC=C(C=C%10)C(F)(F)F</chem>                                                                   |
| 2064 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)CC(=S)N%10CCC(CC%10)C%11CCN(CC%11)C(=O)NC%12C=C%13C(=CC=%12)C(=O)N(C%13)C%14C(=O)NC(=O)CC%14</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                  |
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| 2065 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)CC(=S)N%10CCC%11(CC%10)CCC(CC%11)C(=O)NC%12C=C%13C(=CC=%12)C(=O)N(C%13)C%14C(=O)NC(=O)CC%14</chem> |
| 2066 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(F)CC%10</chem>                                                                                      |
| 2067 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCC(F)(F)CC%10</chem>                                                                                     |
| 2068 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(F)CCC%10</chem>                                                                                     |
| 2069 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(C)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CCC%10</chem>                                                                                     |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2070 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>                    |
| 2071 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>               |
| 2072 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CCC%10</chem>        |
| 2073 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CC(F)F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CCC%10</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                            |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2074 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>                |
| 2075 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>      |
| 2076 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CC%10</chem> |
| 2077 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>      |

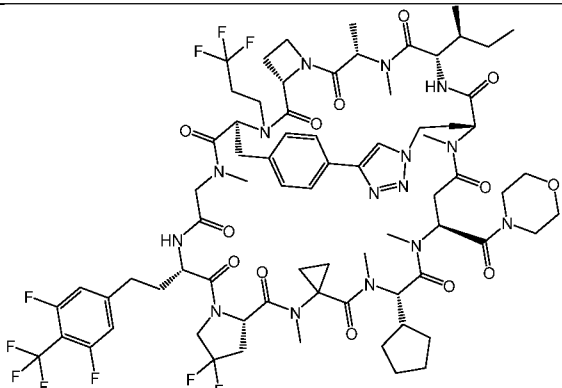
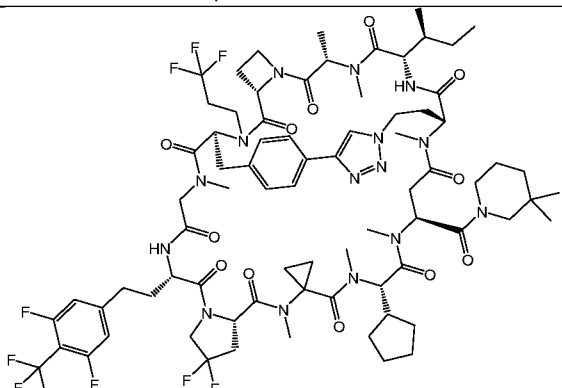
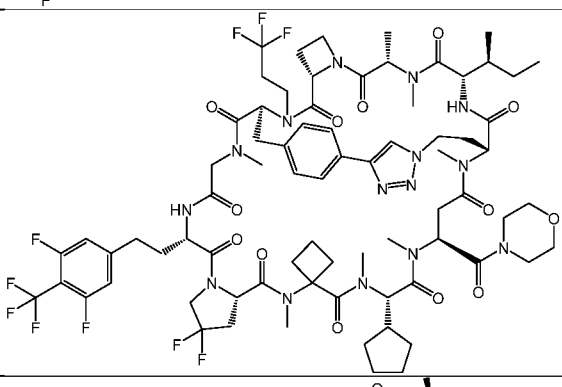
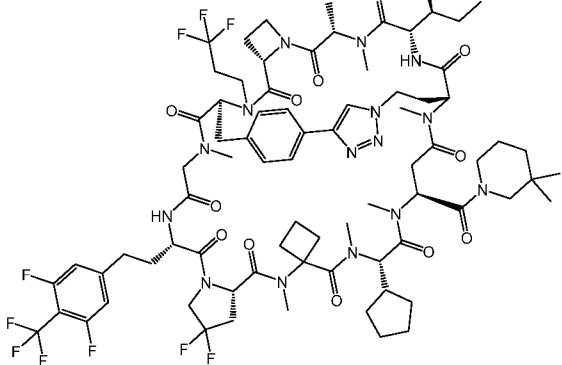
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|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2078 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=C=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(F)C%10</chem>            |
| 2079 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>                   |
| 2080 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem> |
| 2081 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>                   |

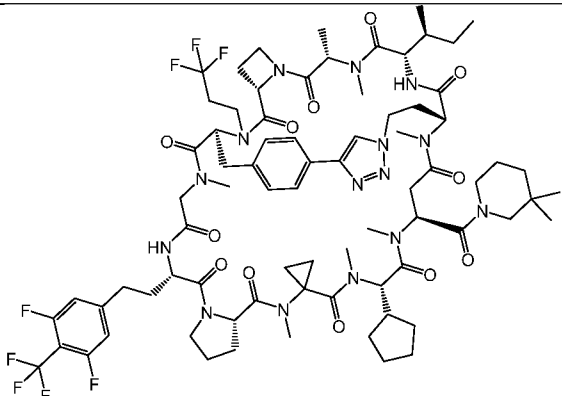
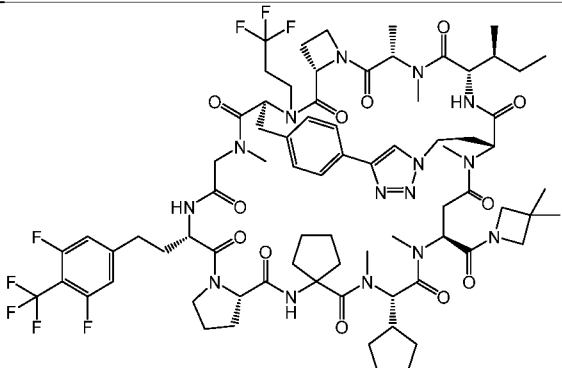
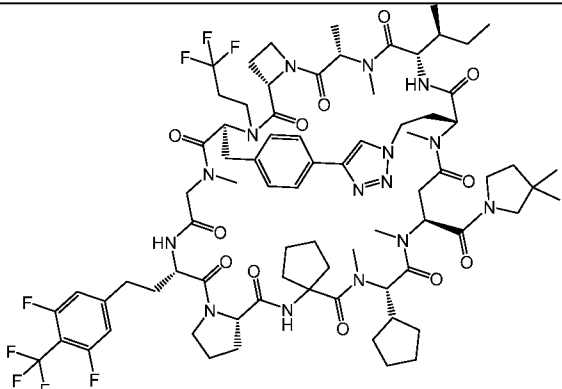
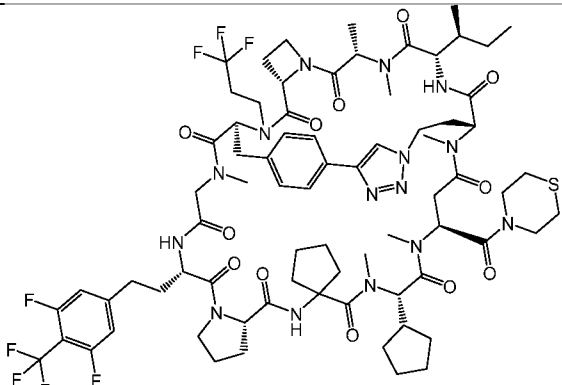
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|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2082 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>       |
| 2083 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem> |
| 2084 |  | <chem>CCC(C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(F)C(F)(F)C%10</chem> |
| 2085 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC(C)C6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(F)(C)C%10</chem>  |

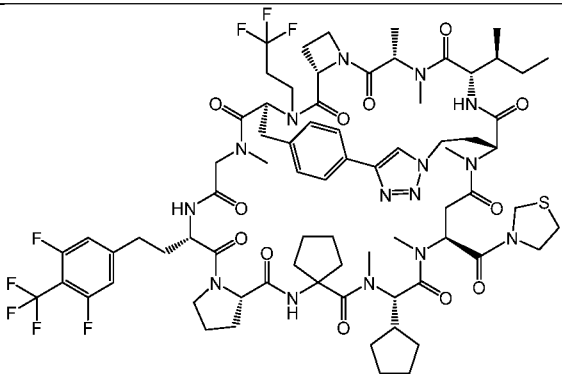
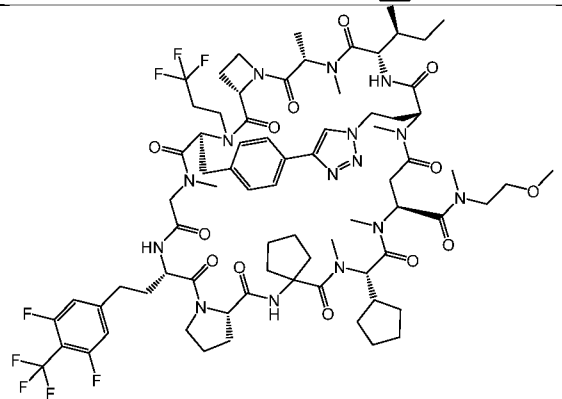
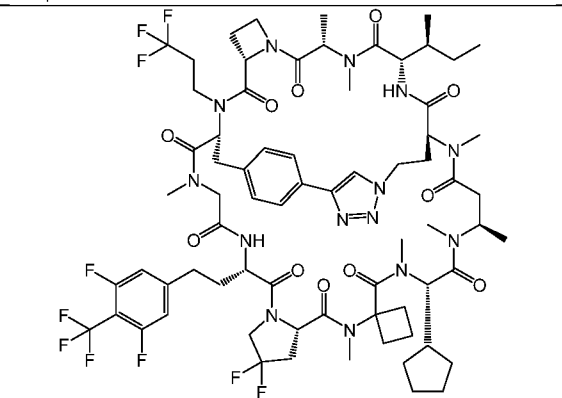
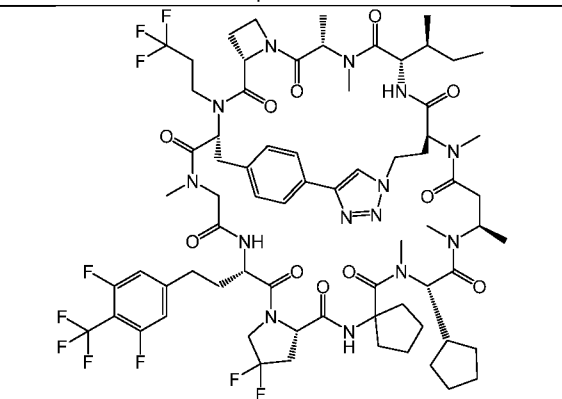
|      |  |                                                                                                                                                                                                                                                                                                        |
|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2086 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)OCC%10</chem> |
| 2087 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)C%10CCOCC%10</chem>   |
| 2088 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)C%10CCCCC%10</chem>   |
| 2089 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)C%10CCCCC%10</chem>   |
| 2090 |  | <chem>CCCN1[C@@H]2C(=O)N(C)CC(=O)N[C@@H](CC(C3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C2)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N9[C@@H](CC9)C1=O)C(=O)N%10CCOCC%10</chem>      |

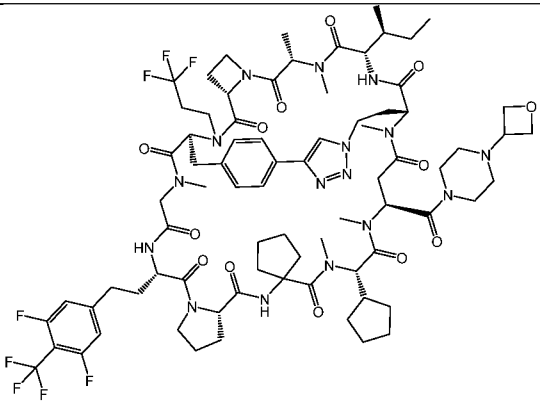
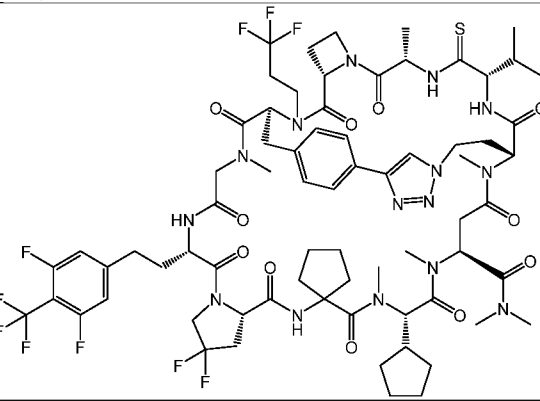
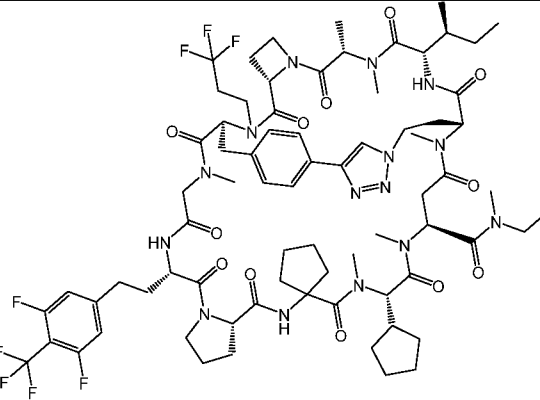
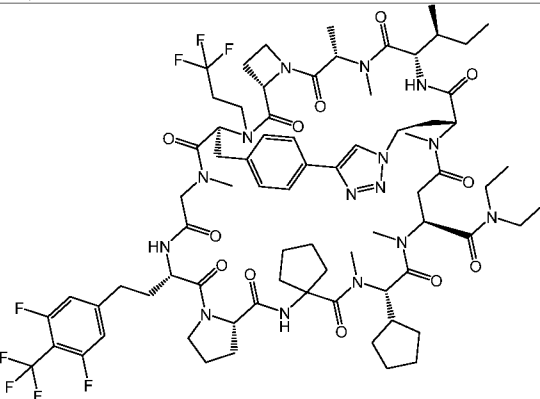
|      |  |                                                                                                                                                                                                                                                                                                 |
|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2091 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>   |
| 2092 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem> |
| 2093 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>    |
| 2094 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=S)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>         |

|      |  |                                                                                                                                                                                                                                                                                                      |
|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2095 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=S)N2[C@@H](CC2)C(=O)N(CCCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCCC=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>       |
| 2096 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)CC(=S)N2[C@@H](CC2)C(=O)N(CCCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCCC=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>    |
| 2097 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCCC=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>      |
| 2098 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCCC=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem> |

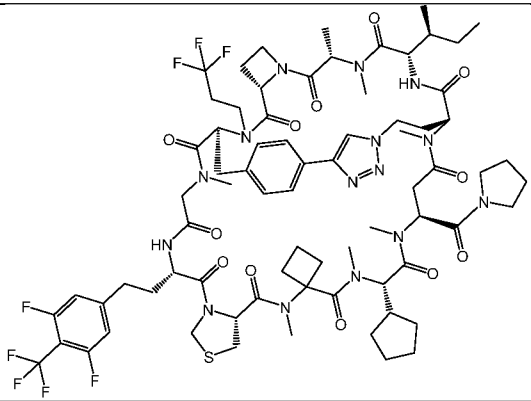
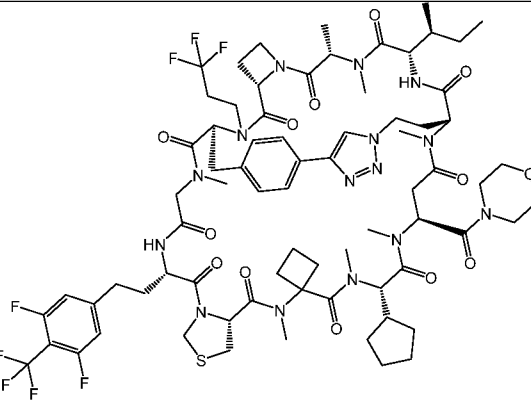
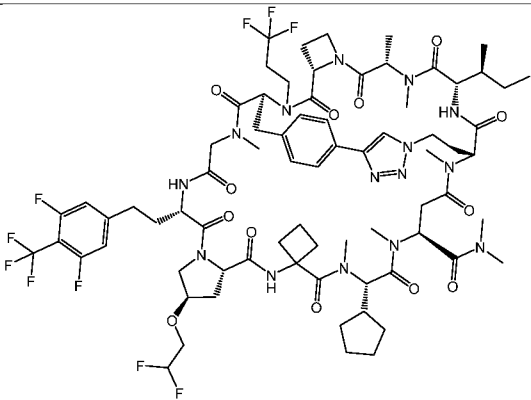
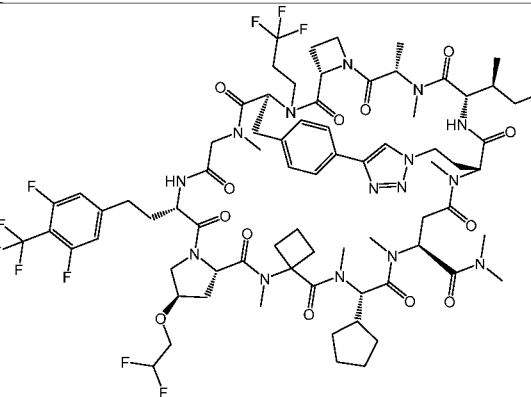
|      |                                                                                     |                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2099 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>        |
| 2100 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CC%10</chem>   |
| 2101 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>       |
| 2102 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CCC%10</chem> |

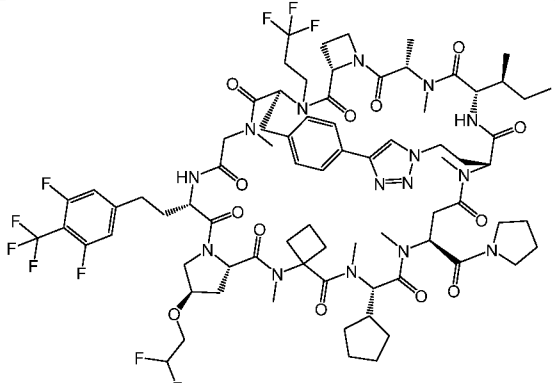
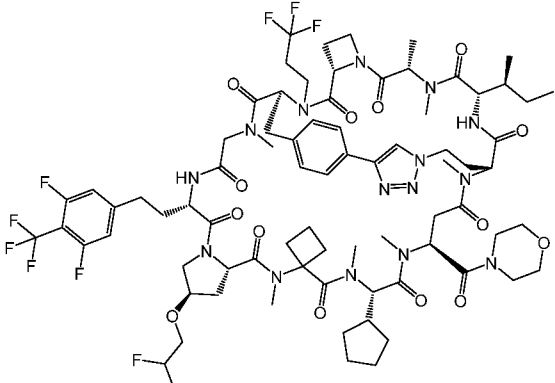
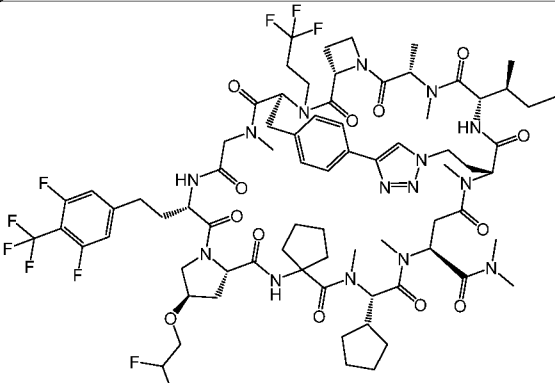
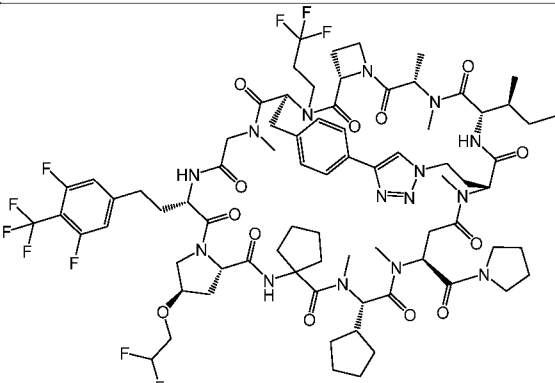
|      |                                                                                     |                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2103 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CCC%10</chem> |
| 2104 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)C%10</chem>       |
| 2105 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CC(C)(C)CC%10</chem>      |
| 2106 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCSCC%10</chem>           |

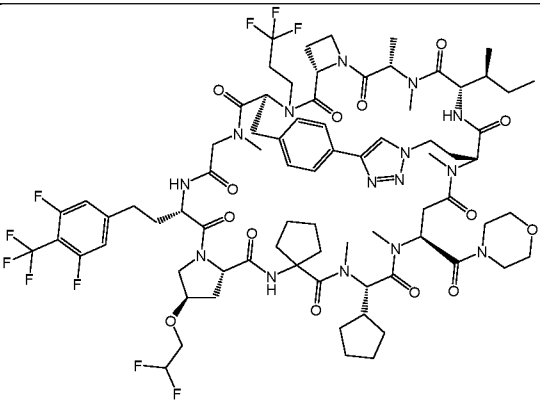
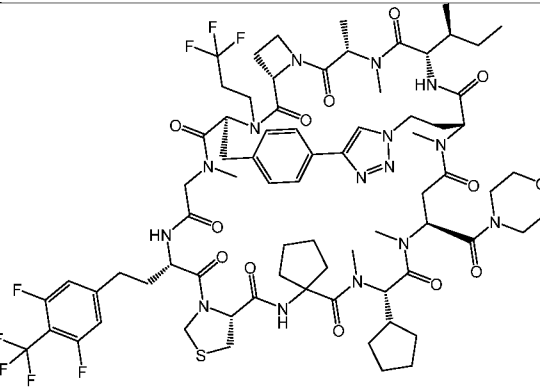
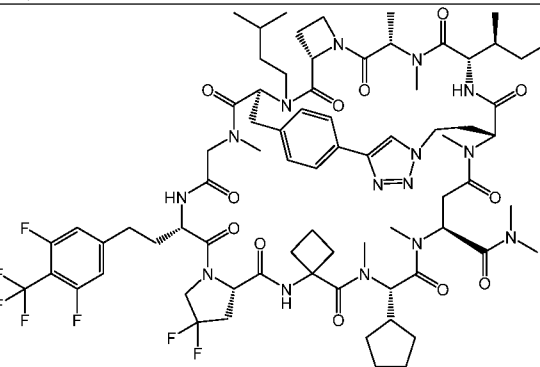
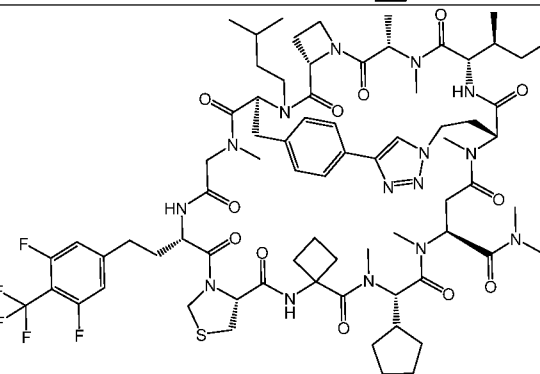
|      |                                                                                     |                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2107 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10SCC%10</chem> |
| 2108 |    | <chem>COCCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCC2)N(C)C(=O)C3(CCCC3)NC(=O)[C@H]4N(CCCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)N(C(=O)CN(C)C(=O)[C@H]6N(CCC(F)(F)F)C(=O)[C@H]7N(CCC7)C(=O)[C@H](C)N(C)C(=O)[C@H](C[C@@H](C)CC)NC(=O)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C6)N(C)C(=O)C1</chem>   |
| 2109 |   | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>      |
| 2110 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>        |

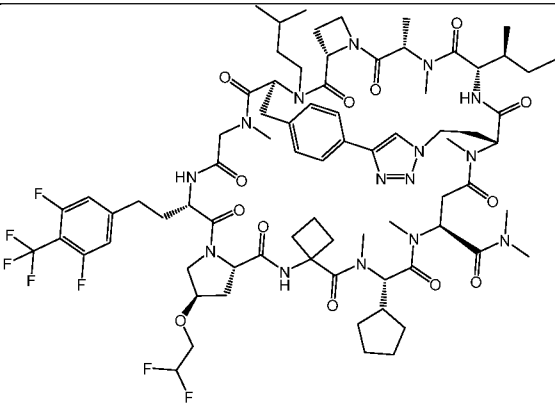
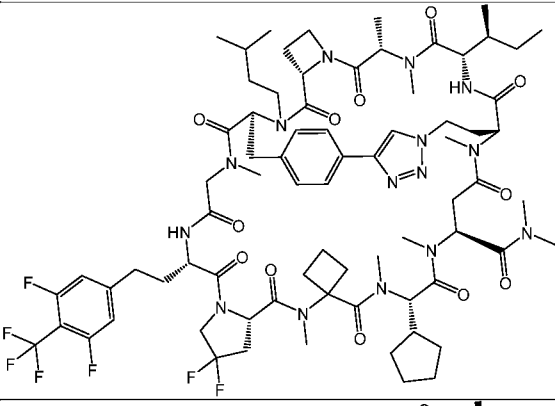
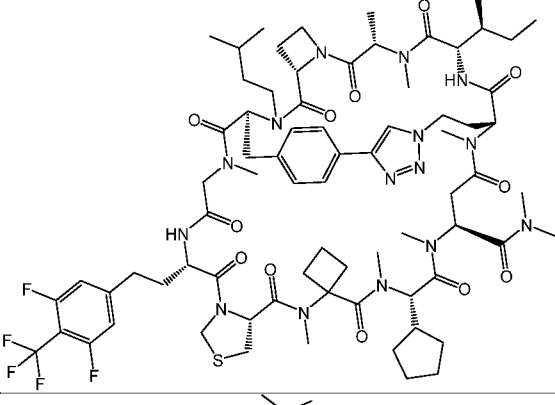
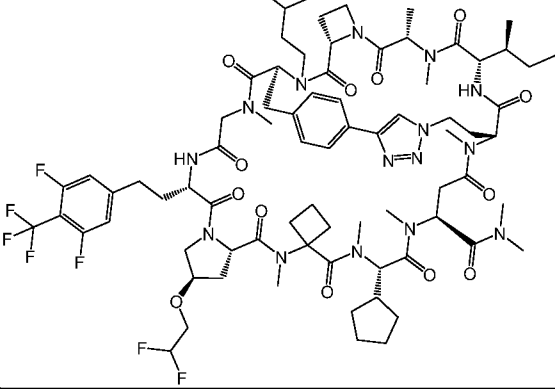
|      |                                                                                     |                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2111 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)F)C(=O)N5[C@@H](CCC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCN(CC%10)C%11COC%11</chem> |
| 2112 |    | <chem>CC(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCCC3)NC(=O)[C@H]4N(CC(F)F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC(F)F)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)NC(=S)[C@H](C(C)C)NC(=O)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C6)N(C)C(=O)C1</chem>                           |
| 2113 |   | <chem>CCN(C)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC(F)F)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H](C@@H)(C)CCNC(=O)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C6)N(C)C(=O)C1</chem>                        |
| 2114 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCCC3)NC(=O)[C@H]4N(CCC4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC(F)F)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H](C@@H)(C)CCNC(=O)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C6)N(C)C(=O)C1</chem>                       |

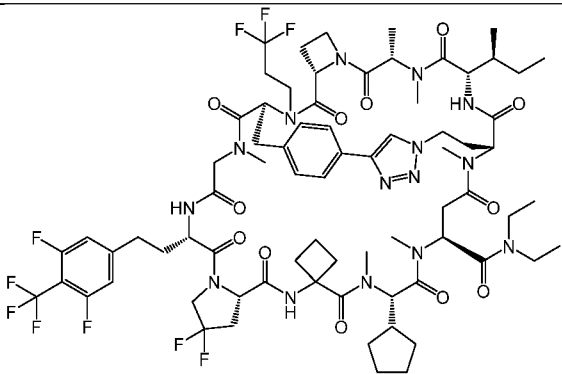
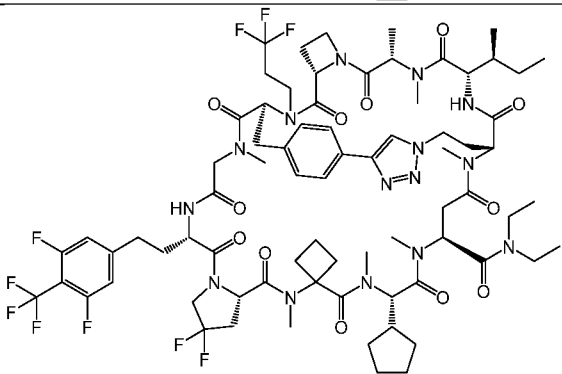
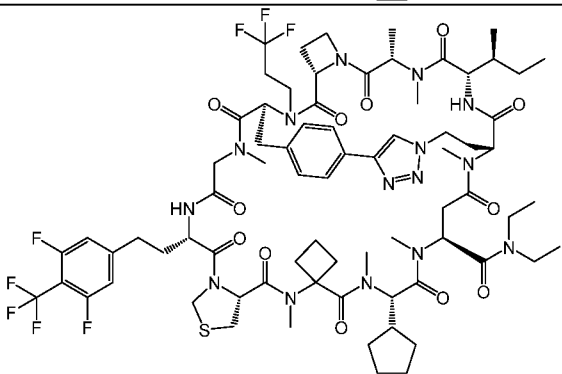
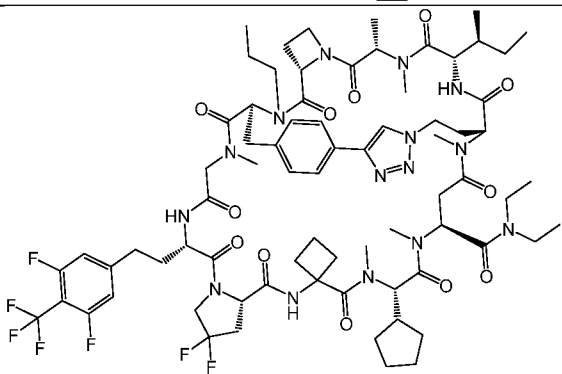
|      |  |                                                                                                                                                                                                                                                                                                |
|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2115 |  | <chem>CC[C@H](C)[C@@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N(C)C</chem>        |
| 2116 |  | <chem>CC[C@H](C)[C@@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>  |
| 2117 |  | <chem>CC[C@H](C)[C@@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem> |
| 2118 |  | <chem>CC[C@H](C)[C@@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>   |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                        |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2119 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=C=C(C=C9)C3)C(=O)N1)C(=O)N%10CCCC%10</chem>        |
| 2120 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=C=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>       |
| 2121 |   | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]3N(CCC(F)(F)F)C(=O)[C@@H]4N(CCC4)C(=O)[C@H](C)N(C)C(=O)[C@H](C[C@@H](C)CC)NC(=O)[C@@H](CCN5C=C(N=N5)C6=CC=C(C=C6)C3)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>   |
| 2122 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H](C[C@@H](C)C)C(=O)N(C)[C@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(F)(F)F)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1)C(=O)N(C)C</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2123 |    | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(F)(F)F)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1)C(=O)N%10CCCC%10</chem>  |
| 2124 |    | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)C)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(F)(F)F)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1)C(=O)N%10CCOCC%10</chem> |
| 2125 |   | <chem>FC(F)CO[C@@H]1C[C@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]3N(CCC(F)(F)F)C(=O)[C@@H]4N(CC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCN5C=C(N=N5)C6=CC=C(C=C6)C3)N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>             |
| 2126 |  | <chem>FC(F)CO[C@H]1CN2[C@@H](C1)C(=O)NC3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](C6)C(=O)N(CCC(F)(F)F)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C2=O)C(=O)N%10CCCC%10</chem>     |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                          |
|------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2127 |    | <chem>FC(F)CO[C@H]1CN2[C@@H](C1)C(=O)NC3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@@H](C@H)(C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](C6)C(=O)N(CCC(F)(F)F)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C2=O)C(=O)N%10CCOCC%10</chem> |
| 2128 |    | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1)C(=O)N%10CCOCC%10</chem>            |
| 2129 |   | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CC(C9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>                  |
| 2130 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@@H](C4CCCC4)N(C)C(=O)C5(CCC5)NC(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>                          |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2131 |    | <chem>FC(F)CO[C@@H]1C[C@@H]2N(C1)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]3N(CCC(C)C)C(=O)[C@H]4N(CCC4)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCN5C=C(N=N5)C6=CC=C(C=C6)C3)N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@H](C7CCCC7)N(C)C(=O)C8(CCC8)NC2=O)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>      |
| 2132 |    | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>            |
| 2133 |   | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>                   |
| 2134 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1</chem> |

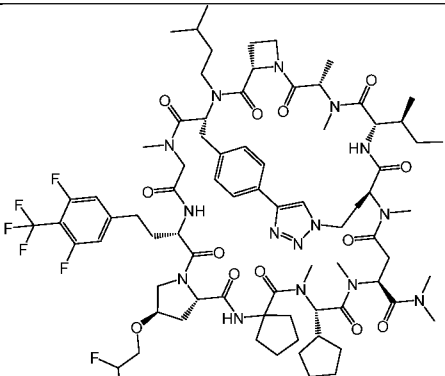
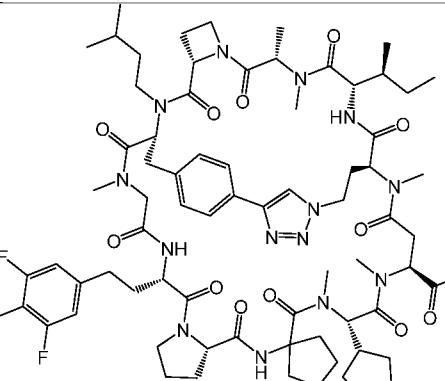
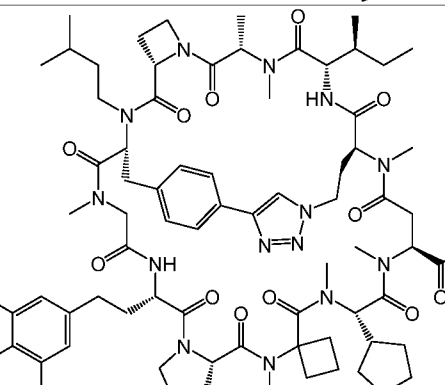
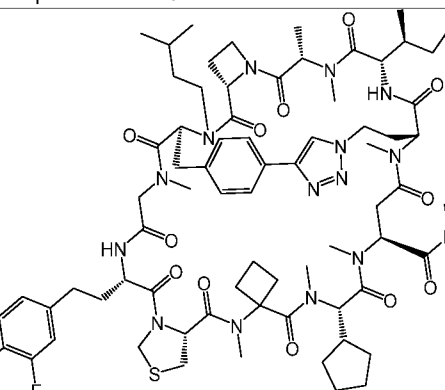
|      |                                                                                     |                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2135 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@H](CCC5=CC(F)=C(C(F)=C5)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]6N(CCC(F)(F)F)C(=O)[C@H]7N(CC7)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C6)N(C)C(=O)C1</chem>      |
| 2136 |    | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CCC(F)(F)F)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C5)N(C)C(=O)C1)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 2137 |   | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)N(C)C(=O)[C@H]4N(CSC4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CCC(F)(F)F)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C5)N(C)C(=O)C1)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>       |
| 2138 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CC)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C5)N(C)C(=O)C1)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>            |

|      |  |                                                                                                                                                                                                                                                                                              |
|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2139 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>      |
| 2140 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)N(C)C(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>            |
| 2141 |  | <chem>CCN(CC)C(=O)[C@H]1N(C)C(=O)[C@H](C2CCCC2)N(C)C(=O)C3(CCCC3)NC(=O)[C@H]4N(CC(F)(F)C4)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]5N(CC(F)(F)F)C(=O)[C@H]6N(CC6)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H](C)CN7C=C(N=N7)C8=CC=C(C=C8)C5)N(C)C(=O)C1)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 2142 |  | <chem>CCCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)NC(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>                  |

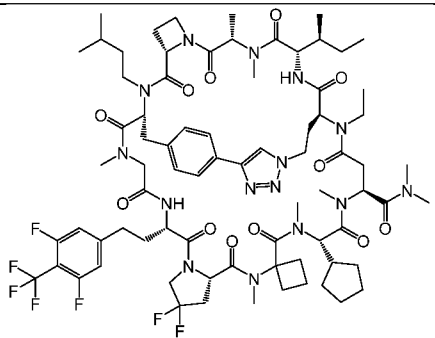
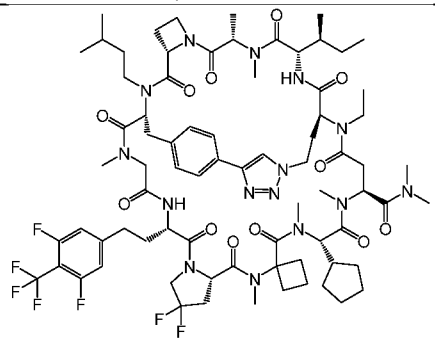
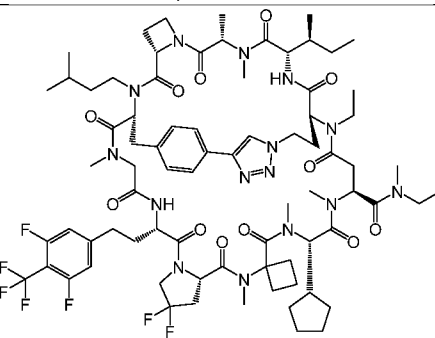
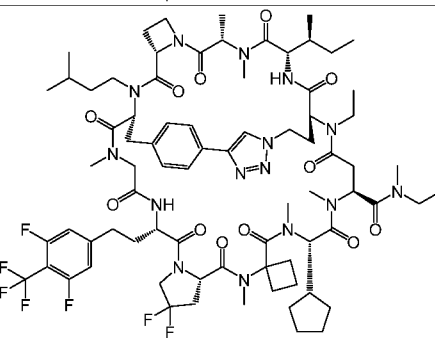
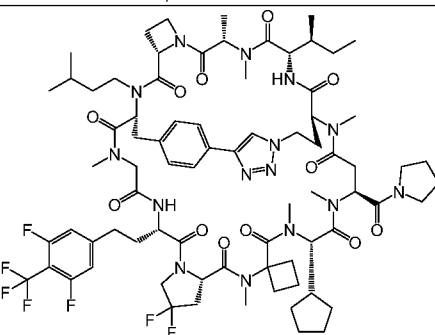
|      |  |                                                                                                                                                                                                                                                                                       |
|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2143 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>    |
| 2144 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC(C6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N8[C@H](CCC8)CC(=O)N(C)[C@@H](CCN9C=C(N=N9)C%10=CC=C(C=C%10)C3)C(=O)N1</chem> |
| 2145 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)N(C)C6(CCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N8[C@H](CCC8)CC(=O)N(C)[C@@H](CCN9C=C(N=N9)C%10=CC=C(C=C%10)C3)C(=O)N1</chem>        |
| 2146 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CCCC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N8[C@H](CCC8)CC(=O)N(C)[C@@H](CCN9C=C(N=N9)C%10=CC=C(C=C%10)C3)C(=O)N1</chem>          |

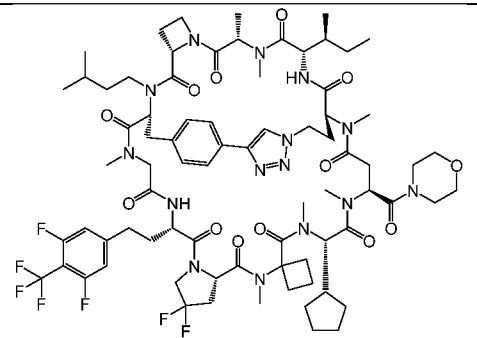
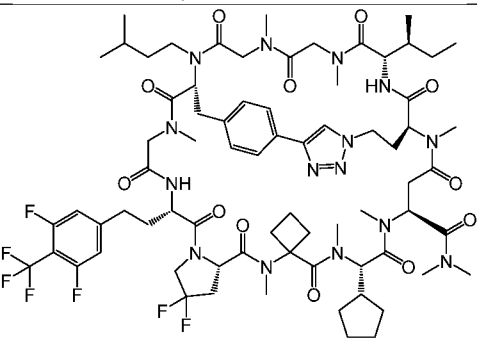
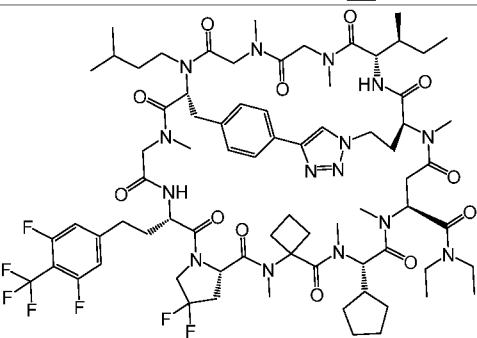
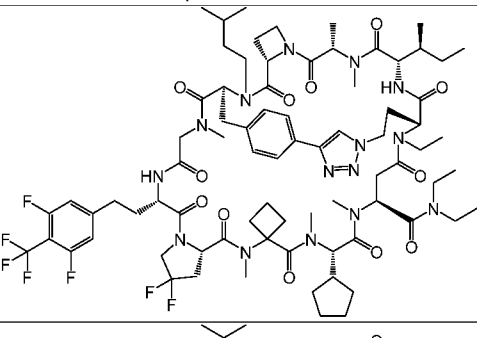
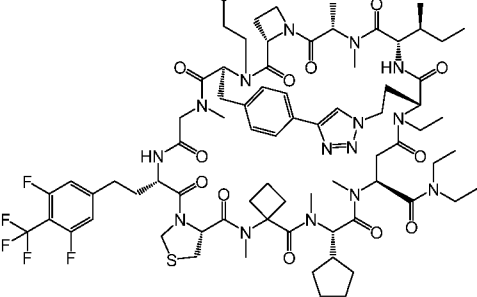
|      |  |                                                                                                                                                                                                                                                                                                   |
|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2147 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC(F)F)[C@@H]3C(=O)N(C)CC(=O)N[C@@H](CCCC=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CSC5)C(=O)NC6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N8[C@H](CCC8)CC(=O)N(C)[C@@H](CCN9C=C(N=N9)C%10=CC=C(C=C%10)C3)C(=O)N1</chem>                   |
| 2148 |  | <chem>CC[C@H](C)[C@H]1C(=O)N(C)[C@@H](C)C(=O)N2[C@@H](CC2)C(=O)N(CCC)[C@H]3C(=O)N(C)CC(=O)N[C@@H](CCC4=CC(F)=C(C(F)=C4)C(F)(F)F)C(=O)N5[C@@H](CC(F)(F)C5)C(=O)N(C)C6(CC6)C(=O)N(C)[C@@H](C7CCCC7)C(=O)N(C)[C@@H](C(=O)N(C)C)CC(=O)N(C)[C@@H](CCN8C=C(N=N8)C9=CC=C(C=C9)C3)C(=O)N1</chem>          |
| 2149 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@@H]5C(=O)N[C@@H](C[C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC)[C@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](C)CC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1</chem> |
| 2150 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>              |

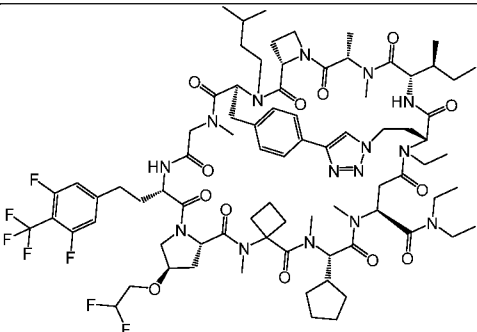
|      |  |                                                                                                                                                                                                                                                                                                    |
|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2151 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(CC)CC)N(C)C(=O)[C@H]([C@@H](C4CCCC4)N(C)C(=O)C5(CCCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CC7=CC=C(C=C7)C8=CN(N=N8)C(C3)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem> |
| 2152 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@H]5C(=O)N[C@H]([C@@H](C)C)C(=O)N(C)[C@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem>  |
| 2153 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)NC(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>               |
| 2154 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)NC(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>                     |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2155 |    | <chem>FC(F)CO[C@H]1CN2[C@@H](C1)C(=O)NC3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@H](CC7=C=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C2=O</chem> |
| 2156 |    | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCCC5)NC(=O)[C@H]6N(CCC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>                   |
| 2157 |   | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@H](C(=O)N(C)C)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CCC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>                 |
| 2158 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CSC6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>               |

|      |  |                                                                                                                                                                                                                                                                                                        |
|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2159 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@@H]3N(C)C(=O)C[C@@H](C(=O)N(C)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@H](CCC7=CC(F)=C(C(F)=C7)C(F)(F)F)NC(=O)CN(C)C(=O)[C@@H]1C8=CC=C(C=C8)C9=CN(N=N9)CC3</chem>               |
| 2160 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(C)[C@@H]5C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@H](CC7=CC=C(C=C7)C8=C(N(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem> |
| 2161 |  | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C2)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N9[C@@H](CC9)C1=O)C(=O)N%10CCCC%10</chem>  |
| 2162 |  | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C2)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N9[C@@H](CC9)C1=O)C(=O)N%10CCOCC%10</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2163 |    | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H]([C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>      |
| 2164 |    | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H]([C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>      |
| 2165 |   | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H]([C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>      |
| 2166 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N(C)C)N(C)C(=O)[C@H]([C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CC(F)(F)C6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CCC9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>      |
| 2167 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@@H](C(=O)N4CCCC4)N(C)C(=O)[C@H]([C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@H](CCC8=CC(F)=C(C(F)=C8)C(F)(F)F)NC(=O)CN(C)C(=O)[C@H]1CC9=CC=C(C=C9)C%10=CN(N=N%10)CC3</chem> |

|      |                                                                                     |                                                                                                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2168 |    | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(C)C(=O)C[C@H](C(=O)N4CCOCC4)N(C)C(=O)[C@H](C5CCCC5)N(C)C(=O)C6(CCC6)N(C)C(=O)[C@H]7N(CC(F)(F)C7)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@H]1CC8=CC=C(C=C8)C9=CN(N=N9)CC3)CCC%10=CC(F)=C(C(F)=C%10)C(F)(F)F</chem>    |
| 2169 |    | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@H](C(=O)N(C)C)CC(=O)N(C)[C@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C2)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)CC1=O</chem>                     |
| 2170 |   | <chem>CC(C)CCN1[C@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@H](CC(=O)N(C)[C@@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C2)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)CC(=O)N(C)C1=O)C(=O)N(CC)CC</chem>                   |
| 2171 |  | <chem>CC(C)CCN1[C@@H]2C(=O)N(C)CC(=O)N[C@@H](C)CC3=CC(F)=C(C(F)=C3)C(F)(F)F)C(=O)N4[C@@H](CC(F)(F)C4)C(=O)N(C)C5(CCC5)C(=O)N(C)[C@@H](C6CCCC6)C(=O)N(C)[C@H](CC(=O)N(CC)[C@H](CCN7C=C(N=N7)C8=CC=C(C=C8)C2)C(=O)N[C@@H]([C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N9[C@@H](CC9)C1=O)C(=O)N(CC)CC</chem> |
| 2172 |  | <chem>CC(C)CCN1C(=O)[C@H]2N(CC2)C(=O)[C@H](C)N(C)C(=O)[C@H]([C@@H](C)CC)NC(=O)[C@H]3N(CC)C(=O)C[C@H](C(=O)N(CC)CC)N(C)C(=O)[C@H](C4CCCC4)N(C)C(=O)C5(CCC5)N(C)C(=O)[C@H]6N(CSC6)C(=O)[C@@H](NC(=O)CN(C)C(=O)[C@@H]1CC7=CC=C(C=C7)C8=CN(N=N8)CC3)CC(C9=CC(F)=C(C(F)=C9)C(F)(F)F</chem>            |

|      |                                                                                   |                                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2173 |  | <chem>FC(F)CO[C@@H]1C[C@H]2C(=O)N(C)C3(CCC3)C(=O)N(C)[C@@H](C4CCCC4)C(=O)N(C)[C@@H](CC(=O)N(CC)[C@H]5C(=O)N[C@@H](C[C@@H](C)CC)C(=O)N(C)[C@@H](C)C(=O)N6[C@@H](CC6)C(=O)N(CCC(C)C)[C@@H](CC7=CC=C(C=C7)C8=CN(N=N8)CC5)C(=O)N(C)CC(=O)N[C@@H](CCC9=CC(F)=C(C(F)=C9)C(F)(F)F)C(=O)N2C1)C(=O)N(C)CC</chem> |
|------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Compound # | LCMS RT (min) | MS (m/z) | LCMS Method | MS Polarity          | KRAS G12V GDP KD (nM) |
|------------|---------------|----------|-------------|----------------------|-----------------------|
| 1036       | 1.555         | 1,610.9  | 40-100-2MIN | [M+H] <sup>+</sup>   | 9.771                 |
| 1037       | 1.554         | 1,618.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 34.128                |
| 1038       | 1.579         | 1,624.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 5.505                 |
| 1039       | 1.579         | 1,650.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 5.426                 |
| 1040       | 1.406         | 1,558.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.186                 |
| 1041       | 1.456         | 1,586.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.343                 |
| 1042       | 1.48          | 1,592.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.809                 |
| 1043       | 1.505         | 1,618.9  | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.974                 |
| 1044       | 1.505         | 1,589.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 8.000                 |
| 1045       | 1.505         | 1,615.7  | 40-100-2MIN | [M+H] <sup>+</sup>   | 12.873                |
| 1046       | 1.491         | 537.6    | 40-100-2MIN | [M+3H] <sup>3+</sup> | 1.778                 |
| 1047       | 5.998         | 1,596.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.430                 |
| 1048       | 1.53          | 1,610.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.076                 |
| 1049       | 2.588         | 1,558.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.665                 |
| 1050       | 1.604         | 1,544.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.424                 |
| 1051       | 4.315         | 1,556.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.899                 |
| 1052       | 1.548         | 1,624.9  | 40-100-2MIN | [M+H] <sup>+</sup>   | 14.898                |
| 1053       | 1.505         | 1,556.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.233                 |
| 1054       | 1.406         | 1,528.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.161                 |
| 1055       | 1.53          | 1,598.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 55.907                |
| 1056       | 1.505         | 1,596.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.908                 |
| 1057       | 1.52          | 1,610.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 20.227                |
| 1058       | 1.48          | 1,542.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.271                 |
| 1059       | 1.456         | 1,564.7  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.040                 |
| 1060       | 1.505         | 1,578.7  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.187                 |
| 1061       | 0.361         | 1,542.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 6.194                 |
| 1062       | 0.344         | 1,564.7  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.678                 |
| 1063       | 3.249         | 1,578.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.899                 |
| 1064       | 5.761         | 1,592.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.308                 |
| 1065       | 1.53          | 1,618.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.683                 |
| 1066       | 2.04          | 1,634.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.654                 |
| 1067       | 2.746         | 1,660.9  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.853                 |
| 1068       | 3.15          | 1,578.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.160                 |
| 1069       | 0.358         | 1,556.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.347                 |
| 1070       | 2.506         | 1,604.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.150                 |
| 1071       | 1.881         | 1,620.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.255                 |
| 1072       | 2.675         | 1,646.9  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.246                 |
| 1073       | 2.798         | 1,590.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.082                 |
| 1074       | 2.043         | 1,606.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.108                 |
| 1075       | 2.628         | 1,632.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.324                 |
| 1076       | 0.612         | 1,604.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.265                 |
| 1077       | 0.966         | 1,620.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.930                 |
| 1078       | 1.545         | 1,646.9  | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.564                 |
| 1079       | 5.395         | 1,542.8  | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.298                 |

|      |       |         |               |                    |        |
|------|-------|---------|---------------|--------------------|--------|
| 1080 | 1.53  | 1,592.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 3.345  |
| 1081 | 5.921 | 1,592.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.348  |
| 1082 | 0.449 | 1,606.8 | 30-100-0.8MIN | [M+H] <sup>+</sup> | 1.054  |
| 1083 | 1.53  | 1,606.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.668  |
| 1084 | 0.243 | 1,606.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.494  |
| 1085 | 1.555 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.161  |
| 1086 | 1.579 | 1,521.7 | 40-100-2MIN   | [M+H] <sup>+</sup> | 11.026 |
| 1087 | 1.592 | 1,535.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 12.246 |
| 1088 | 2.339 | 1,564.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.817  |
| 1089 | 0.804 | 1,590.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.541  |
| 1090 | 2.504 | 1,604.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.492  |
| 1091 | 1.57  | 1,606.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.164  |
| 1092 | 1.594 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.232  |
| 1093 | 1.554 | 1,606.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.959  |
| 1094 | 2.013 | 1,546.7 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.169  |
| 1095 | 1.663 | 1,560.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.281  |
| 1096 | 1.48  | 1,560.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.064  |
| 1097 | 1.503 | 1,574.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.435  |
| 1098 | 1.48  | 1,608.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.171  |
| 1099 | 1.694 | 1,622.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.248  |
| 1100 | 1.52  | 1,622.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.626  |
| 1101 | 1.838 | 1,636.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.727  |
| 1102 | 1.554 | 1,636.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.579  |
| 1103 | 1.999 | 1,650.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.591  |
| 1104 | 1.529 | 1,574.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.034  |
| 1105 | 1.53  | 1,588.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.300  |
| 1106 | 1.619 | 1,549.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 9.849  |
| 1107 | 1.598 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.404  |
| 1108 | 1.583 | 1,602.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.092  |
| 1109 | 1.579 | 1,664.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 4.555  |
| 1110 | 1.944 | 1,634.9 | 10-80-2MIN    | [M+H] <sup>+</sup> | 1.343  |
| 1111 | 1.604 | 1,617.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.335  |
| 1112 | 1.624 | 1,678.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.020  |
| 1113 | 1.573 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.071  |
| 1114 | 1.551 | 1,588.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.601  |
| 1115 | 1.58  | 1,602.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.195  |
| 1116 | 2.048 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.403  |
| 1117 | 2.204 | 1,634.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.318  |
| 1118 | 1.692 | 1,602.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.836  |
| 1119 | 2.102 | 1,617.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.542  |
| 1120 | 1.579 | 1,634.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.318  |
| 1121 | 1.604 | 1,648.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.146  |
| 1122 | 1.62  | 1,678.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.674  |
| 1123 | 2.799 | 1,570.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.865  |
| 1124 | 1.599 | 1,598.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.183  |
| 1125 | 1.656 | 1,662.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 2.389  |
| 1126 | 2.362 | 1,648.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 6.632  |
| 1127 | 1.585 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.958  |
| 1128 | 1.579 | 1,620.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.739  |
| 1129 | 1.845 | 1,634.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.771  |
| 1130 | 1.609 | 1,561.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 28.386 |
| 1131 | 1.674 | 1,602.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.032  |
| 1132 | 1.724 | 1,664.9 | 40-100-2MIN   | [M+H] <sup>+</sup> | 1.063  |
| 1133 | 1.554 | 1,648.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.409  |
| 1134 | 1.571 | 1,632.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.287  |
| 1135 | 1.584 | 1,605.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.382  |
| 1136 | 1.501 | 1,580.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.109  |
| 1137 | 1.53  | 1,580.8 | 40-100-2MIN   | [M+H] <sup>+</sup> | 0.102  |

|      |       |         |             |                      |         |
|------|-------|---------|-------------|----------------------|---------|
| 1138 | 1.636 | 1,704.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 4.220   |
| 1139 | 1.62  | 1,720.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.320   |
| 1140 | 1.684 | 1,718.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 9.783   |
| 1141 | 1.641 | 1,692.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 4.650   |
| 1142 | 1.586 | 1,676.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.480   |
| 1143 | 1.57  | 1,692.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.958   |
| 1144 | 1.629 | 1,690.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.912   |
| 1145 | 1.601 | 1,664.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.925   |
| 1146 | 1.727 | 1,608.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.528   |
| 1147 | 1.723 | 1,608.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.351   |
| 1148 | 1.678 | 1,624.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.279   |
| 1149 | 1.678 | 1,624.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.476   |
| 1150 | 1.727 | 1,652.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.518   |
| 1151 | 1.727 | 1,652.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.601   |
| 1152 | 1.727 | 1,652.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.704   |
| 1153 | 0.531 | 1,635.2 | 30-90-1MIN  | [M+H] <sup>+</sup>   | 0.740   |
| 1154 | 1.531 | 1,664.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.355   |
| 1155 | 1.529 | 1,678.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.380   |
| 1156 | 1.556 | 1,634.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.247   |
| 1157 | 1.598 | 1,650.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.971   |
| 1158 | 1.53  | 1,570.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.426   |
| 1159 | 1.585 | 1,620.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.633   |
| 1160 | 1.604 | 1,664.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 4.324   |
| 1161 | 0.562 | 1,584.7 | 30-90-1MIN  | [M+H] <sup>+</sup>   | 3.392   |
| 1162 | 1.921 | 1,678.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.671   |
| 1163 | 1.53  | 1,556.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.668   |
| 1164 | 1.589 | 1,650.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.330   |
| 1165 | 1.562 | 1,570.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.673   |
| 1166 | 1.621 | 1,664.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 3.111   |
| 1167 | 1.594 | 1,584.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 1.128   |
| 1168 | 0.557 | 1,606.7 | 30-90-1MIN  | [M+H] <sup>+</sup>   | 1.461   |
| 1169 | 1.557 | 1,584.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.968   |
| 1170 | 1.606 | 1,634.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.077   |
| 1171 | 1.603 | 840.1   | 40-100-2min | [M+2H] <sup>2+</sup> | 2.055   |
| 1172 | 1.579 | 1,598.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.607   |
| 1173 | 1.604 | 1,584.9 | 40-100-2min | [M+H] <sup>+</sup>   | 5.047   |
| 1174 | 1.976 | 1,592.8 | 40-100-2min | [M+H] <sup>+</sup>   | 13.408  |
| 1175 | 1.561 | 1,618.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.421   |
| 1176 | 1.876 | 1,690.9 | 40-100-2min | [M+H] <sup>+</sup>   | 3.707   |
| 1177 | 1.638 | 1,662.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.918   |
| 1178 | 1.531 | 1,648.8 | 40-100-2min | [M+H] <sup>+</sup>   | 1.324   |
| 1179 | 1.759 | 1,598.9 | 40-100-2min | [M+H] <sup>+</sup>   | 5.651   |
| 1180 | 1.600 | 1,620.8 | 40-100-2min | [M+H] <sup>+</sup>   | 2.555   |
| 1181 | 1.579 | 1,620.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.604   |
| 1182 | 1.554 | 1,606.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.526   |
| 1183 | 1.637 | 1,606.8 | 40-100-2min | [M+H] <sup>+</sup>   | 178.260 |
| 1184 | 1.593 | 1,664.9 | 40-100-2min | [M+H] <sup>+</sup>   | 3.900   |
| 1185 | 1.562 | 1,584.8 | 40-100-2min | [M+H] <sup>+</sup>   | 2.540   |
| 1186 | 1.588 | 1,650.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.846   |
| 1187 | 1.554 | 1,570.8 | 40-100-2min | [M+H] <sup>+</sup>   | 1.436   |
| 1188 | 1.577 | 1,602.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.414   |
| 1189 | 1.727 | 1,608.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.318   |
| 1190 | 1.702 | 1,608.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.710   |
| 1191 | 1.727 | 1,572.8 | 40-100-2min | [M+H] <sup>+</sup>   | 1.209   |
| 1192 | 1.749 | 1,572.8 | 40-100-2min | [M+H] <sup>+</sup>   | 1.668   |
| 1193 | 1.894 | 1,636.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.082   |
| 1194 | 1.875 | 1,636.9 | 40-100-2min | [M+H] <sup>+</sup>   | 3.204   |
| 1195 | 1.662 | 1,696.9 | 40-100-2min | [M+H] <sup>+</sup>   | 12.656  |

|      |       |         |             |                      |         |
|------|-------|---------|-------------|----------------------|---------|
| 1196 | 1.647 | 1,660.9 | 40-100-2min | [M+H] <sup>+</sup>   | 6.362   |
| 1197 | 1.549 | 1,678.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.709   |
| 1198 | 1.520 | 1,598.8 | 40-100-2min | [M+H] <sup>+</sup>   | 3.159   |
| 1199 | 1.544 | 1,606.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.778   |
| 1200 | 1.557 | 1,620.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.445   |
| 1201 | 1.568 | 1,620.9 | 40-100-2min | [M+H] <sup>+</sup>   | 2.242   |
| 1202 | 1.589 | 1,634.9 | 40-100-2min | [M+H] <sup>+</sup>   | 2.473   |
| 1203 | 1.553 | 1,606.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.386   |
| 1204 | 1.530 | 1,570.9 | 40-100-2min | [M+H] <sup>+</sup>   | 3.530   |
| 1205 | 1.517 | 1,588.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.326   |
| 1206 | 1.563 | 1,650.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.517   |
| 1207 | 1.603 | 1,622.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.329   |
| 1208 | 1.624 | 1,622.9 | 40-100-2min | [M+H] <sup>+</sup>   | 21.478  |
| 1209 | 1.513 | 1,612.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.182   |
| 1210 | 1.549 | 1,596.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.770   |
| 1211 | 1.600 | 1,612.9 | 40-100-2min | [M+H] <sup>+</sup>   | 3.724   |
| 1212 | 1.602 | 1,634.8 | 40-100-2min | [M+H] <sup>+</sup>   | 1.896   |
| 1213 | 1.582 | 1,620.8 | 40-100-2min | [M+H] <sup>+</sup>   | 0.771   |
| 1214 | 1.568 | 1,618.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.380   |
| 1215 | 1.578 | 1,646.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.101   |
| 1216 | 1.597 | 1,610.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.708   |
| 1217 | 1.529 | 1,582.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.815   |
| 1218 | 1.547 | 1,618.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.245   |
| 1219 | 1.560 | 1,614.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.527   |
| 1220 | 1.702 | 1,628.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.701   |
| 1221 | 1.613 | 1,643.0 | 40-100-2min | [M+H] <sup>+</sup>   | 1.859   |
| 1222 | 1.543 | 1,632.9 | 40-100-2min | [M+H] <sup>+</sup>   | 5.962   |
| 1223 | 1.576 | 1,632.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.247   |
| 1224 | 1.593 | 1,692.9 | 40-100-2min | [M+H] <sup>+</sup>   | 12.003  |
| 1225 | 1.627 | 1,692.9 | 40-100-2min | [M+H] <sup>+</sup>   | 3.036   |
| 1226 | 1.579 | 1,668.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.784   |
| 1227 | 2.132 | 1,660.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.123   |
| 1228 | 1.624 | 1,646.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.792   |
| 1229 | 1.568 | 1,640.9 | 40-100-2min | [M+H] <sup>+</sup>   | 0.652   |
| 1230 | 1.702 | 1,656.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.377   |
| 1231 | 2.130 | 1,654.9 | 40-100-2min | [M+H] <sup>+</sup>   | 1.880   |
| 2035 | 1.505 | 1,542.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.159   |
| 2036 | 1.042 | 1,462.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.036   |
| 2037 | 1.258 | 1,490.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.030   |
| 2038 | 1.209 | 1,028.8 | 40-100-2MIN | [M+2H] <sup>2+</sup> | 0.226   |
| 2039 | 1.283 | 1,035.7 | 40-100-2MIN | [M+2H] <sup>2+</sup> | 0.333   |
| 2040 | 1.16  | 1,077.4 | 40-100-2MIN | [M+2H] <sup>2+</sup> | 0.163   |
| 2041 | 1.209 | 1,084.3 | 40-100-2MIN | [M+2H] <sup>2+</sup> | 0.229   |
| 2042 | 1.357 | 1,522.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.108   |
| 2043 | 1.431 | 1,590.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.104   |
| 2044 | 1.456 | 1,617.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.934   |
| 2045 | 1.357 | 1,528.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.060   |
| 2046 | 1.368 | 770.1   | 40-100-2MIN | [M+2H] <sup>2+</sup> | 1.359   |
| 2047 | 1.456 | 1,568.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.122   |
| 2048 | 1.332 | 1,570.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.083   |
| 2049 | 1.444 | 1,569.2 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.124   |
| 2050 | 1.332 | 1,541.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.046   |
| 2051 | 1.332 | 1,526.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.088   |
| 2052 | 1.391 | 1,529.2 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.078   |
| 2053 | 1.456 | 1,570.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 880.256 |
| 2054 | 1.456 | 1,570.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.403   |
| 2055 | 1.456 | 1,556.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.084   |
| 2056 | 1.382 | 1,590.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.162   |

|      |       |         |             |                      |         |
|------|-------|---------|-------------|----------------------|---------|
| 2057 | 1.431 | 1,604.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.209   |
| 2058 | 1.431 | 1,604.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.279   |
| 2059 | 1.48  | 1,597.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.209   |
| 2060 | 1.332 | 1,540.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.037   |
| 2061 | 1.382 | 1,554.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.043   |
| 2062 | 1.456 | 1,596.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.136   |
| 2063 | 1.522 | 1,672.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.694   |
| 2064 | 1.357 | 1,012.7 | 40-100-2MIN | [M+2H] <sup>2+</sup> | 0.235   |
| 2065 | 1.382 | 1,005.2 | 40-100-2MIN | [M+2H] <sup>2+</sup> | 0.814   |
| 2066 | 1.727 | 1,523.2 | 10-80-2MIN  | [M+H] <sup>+</sup>   | 0.168   |
| 2067 | 1.332 | 1,536.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.392   |
| 2068 | 1.357 | 1,536.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.194   |
| 2069 | 1.431 | 1,528.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.244   |
| 2070 | 1.406 | 1,546.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.022   |
| 2071 | 1.431 | 1,572.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.054   |
| 2072 | 1.53  | 1,614.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 5.428   |
| 2073 | 1.554 | 1,633.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.135   |
| 2074 | 1.431 | 1,564.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.083   |
| 2075 | 1.406 | 1,606.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.176   |
| 2076 | 1.579 | 1,646.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.513   |
| 2077 | 1.431 | 1,620.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.129   |
| 2078 | 1.382 | 1,576.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.245   |
| 2079 | 1.332 | 1,514.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.030   |
| 2080 | 1.678 | 1,556.9 | 10-80-2MIN  | [M+H] <sup>+</sup>   | 0.208   |
| 2081 | 1.357 | 1,528.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.066   |
| 2082 | 1.406 | 1,564.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.280   |
| 2083 | 1.456 | 1,590.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.357   |
| 2084 | 1.48  | 1,626.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 2.651   |
| 2085 | 1.406 | 1,572.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.114   |
| 2086 | 1.406 | 1,599.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.180   |
| 2087 | 1.382 | 1,598.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.097   |
| 2088 | 1.53  | 1,596.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.297   |
| 2089 | 1.48  | 1,582.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 6.164   |
| 2090 | 1.382 | 1,552.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.089   |
| 2091 | 1.382 | 1,552.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.076   |
| 2092 | 1.431 | 1,536.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.075   |
| 2093 | 1.431 | 1,536.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.076   |
| 2094 | 1.423 | 1,572.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.372   |
| 2095 | 1.456 | 1,608.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.080   |
| 2096 | 1.445 | 1,608.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 138.678 |
| 2097 | 1.431 | 1,550.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.059   |
| 2098 | 1.431 | 1,576.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.079   |
| 2099 | 1.406 | 1,592.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.045   |
| 2100 | 1.554 | 1,619.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.187   |
| 2101 | 1.431 | 1,606.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.084   |
| 2102 | 1.579 | 1,633.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.174   |
| 2103 | 1.48  | 1,582.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.134   |
| 2104 | 1.431 | 1,569.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.901   |
| 2105 | 1.456 | 1,583.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.172   |
| 2106 | 1.431 | 1,586.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.123   |
| 2107 | 1.431 | 1,572.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.107   |
| 2108 | 1.456 | 1,572.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.084   |
| 2109 | 1.505 | 1,508.0 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.100   |
| 2110 | 1.48  | 1,507.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.084   |
| 2111 | 1.678 | 1,626.0 | 10-80-2MIN  | [M+H] <sup>+</sup>   | 0.294   |
| 2112 | 1.431 | 1,552.8 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.050   |
| 2113 | 1.406 | 1,542.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.075   |
| 2114 | 1.48  | 1,556.9 | 40-100-2MIN | [M+H] <sup>+</sup>   | 0.135   |

|      |       |         |             |                    |       |
|------|-------|---------|-------------|--------------------|-------|
| 2115 | 1.406 | 1,532.8 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.054 |
| 2116 | 1.431 | 1,558.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.071 |
| 2117 | 1.406 | 1,574.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.041 |
| 2118 | 1.431 | 1,546.8 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.006 |
| 2119 | 1.456 | 1,572.8 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.042 |
| 2120 | 1.406 | 1,588.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.013 |
| 2121 | 1.431 | 1,594.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.388 |
| 2122 | 1.456 | 1,609.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.131 |
| 2123 | 1.48  | 1,634.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.202 |
| 2124 | 1.456 | 1,650.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.195 |
| 2125 | 1.456 | 1,608.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.022 |
| 2126 | 1.48  | 1,634.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.039 |
| 2127 | 1.431 | 1,650.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.073 |
| 2128 | 1.426 | 1,588.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.110 |
| 2129 | 1.456 | 1,524.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.045 |
| 2130 | 1.406 | 1,506.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.028 |
| 2131 | 1.456 | 1,568.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.063 |
| 2132 | 1.456 | 1,538.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.024 |
| 2133 | 1.456 | 1,520.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.039 |
| 2134 | 1.48  | 1,583.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.037 |
| 2135 | 1.505 | 1,578.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.071 |
| 2136 | 1.53  | 1,592.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.075 |
| 2137 | 1.505 | 1,574.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.051 |
| 2138 | 1.456 | 1,525.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.089 |
| 2139 | 1.48  | 1,538.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.331 |
| 2140 | 1.456 | 1,520.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.077 |
| 2141 | 1.505 | 1,592.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.067 |
| 2142 | 1.382 | 1,478.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.043 |
| 2143 | 1.406 | 1,492.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.138 |
| 2144 | 1.48  | 1,465.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.180 |
| 2145 | 1.456 | 1,447.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.119 |
| 2146 | 1.456 | 1,447.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.076 |
| 2147 | 1.505 | 1,501.8 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.112 |
| 2148 | 1.406 | 1,510.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.038 |
| 2149 | 1.431 | 1,554.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.106 |
| 2150 | 1.53  | 1,548.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.068 |
| 2151 | 1.554 | 1,567.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.074 |
| 2152 | 1.554 | 1,611.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.077 |
| 2153 | 1.456 | 1,539.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.103 |
| 2154 | 1.48  | 1,520.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.141 |
| 2155 | 1.48  | 1,583.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.033 |
| 2156 | 1.431 | 1,502.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.105 |
| 2157 | 1.431 | 1,503.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.025 |
| 2158 | 1.48  | 1,534.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.124 |
| 2159 | 1.505 | 1,552.9 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.062 |
| 2160 | 1.524 | 1,597.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.271 |
| 2161 | 1.53  | 1,579.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.084 |
| 2162 | 1.505 | 1,595.0 | 40-100-2MIN | [M+H] <sup>+</sup> | 0.111 |
| 2163 | 1.484 | 1,553.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.076 |
| 2164 | 1.606 | 1,553.0 | 40-100-2min | [M+H] <sup>+</sup> | 1.265 |
| 2165 | 1.545 | 1,566.9 | 40-100-2min | [M+H] <sup>+</sup> | 0.093 |
| 2166 | 1.602 | 1,567.0 | 40-100-2min | [M+H] <sup>+</sup> | 1.114 |
| 2167 | 1.506 | 1,564.9 | 40-100-2min | [M+H] <sup>+</sup> | 0.113 |
| 2168 | 1.480 | 1,580.9 | 40-100-2min | [M+H] <sup>+</sup> | 0.080 |
| 2169 | 1.456 | 1,513.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.041 |
| 2170 | 1.527 | 1,541.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.056 |
| 2171 | 1.554 | 1,581.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.423 |
| 2172 | 1.530 | 1,563.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.129 |

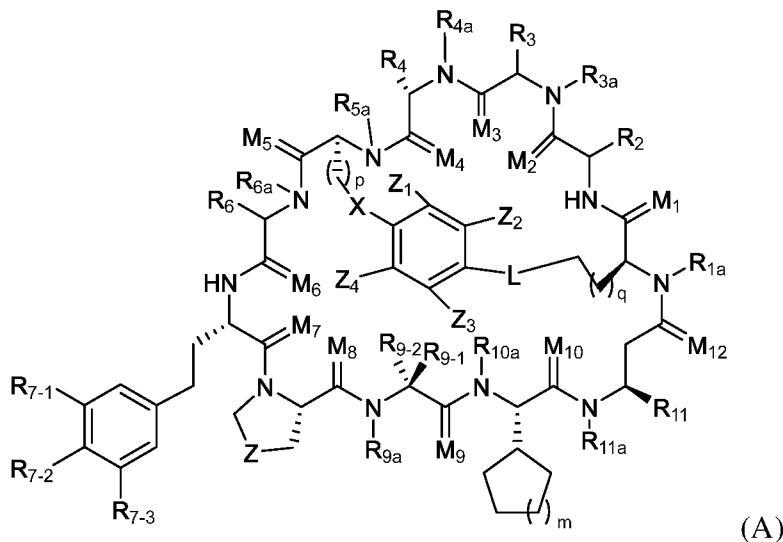
|      |       |         |             |                    |       |
|------|-------|---------|-------------|--------------------|-------|
| 2173 | 1.549 | 1,625.0 | 40-100-2min | [M+H] <sup>+</sup> | 0.130 |
|------|-------|---------|-------------|--------------------|-------|

A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. Accordingly, other embodiments are within the scope of the following claims.

## CLAIMS

## WHAT IS CLAIMED IS:

1. A compound of Formula (A):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

R<sub>1a</sub>, R<sub>9a</sub>, R<sub>10a</sub>, and R<sub>11a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3a</sub>, R<sub>4a</sub>, R<sub>5a</sub>, R<sub>6a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>2-6</sub> alkenyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>2</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>3-8</sub> cycloalkyl, or 4- to 8-membered heterocyclyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

Z is O, S, or C(R<sub>8-1</sub>)(R<sub>8-2</sub>);

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

R<sub>11</sub> is selected from H, C<sub>1-4</sub> alkyl, C<sub>1-4</sub> haloalkyl, or C(=M<sub>11</sub>)N(R<sub>12a</sub>)(R<sub>12b</sub>); or R<sub>11</sub> and R<sub>11a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

R<sub>12a</sub>, and R<sub>12b</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>12a</sub> and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with 1, or more halo, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>1-6</sub> alkoxy, C<sub>1-6</sub> haloalkoxy, C<sub>3-8</sub> cycloalkyl, or 4- to 8-membered heterocyclyl;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or -NR<sub>N</sub>C(=Y)-;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or -C(=Y)NR<sub>N</sub>-;

wherein Y is O or S, R<sub>N</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

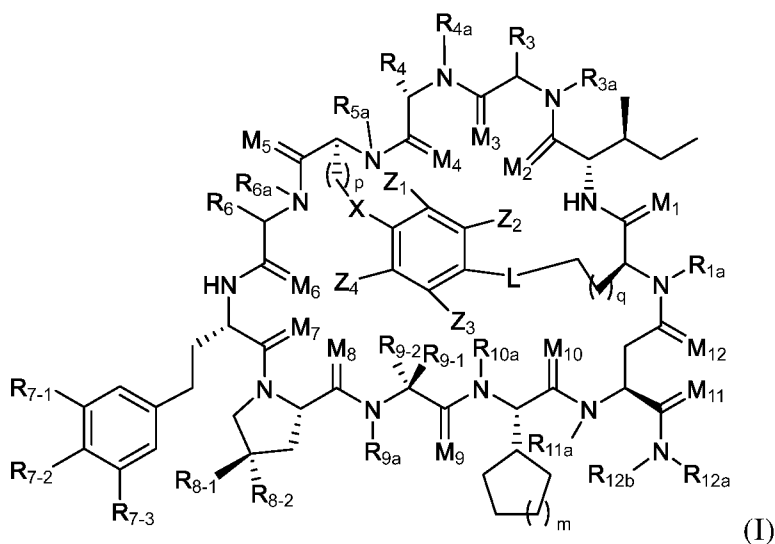
Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

p = 1, 2, or 3;

q = 0, 1, 2, or 3;

M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are independently selected from O or S.

2. A compound of Formula (I):



or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein:

R<sub>1a</sub>, R<sub>9a</sub>, R<sub>10a</sub>, and R<sub>11a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3a</sub>, R<sub>4a</sub>, R<sub>5a</sub>, R<sub>6a</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl;

R<sub>12a</sub>, and R<sub>12b</sub> are independently selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>12a</sub> and R<sub>12b</sub> together with the atom to which they are connected form a 4- to 8-membered heterocyclyl; which is optionally substituted with  
 5 1, or more halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>3</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, -C<sub>0-6</sub> alkylene-C<sub>3-8</sub> cycloalkyl, -C<sub>0-6</sub> alkylene-4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; or R<sub>3</sub> and R<sub>3a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

10 R<sub>4</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl; or R<sub>4</sub> and R<sub>4a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl; which is optionally substituted with -OC<sub>1-6</sub> alkyl, or -OC<sub>1-6</sub> haloalkyl;

R<sub>6</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

R<sub>7-1</sub> and R<sub>7-3</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

15 R<sub>7-2</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>8-1</sub> and R<sub>8-2</sub> are independently selected from H, halo, C<sub>1-6</sub> alkoxyl, or C<sub>1-6</sub> haloalkoxyl;

R<sub>9-1</sub> is selected from C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

R<sub>9-2</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl;

or R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl,  
 20 or 4- to 7-membered heterocyclyl;

m = 0, 1, 2, or 3;

X is selected from a bond, O, S, NH, 5- to 6-membered heteroarylene, or -NR<sub>N</sub>C(=Y)-;

L is selected from a bond, O, S, NH, phenylene, 5- to 6-membered heteroarylene, or -  
 C(=Y)NR<sub>N</sub>-;

25 wherein Y is O or S, R<sub>N</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, halo, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl;

p = 1, 2, or 3;

q = 0, 1, 2, or 3;

M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are independently selected from  
 30 O or S.

3. The compound according to claim 1 or 2, or a pharmaceutically acceptable salt, a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, wherein R<sub>1a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

35 4. The compound according to any one of claims 1 to 3, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>3a</sub> is selected from H, C<sub>1-6</sub> alkyl, C<sub>1-6</sub> haloalkyl, C<sub>3-8</sub> cycloalkyl, 4- to 8-membered heterocyclyl, -C<sub>0-6</sub> alkylene-C<sub>6-10</sub> aryl, or -C<sub>0-6</sub> alkylene-5- to 10-membered heteroaryl; preferably, C<sub>1-4</sub> alkyl, C<sub>1-</sub>

4 haloalkyl, C<sub>3-6</sub> cycloalkyl, or -C<sub>1-6</sub> alkylene-phenyl; preferably, H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

5 5. The compound according to any one of claims 1 to 3, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>3a</sub> is selected from cyclopropyl, phenyl, or benzyl; preferably, cyclopropyl, or benzyl.

6. The compound according to any one of claims 1 to 5, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>4a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl, preferably, Me, Et, n-Pr, or 3,3,3-trifluoropropyl.

10 7. The compound according to any one of claims 1 to 5, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>4a</sub> is selected from cyclopropyl, phenyl, or benzyl; preferably, cyclopropyl, or benzyl.

8. The compound according to any one of claims 1 to 7, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>5a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, i-Amyl, or 3,3,3-trifluoropropyl; preferably, n-Pr.

15 9. The compound according to any one of claims 1 to 7, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>5a</sub> is selected from phenyl, or benzyl.

20 10. The compound according to any one of claims 1 to 9, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>6a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, Et, n-Pr, i-Pr, or 3,3,3-trifluoropropyl; preferably, Me, Et, or 3,3,3-trifluoropropyl; preferably, Me.

25 11. The compound according to any one of claims 1 to 9, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>6a</sub> is selected from cyclopropyl, phenyl, or benzyl; preferably, cyclopropyl.

12. The compound according to any one of claims 1 to 11, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>9a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

30 13. The compound according to any one of claims 1 to 11, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>9a</sub> is H.

35 14. The compound according to any one of claims 1 to 13, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>10a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

15. The compound according to any one of claims 1 to 14, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>11a</sub> is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; preferably, C<sub>1-4</sub> alkyl, such as Me.

16. The compound according to any one of claims 1 to 15, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{11}$  is  $C(=M_{11})N(R_{12a})(R_{12b})$ .

17. The compound according to any one of claims 1 to 16, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{12a}$  and  $R_{12b}$  are independently selected from  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably,  $C_{1-4}$  alkyl, such as Me or Et; or  $R_{12a}$  and  $R_{12b}$  together with the atom to which they are connected form a 5- to 6-membered heterocyclyl, such as piperidinyl, pyrrolidine, or morpholine.

18. The compound according to any one of claims 1 to 17, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_3$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably, H, or  $C_{1-4}$  alkyl, such as Me; preferably,  $R_3$  has a S chirality.

19. The compound according to any one of claims 1 to 17, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_3$  is selected from  $-C_{0-6}$  alkylene- $C_{3-8}$  cycloalkyl,  $-C_{0-6}$  alkylene-4- to 8-membered heterocyclyl,  $-C_{0-6}$  alkylene- $C_{6-10}$  aryl, or  $-C_{0-6}$  alkylene-5- to 10-membered heteroaryl; or  $R_3$  and  $R_{3a}$  together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl.

20. The compound according to any one of claims 1 to 19, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_4$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl; preferably, H, or  $C_{1-4}$  alkyl, such as Me; or  $R_4$  and  $R_{4a}$  together with the atoms to which they are connected form a 4- to 6-membered heterocyclyl, which is optionally substituted with  $-OC_{1-4}$  alkyl, or  $-OC_{1-4}$  haloalkyl; preferably,  $R_4$  and  $R_{4a}$  together with the atoms to which they are connected form an azetidiny.

21. The compound according to any one of claims 1 to 20, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_6$  is selected from H, or  $C_{1-4}$  alkyl; preferably,  $R_6$  has a R chirality; preferably,  $R_6$  is H.

22. The compound according to any one of claims 1 to 21, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{7-1}$  and  $R_{7-3}$  are independently selected from H, F, or Cl; preferably, F.

23. The compound according to any one of claims 1 to 21, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{7-1}$  and  $R_{7-3}$  are independently selected from  $C_{1-6}$  alkoxy, or  $C_{1-6}$  haloalkoxy.

24. The compound according to any one of claims 1 to 23, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $R_{7-2}$  is selected from  $C_{1-4}$  haloalkyl, such as  $CF_3$ .

25. The compound according to any one of claims 1 to 24, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein Z is O, or S.

26. The compound according to any one of claims 1 to 24, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein Z is C(R<sub>8-1</sub>)(R<sub>8-2</sub>).

27. The compound according to any one of claims 1 to 26, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>8-1</sub> and R<sub>8-2</sub> are H.

28. The compound according to any one of claims 1 to 26, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>8-1</sub> and R<sub>8-2</sub> are F.

29. The compound according to any one of claims 1 to 26, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>8-1</sub> is OEt and R<sub>8-2</sub> is H; or R<sub>8-1</sub> is H and R<sub>8-2</sub> is OEt.

30. The compound according to any one of claims 1 to 29, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>9-1</sub> is selected from C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl, such as Me, or isopropyl; and R<sub>9-2</sub> is H, or C<sub>1-4</sub> alkyl, such as Me.

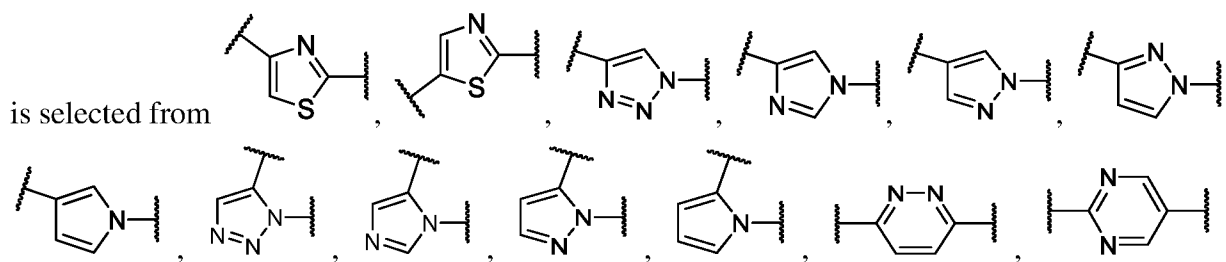
31. The compound according to any one of claims 1 to 30, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein R<sub>9-1</sub> and R<sub>9-2</sub> together with the atom to which they are connected form a C<sub>3-6</sub> cycloalkyl, such as cyclopropyl, cyclobutyl, or cyclopentyl; preferably, cyclobutyl; preferably, cyclopentyl.

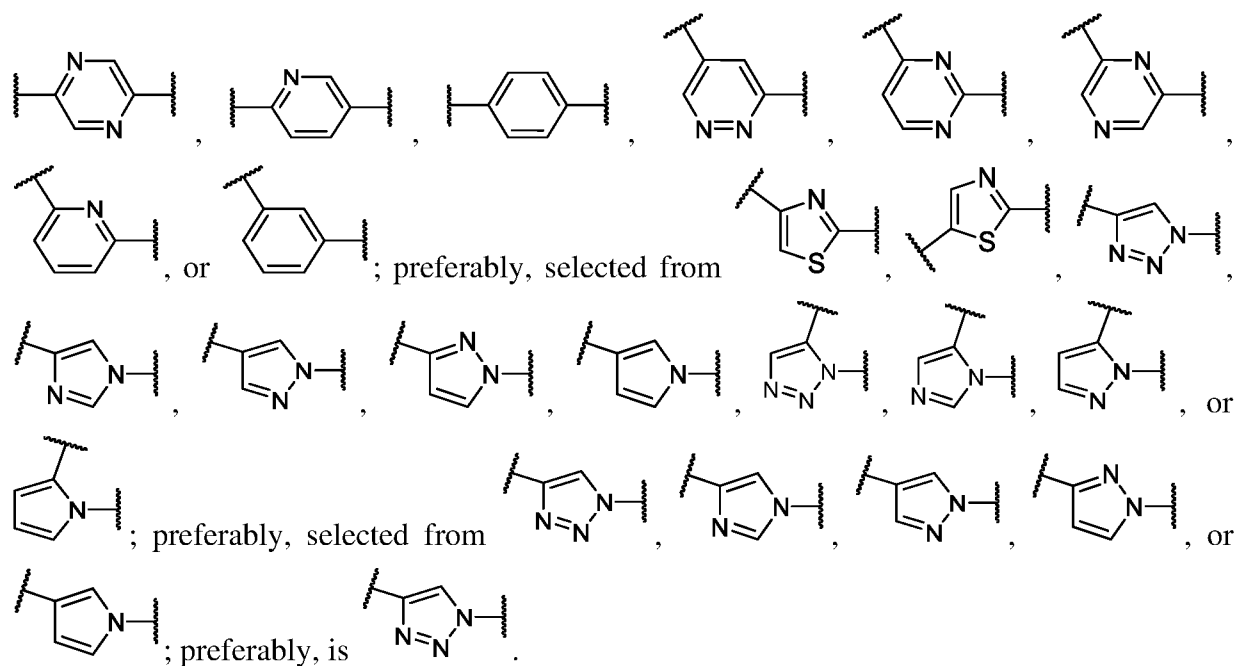
32. The compound according to any one of claims 1 to 31, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein m = 1.

33. The compound according to any one of claims 1 to 32, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein X is a bond, or S; preferably, X is a bond; preferably, X is S.

34. The compound according to any one of claims 1 to 32, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein X is -NHC(=O)-, or -NHC(=S)-.

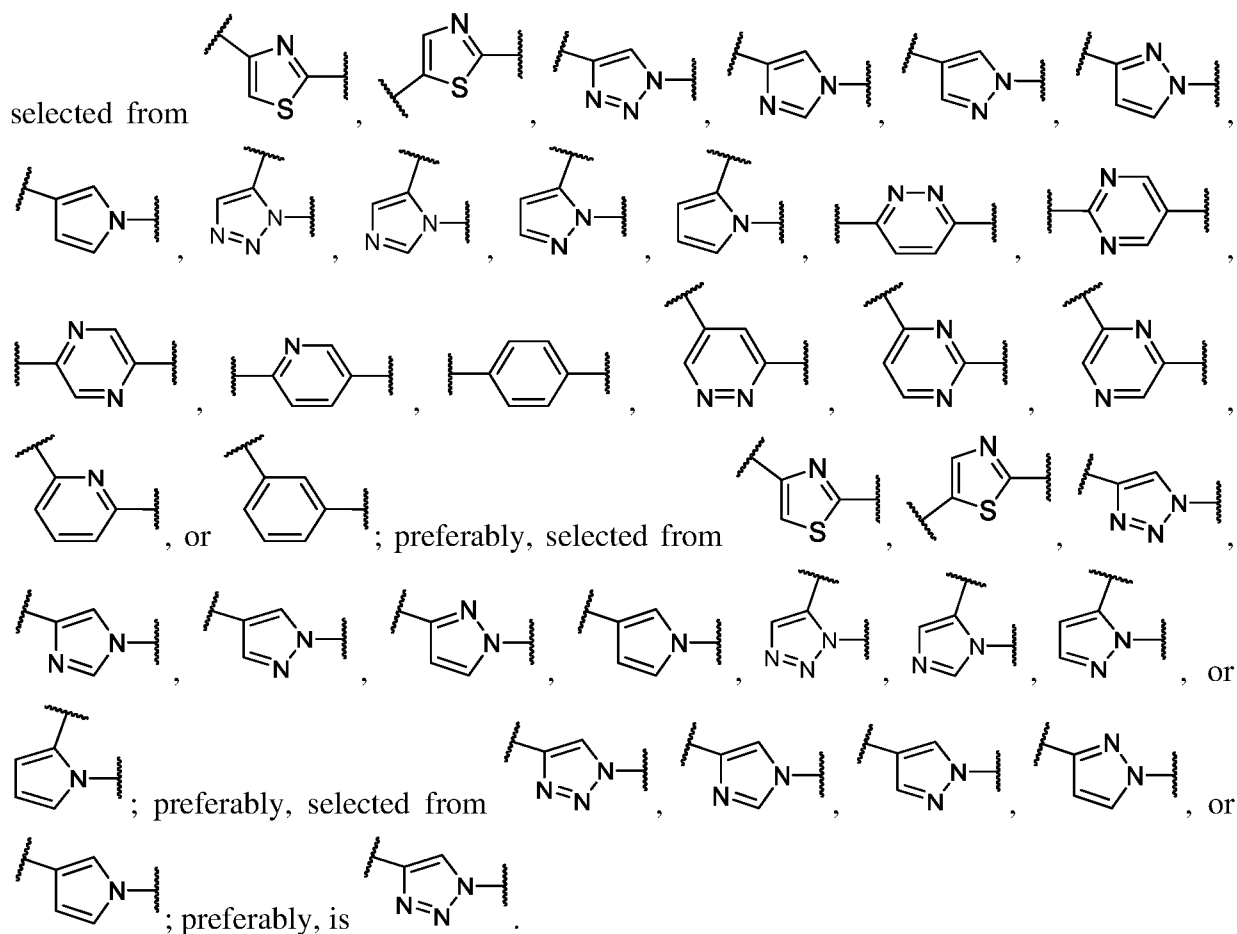
35. The compound according to any one of claims 1 to 32, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein X





36. The compound according to any one of claims 1 to 35, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L is -C(=O)NH-, or -C(=S)NH-; preferably, -C(=S)NH-.

37. The compound according to any one of claims 1 to 35, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L is



38. The compound according to any one of claims 1 to 35, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L is a bond, or S.

39. The compound according to any one of claims 1 to 38, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein Z<sub>1</sub>, Z<sub>2</sub>, Z<sub>3</sub>, and Z<sub>4</sub> are independently selected from H, or halo, such as F, Cl, Br; preferably, F.

40. The compound according to any one of claims 1 to 39, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein Z<sub>1</sub>, and Z<sub>4</sub> are Cl, and Z<sub>2</sub>, and Z<sub>3</sub> are F; or Z<sub>1</sub>, and Z<sub>4</sub> are F, and Z<sub>2</sub>, and Z<sub>3</sub> are Cl.

41. The compound according to any one of claims 1 to 39, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein p = 1.

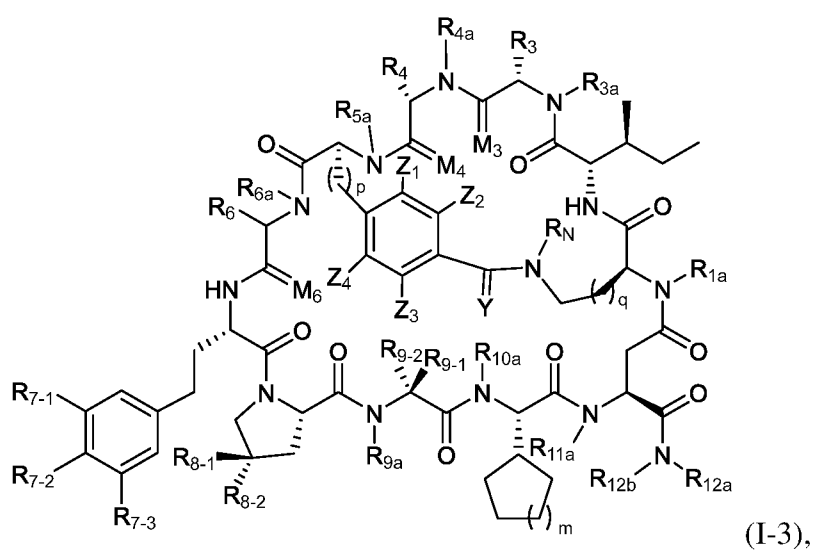
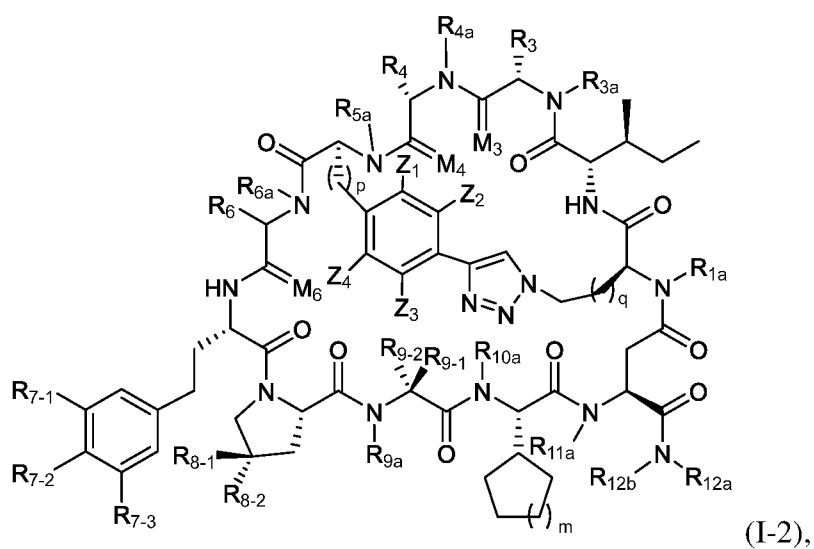
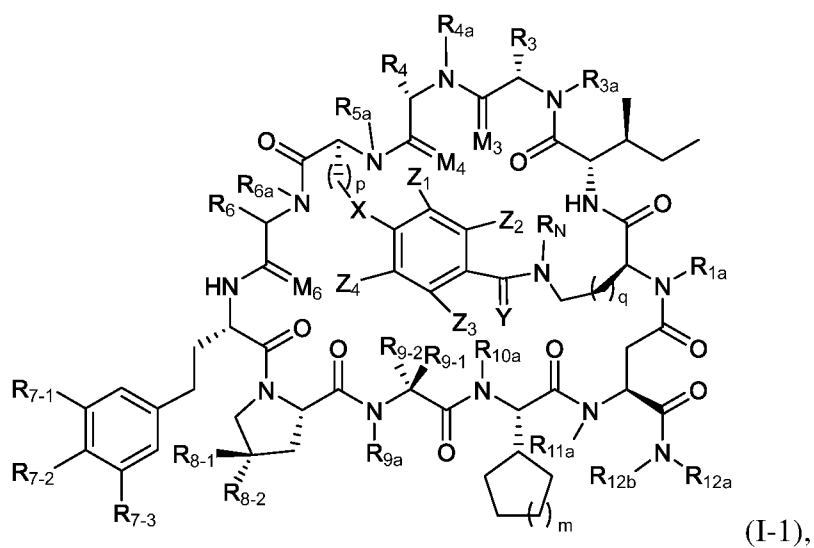
42. The compound according to any one of claims 1 to 41, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein q = 1, 2, or 3; preferably, q = 1.

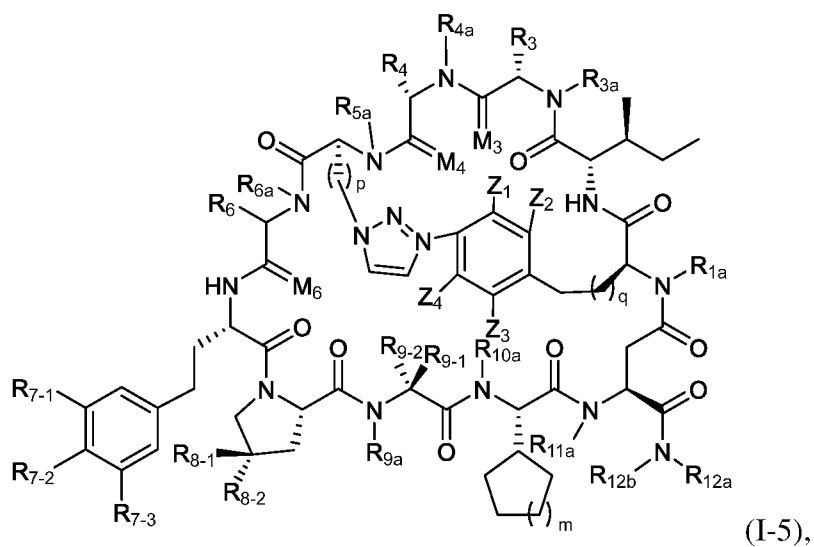
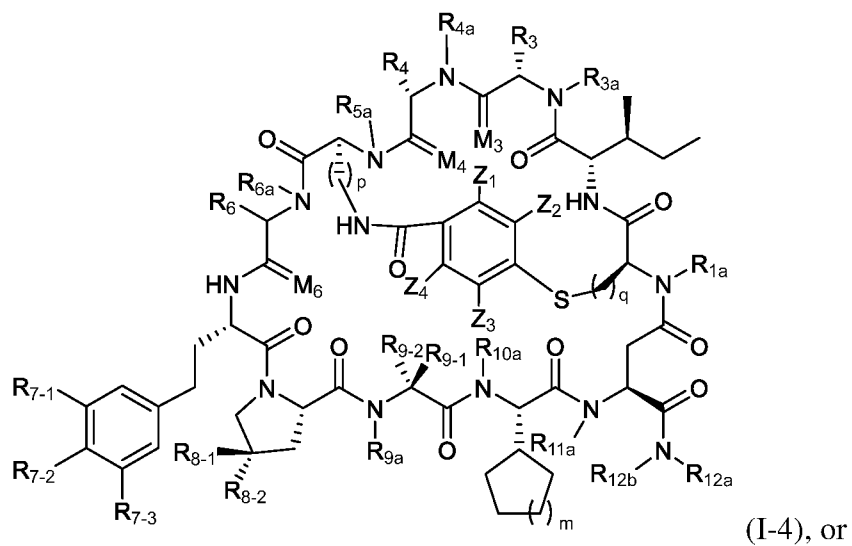
43. The compound according to any one of claims 1 to 42, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein M<sub>1</sub>, M<sub>2</sub>, M<sub>3</sub>, M<sub>4</sub>, M<sub>5</sub>, M<sub>6</sub>, M<sub>7</sub>, M<sub>8</sub>, M<sub>9</sub>, M<sub>10</sub>, M<sub>11</sub>, and M<sub>12</sub> are O.

44. The compound according to any one of claims 1 to 42, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein one or more M<sub>3</sub>, M<sub>4</sub>, M<sub>6</sub> are S.

45. The compound according to any one of claims 1 to 42, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein one or more M<sub>2</sub> is S.

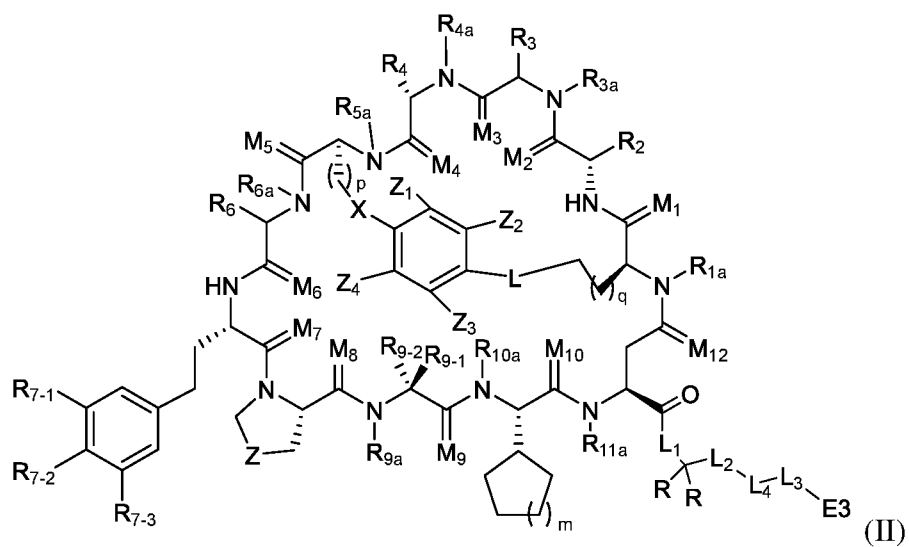
46. The compound according to any one of claims 1 to 45, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the compound has a formula selected from the following:





wherein the other variables are as defined in any one of claims 1 to 45.

47. A compound of Formula (II):



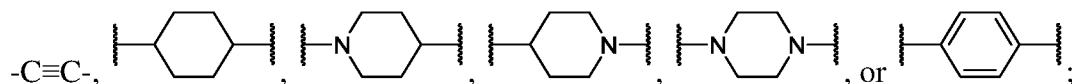
wherein,

L<sub>1</sub> is a divalent group selected from -amine group-, -diamine group-, -AA-amine group-, or -AA-diamine group-, wherein the amine group or diamine group has one amino linked to the carbonyl group on the cyclic peptide, and AA is an amino acid having the formula of -N(R<sub>13a</sub>)CH(R<sub>13</sub>)C(=Q)-, wherein Q is O, or S, R<sub>13a</sub> is selected from H, C<sub>1-6</sub> alkyl, or C<sub>1-6</sub> haloalkyl, and R<sub>13</sub> is the side chain of an amino acid optionally in combination with R<sub>13a</sub>, or R<sub>13</sub> and R<sub>13a</sub> together with the atoms to which they are connected form a 4- to 7-membered heterocyclyl;

L<sub>2</sub> is -C<sub>0-8</sub> alkylene-;

L<sub>3</sub> is -C<sub>0-8</sub> alkylene-;

L<sub>4</sub> is selected from bond, -CH<sub>2</sub>CH<sub>2</sub>-, -OCH<sub>2</sub>-, -CH<sub>2</sub>O-, -NHCH<sub>2</sub>-, -CH<sub>2</sub>NH-, -CH=CH-,



each R is selected from H, C<sub>1-4</sub> alkyl, or C<sub>1-4</sub> haloalkyl; or CRR forms C=O or C=S;

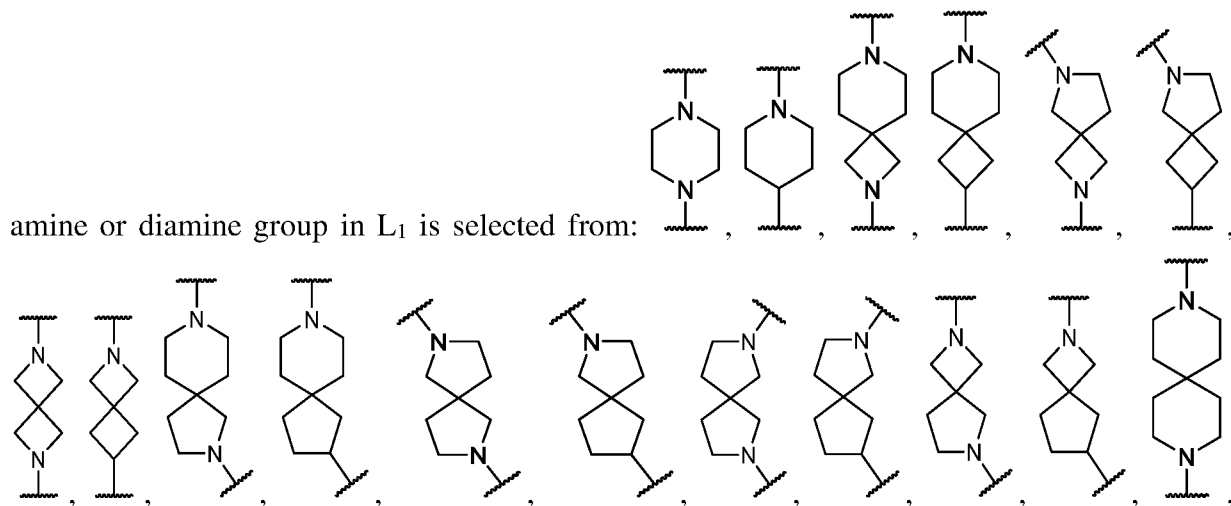
E3 is a ligand of E3 ligase, and

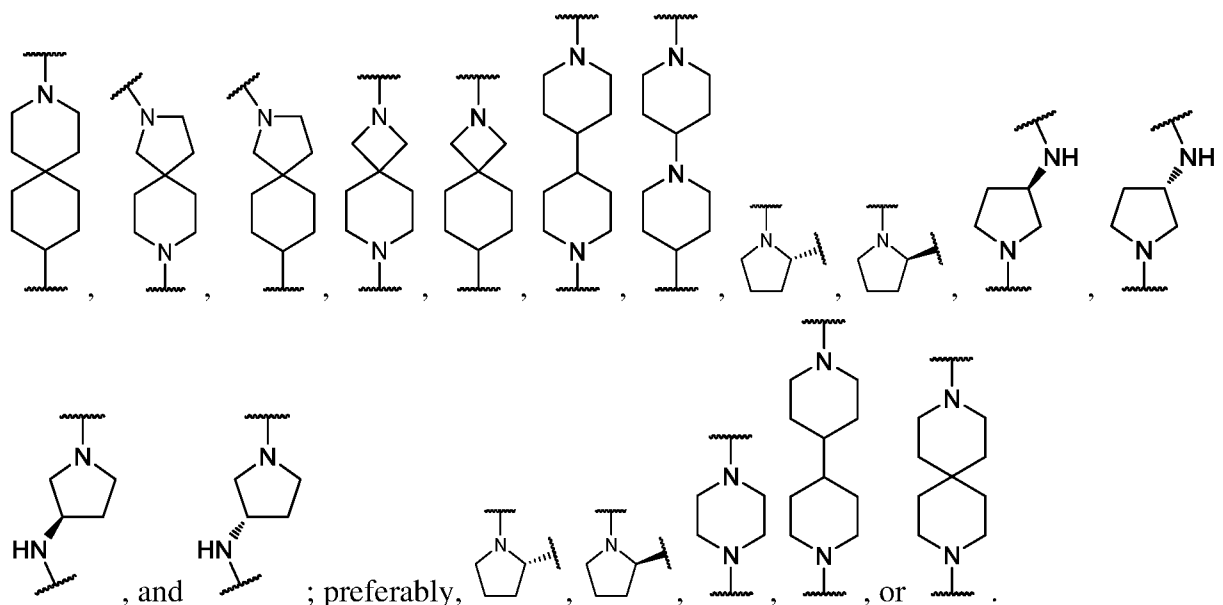
and variables are as defined in any one of claims 1 to 45.

48. The compound according to claim 47, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L<sub>1</sub> is -amine group-, or -diamine group-.

49. The compound according to claim 47 or 48, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein L<sub>1</sub> is -AA-amine group-, or -AA-diamine group-.

50. The compound according to any one of claims 47 to 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the





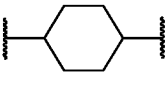
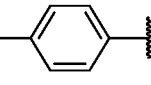
51. The compound according to any one of claims 47 to 50, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the amine group in  $L_1$  is  $-N(R_{N1})-$ , wherein  $R_{N1}$  is selected from H,  $C_{1-4}$  alkyl, or  $C_{1-4}$  haloalkyl.

52. The compound according to any one of claims 47 and 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein in  $-N(R_{13a})CH(R_{13})C(=Q)-$  of AA, Q is O;  $R_{13a}$  is selected from  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl, such as Me; and  $R_{13}$  is H, or Me; or  $R_{13}$  and  $R_{13a}$  together with the atoms to which they are connected form a 4- to 6-membered heterocyclyl, such as azetidiny, piperaziny, or pyrrolidinyl.

53. The compound according to any one of claims 47 and 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein in  $-N(R_{13a})CH(R_{13})C(=Q)-$  of AA, Q is O or S;  $R_{13a}$  is selected from  $C_{1-6}$  alkyl, or  $C_{1-6}$  haloalkyl, such as Me; and  $R_{13}$  is H.

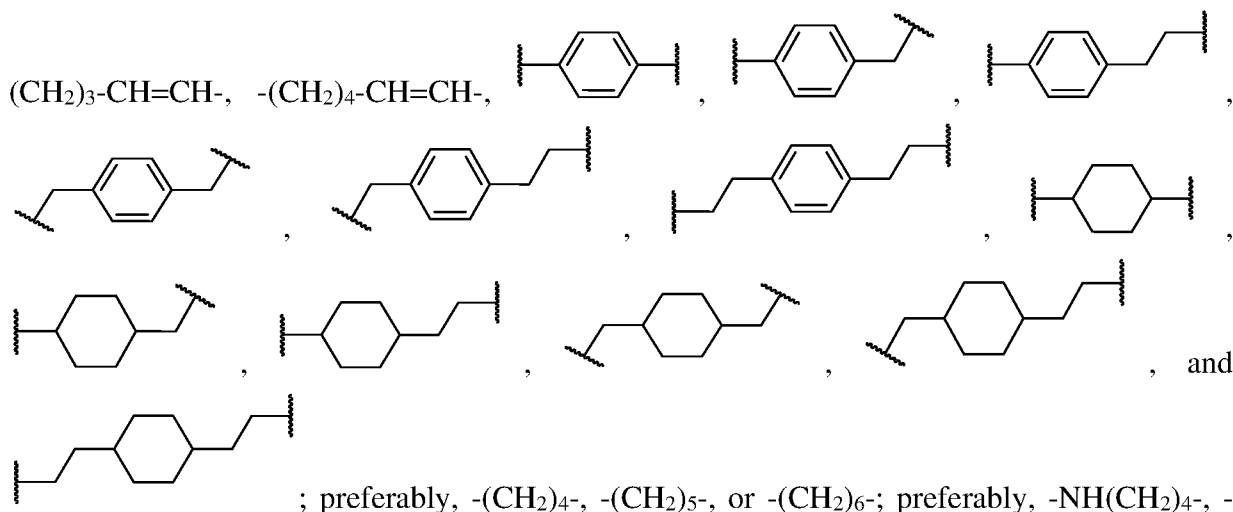
54. The compound according to any one of claims 47 and 49, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein in  $-N(R_{13a})CH(R_{13})C(=Q)-$  of AA, Q is O or S;  $R_{13}$  is the side chain of a natural amino acid (optionally in combination with  $R_{13a}$ ) selected from Gly, Ala, Val, Leu, Ile, Phe, Trp, Tyr, Asp, Asn, Glu, Lys, Gln, Met, Ser, Thr, Cys, Pro, His, and Arg; preferably Pro in combination with  $R_{13a}$ .

55. The compound according to any one of claims 47 to 54, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $L_4$

is selected from  $-CH_2CH_2-$ ,  $-NHCH_2-$ ,  $-CH=CH-$ , , or ; preferably,  $-CH_2CH_2-$  or  $-NHCH_2-$ .

56. The compound according to any one of claims 47 to 55, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein  $-L_2-$

L<sub>4</sub>-L<sub>3</sub>- is selected from bond,  $-(CH_2)_4-$ ,  $-(CH_2)_5-$ ,  $-(CH_2)_6-$ ,  $-NH(CH_2)_4-$ ,  $-NH(CH_2)_5-$ ,  $-NH(CH_2)_6-$ ,  $-CH=CH-(CH_2)_2-$ ,  $-CH=CH-(CH_2)_3-$ ,  $-CH=CH-(CH_2)_4-$ ,  $-(CH_2)_2-CH=CH-$ ,  $-(CH_2)_3-CH=CH-$ ,  $-(CH_2)_4-CH=CH-$ ,

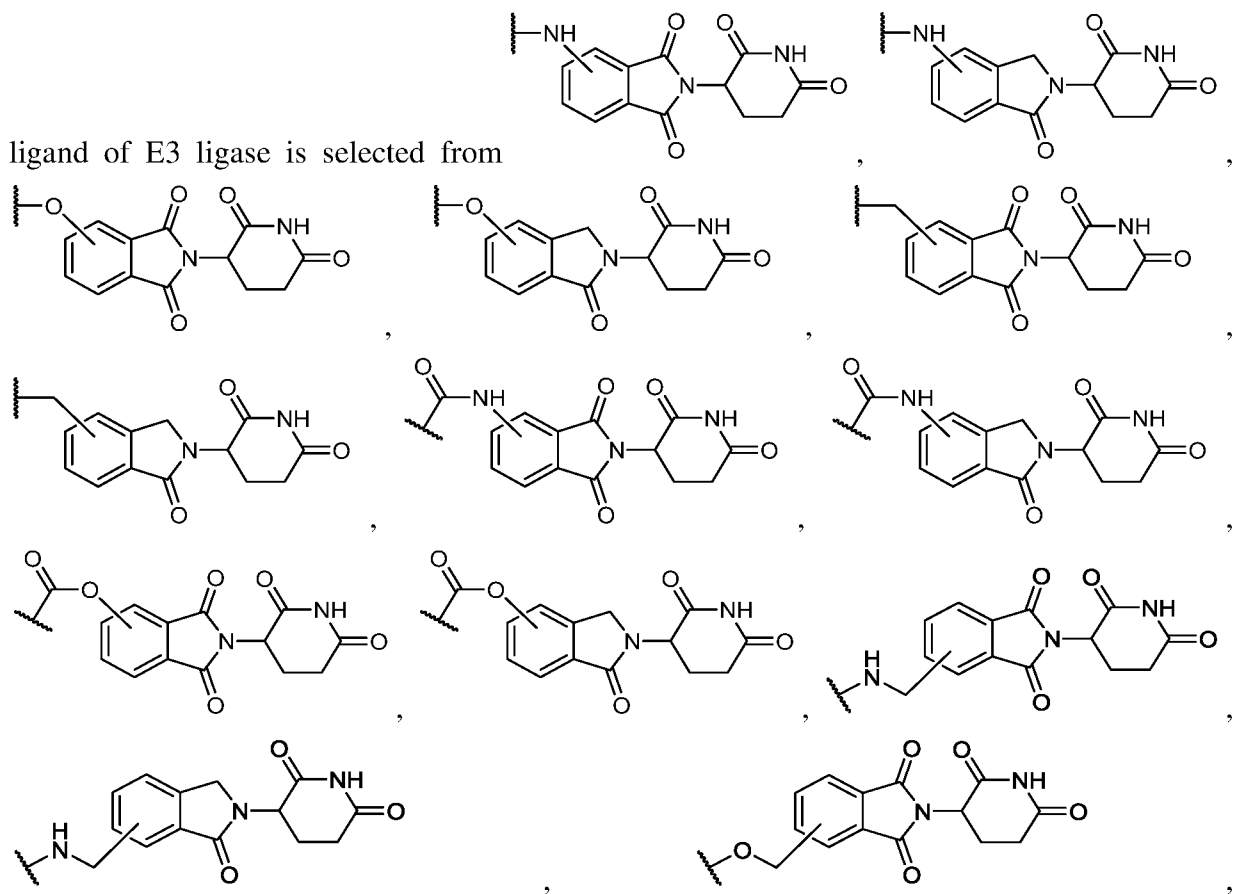


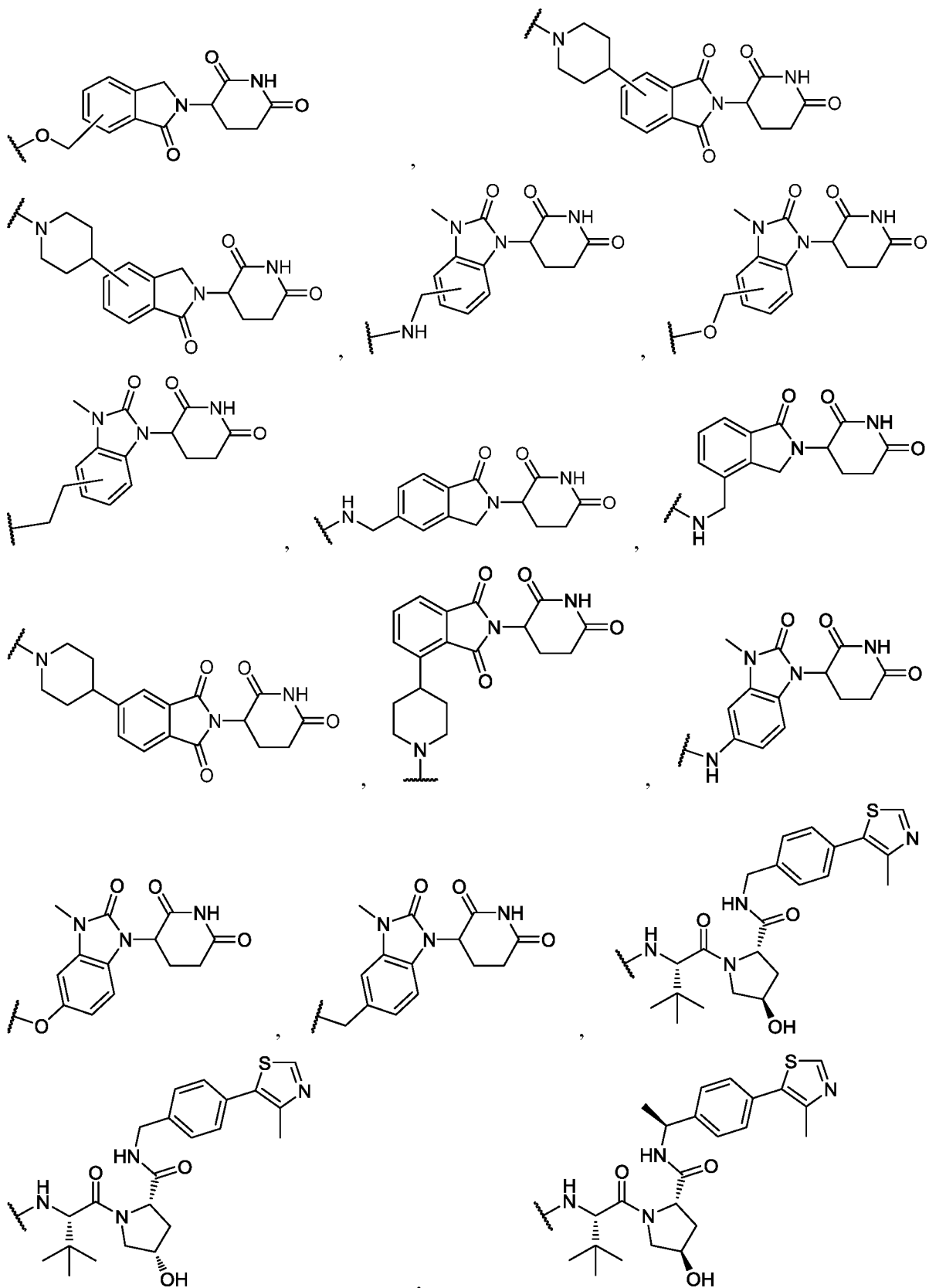
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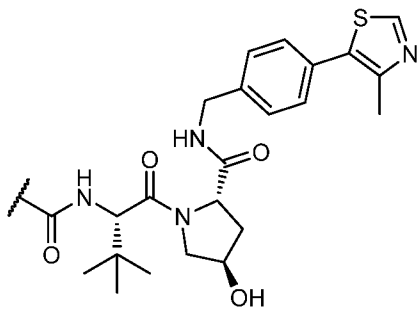
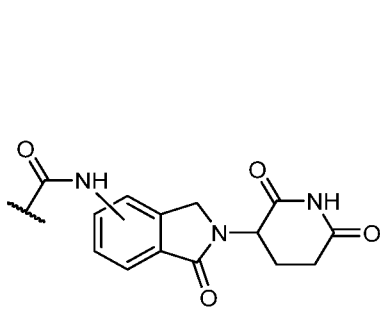
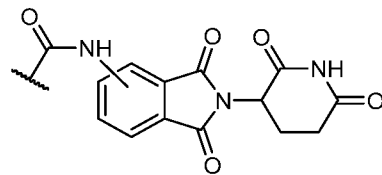
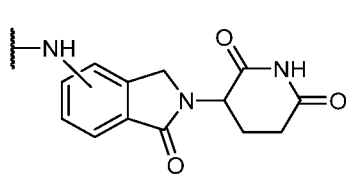
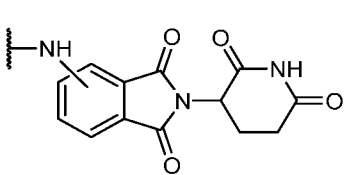
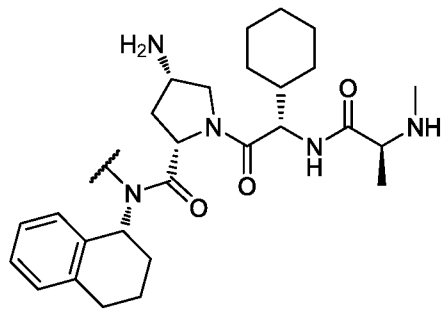
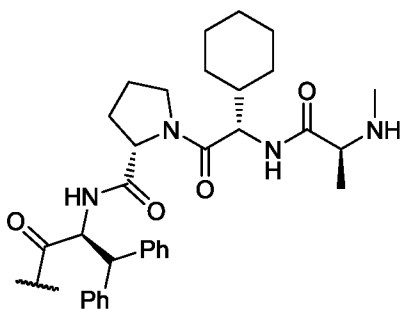
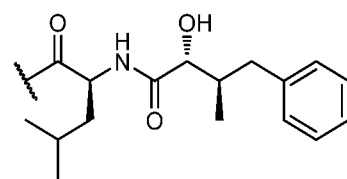
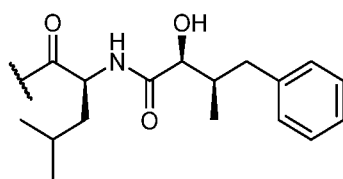
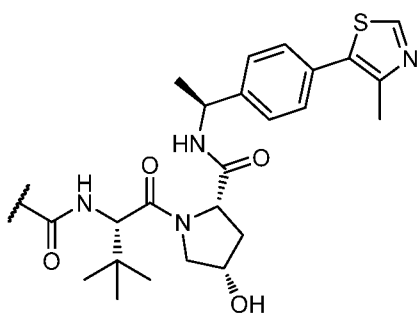
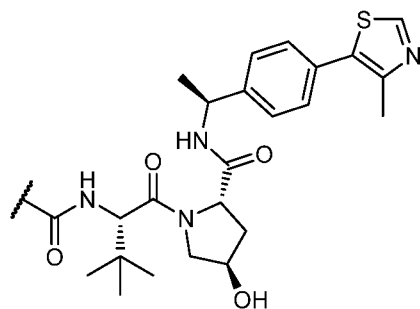
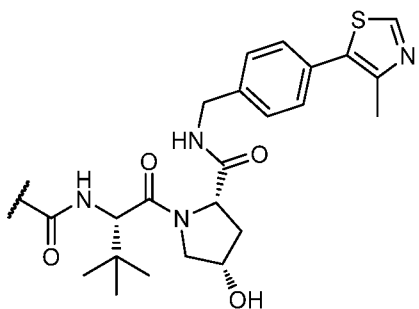
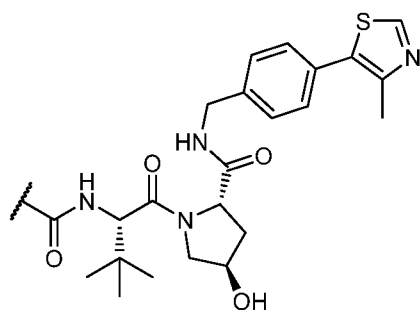
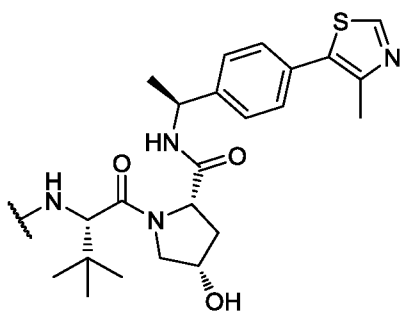
; preferably,  $-(CH_2)_4-$ ,  $-(CH_2)_5-$ , or  $-(CH_2)_6-$ ; preferably,  $-NH(CH_2)_4-$ ,  $-NH(CH_2)_5-$ ,  $-NH(CH_2)_6-$ ; preferably,  $-(CH_2)_5-$ , or  $-(CH_2)_6-$ ; preferably,  $-(CH_2)_6-$ .

57. The compound according to any one of claims 47 to 56, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the

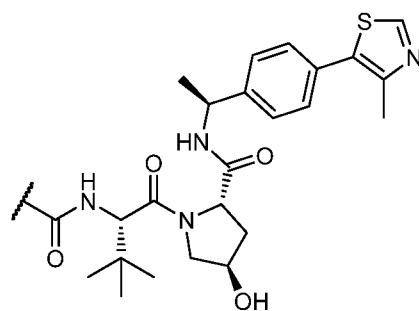
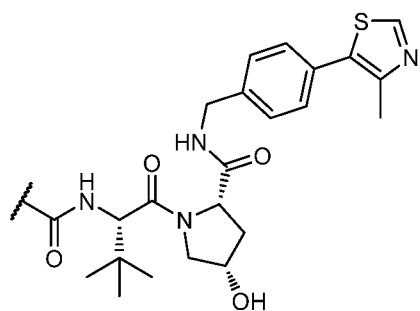
10 ligand of E3 ligase is selected from







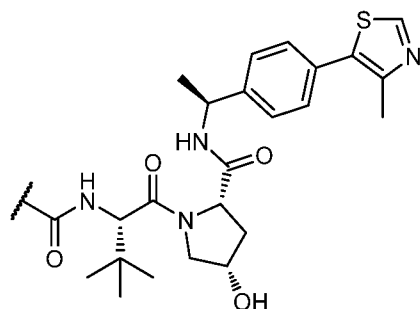
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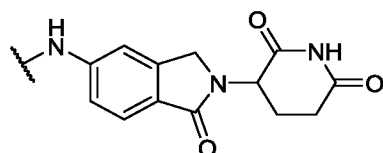
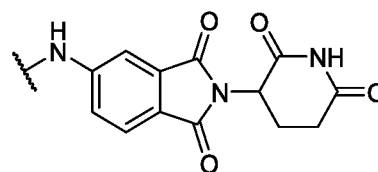
or



;

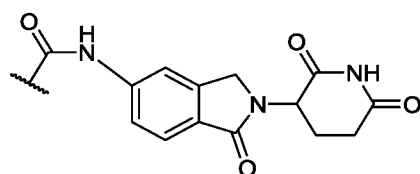
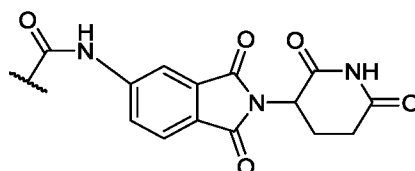
preferably,

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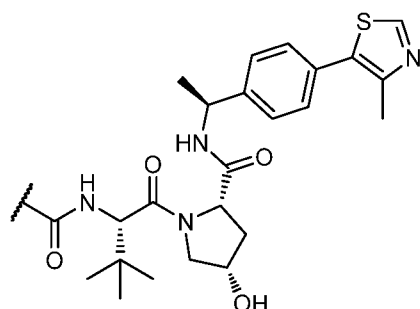
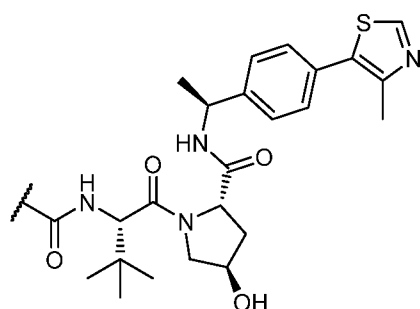
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or



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58. The compound according to any one of claims 47 to 57, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein CRR forms C=O.

59. The compound according to any one of claims 1 to 58, or a stereoisomer, a tautomer, a stable isotope, or a prodrug thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is selected from compounds 1001 to 1231, 2001 to 2173, and 3001 to 3003.

60. A pharmaceutical composition, comprising:

the compound, or the stereoisomer, the tautomer, the stable isotopic variant, or the prodrug thereof, or the pharmaceutical acceptable salt thereof according to any one of claims 1 to 59;

pharmaceutically acceptable excipient(s); and

optionally, one or more other therapeutic agents, such as RAS inhibitors, RTK inhibitors, EGFR inhibitors, immune checkpoint inhibitors for example PD-1 or PD-L1 inhibitors, and the like.

61. A kit, comprising:

a first container which contains the compound, or the stereoisomer, the tautomer, the stable isotopic variant, or the prodrug thereof, or the pharmaceutical acceptable salt thereof according to any one of claims 1 to 59; and

optionally, a second container which contains one or more other therapeutic agents; and

optionally, a third container which contains pharmaceutically acceptable excipient(s) for diluting or suspending the said compound and/or other therapeutic agent(s), such as RAS inhibitors, RTK inhibitors, EGFR inhibitors, immune checkpoint inhibitors for example PD-1 or PD-L1 inhibitors, and the like.

62. Use of a compound, or a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof, or a pharmaceutical acceptable salt thereof according to any one of claims 1 to 59, in the manufacture of a medicament for treating a disease mediated by Ras kinase.

63. The compound, or a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof, or a pharmaceutical acceptable salt thereof according to any one of claims 1 to 59, for use in treating a disease mediated by Ras kinase.

64. A method of treating a disease mediated by Ras kinase in a subject in need thereof, comprising administering to the subject a compound, or a stereoisomer, a tautomer, a stable isotopic variant, or a prodrug thereof, or a pharmaceutical acceptable salt thereof according to any one of claims 1 to 59.

65. The use of claim 62, or the compound for use of claim 63, or the method of claim 64, wherein the disease mediated by Ras kinase is cancer.

66. The use, or the compound for use, or the method of claim 65, wherein the cancer is solid cancer or blood cancer.

67. The use, or the compound for use, or the method of claim 65 or 66, wherein the cancer is selected from lung cancer, esophageal cancer, stomach cancer, colon cancer, uterine cancer, ovarian cancer, pancreatic cancer, bladder cancer, breast cancer, brain cancer, glioblastoma (GBM), kidney cancer, liver cancer, mesothelioma, prostate cancer, sarcoma, testicular cancer, thyroid cancer, uterine cancer, cholangiocarcinoma, thyroid cancer, skin cancer, uveal melanoma; leukemia, malignant lymphoma, and multiple myeloma.

68. The use, or the compound for use, or the method of claim 65 or 66, wherein the cancer is head and neck cancer, biliary tract cancer, Nasopharyngeal cancer, non-small cell lung cancer, small cell lung cancer, colon cancer, rectal cancer, endometrial cancer, cervical cancer, AML (acute myeloid leukemia), CML (chronic myelogenous leukemia), ALL (acute lymphocytic leukemia), CLL (chronic lymphocytic leukemia), Hodgkin lymphoma, and Non-Hodgkin lymphoma.

69. The use, or the compound for use, or the method of any one of claims 65 to 68, wherein the cancer is associated with Ras gene abnormalities or RAS pathway overactivation.

70. The use, or the compound for use, or the method of any one of claims 65 to 69, wherein the cancer is resistant to known KRAS inhibitors, such as sotorasib or adagrasib.

71. The use, or the compound for use, or the method of any one of claims 62 to 70, which comprises using other therapeutic agent(s), such as RAS inhibitors, RTK inhibitors, EGFR inhibitors, immune checkpoint inhibitors for example PD-1 or PD-L1 inhibitors, and the like.

## INTERNATIONAL SEARCH REPORT

International application No  
PCT/CN2025/075389

## A. CLASSIFICATION OF SUBJECT MATTER

INV. C07K7/54 A61K38/12  
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

## B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

C07K A61K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, CHEM ABS Data, BIOSIS, EMBASE, WPI Data

## C. DOCUMENTS CONSIDERED TO BE RELEVANT

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Further documents are listed in the continuation of Box C.



See patent family annex.

\* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&amp;" document member of the same patent family

Date of the actual completion of the international search

14 May 2025

Date of mailing of the international search report

14/07/2025

Name and mailing address of the ISA/

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Fax: (+31-70) 340-3016

Authorized officer

Pinheiro Vieira, E

## INTERNATIONAL SEARCH REPORT

International application No

PCT/CN2025/075389

| C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT |                                                                                                                                                                                                                                                                                                                              |                       |
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| Y                                                    | THI B. TRINH ET AL: "Discovery of a Direct Ras Inhibitor by Screening a Combinatorial Library of Cell-Permeable Bicyclic Peptides", ACS COMBINATORIAL SCIENCE, vol. 18, no. 1, 11 January 2016 (2016-01-11), pages 75-85, XP055431276, US<br>ISSN: 2156-8952, DOI: 10.1021/acscombsci.5b00164<br>the whole document<br>----- | 1-46,<br>59-71        |

# INTERNATIONAL SEARCH REPORT

International application No.  
PCT/CN2025/075389

## Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:  
because they relate to subject matter not required to be searched by this Authority, namely:
  
2. ☐ Claims Nos.:  
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:
  
3. ☐ Claims Nos.:  
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

## Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

**see additional sheet**

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.
2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.
3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:
4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims;; it is covered by claims Nos.:  
**1-46, 59-71 (all partially)**

### Remark on Protest

- ☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.
- ☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.
- ☐ No protest accompanied the payment of additional search fees.

## FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-46, 59-71(all partially)

Compounds of formula (I-1) and their use in the treatment of disorders mediated by ras, e.g. cancer

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2-5. claims: 1-46, 59-71(all partially)

Compounds of formula (I-2) or (I-3) or (I-4) or (I-5) and their use in the treatment of disorders mediated by ras, e.g. cancer, with each markush formula representing an invention.

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6. claims: 47-58(completely); 59-71(partially)

Compounds of formula (II) and their use in the treatment of disorders mediated by ras, e.g. cancer

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7. claims: 1-45, 60-71(all partially)

Compounds of formula (A) or (I) not covered by any of the inventions 1-6 and their use in treatment.

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# INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/CN2025/075389

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# INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/CN2025/075389

| Patent document<br>cited in search report | Publication<br>date | Patent family<br>member(s) | Publication<br>date |
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| <div></div>                               |                     |                            |                     |